CH151

Advanced General Chemistry I

CH101 GENERAL CHEMISTRY I

ABET eCOURSEBOOK AY2021-2026

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DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
600 THAYER ROAD
WEST POINT, NEW YORK 10996

REPLY TO
ATTENTION OF

MADN-CBE 27 January 2026

MEMORANDUM FOR ALL CADETS TAKING ADVANCED GENERAL CHEMISTRY

SUBJECT: General Orientation and Standing Instructions for Students

SECTION I. COURSE DESCRIPTION

Foreword: This semester, instruction remained fully in person. Consistent with 25-1, two sections per lab section were executed, which requires a significantly reduced instructor burden. Dr. Kubiak, Mr. Priedemann, or the second CH151 instructor, always remained present in the lab to satisfy ACS requirements and ensure student safety. Consolidating labs created increased time for other course and department duties. Items were turned in via hard copy only. Lab notebooks remained due at the end of lab blocks or 24-48 hours later, at instructor discretion, which resulted in cadets being as prepared as in typical semesters. Enrichments were eliminated, with CH101 Lab 8 (Lab 10 in this course) and an additional lesson on Lewis Dot Structures taking their place. The elimination of enrichment periods was done to increase the hands-on laboratory time as well as the time spent by students learning and mastering introductory chemistry concepts. The exposure to department research and advanced scientific topics was integrated into the classroom instructional time by each CH151 instructor. In 26-1 CH151 deviated from previous years by using CH101 end of hour quizzes and the bulk of the CH101 WPR to assess and benchmark student learning throughout the semester. The WPRs were augmented with an additional page of questions going deeper into an area unassessed by the CH101 draft of the WPR. For block 1 the extra WPR page focused on assessing student's understanding of IMFs, and block 2 went deeper on acid-base neutralizations incorporating concepts from limiting reactants as well. The common TEE with 101 was retained. To reduce instructor grading time, the labs were taught using CH101-style lab reports which offered specific questions to generate student thinking and problem solving. We eliminated the full lab report requirement in CH151, recognizing that all students in CH102 will be taught how to build a lab report. Instructor grades were assigned based on daily in class quizzes, each containing two questions. The first quiz question was an in-depth question normally taken from the previous night's homework and modified with new numbers. The second quiz question was always a relatively straight forward concept question from the upcoming lessons reading, designed to be an easy points scoring opportunity for any student who did the bare minimum to prepare for class. These quizzes were scored out of 25 points per quiz, with a total of 30 points available on each quiz to allow for students who are slower at adapting to the rigors of college to “come back” from a slow start on instructor quizzes. The course was fully in Canvas this AY. 4 sections remains the right number for CH151 to ensure a high performing cohort that can handle the more rapid pace relative to CH101.

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY26-1 = 70 (4 sections; 2x MAJ Lowell, 2x LTC Toole)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: Same as previous iterations.

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74
AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60
AY20-1: 3.63
AY21-1: 3.81
AY22-1: 3.90
AY23-1: 3.82
AY24-1: 3.84
AY25-1: 3.97
AY26-1: 3.89

AY26 Final Grades	AY25 Final Grades

A+	12	23
A	35	26
A-	14	5
B+	1	4
B	1	0
B-	2	1
C+	0	2
C	0	0
C-	0	0

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%
AY24-1: 91.3%
AY25-1: 90.4%
AY26-1: 93.5%

7. Course Processes:

a. Textbooks
   1) <u>Chemistry: Structure and Dynamics</u>, Tro, 3rd Ed.

b. Course Content
   1) Atomic Structure
   2) Structure, Bonding, and States of Matter
   3) Stoichiometry, Chemical Reactions and Gases
   4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

Month	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
August	3	4	5	6	7	8	9
August	10	11 March Back Reorgy Week	12 Reorgy Week	13 Reorgy Week	14 Reorgy Week	15 Reorgy Week	16
August	17	18 1-1	19 2-1	20 1-2	21 2-2 R Lab 1	22 2-3 DROP S Lab 1	23
August	24	25 1-3	26 2-4	27 1-4	28 2-5 EOH 1 Lab MU	29 1-5	30
September	31	1 Labor Day	2 2-6 R Lab 2	3 1-6	4 2-7 S Lab 2	5 1-7	6
September	7	8 1-8	9 2-8	10 1-9	11 2-9 EOH 2 Lab 3	12 2-10 DROP	13
September	14	15 1-10	16 2-11 S Lab 3	17 1-11	18 2-12	19 1-12	20
September	21	22 1-13	23 2-13 R Lab 4	24 1-14	25 2-14 S Lab 4	26 2-15 DROP	27
October	28	29 1-15	30 2-16	1 1-16	2 2-17 R Lab 5	3 1-17	4
October	5	6 1-18	7 2-18 S Lab 5	8 1-19	9 2-19 DROP	10 2-20 WPR 1 DEANS HOUR	11
October	12	13 Columbus Day	14 1-20	15 2-21 R Lab 6	16 1-21	17 2-22 S Lab 6	18
October	19	20 1-22	21 2-23 EOH 3	22 1-23	23 2-24 R Lab 7	24 1-24	25
October	26	27 1-25	28 2-25 S Lab 7	29 1-26	30 2-26 DROP	31 2-27 EOH 4	1
November	2	3 1-27	4 2-28 R Lab 8	5 1-28	6 2-29 S Lab 8	7 1-29	8
November	9	10	11 Veterans Day	12 2-30	13 1-30	14 2-31 R Lab 9	15
November	16	17 1-31	18 2-32 DROP S Lab 9	19 1-32	20 2-33 DROP	21 2-34 WPR 2 DEANS HOUR	22
November	23	24 1-33	25 2-35 Lab MU	26 1-34	27 Thanksgiving	28 No Class	29
December	30	1 No Class	2 1-35	3 2-36 EOH 5 R Lab 10	4 1-36	5 2-37 DROP S Lab 10	6
December	7	8 1-37	9 2-38	10 1-38	11 2-39 DROP Lab MU	12 1-39	13
December	14	15 1-40	16 TEE	17 TEE	18 TEE	19 TEE	20 TEE
December	21	22	23	24	25	26	27

CH151 BUFF Card

	Block #1: STRUCTURE, BONDING, & STATES OF MATTER
1	Chemistry Basics
2	The Atom, Atomic Number, and Atomic Mass
4	Electromagnetic Radiation/Spectroscopy
5	Quantum Mechanical Model of the Atom & **EOH 1**
6	Electron Configuration and Spin
7	Atomic Forces
8	Periodic Trends
9	Bonding Basics & **EOH 2**
11	Electronegativity, Partial Charge & Nomenclature
12	Lewis Structures and Covalent Bonds
13	Lewis Structures and Covalent Bonds II
14	Molecular Shape & Hybridization
16	Intermolecular Forces
17	Empirical Formula & Combustion Analysis
18	Chemical Reactions and Stoichiometry
20	WPR 1 (Dean's Hour)
	Block #2: STOICHIOMETRY/CHEM RXNS/THERMO
21	Limiting Reactant
22	Solution Chemistry
23	Solubility and Precipitation Reactions & **EOH 3**
24	Acid/Base Neutralization Reactions
25	Oxidation-Reduction Reactions
27	Applications of Chemical Reactions & **EOH 4**
28	Heat and Calorimetry
29	Enthalpy of Reaction
30	Heating/Cooling Curves & Phase Diagrams
31	Ideal Gas Law
32	Backup Lesson Block/ DROP
34	WPR 2 (Dean's Hour)
35	Colligative properties
36	IMF II /Metallic Bonding/Solids & **EOH 5**
38	Fundamental Chemistry of Explosives

Lab #1: Lab techniques and measuring instruments
Lab #2: Spectrophotometry
Lab #3: DROP
Lab #4: DROP
Lab #5: IMFs
Lab #6: Stoichiometry*
Lab #7: Lab Practical
Lab #8: Heat & Calorimetry
Lab #9: Stoichiometry & Gas Laws
Lab #10: Colligative Properties

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	600	1200	24.0%
5	Quizzes	200	1000	20.0%
8	Labs	125	1000	20.0%
2	Instructor Grade	250	500	10.0%
		Total	**5000**	**100%**

Points were changed from previous AY. WPR1 and 2 were common exams this year with CH101, with an extra page worth 100 points added to each. A shared TEE with 101 was also used, with small modifications from the previous years TEE (modifications focusing on presentation and wording rather than assessed material) and cadets performed at a similar level to 25-1. EOH Quizzes that matched the CH101 material (with an exception of slight differences for EOH 5) were used to assess learning between WPRs. Homework completion was not checked directly, but assessed through in class quizzes and captured in the instructor grade worth 10% of the total course grade.

e. Areas of Special Emphasis:

1) Encouraged curiosity and engagement with chemistry concepts with an eye to recruiting future CBSE majors.

2) Used CH151 specific lab program to reinforce concepts learned in class using lab techniques that were more advanced than those used in CH101.

3) Changed graded event structure to ensure Cadets were challenged but still baselined against the CH101 population.

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.

b. Explain and apply foundational chemistry concepts.

c. Develop practices for learning independently (intellectual self-reliance).

d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158

c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits

d. Technician Support: Lab support team provided excellent lab preparation and coverage, as always. Having them present in the lab was excellent, as Cadets gained exposure to additional members of the department and it reduced instructor labs from 2 sessions to 1 session.

e. Supplies: LoggerPro was used in both classrooms and labs for analytical inquiry.

f. Additional Facilities: None

g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Sustained the removal of Mastering Chemistry homework as a requirement in AY22, which was effective.

b. Further increased the collective grade value of quizzes (EOH and Instructor) to give them a higher degree of importance.

c. Used common WPRs and EOHs relative to CH101, but with supplemental questions.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: Enrollment was approximately the same as in AY25, with approximately 17.5 students per section for 4 sections. Recommend we sustain only 4 sections of CH151 in future. The course had strong performers with only a few that struggled, whereas more struggle with the enhanced pace if we open more sections.

3. Course Objectives:

a. Coverage:

1) Atomic Structure: EOH1, Lab 2, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 5, 10, EOH5
4) Stoichiometry, Chemical Reactions: Labs 6, 7, EOH3, EOH5 WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	EOH 1, 2
	Lab 2
	WPR I, TEE
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	EOH 2, 3
	WPR I, TEE
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	EOH 5, Labs 5, 10
	WPR I, TEE
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 6, 7
	EOH 3, 5
	WPR II, TEE
Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 8, 9
	EOH4 WPR II, TEE

4. Course QPA: The QPA was consistent with previous terms.

6. TEE Grade: The TEE executed its typical format this year, with the ACS exam (70 questions over 110-minutes) the at least 100-minute free response part II portion of the test. TEE grades were similar to past years.

AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%
AY24-1: 91.3%
AY25-1: 90.4%
AY26-1: 93.5%

7. Course Processes
a. Textbook: The Tro text is appropriate for the course.
b. Course Content: Lesson/lab sequence was consistent with previous AY.
c. Lessons and labs: Lessons are appropriate. Labs reinforce learning well.
d. Summary of Graded Requirements: Graded events were changed as described above, and performance remained high.

8. Resources and Laboratories
a. Laboratories: No issues.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: Excellent support, as always.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course by offering 4 sections.

3. Course Objectives:
a. Coverage: No recommended changes.
b. Performance: No recommended changes. Cadet performed at a high level.

4. Survey Questions:
a. Survey Results: Do not change them from year to year to facilitate comparisons.

5. Course QPA: The QPA was up consistent with the previous academic year.

6. TEE Grade: Sustain current practices.

7. Course Process

a. Textbook: The Tro text is the appropriate text for this course and should be the only text.

b. Course Content: No changes.

c. List of lessons and labs: Consider using Lab 6 (Stoichiometry) as the introduction to the Cu-Zn replacement reaction and the HCl-Zn Gas evolution reaction, which can then be revisited over the next few labs. Lab 6 could use the Fisher Burner to dry the sample with a rinsing process to remove the excess HCl, Lab 7 could then incorporate an acid base neutralization to eliminate excess acid prior to using vacuum filtration to capture and weigh the solid copper. Finally, Lab 9 could use the evolution of hydrogen gas in a displacement apparatus to determine the yield. All three methods can then be used to calculate the copper content in the original mixture and then Cadets can compare methodologies.

d. Summary of Graded Requirements: Continue using same EOHs and WPRs to reinforce CH151 learning. Add supplemental questions to as desired.

e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes

a. Coverage: No recommended changes.

b. Performance: No recommended changes.

9. Resources and Laboratories

a. Laboratories: No issues. Continue to use BH158.

b. Computer Labs: N/A

c. Physical Models & Demos: No issues.

d. Technician Support: No issues.

e. Supplies: No issues.

f. Additional Facilities: N/A

g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

This information can be found in the AY26-1 CH151 folder on Sharepoint.

SAMUEL LOWELL
Major, Senior Instructor
CH151 Course Director, Department of Chemical and Biological Sciences and Engineering

TAB B AY2025 COURSE ASSESSMENT

SECTION I. COURSE DESCRIPTION

Foreword: This semester, instruction remained fully in person. Two sections per lab section was executed, different from AY21-1, AY22-1, and AY23-1, which requires reduced instructor burden significantly. Mr. Treus remained present in the lab at all times to satisfy ACS requirements and ensure student safety, which he really enjoyed. This is a great option if resourcing allows it. Items were turned in via hard copy only. Lab notebooks remained due at the end of lab blocks or 24-48 hours later, at instructor discretion, which resulted in cadets being as prepared as in typical semesters. Enrichments were sustained at 2 sessions, including a research overview enrichment executed during lab 2. These enrichments remain a big sustain, as are “Science of the Day” presentations. We executed both separate and common graded events (unique BOHs, common WPRs) for CH151 to make it more challenging and benchmark to 101, but in the end, final grades were similar to past AYs. The common TEE with 101 was retained. We attempted to reduce instructor grading, as it was high last term, by executing one full lab report instead of two reports covering half a report each. We did not allow groups for the report, which was good for cadet learning and accountability, but bad for instructor time. We tried pre-lab quizzes in lieu of pre-lab worksheets, but they took too much time to execute and were discontinued after lab 3. We also replaced the 4 EOH quizzes with 10 BOH CD quizzes to ensure cadets kept up with homework/reading, and continued to spot-check their homework notebooks for their homework grade (still no Mastering Chemistry). Homework grade was reduced to 3% of the course grade to make room for this. Outcomes were very good, as indicated by TEE performance and final grades. The course was fully in Canvas this AY. 4 sections remains the right number for CH151 to ensure a high performing cohort that can handle the more rapid pace relative to CH101.

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY25-1 = 66 (4 sections; 2x Limbocker, 2x Burpo)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: Same as previous iterations.

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74
AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60
AY20-1: 3.63
AY21-1: 3.81
AY22-1: 3.90
AY23-1: 3.82
AY24-1: 3.84 (see below for breakdown)
AY25-1: 3.97

	AY24 Final Grades	AY25 Final Grades
A+	21	23
A	16	26
A-	20	5
B+	5	4
B	2	0
B-	3	1
C+	1	2
C	0	0.0
C-	0	0.0

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)

CH151 AY25 Course Assessment Package

AY23-1: 92.3%
AY24-1: 91.3%
AY25-1: 90.4%

7. Course Processes:

a. Textbooks

1) <u>Chemistry: Structure and Dynamics</u>, Tro, 3rd Ed.

b. Course Content

1) Atomic Structure
2) Structure, Bonding, and States of Matter
3) Stoichiometry, Chemical Reactions and Gases
4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

Month	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
August	7	8 REORGY Week	9 REORGY Week	10 REORGY Week	11 REORGY Week	12 REORGY Week	13 A-Day
August	14 WK 1	15 1-1	16 SD	17 1 Day Drop	18 2-1 R/U Lab 1	19 1-2	20 A/C
August	21 WK 2	22 1 Day Drop	23 2-2 S/V Lab 1	24 1-3	25 2-3 T/W Lab 1	26 1 Day Drop	27 Ring Wknd
August	28 WK 3	29 1-4	30 2-4 R/U Lab 2	31 SD	1 1-5	2 2-5 S/V Lab 2	3
September	4 WK 4	5 Labor Day (no classes)	6 2-6 T/W Lab 2	7 1-6	8 2-7 R/U Lab 3	9 1-7	10
September	11 WK 5	12 1-8	13 2-8 S/V Lab 3	14 SD	15 1-9	16 2-9 Lab drop	17
September	18 WK 6	19 1-10	20 2-10 Lab drop	21 1-11	22 2-11 Lab drop	23 1 Day Drop	24
September	25 WK 7	26 1-12	27 2-12 T/W Lab 3	28 SD	29 1-13	30 2-13	1
October	2 WK 8	3 1 Day Drop	4 2-14 Lab drop	5 1-14	6 2-15 Lab drop WPR 1	7 1-15 WPR 1	8
October	9 WK 9	10 Columbus Day (no classes)	11 2-16 R/U Lab 4	12 1-16	13 2-17 S/V Lab 4	14 1-17	15
October	16 WK 10	17 1-18	18 2-18 T/W Lab 4	19 SD	20 1-19	21 2-19 R/U Lab 5	22
October	23 WK 11	24 1-20	25 2-20 S/V Lab 5	26 1-21	27 2-21 T/W Lab 5	28 1 Day Drop	29
November	30 WK 12	31 1-22	1 2-22 R/U Lab 6	2 SD	3 1-23	4 2-23 S/V Lab 6	5
November	6 WK 13	7 1-24	8 2-24 T/W Lab 6	9 1-25	10 2-25 R/U Lab 7	11 Veterans Day No classes	12
November	13 WK 14	14 1-26	15 2-26 S/V Lab 7	16 SD	17 1-27	18 2-27 T/W Lab 7	19
November	20 WK 15	21 1 Day Drop	22 2-28 R/U Lab 8	23 No Classes	24 Thanksgiving	25 No Classes	26
November	27 WK 16	28 1-28	29 2-29 S/V Lab 8	30 1-29	1 2-30 T/W Lab 8	2 1 Day Drop	3
December	4	5 1-30	6 SD WPR 2 MU	7 1 Day Drop	8 SD	9 1 Day Drop	10 BEAT NAVY
December	11	12 SD TEE	13 TEE	14 TEE	15 TEE	16 TEE	17 TEE

BUFF Card		
Block #1: STRUCTURE, BONDING, & STATES OF MATTER		
1	Chemistry Basics	
2	The Atom, Atomic Number, and Atomic Mass	
3	Electromagnetic Radiation/Spectroscopy	Announced Quiz
4	Quantum Mechanical Model of the Atom	
5	Electron Configuration and Spin	POP Quiz
6	Atomic Forces	
7	Periodic Trends	
8	Bonding Basics	POP Quiz
9	Electronegativity and Partial Charge	
10	Empirical Formula & Combustion Analysis	
11	Lewis Structures and Covalent Bonds	POP Quiz
12	Molecular Shape & Hybridization	
13	Intermolecular Forces	
14	Nomenclature	POP Quiz
15	WPR 1	
Block #2: STOICHIOMETRY/CHEM RXNS/THERMO		
16	Chemical Reactions and Stoichiometry, and Limiting Reactant	
17	Solution Chemistry	POP Quiz
18	Solubility and Precipitation Reactions	
19	Acid/Base Neutralization Reactions	POP Quiz
20	Oxidation-Reduction Reactions	
21	Applications of Chemical Reactions	POP Quiz
22	Heat and Calorimetry	
23	Enthalpy of Reaction	
24	Heating/Cooling Curves & Phase Diagrams	POP Quiz
25	Ideal Gas Law	
26	Enrichment 2: Gas Laws (Here comes the boom)	
27	WPR 2	
28	IMF II /Metallic Bonding/Solids	
29	Colligative properties	POP Quiz
30	Fundamental Chemistry of Explosives	

Lab #1: Lab techniques and measuring instruments
Lab #2: Enrichment 1: CLS Research Day
Lab #3: Spectrophotometry
Lab #4: IMFs (Report Assigned, due 4 NOV)
Lab #5: Stoichiometry
Lab #6: Lab Practical
Lab #7: Heat & Calorimetry
Lab #8: Stoichiometry & Gas Laws

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	600	1200	20.0%
10	Quizzes	100	1000	20.0%
8	Labs	150	1200	24.0%
2	Homework	75	150	3.0%
2	Instructor Grade	75	150	3.0%
		Total	**5000**	**100%**

Points were changed from previous AY. WPR1 and 2 were common exams this year with CH101. A shared TEE with 101 was also used (same TEE as usual), and cadets performed above average compared to pre-covid exams. BOH quizzes were increased to 10 (last AY 4 EOH quizzes) and worth more points, with good course outcomes. Students seemed to keep up with the material better given this forcing function. Homework completion rates were typical, as assessed by periodic homework notebook checks, despite being worth a lower percentage of their final grade.

e. Areas of Special Emphasis:

1) Required each Cadet to conduct one 5-minute ‘Science of the Day’ brief prior to class to encourage scientific literacy.

2) In AY21, made the first two labs notebook requirement as a non-graded turn in but graded them for the remaining labs so that cadets were forced to do some preparation prior to coming to lab. This year same as last year, we dedicated time during Lab 2 (enrichment) to discuss execution of a lab notebook, which began in lab 3. This was effective.

3) Continued to use the re-focused enrichments, which is a major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom. These were focused on primary literature and research to broader the general chemistry experience for our advanced cadets, and on applications of explosives to Chemistry/gas laws/thermodynamics.

4) Changed graded event structure to encourage cadets to keep up with course materials (more quizzes).

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.

b. Explain and apply foundational chemistry concepts.

c. Develop practices for learning independently (intellectual self-reliance).

d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158

c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits

d. Technician Support: Manuel provided excellent lab support, as always. Having him present in the lab was wonderful, as cadet enjoy working with him and it reduced instructor labs from 2 sessions to 1 session.

e. Supplies: LoggerPro was used in both classrooms and labs for analytical inquiry.

f. Additional Facilities: None

g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Sustained the removal of Mastering Chemistry homework as a requirement in AY22, which was effective.

b. Further increased the collective grade value of quizzes to give them a higher degree of importance.

c. Used common WPRs relative to CH101, but unique EOH. Still no molecular orbital theory, which is in my opinion appropriate. It was superficial in an already full lesson sequence, and the chemists will get it later in much better depth.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: Enrollment was approximately the same as in AY24, with approximately 16.5 students per section for 4 sections. Recommend we sustain only 4 sections of CH151 in future. The course had strong performers with only a few that struggled, whereas more struggle with the enhanced pace as we open up more sections.

3. Course Objectives:

a. Coverage:

1) Atomic Structure: EOH1, Lab 3, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 6
4) Stoichiometry, Chemical Reactions: Labs 7, 7, EOH3, WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	Quizzes 1,2, 3
	Lab 3
	WPR I, TEE
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	Quiz 3, 4, 5
	WPR I, TEE
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	Quiz 5, 6 Labs 4,5
	WPR II, TEE
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 6
	Quizzes 6, 7, 8
	WPR II, TEE

Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 7-8 Quiz 8 WPR II, TEE

4. Course QPA: The QPA was higher than past years. Likely a result of having the less challenging WPRs (since we used shared ones with 101).

6. TEE Grade: The TEE executed its typical format this year, with the ACS exam (70 questions over 110-minutes) the at least 100-minute free response part II portion of the test. TEE grades were similar to past years, if not a little lower but still high.

AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%
AY24-1: 91.3%
AY25-1: 90.4%

7. Course Processes

a. Textbook: The Tro text is appropriate for the course. Updated lesson objective sheets and course content to 3rd edition.

b. Course Content: Lesson/lab sequence was consistent with previous AY.

c. Lessons and labs: Lessons are appropriate. Labs reinforce learning well.

d. Summary of Graded Requirements: Graded events were changed as described above, and performance remained high.

8. Resources and Laboratories

a. Laboratories: No issues.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: Excellent support, as always.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course by offering 4 sections.

3. Course Objectives:

a. Coverage: No recommended changes.

b. Performance: No recommended changes. Cadet performed at a high level.

4. Survey Questions:

a. Survey Results: Do not change them from year to year to facilitate comparisons.

5. Course QPA: The QPA was up consistent with the previous academic year.

6. TEE Grade: Sustain current practices.

7. Course Process

a. Textbook: The Tro text is the appropriate text for this course and should be the only text.

b. Course Content: We also sustained the removal MO theory in lieu of more hybridization, and we continued to clean up the lab documents.

c. List of lessons and labs: Recommend to implement the same lesson and lab sequences next year.

d. Summary of Graded Requirements: Continue using separate EOHs and possibly also WPRs to reinforce CH151 learning. MO theory omitted with good effect.

e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes

a. Coverage: No recommended changes.

b. Performance: No recommended changes.

9. Resources and Laboratories

a. Laboratories: No issues. Continue to use BH158.

b. Computer Labs: N/A

c. Physical Models & Demos: No issues.

d. Technician Support: No issues.

e. Supplies: No issues.

f. Additional Facilities: N/A

g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

This information can be found in the AY25-1 CH151 folder on Sharepoint.

UNITED STATES MILITARY ACADEMY
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TAB C AY2024 COURSE ASSESSMENT

CH151 AY24 Course Assessment Package

SECTION I. COURSE DESCRIPTION

Foreword: This semester, instruction remained fully in person. One section per lab section was sustained from AY21-1, AY22-1, and AY23-1, which requires greater instructor burden. Items were turned in via hard copy or digitally, depending on instructor preferences. Lab notebooks were due at the end of lab blocks or 24-48 hours later, at instructor discretion, which resulted in cadets being as prepared as in typical semesters. Enrichments were reduced from 3 to 2 sessions, including a research overview enrichment executed during lab 2. These enrichments are a big sustain, as are “Science of the Day” presentations. We attempted a journal club, but it was not well received. It is likely too early in their cadet-career for this type of activity, which is better suited for upper level classes. We executed separate graded events (EOHs and WPRs) for CH151 to make it more challenging, but in the end, final grades were similar to past AYs. The common TEE was 101 was retained.

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY24-1 = 68 (4 sections; 2x Limbocker, 1x Kick, 1x Lachance)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: Same as previous iterations.

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74

AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60
AY20-1: 3.63
AY21-1: 3.81
AY22-1: 3.90
AY23-1: 3.82
AY24-1: 3.82 (see below for breakdown)

AY23 Final Grades

	total	%
A+	26	28.3
A	31	33.7
A-	17	18.5
B+	10	10.9
B	4	4.3
B-	1	1.1
C+	1	1.1
C	1	1.1
C-	1	1.1

AY24 Final Grades

	total	%
A+	21	30.9
A	16	23.5
A-	20	29.4
B+	5	7.4
B	2	2.9
B-	3	4.4
C+	1	1.5
C	0	0.0
C-	0	0.0

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%
AY24-1: 91.3%

7. Course Processes:

a. Textbooks
   1) Chemistry: Structure and Dynamics, Tro, 2nd Ed.

b. Course Content
   1) Atomic Structure
   2) Structure, Bonding, and States of Matter
   3) Stoichiometry, Chemical Reactions and Gases
   4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

CH151 AY24 Course Assessment Package

Month	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
August	Summer LV	Summer LV	1 Summer LV	2 Summer LV	3 Summer LV	4 Summer LV	5
	6	7 REORGY Week	8 REORGY Week	9 REORGY Week	10 REORGY Week	11 REORGY Week	12
	13 WK 1	14 1-1	15 2-1 R/U lab 1	16 1-2	17 2-2 S lab 1	18 2 Day Drop T lab 1	19 A
	20 WK 2	21 1-3	22 2-3	23 1-4	24 2-4 R/U lab 2	25 1 Day Drop	26 B Ring WKND
September	27 WK 3	28 1-5 BOH Quiz 1	29 2-5 BOH Quiz 1 S lab 2	30 1-6	31 2-6 T lab 2	1 2-7	2 B
	3 WK 4	4 Labor Day	5 1-7	6 2-8 R/U lab 3	7 1-8	8 2 Day Drop S lab 3	9 F Delaware St
	10 WK 5	11 1-9	12 2-9 T lab 3	13 1-10 BOH Quiz 2	14 2-10 BOH Quiz 2	15 1 Day Drop	16 B
	17 WK 6	18 1-11	19 2-11 R/U lab drop	20 1-12	21 2-12 S lab drop	22 2 Day Drop T lab drop	23 A
	24 WK 7	25 1 Day Drop	26 2-13 R/U lab drop	27 1-13	28 2-14	29 1-14	30 B
October	1 WK 8	2 1 Day Drop	3 2 Day Drop S lab drop	4 1-15 WPR 1	5 2-15 WPR 1 T lab drop	6 2 Day Drop	7 F BC
	8 WK 9	9 Columbus Day	10 1-16	11 2-16 R/U lab 4	12 1-17	13 2-17 S lab 4	14 F Troy
	15 WK 10	16 1-18 BOH Quiz 3	17 2-18 BOH Quiz 3 T lab 4	18 1-19	19 2-19	20 1 Day Drop	21 A
	22 WK 11	23 1-20	24 2-20 R/U lab 5	25 1-21	26 2-21 S lab 5	27 2 Day Drop T lab 5	28 F Umass
November	29 WK 12	30 1-22	31 2-22	1 1-23 BOH Quiz 4	2 2-23 BOH Quiz 4 R/U lab 6	3 1 Day Drop	4 B @AirForce
	5 WK 13	6 1-24	7 2-24 S lab 6	8 1-25	9 2-25 T lab 6	14 Veteran's Day	11 F Holy Cross
	12 WK 14	13 1-26 WPR 2	14 2-26 WPR 2	15 1-27	16 2 Day Drop R/U lab 7	17 2-27 S lab 7	18 F Coastal Car.
	19 WK 15	20 1 Day Drop	21 2 Day Drop T lab 7	22 Thanksgiving LV	23 Thanksgiving LV	24 Thanksgiving LV	25
December	26 WK 16	27 Thanksgiving LV	28 1-28	29 2-28	30 1-29	1 2-29 R/U lab 8	2 B
	3 WK 17	4 1-30	5 2-30 S lab 8	6 1 Day Drop	7 2 Day Drop T lab 8	8 1 Day Drop	9 Army-Navy
	10	11 1 Day Drop	12 2 Day Drop	13 TEE	14 TEE	15 TEE	16 TEE
	17	18 TEE	19 TEE	20 Winter LV	21 Winter LV	22 Winter LV	23 Winter LV

BUFF Card		
Block #1: STRUCTURE, BONDING, & STATES OF MATTER		
1	Chemistry Basics	
2	The Atom, Atomic Number, and Atomic Mass	
3	Electromagnetic Radiation/Spectroscopy	
4	Quantum Mechanical Model of the Atom	
5	Electron Configuration and Spin	BOH Quiz 1
6	Atomic Forces	
7	Periodic Trends	
8	Bonding Basics	
9	Electronegativity and Partial Charge	
10	Empirical Formula & Combustion Analysis	EOH Quiz 2
11	Lewis Structures and Covalent Bonds	
12	Molecular Shape & Hybridization	
13	Intermolecular Forces	
14	Nomenclature	
15	WPR 1	
Block #2: STOICHIOMETRY/CHEM RXNS/THERMO		
16	Chemical Reactions and Stoichiometry, and Limiting Reactant	
17	Solution Chemistry	
18	Solubility and Precipitation Reactions	BOH Quiz 3
19	Acid/Base Neutralization Reactions	
20	Oxidation-Reduction Reactions	
21	Applications of Chemical Reactions	
22	Heat and Calorimetry	
23	Enthalpy of Reaction	BOH Quiz 4
24	Heating/Cooling Curves & Phase Diagrams	
25	Ideal Gas Law	
26	WPR 2	
27	Enrichment 2: Gas Laws (Here comes the boom)	
28	IMF II /Metallic Bonding/Solids	
29	Colligative properties	
30	Fundamental Chemistry of Explosives	

Lab #1: Lab techniques and measuring instruments
Lab #2: Enrichment 1: CLS Research Day
Lab #3: Spectrophotometry
Lab #4: IMFs (Report 1 Assigned)
Lab #5: Stoichiometry
Lab #6: Lab Practical
Lab #7: Heat & Calorimetry (Report 2 Assigned)
Lab #8: Stoichiometry & Gas Laws

Lab Report 1 (Intro+Materials/Methods) due 29 Oct 2359

Lab Report 2 (Results/Discussion+Conclusion+Abstract) due 3 Dec 2359

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	500	1000	20.0%
5	Quizzes	200	1000	20.0%
8	Labs	150	1200	24.0%
2	Homework	200	400	8.0%
2	Instructor Grade	50	100	2.0%
		Total	**5000**	**100%**

Points were maintained from previous AY. WPR1 and 2 were written separately for CH151 (CH101 WPRs not used). Although they were made more difficult, cadets performed very well on these exams which likely enhanced learning. A shared TEE with 101 was used (same TEE as usual), and cadets performed above average compared to pre-covid exams.

e. Areas of Special Emphasis:

1) Required each Cadet to conduct one 5-minute ‘Science of the Day’ brief prior to class to encourage scientific literacy.

2) In AY21, made the first two labs notebook requirement as a non-graded turn in but graded them for the remaining labs so that cadets were forced to do some preparation prior to coming to lab. This year same as last year, we dedicated time during Lab 2 (enrichment) to discuss execution of a lab notebook, which began in lab 3. This was effective.

3) Re-focused enrichments, which is a major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom. These were focused on primary literature and research to broader the general chemistry experience for our advanced cadets, and on applications of explosives to Chemistry/gas laws/thermodynamics. We planned 4 enrichments in the core course brief, including in depth group projects and a journal club, but removed these to focus on learning objectives given cadets’ pace/performance in the course at that time. This was the right thing to do.

4) The Stoichiometry lab was restructured to promote safety. All control measures were maintained, but 12 M HCl was reduced to 6 M HCl without detriment to experimental outcomes. This concentration stock was immediately diluted by the cadets in a fume hood, which mitigated risk appropriately. Always working concentration was ~1 M, which is similar to what they work with in past labs. Further reducing the concentration may lead to worse experimental outcomes due to dilution effects.

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.

b. Explain and apply foundational chemistry concepts.

c. Develop practices for learning independently (intellectual self-reliance).

d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158

c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits

d. Technician Support: Glenn and Manuel provided excellent lab support, as always.

e. Supplies: Vernier was used in both classrooms and labs for analytical inquiry.

f. Additional Facilities: None

g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Sustained the removal of Mastering Chemistry homework as a requirement in AY22, which was effective.

b. Sustained an increase in the points per quiz to give them a higher degree of importance.

c. Used unique WPRs relative to CH101.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: Enrollment was lower than in AY23 (86 students past AY), which is approximately 16 students per section for 6 sections. Recommend we sustain only 4 sections of CH151. We did not have the need to fill six sections in AY23, and some people that were included in that AY would have likely been better off in CH101. The course had strong performers with only a few that struggled, whereas more struggle with the enhanced pace as we open up more sections.

3. Course Objectives:

a. Coverage:

1) Atomic Structure: EOH1, Lab 3, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 6
4) Stoichiometry, Chemical Reactions: Labs 7, 7, EOH3, WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	Quizzes 1,2
	Lab 3
	WPR I, TEE
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	Quiz 3
	WPR I, TEE
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	Quiz 3, Labs 4,5
	WPR II, TEE
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 6
	Quiz 3
	WPR II, TEE
Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 7-8
	Quiz 4
	WPR II, TEE

4. Course QPA: The QPA was consistent with past years, which is higher than historical norms.

6. TEE Grade: The TEE returned to its typical format this year, with the ACS exam (70 questions over 110-minutes) the at least 100-minute free response part II portion of the test. TEE grades were higher than past years despite giving the same exam, indicating a likely positive difference in student outcomes in this year's cohort.

AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%
AY24-1: 91.3%

7. Course Processes

a. Textbook: The Tro text is appropriate for the course. Consider updating to 3rd edition.

b. Course Content: Lesson/lab sequence was consistent with previous AY.

c. Lessons and labs: Lessons are appropriate. Labs reinforce learning well. 12 M HCl switched to 6 M to mitigate risk.

d. Summary of Graded Requirements: Graded events were made to be more challenging given last year's higher grades, but performance remained high.

8. Resources and Laboratories

a. Laboratories: No issues.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: Excellent support, as always.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course by offering 4 sections.

3. Course Objectives:

a. Coverage: No recommended changes.
b. Performance: No recommended changes. Cadet performed at a high level.

4. Survey Questions:

a. Survey Results: Do not change them from year to year to facilitate comparisons.

5. Course QPA: The QPA was up consistent with the previous academic year.

6. TEE Grade: Sustain current practices.

7. Course Process

a. Textbook: The Tro text is the appropriate text for this course and should be the only text.
b. Course Content: We also sustained the removal MO theory in lieu of more hybridization, and we continued to clean up the lab documents.
c. List of lessons and labs: Plan to implement the same lesson and lab sequences next year.
d. Summary of Graded Requirements: Continue using separate EOHs and WPRs to reinforce CH151 learning. MO theory omitted this year with good effect.
e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes

a. Coverage: No recommended changes.
b. Performance: No recommended changes.

9. Resources and Laboratories

a. Laboratories: No issues. Continue to use BH158.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: No issues.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

This information can be found in the AY24-1 CH151 folder on Sharepoint.

TAB D AY2023 COURSE ASSESSMENT

SECTION I. COURSE DESCRIPTION

Foreword: This semester, instruction was fully in person. One section per lab section was sustained from AY21-1 and AY22-1, which requires greater instructor burden but was better for student learning. Items were turned in via hard copy or digitally, depending on instructor preferences. Lab notebooks were due at the end of lab blocks or 24-48 hours later, a instructor discretion, which resulted in cadets being as prepared as in typical semesters. Enrichments were kept at 3 sessions, including a research overview enrichment executed during lab 2. This is a big sustain. Steam plant was dropped, which is also a sustain as it is more appropriate for CH102 (if at all).

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY22-1 = 102 (6 sections)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: Same as previous iterations.

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74
AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60

AY20-1: 3.63
AY21-1: 3.81
AY22-1: 3.90
AY23-1: 3.82

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%

7. Course Processes:

a. Textbooks
   1) <u>Chemistry: Structure and Dynamics</u>, Tro, 2nd Ed.

b. Course Content
   1) Atomic Structure
   2) Structure, Bonding, and States of Matter
   3) Stoichiometry, Chemical Reactions and Gases
   4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

CH151 AY23 Course Assessment Package

Month	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
August	7	8 REORGY Week	9 REORGY Week	10 REORGY Week	11 REORGY Week	12 REORGY Week	13 A-Day
August	14 WK 1	1-1 15	SD 16	17 1 Day Drop	2-1 18 R/U Lab 1	1-2 19	20 A/C
August	21 WK 2	22 1 Day Drop	2-2 23 S/V Lab 1	1-3 24	2-3 25 T/W Lab 1	26 1 Day Drop	27 Ring Wknd
August	28 WK 3	1-4 29 EoH 1	2-4 30 R/U Lab 2 EoH 1	SD 31	1-5 1	2-5 2 S/V Lab 2	3
September	4 WK 4	5 Labor Day (no classes)	2-6 6 T/W Lab 2	1-6 7	2-7 8 R/U Lab 3	1-7 9	10
September	11 WK 5	1-8 12 EoH 2	2-8 13 S/V Lab 3 EoH 2	SD 14	1-9 15	2-9 16 Lab drop	17
September	18 WK 6	1-10 19	2-10 20 Lab drop	1-11 21	2-11 22 Lab drop	23 1 Day Drop	24
September	25 WK 7	1-12 26	2-12 27 T/W Lab 3	SD 28	1-13 29 WPR 1	2-13 30 WPR 1	1
October	2 WK 8	3 1 Day Drop	2-14 4 Lab drop	1-14 5	2-15 6 Lab drop	1-15 7 WPR 1 MU	8
October	9 WK 9	10 Columbus Day (no classes)	2-16 11 R/U Lab 4	1-16 12	2-17 13 S/V Lab 4 EoH 3	1-17 14 EoH 3	15
October	16 WK 10	1-18 17	2-18 18 T/W Lab 4	SD 19	1-19 20	2-19 21 R/U Lab 5	22
October	23 WK 11	1-20 24	2-20 25 S/V Lab 5	1-21 26 EoH 4	2-21 27 T/W Lab 5 EoH 4	28 1 Day Drop	29
November	30 WK 12	1-22 31	2-22 1 R/U Lab 6	SD 2	1-23 3	2-23 4 S/V Lab 6	5
November	6 WK 13	1-24 7 EoH 5	2-24 8 T/W Lab 6 EoH 5	1-25 9	2-25 10 R/U Lab 7	11 Veterans Day No classes	12
November	13 WK 14	1-26 14	2-26 15 S/V Lab 7	SD 16	1-27 17	2-27 18 T/W Lab 7	19
November	20 WK 15	21 1 Day Drop Dean's Hr WPR 2	2-28 22 R/U Lab 8	23 No Classes	24 Thanksgiving	25 No Classes	26
November	27 WK 16	1-28 28	2-29 29 S/V Lab 8	1-29 30	2-30 1 T/W Lab 8	2 1 Day Drop	3
December	4	1-30 5	SD 6 WPR 2 MU	7 1 Day Drop	SD 8	9 1 Day Drop	10 BEAT NAVY
December	11	SD 12 TEE	13 TEE	14 TEE	15 TEE	16 TEE	17 TEE

BUFF Card		
Block #1: STRUCTURE, BONDING, & STATES OF MATTER		
1	Chemistry Basics	
2	The Atom, Atomic Number, and Atomic Mass	
3	Electromagnetic Radiation/Spectroscopy	
4	Quantum Mechanical Model of the Atom	EOH Quiz 1
5	Electron Configuration and Spin	
6	Atomic Forces	
7	Periodic Trends	
8	Bonding Basics	EOH Quiz 2
9	Electronegativity and Partial Charge	
10	Empirical Formula & Combustion Analysis	
11	Lewis Structures and Covalent Bonds	
12	Molecular Shape & Hybridization	
13	WPR 1	
14	Intermolecular Forces	
15	Nomenclature	
Block #2: STOICHIOMETRY/CHEM RXNS/THERMO		
16	Chemical Reactions and Stoichiometry, and Limiting Reactant	
17	Solution Chemistry	EOH Quiz 3
18	Solubility and Precipitation Reactions	
19	Acid/Base Neutralization Reactions	
20	Oxidation-Reduction Reactions	
21	Applications of Chemical Reactions	EOH Quiz 4
22	Heat and Calorimetry	
23	Enthalpy of Reaction	
24	Heating/Cooling Curves & Phase Diagrams	EOH Quiz 5
25	Ideal Gas Law	
26	Enrichment 2: Gas Laws (Here comes the boom)	
WPR 2 (LSN 1-26)		
27	IMF II /Metallic Bonding/Solids	
28	Colligative properties	
29	Enrichment 3: Group Research Talks	
30	Fundamental Chemistry of Explosives	

Lab #1: Lab techniques and measuring instruments
Lab #2: Enrichment 1: CLS Research Day
Lab #3: Spectrophotometry
Lab #4: IMFs (Report 1 Assigned)
Lab #5: Stoichiometry
Lab #6: Lab Practical
Lab #7: Heat & Calorimetry (Report 2 Assigned)
Lab #8: Stoichiometry & Gas Laws

Report 1 (Intro+Materials/Methods) due 28 Oct 2359
Report 2 (Results/Discussion+Conclusion+Abstract) due 2 Dec 2359

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	500	1000	20.0%
5	Quizzes	200	1000	20.0%
8	Labs	150	1200	24.0%
2	Homework	200	400	8.0%
2	Instructor Grade	50	100	2.0%
		Total	**5000**	**100%**

Points were maintained from previous AY. WPR1 and 2 were written separately for CH151 (CH101 WPRs not used). Although they were made more difficult, cadets performed very well on these exams which likely enhanced learning. A shared TEE with 101 was used (same TEE as usual), and cadets performed above average compared to pre-covid exams.

e. Areas of Special Emphasis:

1) Required each Cadet to conduct one 5-minute 'Science of the Day' brief prior to class to encourage scientific literacy.

2) Continued to remove detailed lab instructions from the lab assignments to make the cadets do some critical thinking about what they had to do in lab prior to coming to the lab.

3) In AY21, made the first two labs notebook requirement as a non-graded turn in but graded them for the remaining labs so that cadets were forced to do some preparation prior to coming to lab. This year same as last year, we dedicated time during Lab 2 (enrichment) to discuss execution of a lab notebook, which began in lab 3. This was effective.

4) Expanded enrichments, which is the major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom. These expansions were focused on primary literature and research to broader the general chemistry experience for our advanced cadets.

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.
b. Explain and apply foundational chemistry concepts.
c. Develop practices for learning independently (intellectual self-reliance).
d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158
c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits
d. Technician Support: Glenn and Manuel provided excellent lab support, as always.
e. Supplies: Vernier was used in both classrooms and labs for analytical inquiry.
f. Additional Facilities: None
g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Removed Mastering Chemistry homework as a requirement in AY22, which was sustained and effective.
b. Sustained an increase in the points per quiz to give them a higher degree of importance.
c. Used unique WPRs relative to CH101.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: Enrollment was increased again in AY23 (92 students past AY), which is approximately 16 students per section for 6 sections. Recommend we consider returning to 5 sections (95 cadets). We did not have the need to fill six sections, and some people that were included this AY would have likely been better off in CH101.

3. Course Objectives:

a. Coverage:

1) Atomic Structure: EOH1, Lab 3, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 6
4) Stoichiometry, Chemical Reactions: Labs 7, 7, EOH3, WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	Quizzes 1,2
	Lab 3
	WPR I
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	Quiz 3
	WPR I
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	Labs 4,5
	WPR II
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 6
	Quiz 4
	WPR II
Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 7-8
	Quiz 5
	WPR II, TEE

4. Course QPA: The QPA was consistent with past years, which is higher than historical norms.

6. TEE Grade: The TEE returned to its typical format this year, with the ACS exam (70 questions over 110-minutes) the at least 100-minute free response part II portion of the test. TEE grades were higher than past years despite giving the same exam, indicating a likely positive difference in student outcomes in this year’s cohort.

AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)

AY22-1: 93.4% (return to ACS and open response part 2)
AY23-1: 92.3%

7. Course Processes

a. Textbook: The Tro text is appropriate for the course.
b. Course Content: Lesson/lab sequence was consistent with previous AY.
c. Lessons and labs: Lessons are appropriate. Labs reinforce learning well. 12 M HCl switched to 6 M to mitigate risk, with reassessment to be implemented next AY.
d. Summary of Graded Requirements: Graded events were made to be more challenging given last year's higher grades, but performance remained high.

8. Resources and Laboratories

a. Laboratories: No issues.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: Excellent support, as always.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course.

3. Course Objectives:

a. Coverage: No recommended changes.
b. Performance: No recommended changes. Cadet performed better than expected.

4. Survey Questions:

a. Survey Results: Do not change them from year to year to facilitate comparisons.

5. Course QPA: The QPA was up consistent with the previous academic year.

6. TEE Grade: Sustain current practices.

7. Course Process

a. Textbook: The Tro text is the appropriate text for this course and should be the only text.
b. Course Content: Only planned major change is to replace the Steam plant enrichment, to remove MO theory in lieu of more hybridization, and to continue to clean up the lab documents.
c. List of lessons and labs: Plan to implement the same lesson and lab sequences next year.
d. Summary of Graded Requirements: Continue using separate EOHs and WPRs to reinforce CH151 learning. MO theory omitted this year with good effect.
e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes

a. Coverage: No recommended changes.
b. Performance: No recommended changes.

9. Resources and Laboratories

a. Laboratories: No issues. Continue to use BH158.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: No issues.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

All information can be found in the AY23-1 CH151 folder on Sharepoint.

TAB E AY2022 COURSE ASSESSMENT

SECTION I. COURSE DESCRIPTION

Foreword: This semester, instruction returned to fully in person. One section per lab section was sustained from AY21-1, which requires greater instructor burden but was better for student learning. Items were turned in via hard copy or digitally, depending on instructor preferences. Lab notebooks were due at the end of lab blocks rather than 24 hours later, as in AY21-1, which resulted in cadets being noticeable better prepared than in AY21-1. Enrichments were returned from 2 sessions to 3, with the addition of a research overview enrichment executed during lab 2. This is a big sustain. We are going to look into alternatives for the Steam Plant enrichment for AY23-1, as this was not very value added this year.

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY22-1 = 91 (5 sections)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: (New for 21-1) Second year in a row with a ‘New’ survey.

CH151 AY22 Course Assessment Package

Q#	Question	Mean	rel to academy
Q1	My fellow students contributed to my learning in this course.	3.8	--
Q2	My motivation to learn and to continue learning has increased because of this course.	4	++
Q3	My personal schedule allows me enough time to reflect on the material I have learned in class.	4.3	++
Q4	My personal schedule allows me enough time to adequately prepare for my optimum academic performance.	4.2	++
Q5	In this course, my critical thinking ability increased.	4.1	--
Q6	The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a2:1 ratio of out-of-class time	4	++
Q7	The classroom environment (e.g. desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn.	4.1	++
Q8	In this course, my ability to think creatively and take intellectual risks increased.	4.3	++
Q9	Developing skills in written and oral communication.	2.8	=
Q10	Constructing a scientific argument.	3.4	++
Q11	Asking questions and explaining my knowledge to my peers.	3.6	++
Q12	Relating to and leading my peers.	3.4	++
Q13	Being open to alternative points of view which may initially conflict with my own.	3.5	++
Q14	Thinking scientifically.	3.7	+
Q15	Developing intellectual self-reliance.	3.8	++
Q16	Developing teamwork skills.	3.3	-
Q17	Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the D	3.9	++
Q18	The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content.	4.1	++
Q19	The course text and references were very helpful in my gaining understanding of the course material.	4.1	++
Q20	The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me in order to do well in this cours	4.2	++
Q21	The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material.	4.5	++
Q22	This instructor encouraged students to be responsible for their own learning.	4.6	++
Q23	This instructor used effective techniques for learning, both in class and for out-of-class assignments.	4.6	++
Q24	My instructor cared about my learning in this course.	4.7	++
Q25	My instructor demonstrated respect for cadets as individuals.	4.6	+
Q26	This instructor stimulated my thinking.	4.5	++
Q27	This instructor encouraged students to think creatively and take intellectual risks.	4.5	++
Q28	Understanding the classical model of the atom and how atomic properties determine chemical properties.	3.7	+
Q29	Using the nature of bonding forces in ionic, covalent, and metallic compounds to predict and explain the behavior of substances.	3.8	+
Q30	Describing the three states of matter and explaining the behavior of matter at the molecular level.	3.6	=
Q31	Writing balanced equations for different types of chemical reactions and using these equations to calculate amounts of reactants and products.	3.7	=
Q32	Using the First Law of Thermodynamics to determine the enthalpy change of a reaction, the mass of an object, or a specific heat capacity.	3.8	+

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74
AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60
AY20-1: 3.63
AY21-1: 3.81
AY22-1: 3.90

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)

7. Course Processes:

a. Textbooks
   1) Chemistry: Structure and Dynamics, Tro, 2nd Ed.

b. Course Content
   1) Atomic Structure
   2) Structure, Bonding, and States of Matter
   3) Stoichiometry, Chemical Reactions and Gases
   4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

CH151 AY22 Course Assessment Package

Month	Week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
August		8	9 REORGY Week	10 REORGY Week	11 REORGY Week	12 REORGY Week	13 REORGY Week	14 A-Day
August	WK 1	15	16 1-1	17 2-1 R/U - 1	18 1-2	19 2-2 S/V - 1	20 1-DAY DROP	21 A/C
August	WK 2	22	23 2-3 T/W - 1	24 1-3	25 SD	26 1-4	27 2-4 R/U - 2	28 Ring Wknd
August	WK 3	29	30 1-5 EoH 1	31 2-5 S/V - 2 EoH 1	1 1-6	2 2-6 T/W - 2	3 1-7	4 FB Away B
September	WK 4	5	6 Labor Day (no classes)	7 2-7 R/U - 3	8 MOD COM 1-DAY DROP	9 2-8 S/V - 3	10 1-8	11 FB Home F
September	WK 5	12	13 1-9 EoH 2	14 2-9 T/W-3 EoH 2	15 SD	16 1-10	17 2-10 R/U - 4	18 FB Home F
September	WK 6	19	20 1-11	21 2-11 S/V - 4	22 SD	23 2-12 T/W - 4	24 1-12	25 FB Home F
September	WK 7	26	27 2-13 R/U - 5 EoH 3	28 HLD	29 1-13 EoH 3	30 2-14 S/V - 5	1 1-14	2 FB Away A/C
October	WK 8	3	4 1-15	5 2-15 T/W - 5	6 SD	7 1-16 Dean's HR WPR #1	8 2-16 R/U-6	9 B
October	WK 9	10	11 Columbus Day	12 2-17 S/V - 6	13 1-17	14 2-18 T/W - 6	15 1-18	16 FB Away A/C
October	WK 10	17	18 2-19 R/U - 7	19 1-19	20 SD	21 1-20 EoH 4	22 2-20 S/V - 7 EoH 4	23 FB Home
October	WK 11	24	25 1-21	26 2-21 T/W - 7	27 1-22	28 2-22 R/U - 8	29 1 - DAY DROP	30 Fall SH A/C
November	WK 12	31	1 1-23	2 SD	3 2-23 S/V-8	4 1-DAY DROP	5 2-24 T/W-8	6 FB @ USAFA A/D
November	WK 13	7	8 1-24	9 2-25 R/U - 9	10 1-25	11 Veterans Day	12 1-DAY DROP	13 FB Home F
November	WK 14	14	15 1-26 EoH 5	16 2-26 S/V - 9 EoH 5	17 SD	18 2-27 T/W - 9	19 1-27	20 FB Home F
November	WK 15	21	22 DROP Dean's HR WPR #2	23 2-28 R/U - 10	24 1-DAY DROP	25 Thanksgiving	26 No Classes	27 FB Away B
November	WK 16	28	29 1-28	30 SD	1 1-29 Dean	2 2-29 S/V - 10	3 1-DAY DROP	4 A/D
December		5	6 1-30 Dean	7 2-30 T/W-10	8 SD	9 1-DAY DROP	10 1-DAY DROP	11 BEAT NAVY
December		12	13 Reading Day	14 TEE	15 TEE	16 TEE	17 TEE	18 TEE

	Block #1: STRUCTURE, BONDING, & STATES OF MATTER		
1	Chemistry Basics		
2	The Atom		
3	Atomic Number and Atomic Mass		
4	Electromagnetic Radiation/Spectroscopy		
5	Quantum Mechanical Model of the Atom		**EOH Quiz 1**
6	Electron Configuration and Spin		
7	Atomic Forces		
8	Periodic Trends		
9	Bonding Basics		**EOH Quiz 2**
10	Electronegativity and Partial Charge		
11	Empirical Formula & Combustion Analysis		
12	Lewis Structures and Covalent Bonds		
13	Molecular Shape and Hybridization		**EOH Quiz 3**
14	Enrichment 2: Steam Plant Tour		
15	Intermolecular Forces		
	WPR I		
	Block #2: STOICHIOMETRY/CHEM RXNS/THERMO/MIL APPS		
16	Nomenclature/Molecular Orbital Theory		
17	Chemical Reactions/Stoichiometry/Limiting Reactant		
18	Solution Chemistry		
19	Solubility and Precipitation Reactions		
20	Acid/Base Neutralization Reactions		**EOH Quiz 4**
21	Oxidation-Reduction Reactions		
22	Chemical Reaction Applications		
23	Heat and Calorimetry		
24	Enthalpy of Reaction		
25	Heating/Cooling Curves & Phase Diagrams		
26	Ideal Gas Law		**EOH Quiz 5**
27	Enrichment 3: Gas Laws		
	WPR 2		
28	IMF II /Metallic Bonding/Solids		
29	Colligative Properties		
30	Fundamental Chemistry of Explosives		

Lab #1: Lab Techniques and Measuring Instruments
Lab #2: Enrichment 1: CLS Research Tour and Publications
Lab #3: Spectrophotometry
Lab #4: Drop
Lab #5: IMFs
Lab #6: Drop
Lab #7: Stoichiometry
Lab #8: Lab Practical
Lab #9: Heat & Calorimetry
Lab #10: Stoichiometry & Gas Laws

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	500	1000	20.0%
5	Quizzes	200	1000	20.0%
8	Labs	150	1200	24.0%
2	Homework	200	400	8.0%
2	Instructor Grade	50	100	2.0%
		Total	**5000**	**100%**

Allocation of points were adjusted from last year to reduce the value of each EOH and add one more EOH quiz. Instructor grade was reduced by 1%, homework from 6% to 8%, and lab grade was increased by 4% to accommodate. The overall points remained at 5000. Major graded event points matched CH101 for equity.

e. Areas of Special Emphasis:

1) Required each Cadet to conduct one 5-minute 'Science of the Day' brief prior to class to encourage scientific literacy.

2) Continued to remove detailed lab instructions from the lab assignments to make the cadets do some critical thinking about what they had to do in lab prior to coming to the lab.

3) In AY21, made the first two labs notebook requirement as a non-graded turn in but graded them for the remaining labs so that cadets were forced to do some preparation prior to coming to lab. This year, we dedicated time during Lab 2 (enrichment) to discuss execution of a lab notebook, which began in lab 3. This was effective.

4) Continued to give the same graded events as CH101, with the exception of MO theory in EOH3. Plan is to remove MO theory and focus on hybridization for AY23-1 given feedback from COL Burpo.

5) Continued and even expanded enrichments, which is the major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom.

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.
b. Explain and apply foundational chemistry concepts.
c. Develop practices for learning independently (intellectual self-reliance).
d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158
c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits
d. Technician Support: Mr. Treus provided excellent lab support, as always.
e. Supplies: Vernier was used in both classrooms and labs for analytical inquiry.
f. Additional Facilities: None
g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Removed Mastering Chemistry homework as a requirement.
b. Increased the points per quiz to give them a higher degree of importance.
c. Continued using the same grades events as CH101.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: Enrollment was similar to AY21 at circa 92 cadets, which is approximately the maximum allowed for five instructors. Recommend we consider returning to 6 sections (~120 cadets).

3. Course Objectives:

a. Coverage:
1) Atomic Structure: EOH1, Lab 3, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 6
4) Stoichiometry, Chemical Reactions: Labs 7, 7, EOH3, WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	Quizzes 1,2
	Lab 3
	WPR I
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	Quiz 3
	WPR I
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	Labs 5,7
	WPR II
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 7
	Quiz 4
	WPR II
Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 8-9
	Quiz 5
	WPR II, TEE

4. Survey Questions:

a. Survey Results: Unfortunately, the survey system was changed, resulting in a loss of historical data. Now the only item that can be reported is the average answer based out of 5. 5=SA, 4=A, 3=N, 2=D, 1=SD. (SA: Strongly Agree, A: Agree, N: Neutral, D: Disagree, SD: Strongly Disagree)

1) My personal schedule allows me enough time to reflect on the material I have learned in class. (4.3)

2) My personal schedule allows me enough time to adequately prepare for my optimum academic performance. (4.2)

3) The course text and references were very helpful in my gaining understanding of the course material. (4.1)

5. Course QPA: The QPA was higher than average. This is due to easier graded events that AY21, as mirror in CH101's grade distribution, and the addition of another 50 point extra credit assignment. The second extra credit assignment can be removed in AY23.

6. TEE Grade: The TEE returned to its typical format this year, with the ACS exam (70 questions over 110-minutes) the at least 100-minute free response part II portion of the test. TEE grades were higher than past years despite giving the same exam, indicating a likely positive difference in student outcomes in this year's cohort.

AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (just ACS part 1 due to COVID)
AY22-1: 93.4% (return to ACS and open response part 2)

7. Course Processes

a. Textbook: The Tro text is appropriate for the course.

b. Course Content: Lesson/lab sequence was modified slightly to align with Tro text.

c. Lessons and labs: Lessons are appropriate.

d. Summary of Graded Requirements: Graded events can be more challenging next year given this year's higher grades.

e. Areas of Special Emphasis:

1) Sustain making key labs more challenging than CH101. Facilitate cadets to develop their own procedures to assist cadets as they learn to develop critical thinking skills.

2) Continue to use lab notebooks as it forces cadets to do some preparation and critical thinking prior to coming to lab

3) For AY22-1, consider creating CH151 specific EOHs, and WPRs, while using the same TEE as CH101. This would make the course more challenging and the grades could be adjusted accordingly.

4) Continue enrichments, which is the major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom.

8. Resources and Laboratories

a. Laboratories: No issues.

b. Computer Labs: N/A

c. Physical Models & Demos: No issues.

d. Technician Support: Excellent support, as always.

e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course. If there is capacity to do so, consider increasing the number of sections to accommodate more cadets in the fall due to dropping the spring semester but as seen from this year, 5 sections (~90 cadet) should be increased to 6 sections to maintain the model of approximately 10% of cadets take CH151.

3. Course Objectives:
a. Coverage: No recommended changes.
b. Performance: No recommended changes. Cadet performed better than expected.

4. Survey Questions:
a. Survey Results: Do not change them from year to year to facilitate comparisons.

5. Course QPA: The QPA was up slightly from the previous academic year. Suggest creating harder EOHs and WPRs specific to CH151.

6. TEE Grade: Sustain current practices.

7. Course Process
a. Textbook: The Tro text is the appropriate text for this course and should be the only text.
b. Course Content: Only planned major change is to replace the Steam plant enrichment, to remove MO theory in lieu of more hybridization, and to continue to clean up the lab documents.
c. List of lessons and labs: Plan to implement the same lesson and lab sequences next year.
d. Summary of Graded Requirements: Consider updating EOHs and WPRs to be more challenging than CH101
e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes
a. Coverage: No recommended changes.
b. Performance: No recommended changes.

9. Resources and Laboratories
a. Laboratories: No issues. Continue to use BH158.
b. Computer Labs: N/A
c. Physical Models & Demos: No issues.
d. Technician Support: No issues.
e. Supplies: No issues.
f. Additional Facilities: N/A
g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

1. Lessons:
https://usarmywestpoint.sharepoint.com/sites/cls.ch151/Shared%20Documents/Forms/AllItems.aspx?id=/sites/cls.ch151/Shared%20Documents/Lesson%20Assignments

2. Labs:
https://usarmywestpoint.sharepoint.com/sites/cls.ch151/Shared%20Documents/Forms/AllItems.aspx?id=/sites/cls.ch151/Shared%20Documents/Laboratories

3. Graded events:
https://usarmywestpoint.sharepoint.com/sites/cls.gradedevents/SitePages/Home.aspx

4. Supplemental Material:
https://usarmywestpoint.sharepoint.com/sites/cls.ch151/Shared%20Documents/Forms/AllItems.aspx?id=%2Fsites%2Fcls%2Ech151%2FShared%20Documents%2FSupporting%20Documents

5. Other Resources: Enrichments 2 and 3
https://usarmywestpoint.sharepoint.com/sites/cls.ch151/Shared%20Documents/Forms/AllItems.aspx?id=%2Fsites%2Fcls%2Ech151%2FShared%20Documents%2FEnrichments

6. Other historical information:
https://usarmywestpoint.sharepoint.com/sites/cls.courses/SitePages/Home.aspx

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TAB F AY2021 COURSE ASSESSMENT

SECTION I. COURSE DESCRIPTION

Foreword: Due to the global pandemic many items for instruction of AY21-1 were forced to be modified to support the risk of transmission. Below is short list of items changed with comments regarding their effectiveness and is by no means meant to cover every small change.

1. Class was taught in a hybrid fashion. This was an effective method with the students rotating lesson between in-person and remote. I would prefer in the future to be full in-person.
2. Lab execution was changed from two sections in lab at a time, to only one section in lab with empty benches between groups. This was effective; however, it increased the instructor time burden.
3. All items were turned in digitally to avoid contact. This was effective but burdensome, I would suggest that this is 'over' kill and to allow students to turn in items in paper form with proper cover letters.
4. Lab notebooks due to digital turn-in were allowed to be turned in 24hrs after the lab. This was a mistake, cadet prep for the labs was noticeably lacking.
5. Enrichments were reduced from 3 to 2, because we didn't have a safe way for Cadets to share the equipment for the gas laws enrichment. This was needed and effective.

1. Red Book Description: An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program.
CH101: This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.

2. Enrollment: AY21-1 = 92 (5 sections)

3. Course Objectives:

a. Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).

b. Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.

c. States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).

d. Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.

e. Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.

4. Survey Questions: (New for 21-1) Second year in a row with a 'New' survey.

Q#	Question	Average (5pts)
1	**Asking questions and explaining my knowledge to my peers**	3.37
2	**Being open to alternative points of view which may initially conflict with my own.**	2.91
3	**Consider the departments responsible for the courses which you have taken this semester.**	4.34
4	**Constructing a scientific argument**	3.20
5	**Describing the three states of matter and explaining the behavior of matter at the molecular level**	3.60
6	**Developing intellectual self-reliance**	3.64
7	**Developing skills in written and oral communication**	2.89
8	**Developing teamwork skills**	3.43
9	**Graded events (WPR's, Projects, Papers, TEE's, etc.) in this course were effective and fair**	4.53
10	**I had to spend more time in this course to do the work required in the remote environment**	3.07
11	**I had to work harder in this course to achieve a similar amount of learning because of the transition to remote instruction**	3.36
12	**I took more responsibility for my own learning in this course as a result of switching to a remote**	3.91
13	**I was able to maintain my attention in this course during remote class sessions to the same extent a**	3.11
14	**In this course, my ability to think creatively and take intellectual risks increased**	3.93
15	**In this course, my critical thinking ability increased.**	4.09
16	**My fellow students contributed to my learning in this course**	4.06
17	**My motivation to learn and to continue learning has increased because of this course**	4.24
18	**My personal schedule allows me enough time to adequately prepare for my optimum academic performance**	3.97
19	**My personal schedule allows me enough time to reflect on the material I have learned in class**	3.89
20	**Participating in remote learning made me want to continue to learn on my own at USMA and beyond.**	3.31
21	**Relating to and leading my peers.**	3.13
22	**The classroom environment (e.g. desk setup, boards, technology, lights, etc.) had a positive impact**	4.17
23	**The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected**	4.27
24	**The course text and references were very helpful in my gaining understanding of the course material.**	4.07
25	**The homework assignments, papers, and projects in this course could be completed within the USMA time**	3.97
26	**The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course**	4.01
27	**The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding**	4.44
28	**Thinking scientifically**	3.59
29	**Understanding the classical model of the atom and how atomic properties determine chemical properties**	3.59
30	**Using the First Law of Thermodynamics to determine the enthalpy change of a reaction**	3.66
31	**Using the nature of bonding forces in ionic, covalent, and metallic compounds to predict and explain**	3.69

32	**Writing balanced equations for different types of chemical reactions and using these equations**	3.59

This new form of evaluation lacks the ability to compare to program or the department.

5. Course GPA:

AY15-1: 3.79
AY16-1: 3.74
AY16-2: 3.91
AY17-1: 3.65
AY17-2: 3.62
AY18-1: 3.67
AY19-1: 3.60
AY20-1: 3.63
AY21-1: 3.81; CH151 AVG: 3.71

6. TEE Grade:

AY15-1: 89.7%
AY16-1: 88.8%
AY16-2: 93.2%
AY17-1: 87.9%
AY17-2: 87.4%
AY18-1: 89.2%
AY19-1: 88.8%
AY20-1: 86.2%
AY21-1: 90.1% (ACS Only due to COVID, additional the questions were hand selected to ensure they were covered by course learning objectives)

7. Course Processes:

a. Textbooks
   1) Chemistry: Structure and Dynamics, Tro, 2nd Ed.

b. Course Content
   1) Atomic Structure
   2) Structure, Bonding, and States of Matter
   3) Stoichiometry, Chemical Reactions and Gases
   4) Thermodynamics and Military Applications of Explosives

c. Lessons and Labs

Block #1: STRUCTURE, BONDING, & STATES OF MATTER		Points
1	Chemistry Basics	
2	The Atom	
3	Atomic Number and Atomic Mass	
4	Electromagnetic Radiation/Spectroscopy	
5	Quantum Mechanical Model of the Atom	
6	Electron Configuration and Spin	
7	Atomic Forces **(EOH Quiz 1)**	**300**
8	Periodic Trends	

9	Basics of Ionic and Covalent Bonding	
10	Empirical Formula & Combustion Analysis	
11	Electronegativity and Partial Charge **(EOH Quiz 2)**	**300**
12	Lewis Structures and Covalent Bonds	
13	Enrichment 1: Steam Plant Tour	
14	Molecular Shape & Hybridization	
15	Intermolecular Forces	
	WPR I	**500**
	Block #2: STOICHIOMETRY/CHEM RXNS/THERMO/MIL APPS	
16	Chemical Reactions/Stoichiometry/Limiting Reactant	
17	Molecular Theory / Nomenclature Handout	
18	Solution Chemistry	
19	Solubility and Precipitation Reactions	
20	Acid/Base Neutralization Reactions **(EOH Quiz 3)**	**300**
21	Oxidation-Reduction Reactions	
22	Heat and Calorimetry	
23	Enthalpy of Reaction	
24	Ideal Gas Law	
25	Heating/Cooling Curves & Phase Diagrams **(EOH 4)**	**300**
26	Enrichment 3: Gas Laws	
27	IMFs II/Metallic Bonding/Solids	
	WPR II	**500**
28	Colligative Properties / WPR II (2-Day classes only)	
29	Colligative Properties / WPR II (1-Day classes only)	
30	Fundamental Chemistry of Explosives	

Lab	Title	Points
1	Drop	
2	Laboratory Techniques & Measuring Instruments	**125**
3	Spectrophotometry	**125**
4	Drop	
5	Drop	
6	Intermolecular Forces (IMFs) (Report Due Lab 7)	**160**
7	Stoichiometry (Report Due Lab 8)	**160**

8	Acid Base Titrations (Report Due Lab 9)	**160**
9	Calorimetry (Report Due Lab 10)	**160**
10	Gas Laws (Report Due NLT 07DEC)	**160**
	Instructor Grades: 1 per block (75pts each)	**150**
	Mastering Chemistry: 1 per block (150pts each)	**300**

d. Summary of Graded Requirements

#	Event	Per Event	Possible	% of Possible
1	TEE	1300	1300	26.0%
2	WPRs	500	1000	20.0%
4	Quizzes	300	1200	24.0%
7	Labs	2x125,5x160	1050	21.0%
2	Homework	150	300	6.0%
2	Instructor Grade	75	150	3.0%
		Total	**5000**	**100%**

Allocation of points were adjusted from last year. WPR's were reduced by one event, quizzes were reduced to four from five, but the points per event increased from 150 to 300 points. The final adjustment was the addition of homework grade for 300 points. The overall points remained at 5000.

e. Areas of Special Emphasis:

1) Required each Cadet to conduct one 5-minute 'Science of the Day' brief prior to class to encourage scientific literacy.

2) Continued to remove detailed lab instructions from the lab assignments to make the cadets do some critical thinking about what they had to do in lab prior to coming to the lab.

3) Made the first two labs notebook requirement as a non-graded turn in but graded them for the remaining labs so that cadets were forced to do some preparation prior to coming to lab.

4) Continued to give the same graded events as CH101.

5) Continued enrichments, which is the major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom.

8. Contribution to Program Outcomes:

a. Think scientifically by appropriating scientific language to deduce, predict, and explain scientific observations.
b. Explain and apply foundational chemistry concepts.
c. Develop practices for learning independently (intellectual self-reliance).
d. Build interactive practices to work within teams and lead peers (teamwork).

9. Resources and Laboratories

a. Laboratories: BH158
c. Physical Models and Demos: in addition to the resources available in the demo handbook, the course employed Styrofoam model kits
d. Technician Support: Mr. Treus provided excellent lab support, as always.
e. Supplies: Microlab was used in both classrooms and labs for analytical inquiry.
f. Additional Facilities: None

g. Unfunded Requests: None

10. Implementation of Recommendations from last AY:

a. Added Mastering Chemistry homework as a requirement.
b. Increased the points per quiz to give them a higher degree of importance.
c. Continued using the same grades events as CH101.

SECTION II. COURSE ASSESSMENT

1. Red Book Description: No recommended changes.

2. Enrollment: We increased from 88 to 92, which is approximately the maximum allowed for five instructors. **Recommend we return to 6 sections (~120 cadets)**

3. Course Objectives:

a. Coverage:

1) Atomic Structure: EOH1, Lab 3, WPR1
2) Structure and Bonding: EOH2, WPR2
3) States of Matter: WPR1, Lab 6
4) Stoichiometry, Chemical Reactions: Labs 7, 7, EOH3, WPR2
5) Thermodynamics and Explosives: Labs 8-9, EOH4, WPR2

b. Performance:

Course Objective	Assessment Tool	AY21-1
Atomic Structure: Cadets can interpret the periodic table by using atomic structure to explain atomic properties (like ionization energy and electronegativity).	Quiz 1	92.25%
	Lab 3	96.05%
	WPR I	91.97%
Structure and Bonding: Cadets can construct Lewis structures for molecules from atoms and relate those structures and shapes to intermolecular (nonbonding) forces.	Quiz 2	96.42%
	WPR I	91.97%
States of Matter: Cadets can describe the intramolecular (bonding) and intermolecular (nonbonding) forces within substances in their various physical states (e.g. solid, liquid, gas, and aqueous).	Lab 6	92.79%
	WPR I	91.97%
Stoichiometry, Chemical Reactions: Cadets can construct and balance chemical reactions, to include acid-base and combustion reactions and relate and calculate amounts of products and reactants in chemical reactions.	Lab 7	91.06%
	Quiz 3	96.51%
	WPR II	90.31%
Thermodynamics and Explosives: Cadets can describe heat transfers both qualitatively (conceptually) and quantitatively (using calculations) and relate changes in enthalpy of reactions to energy loss/gain and the bonding and nonbonding forces in reactants and products using enthalpies of atom combination as a reference state.	Lab 8-9	91.56%
	Quiz 4	92.47%
	WPR II	90.31%

4. Survey Questions:

a. Survey Results: Unfortunately, the survey system was changed, resulting in a loss of historical data. Now the only item that can be reported is the average answer based out of 5. 5=SA, 4=A, 3=N, 2=D, 1=SD. (SA: Strongly Agree, A: Agree, N: Neutral, D: Disagree, SD: Strongly Disagree)

1) My personal schedule allows me enough time to reflect on the material I have learned in class. (3.89)

2) My personal schedule allows me enough time to adequately prepare for my optimum academic performance. (3.97)

3) The course text and references were very helpful in my gaining understanding of the course material. (4.07)

4) As a result of this course, my written and verbal communication skills have improved. (Not asked in AY21-1)

5. Course QPA: The QPA was higher than average. This is a result of three key items in my opinion, the addition of Mastering Chemistry, which encourages Cadets to finish their homework. Secondly graded events (EOHs and WPRs) were over 90% from Mastering Chemistry which aided the cadets as the questions were similar to their homework. Third, this year the cadets were offered a 75 point bonus.

6. TEE Grade: The TEE was only the ACS exam, which was itself edited to ensure only questions asked were material covered during the course.

7. Course Processes

a. Textbook: The Tro text is appropriate for the course.

b. Course Content: Lesson sequence was modified slightly to align with Tro text.

c. Lessons and labs: Lessons are appropriate with the addition of the acid base lab.

d. Summary of Graded Requirements: Graded events in my opinion have become too easy for CH151 due to use of Mastering Chemistry test questions.

e. Areas of Special Emphasis:

1) Continue to make labs more challenging than CH101 be requiring the Cadets to develop their own procedures to assist cadets as they learn to develop critical thinking skills.

2) Continue to use lab notebooks as it forces cadets to do some preparation and critical thinking prior to coming to lab

3) For AY22-1, create CH151 specific EOHs, and WPRs, while using the same TEE as CH101.

4) Continue enrichments, which is the major thing that separates CH151 from CH101 and provides cadets hands on experience with what they are learning in the classroom.

8. Resources and Laboratories

a. Laboratories: No issues.

b. Computer Labs: N/A

c. Physical Models & Demos: No issues.

d. Technician Support: Excellent support, as always.

e. Supplies: No issues.

f. Additional Facilities: N/A

g. Unfunded Requests: N/A

SECTION III. RECOMMENDED CHANGES

1. Red Book Description: No change.

2. Enrollment: Continue this course in the fall only. Continue selecting only the top cadets into this course. If there is capacity to do so, increase the number of sections to accommodate more cadets in the fall due to dropping the spring semester but as seen from this year, 5 sections (~90 cadet) should be increased to 6 sections to maintain the model of approximately 10% of cadets take CH151.

3. Course Objectives:

a. Coverage: No recommended changes.

b. Performance: No recommended changes. Cadet performed as expected and don't believe they have enough time to prepare for class.

4. Survey Questions:

a. Survey Results: Do not change them from year to year, the grading is now hard to compare.

b. Survey Freeform Comments: No recommended changes. Continue to ask the same things to gather data over multiple years before making any changes.

5. Course QPA: The QPA was up slightly from the previous academic year. Suggest creating harder EOHs and WPRs specific to CH151.

6. TEE Grade: Keep ACS only, it is a national standard which we can judge our departments performance.

7. Course Process

a. Textbook: The Tro text is the appropriate text for this course and should be the only text.

b. Course Content: Recommend keeping the Analytical enrichment as an enrichment instead of a lab. It is a good a way to showcase the department as the lab does not fit the course content.

c. List of lessons and labs: No change to lessons and labs. Keep the Analytical enrichment as that and not a lab but find a new lab to replace it.

d. Summary of Graded Requirements: Update EOHs and WPRs to be more challenging than CH101

e. Areas of Special Emphasis: No recommended changes.

8. Contribution to Program Outcomes

a. Coverage: No recommended changes.

b. Performance: No recommended changes.

9. Resources and Laboratories

a. Laboratories: No issues. Continue to use BH158.

b. Computer Labs: N/A

c. Physical Models & Demos: No issues.

d. Technician Support: No issues.

e. Supplies: No issues.

f. Additional Facilities: N/A

g. Unfunded Requests: N/A

SECTION IV. GRADED EVENTS AND SUPPLEMENTAL MATERIAL

1. WPRs, Quizzes and Labs:
https://usarmywestpoint.sharepoint.com/sites/cls.gradedevents/SitePages/Home.aspx

2. Supplemental Material:
https://usarmywestpoint.sharepoint.com/sites/cls.courses/SitePages/Home.aspx

3. Other Resources: N/A

4. Other historical information:
https://usarmywestpoint.sharepoint.com/sites/cls.courses/SitePages/Home.aspx

SECTION II

EXAMINATIONS

UNITED STATES MILITARY ACADEMY
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TAB G WPR #1

WPR1 - Handouts

CADET ______________________________ SECTION/HR ______________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 151, AY26-1
Written Partial Review I
10 October 2025
75 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Edition
SCOPE: Lessons 1-17

Authorized References
TI-36X Pro Calculator (or equivalent) & 13th Edition RDC **(No additional markings or notes)**

1. Do not mark on this exam or open it until "begin work" is given.

2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly and answer all questions
   b. ***Show all work***
   c. Use complete sentences for all explanations

3. There are five numbered pages in this WPR, not including this cover page. Report any illegible or missing pages to your instructor.

4. Cadets are not authorized to discuss the WPR with other cadets until it has been returned.

Sign to Acknowledge: ___________________________

(TOTAL WEIGHT 500 POINTS)

THIS SPACE FOR INSTRUCTOR USE ONLY			
PAGE	**MAX**	**CUT**	**POINTS**
1	**95**		
2	**105**		
3	**135**		
4	**165**		
5	**100**		
		TOTAL SCORE	**/600**

Page	Weight
1	95

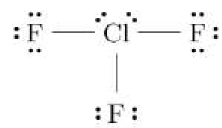
_____ **1.** (15 pts) Which statement best describes how Coulomb's Law (in terms of potential energy) explains the trend in first ionization energy across a period?

A. As nuclear charge increases, potential energy increases, so ionization energy decreases.
B. As nuclear charge increases, potential energy decreases, so ionization energy increases.
C. As atomic size increases, distance *r* increases, so ionization energy increases.
D. Coulomb's Law only applies to ionic compounds, not ionization energy.

_____ **2.** (15 pts) Determine the number of protons and neutrons in the isotope $^{226}_{88}Ra$.

A. $p^{+} = 226$ $n^{\circ} = 88$
B. $p^{+} = 88$ $n^{\circ} = 226$
C. $p^{+} = 88$ $n^{\circ} = 138$
D. $p^{+} = 138$ $n^{\circ} = 88$

_____ **3.** (20 pts) Consider the molecule below and determine the electron geometry, molecular geometry, hybridization, and whether the bond angle is ideal around the central atom.



A. **Electron Geometry:** trigonal bipyramidal
Molecular Geometry: t-shaped
Hybridization: sp^3d
Ideal Bond Angle: No

B. **Electron Geometry:** trigonal bipyramidal
Molecular Geometry: t-shaped
Hybridization: sp^3d
Ideal Bond Angle: Yes

C. **Electron Geometry:** t-shaped
Molecular Geometry: trigonal bipyramidal
Hybridization: sp^2d
Ideal Bond Angle: No

D. **Electron Geometry:** trigonal planar
Molecular Geometry: trigonal planar
Hybridization: sp^2
Ideal Bond Angle: Yes

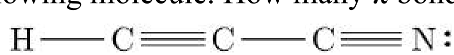
_____ **4.** (15 pts) Select the chemical formula for the compound that forms between magnesium and chlorine.

A. $MgCl$
B. $MgClO_4$
C. $MgCl_2$
D. Mg_2Cl

_____ **5.** (20 pts) Given the following electron configuration for phosphorus, determine the number of core electrons (e-) and valence electrons: $1s^22s^22p^63s^23p^3$.

A. 5 core e^-, 10 valence e^-
B. 10 core e^-, 5 valence e^-
C. 3 core e^-, 12 valence e^-
D. 12 core e^-, 3 valence e^-

_____ **6.** (10 pts) Consider the following molecule. How many π bonds are present?



A. 3 π bonds
B. 2 π bonds
C. 4 π bonds
D. 8 π bonds

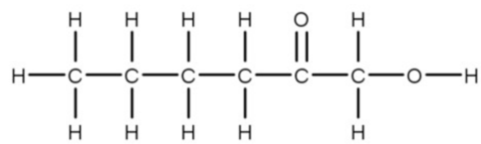
_____ **7.** (15 pts) The electronegativity is 2.1 for H and 3.0 for Cl. Based on these electronegativities, HCl would be expected to

A. have nonpolar covalent bonds and contain H- ions.
B. have ionic bonds and contain H+ ions.
C. have nonpolar covalent bonds with a partial negative charges on the H atoms.
D. have polar covalent bonds with a partial positive charges on the H atoms.

_____ **8.** (15 pts) Determine the type of bond that forms between phosphorus and chlorine.

A. Ionic
B. Metallic
C. Covalent
D. Network

_____ **9.** (20 pts) Which intermolecular forces are present among molecules with the following structure?



A. dispersion
B. dipole-dipole
C. hydrogen bonding
D. ion-dipole
E. A and B
F. A, B, and C
G. all the above
H. none of the above

_____ **10.** (15 pts) Determine the name for $CuSO_4 \cdot 5\ H_2O$

A. copper sulfite hydrate
B. copper(II) sulfide aqueous
C. copper(I) sulfate pentahydrate
D. copper(II) sulfate pentahydrate

_____ **11.** (10 pts) What is a possible set of quantum numbers for an electron in a strontium atom?

A. $n = 5$, $l = 0$, $m_l = 0$, $m_s = +{}^1/_2$

B. $n = 3$, $l = 3$, $m_l = 3$, $m_s = 1$

C. $n = 1$, $l = 0$, $m_l = 1$, $m_s = +{}^1/_2$

D. $n = 6$, $l = 0$, $m_l = 2$, $m_s = -{}^1/_2$

_____ **12.** (15 pts) Which is the larger species, N or N^{3-} ?

A. N
B. N^{3-}
C. Both species are the same size
D. Cannot determine from this information

_____ **13.** (15 pts) Which is the larger species, Cr or Cr^{3+}?

A. Cr
B. Cr^{3+}
C. Both species are the same size
D. Cannot determine from this information

Page	Weight
3	135

13. (70 pts) A 35.00-g sample of a simple hydrocarbon undergoes complete combustion, producing 792.18 g of carbon dioxide and 270.24 g of water. Determine the empirical formula.

14. (30 pts) The Hycon 73-B was a film camera that was mounted in the U-2 spy plane that observed U.S.S.R. R-12 nuclear missiles in Cuba during the Cuban Missile Crisis. Assuming the average wavelength of visible light from the Sun is 555 nm, calculate the energy per photon hitting the film.

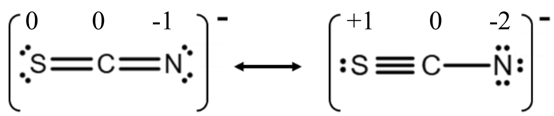
15. Consider the following four molecules:

CH_3CH_2H CH_3CH_2SH $CH_3CH_2CH_3$ $CH_3CH_2CH_2OH$

a. (10 pts) Circle the molecule with the highest boiling point.

b. (25 pts) Explain why the molecule you selected has the highest boiling point.

16. (30 pts) Given the formal charges below, explain which of the provided resonance structures for SCN^- will contribute most to the structure of SCN^-.



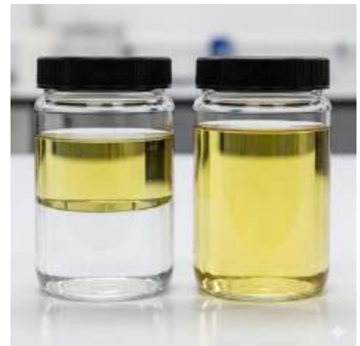
17. (60 pts) Draw the most appropriate Lewis structure for dichloromethane (CH_2Cl_2).

18. (15 pts) Calculate the mass percent composition of lithium in Li_3PO_4.

19. (60 pts) The Chernobyl disaster released a large amount of a certain radioactive cesium isotope, raising the average atomic mass of cesium in the surrounding area to 133.4 amu. Using the data below, calculate the atomic mass (variable x) of the released radioactive cesium isotope.

Isotope	**Mass (amu)**	**Abundance (%)**
Cs-133	132.905	87.82%
Cs-?	x	12.28%

20. Background: Aviation fuel checks rely on the immiscibility of water and jet fuel to allow for visual detection of any water contamination present in a fuel tank. *See image on right for example of contaminated (on left) and clean (on right) fuel checks.* **A researcher proposes** to use CH_3CH_2OH (Ethanol) as an alternative fuel source for Army helicopters.



a. (25 pts) **Draw** a Lewis Dot Structure for use CH_3CH_2OH (Ethanol).

b. (30 pts) **List** all Intermolecular Forces for both *Ethanol* and *Water.*

Ethanol IMFs:

Water IMFs:

c. (25 pts) Is the existing fuel check process feasible for an ethanol-based fuel?

A. **Yes**, because ethanol **is** miscible in water.

B. **Yes**, because ethanol **is not** miscible in water.

C. **No**, because ethanol **is** miscible in water.

D. **No**, because ethanol **is not** miscible in water

d. (20 pts) **Propose** an alternate fuel check method using concepts learned in this course or elsewhere. *Answer this question using only the space provided below.*

WPR1 - Instructor Solution

Approved Solutions

CADET ______________________________ SECTION/HR ______________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 151, AY26-1
Written Partial Review I
10 October 2025
75 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Edition
SCOPE: Lessons 1-17

Authorized References
TI-36X Pro Calculator (or equivalent) & 13th Edition RDC **(No additional markings or notes)**

1. Do not mark on this exam or open it until "begin work" is given.
2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly and answer all questions
   b. ***Show all work***
   c. Use complete sentences for all explanations
3. There are five numbered pages in this WPR, not including this cover page. Report any illegible or missing pages to your instructor.
4. Cadets are not authorized to discuss the WPR with other cadets until it has been returned.

Sign to Acknowledge: ___________________________

(TOTAL WEIGHT 500 POINTS)

THIS SPACE FOR INSTRUCTOR USE ONLY			
PAGE	**MAX**	**CUT**	**POINTS**
1	**90**		
2	**100**		
3	**135**		
4	**165**		
5	**100**		
		TOTAL SCORE	**/600**

Page	Weight
1	90

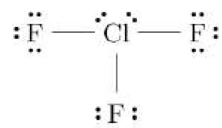
B ____ **1.** (15 pts) Which statement best describes how Coulomb's Law (in terms of potential energy) explains the trend in first ionization energy across a period?

A. As nuclear charge increases, potential energy increases, so ionization energy decreases.
B. As nuclear charge increases, potential energy decreases, so ionization energy increases.
C. As atomic size increases, distance *r* increases, so ionization energy increases.
D. Coulomb's Law only applies to ionic compounds, not ionization energy.

C ____ **2.** (15 pts) Determine the number of protons and neutrons in the isotope ${}^{226}_{88}Ra$.

A. $p^+ = 226$ $n° = 88$
B. $p^+ = 88$ $n° = 226$
C. $p^+ = 88$ $n° = 138$
D. $p^+ = 138$ $n° = 88$

A. ____ **3.** (20 pts) Consider the molecule below and determine the electron geometry, molecular geometry, hybridization, and whether the bond angle is ideal around the central atom.



A. **Electron Geometry:** trigonal bipyramidal
Molecular Geometry: t-shaped
Hybridization: sp^3d
Ideal Bond Angle: No

B. **Electron Geometry:** trigonal bipyramidal
Molecular Geometry: t-shaped
Hybridization: sp^3d
Ideal Bond Angle: Yes

C. **Electron Geometry:** t-shaped
Molecular Geometry: trigonal bipyramidal
Hybridization: sp^2d
Ideal Bond Angle: No

D. **Electron Geometry:** trigonal planar
Molecular Geometry: trigonal planar
Hybridization: sp^2
Ideal Bond Angle: Yes

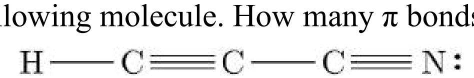
C. ____ **4.** (10 pts) Select the chemical formula for the compound that forms between magnesium and chlorine.

A. $MgCl$
B. $MgClO_4$
C. $MgCl_2$
D. Mg_2Cl

B. ____ **5.** (20 pts) Given the following electron configuration for phosphorus, determine the number of core electrons (e-) and valence electrons: $1s^22s^22p^63s^23p^3$.

A. 5 core e^-, 10 valence e^-
B. 10 core e^-, 5 valence e^-
C. 3 core e^-, 12 valence e^-
D. 12 core e^-, 3 valence e^-

C. ____ **6.** (10 pts) Consider the following molecule. How many π bonds are present?



A. 3 π bonds
B. 2 π bonds
C. 4 π bonds
D. 8 π bonds

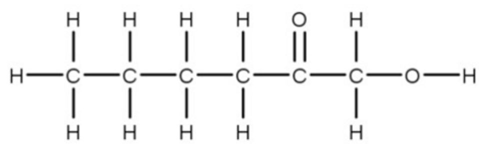
D
______ **7.** (15 pts) The electronegativity is 2.1 for H and 3.0 for Cl. Based on these electronegativities, HCl would be expected to

A. have nonpolar covalent bonds and contain H- ions.
B. have ionic bonds and contain H+ ions.
C. have nonpolar covalent bonds with a partial negative charges on the H atoms.
D. have polar covalent bonds with a partial positive charges on the H atoms.

C.
______ **8.** (10 pts) Determine the type of bond that forms between phosphorus and chlorine.

A. Ionic
B. Metallic
C. Covalent
D. Network

F.
______ **9.** (20 pts) Which intermolecular forces are present among molecules with the following structure?



A. dispersion
B. dipole-dipole
C. hydrogen bonding
D. ion-dipole
E. A and B
F. A, B, and C
G. all the above
H. none of the above

D.
______ **10.** (15 pts) Determine the name for $CuSO_4 \cdot 5\ H_2O$

A. copper sulfite hydrate
B. copper(II) sulfide aqueous
C. copper(I) sulfate pentahydrate
D. copper(II) sulfate pentahydrate

A.
______ **11.** (10 pts) What is a possible set of quantum numbers for an electron in a strontium atom?

A. $n = 5$, $l = 0$, $m_l = 0$, $m_s = +^1/_2$
B. $n = 3$, $l = 3$, $m_l = 3$, $m_s = 1$
C. $n = 1$, $l = 0$, $m_l = 1$, $m_s = +^1/_2$
D. $n = 6$, $l = 0$, $m_l = 2$, $m_s = -^1/_2$

B.
______ **12.** (15 pts) Which is the larger species, N or N^{3-} ?

A. N
B. N^{3-}
C. Both species are the same size
D. Cannot determine from this information

A.
______ **13.** (15 pts) Which is the larger species, Cr or Cr^{3+}?

A. Cr
B. Cr^{3+}
C. Both species are the same size
D. Cannot determine from this information

13. (70 pts) A 35.00-g sample of a simple hydrocarbon undergoes complete combustion, producing 792.18 g of carbon dioxide and 270.24 g of water. Determine the empirical formula.

792.18 g CO_2	1 mol CO_2	1 mol C	= **18.00 mol C**
	44.01 g CO_2	1 mol CO_2	

270.24 g H_2O	1 mol H_2O	2 mol H	= **30.00 mol H**
	18.016 g H_2O	1 mol H_2O	

$C_{18.00}\, H_{30.00}$

$C_{\frac{18.00}{18.00}}\, H_{\frac{30.00}{18.00}}$

$C_1\, H_{1.666} \times 3$

$C_3\, H_5$

14. (30 pts) The Hycon 73-B was a film camera that was mounted in the U-2 spy plane that observed U.S.S.R. R-12 nuclear missiles in Cuba during the Cuban Missile Crisis. Assuming the average wavelength of visible light from the Sun is 555 nm, calculate the energy per photon hitting the film.

$E_{ph} = \frac{hc}{\lambda}$

E_{ph} =	6.626×10^{-34} J·s	2.998×10^{8} m		1 nm	= 3.5577×10^{-19} J/photon
	photon	s	555 nm	1×10^{-9} m	

3.56×10^{-19} J/photon

15. Consider the following four molecules:

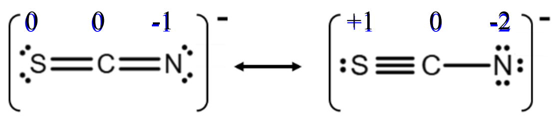
CH_3CH_2H CH_3CH_2SH $CH_3CH_2CH_3$ $CH_3CH_2CH_2OH$

a. (10 pts) Circle the molecule with the highest boiling point.

b. (25 pts) Explain why the molecule you selected has the highest boiling point.

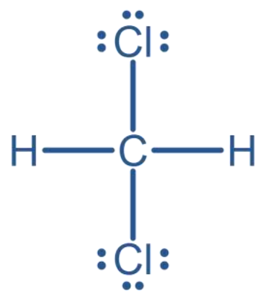
$CH_3CH_2CH_2OH$ has the highest boiling point because it includes hydrogen bonding, the strongest IMF among the four given molecules.

16. (30 pts) Given the formal charges below, explain which of the provided resonance structures for SCN^{-} will contribute most to the structure of SCN^{-}.



The structure on the left will contribute the most to the structure of SCN^{-} because it minimizes formal charges on the atoms.

17. (60 pts) Draw the most appropriate Lewis structure for dichloromethane (CH_2Cl_2).



18. (15 pts) Calculate the mass percent composition of lithium in Li_3PO_4.

$$mass\ \%\ (Li) = \frac{mass\ Li}{mass\ Li_3PO_4} \times 100\%$$

$$= \frac{(3 \times 6.941)\ \text{g/mol}}{115.793\ \text{g/mol}} \qquad = \frac{20.823\ \text{g/mol}}{115.793\ \text{g/mol}} \times 100\% \qquad = 17.98\%$$

19. (60 pts) The Chernobyl disaster released a large amount of a certain radioactive cesium isotope, raising the average atomic mass of cesium in the surrounding area to 133.4 amu. Using the data below, calculate the atomic mass (variable x) of the released radioactive cesium isotope.

Isotope	**Mass (amu)**	**Abundance (%)**
Cs-133	132.905	87.82%
Cs-?	x	12.28%

Atomic mass = $[(\text{mass}_{Cs\text{-}133} \times \text{abundance}_{Cs\text{-}133}) + (x \times \text{abundance}_{Cs\text{-}?})]$

$$x = \frac{average\ atomic\ mass - (mass_{Cs-133} \times abundance_{Cs-133})}{abundance_{Cs-?}}$$

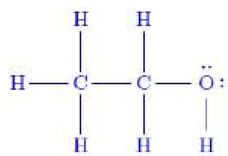
$$= \frac{133.4\ amu - (132.905\ amu\ \times 0.8782)}{0.1228}$$

= 135.853 amu

= 135.9 amu

20. Background: Aviation fuel checks rely on the immiscibility of water and jet fuel to allow for visual detection of any water contamination present in a fuel tank. **A researcher proposes** to use CH_3CH_2OH (Ethanol) as an alternative fuel source for Army helicopters. *See images below for example of contaminated (on left) and clean (on right) fuel checks.*

a. (25 pts) **Draw** a Lewis Dot Structure for use CH_3CH_2OH (Ethanol).



+10 Skeletal Structure
+5 Octet Rule
+5 Formal Charge minimized
+5 No resonance structures

b. (30 pts) **List** all Intermolecular Forces for both *Ethanol* and *Water.*

Ethanol IMFs:	**Water IMFs:**
Dispersion	(Same for Water)
Dipole-Dipole	+10 for each correct IMF
Hydrogen Bonding	-10 for each EXTRA IMF

c. (25 pts) Is the existing fuel check process feasible for an ethanol-based fuel?

+25 AON

a. **Yes**, because ethanol **is** miscible in water.

b. **Yes**, because ethanol **is not** miscible in water.

c. **No**, because ethanol **is** miscible in water.

d. **No**, because ethanol **is not** miscible in water

d. (20 pts) **Propose** an alternate fuel check method using concepts learned in this course or elsewhere. *Answer this question using only the space provided below.*

Full credit for any answer that makes sense. Some possible answers: Specific Gravity/Density or Spectrophotometry

TAB H WPR #2

WPR2 - Handouts

CADET ______________________________ SECTION/HR ______________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 151, AY26-1
Written Partial Review II
21 November 2025
75 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Edition
SCOPE: Lessons 1-30

Authorized References
TI-36X Pro Calculator (or equivalent) & 13th Edition RDC **(No additional markings or notes)**

1. Do not mark on this exam or open it until "begin work" is given.
2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly and answer all questions
   b. ***Show all work***
   c. Use complete sentences for all explanations
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Sign to Acknowledge: ___________________________

(TOTAL WEIGHT 600 POINTS)

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PAGE	MAX	CUT	POINTS
1	75		
2	75		
3	175		
4	175		
5	100		
		TOTAL SCORE	/600

Choose the most appropriate answer for each of the following multiple-choice questions:

_____ **1.** (15 pts) Write a balanced chemical equation for the reaction between solid iron(III) oxide and hydrogen gas to form solid iron and liquid water.

A. $FeO(s) + H_2(g) \rightarrow Fe(s) + H_2O(l)$
B. $Fe_3O_2(s) + 4\ H_2(g) \rightarrow 3\ Fe^{3+}(aq) + 2\ H_2O(l)$
C. $Fe_2O_3(s) + H_2(g) \rightarrow Fe(s) + H_2O(l)$
D. $Fe_2O_3(s) + 3\ H_2(g) \rightarrow 2\ Fe(s) + 3\ H_2O(l)$
E. $Fe_2O_3(s) + 6\ H(g) \rightarrow 2\ Fe(s) + 3\ H_2O(l)$

_____ **2.** (15 pts) If 2.75 L of a 4.8 M $SrCl_2$ solution is diluted to 32 L, what is the molarity of the diluted solution?

A. 0.018 M
B. 0.41 M
C. 0.00041 M
D. 56 M
E. 0.24 M

_____ **3.** (10 pts) Choose the statement below that is FALSE.

A. Weak electrolytes partially dissociate into ions in water.
B. Strong electrolytes completely dissociate into ions in water.
C. A strong acid solution consists of only partially ionized acid molecules.
D. Nonelectrolyte solutions do not conduct electricity.

_____ **4.** (10 pts) Which compound listed below is soluble in water?

A. K_3PO_4
B. $Cu_3(PO_4)_2$
C. AgI
D. $CoCO_3$
E. $Fe(OH)_3$

_____ **2.** (10 pts) Give the complete ionic equation for the reaction (if any) that occurs when aqueous solutions of K_2S and $Fe(NO_3)_2$ are mixed.

A. $K^+(aq) + NO_3^-(aq) \rightarrow KNO_3(s)$
B. $2\ K^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow FeS(s) + 2\ K^+(aq) + 2\ NO_3^-(aq)$
C. $2\ K^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ KNO3(s)$
D. $Fe^{2+}(aq) + S^{2-}(aq) \rightarrow FeS(s)$
E. No reaction occurs.

_____ **3.** (15 pts) Determine the oxidizing agent in the following reaction.

$$Ni(s) + 2\ AgClO4(aq) \rightarrow Ni(ClO4)2(aq) + 2\ Ag(s)$$

A. $AgClO4(aq)$
B. $Ni(s)$
C. $Ag(s)$
D. $Ni(ClO4)2(aq)$
E. This is not an oxidation-reduction reaction

_____ **4.** (10 pts) A voltaic cell

A. produces electrical current from a spontaneous chemical reaction.
B. produces electrical current from electricity.
C. consumes electrical current to drive a spontaneous chemical reaction.
D. produces electrical current from a nonspontaneous chemical reaction.
E. consumes electrical current to drive a nonspontaneous chemical reaction.

_____ **5.** (15 pts) Which of the following statements correctly explains why the breaking of chemical bonds is an endothermic process?

A. Breaking bonds releases potential energy, making it exothermic.
B. Breaking bonds requires energy to overcome attractive forces, making it endothermic.
C. Breaking bonds releases kinetic energy as heat, making it endothermic.
D. Breaking bonds requires energy to decrease potential energy of the system.

_____ **6.** (15 pts) For the generic reactions below, determine the value of ΔH_2 in terms of ΔH_1.

$$2\,A \rightarrow 2\,B + 4\,C \qquad \Delta H_1$$
$$B + 2\,C \rightarrow A \qquad \Delta H_2 = ?$$

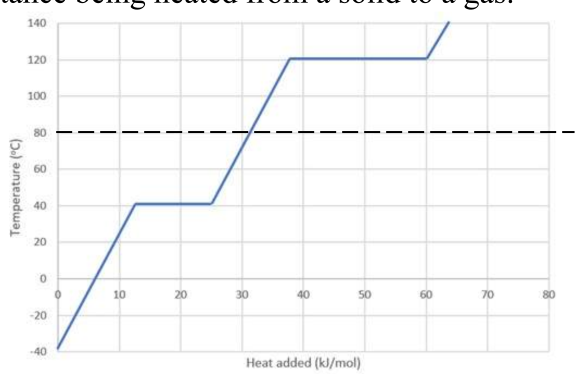
A. $-\Delta H_1$ **C.** $-\frac{1}{2}\,\Delta H_1$ **E.** $-2\,\Delta H_1$
B. $2\,\Delta H_1$ **D.** $\frac{1}{2}\,\Delta H_1$

_____ **7.** (10 pts) In all electrochemical cells, the electrode where oxidation occurs is the anode.

A. True **B.** False

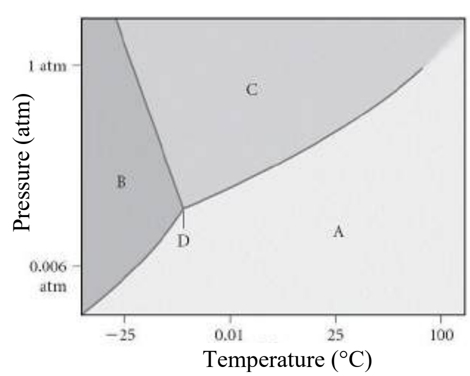
_____ **8.** (10 pts) Consider the heating curve below for a substance being heated from a solid to a gas. Which state of matter exists at 80°C (dashed line)?

A. solid
B. liquid
C. gas
D. plasma
E. liquid crystal



_____ **9.** (15 pts) Assign the appropriate labels to the phase diagram shown below.

A. A = liquid, B = solid, C = gas, D = critical point
B. A = gas, B = solid, C = liquid, D = triple point
C. A = gas, B = liquid, C = solid, D = supercritical fluid
D. A = solid, B = gas, C = liquid, D = triple point
E. A = liquid, B = gas, C = solid, D = critical point



13. A Soldier microwaves a rice-filled cloth bag to use as a reusable hand warmer. The rice bag has a mass of 500.0 g, and a specific heat capacity of approximately 1.6 J/g·°C. The Soldier heats the rice bag from 25.0°C to 45.0°C in the microwave, then immediately uses it.

a. (30 pts) Calculate the amount of heat (in J) absorbed by the rice bag from the microwave.

Answer 13a: ______________________

b. (70 pts) Calculate the final temperature (in °C) of the Soldier's hands if the heated rice bag transfers all its heat energy to the hands. *Assume the following: the mass, initial temperature, and specific heat capacity of the hands is 400.0 g, 25.0°C, and 3.5 J/g·°C, respectively; no heat is lost to the environment.*

Answer 13b: ______________________

14. Many commercial hand warmers rely on the following exothermic reaction to generate heat:

$$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$$

a. (50 pts) Calculate the standard enthalpy of reaction, $\Delta H°_{rxn}$, using standard enthalpies of formation and the balanced equation above.

Answer 14a: ______________________

b. (25 pts) If 1 mole of iron results in $\Delta H_{rxn} = -412.1$ kJ/mol, calculate the heat, q (in kJ), generated by a handwarmer containing 100.0 g of iron.

Answer 14b: ______________________

15. Calcium phosphate is commonly used in bone grafts and dental applications. It can be prepared by mixing aqueous calcium chloride and aqueous sodium phosphate according to the following equation:

$$3\ CaCl_2(aq) + 2\ Na_3PO_4(aq) \rightarrow Ca_3(PO_4)_2(s) + 6\ NaCl(aq)$$

a. (15 pts) Identify whether the reaction above is an acid-base, redox, or precipitation reaction.

Answer 15a: ______________________

b. (80 pts) Calculate the theoretical yield (in g) of calcium phosphate if a lab technician mixes 200.0 mL of 0.250 M $CaCl_2$ with 150.0 mL of 0.150 M Na_3PO_4.

Answer 15b: ______________________

c. (30 pts) Calculate the percent yield if the recovered mass of calcium phosphate is 2.40 g.

Answer 15c: ______________________

16. Nitric acid is a strong acid that is widely used in the production of fertilizers and explosives.

a. (30 pts) **Calculate** the concentration of a nitric acid solution that has a pH is 3.625.

Answer 16a: ______________________

b. (20 pts) **Explain** why nitric acid's identity as an Arrhenius acid also explains why it is a strong electrolyte:

17. 100. mL of an aqueous hydrobromic acid solution is mixed with 55.0 mL of an aqueous rubidium hydroxide solution to demonstrate an acid-base neutralization reaction. The reaction is as follows:

$$HBr(aq) + RbOH(aq) \rightarrow H_2O(l) + RbBr(aq)$$

a. (15 pts) Write the net-ionic equation which would best describe the reaction above:

b. (20 pts) Calculate the pH of the HBr solution if the concentration is 4.13×10^{-2} M.

Answer 17b: ______________________

c. (40 pts) When 100. mL of the hydrobromic acid (4.13×10^{-2} M concentration) is mixed with 55.0 mL of the rubidium hydroxide solution (0.100 M concentration), is the resulting solution acidic or basic? ***Show your work.***

Answer 17c: ______________________

d. (25 pts) What is the pOH of the resulting solution?

Answer 17d: ______________________

WPR2 - Instructor Solution

Cut Scale

CADET ______________________________ SECTION/HR ______________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 151, AY26-1
Written Partial Review II
21 November 2025
75 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Edition
SCOPE: Lessons 1-30

Authorized References
TI-36X Pro Calculator (or equivalent) & 13th Edition RDC **(No additional markings or notes)**

1. Do not mark on this exam or open it until "begin work" is given.

2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly and answer all questions
   b. ***Show all work***
   c. Use complete sentences for all explanations

3. There are five numbered pages in this WPR, not including this cover page. Report any illegible or missing pages to your instructor.

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Sign to Acknowledge: ___________________________

(TOTAL WEIGHT 600 POINTS)

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PAGE	MAX	CUT	POINTS
1	75		
2	75		
3	175		
4	175		
5	100		
		TOTAL SCORE	/600

Page	Weight
1	75

Choose the most appropriate answer for each of the following multiple-choice questions:

_D.__ **1.** (15 pts) Write a balanced chemical equation for the reaction between solid iron(III) oxide and hydrogen gas to form solid iron and liquid water.

A. $FeO(s) + H_2(g) \rightarrow Fe(s) + H_2O(l)$ (-10, wrong rcts, not balanced)
B. $Fe_3O_2(s) + 4\,H_2(g) \rightarrow 3\,Fe^{3+}(aq) + 2\,H_2O(l)$ (-15)
C. $Fe_2O_3(s) + H_2(g) \rightarrow Fe(s) + H_2O(l)$ (-5, not balanced)
D. $Fe_2O_3(s) + 3\,H_2(g) \rightarrow 2\,Fe(s) + 3\,H_2O(l)$ (correct)
E. $Fe_2O_3(s) + 6\,H(g) \rightarrow 2\,Fe(s) + 3\,H_2O(l)$ (-5, monatomic H)

_B.__ **2.** (15 pts) If 2.75 L of a 4.8 M $SrCl_2$ solution is diluted to 32 L, what is the molarity of the diluted solution?

A. 0.018 M (-15)
B. 0.41 M (correct)
C. 0.00041 M (-5, L→mL)
D. 56 M (-10, switched V_1 and V_2)
E. 0.24 M (-15)

_C.__ **3.** (10 pts) Choose the statement below that is FALSE.

A. Weak electrolytes partially dissociate into ions in water.
B. Strong electrolytes completely dissociate into ions in water.
C. A strong acid solution consists of only partially ionized acid molecules. (correct AON)
D. Nonelectrolyte solutions do not conduct electricity.

_A.__ **4.** (10 pts) Which compound listed below is soluble in water?

A. K_3PO_4 (correct)
B. $Cu_3(PO_4)_2$ (-10)
C. AgI (-5 exception)
D. $CoCO_3$ (-10)
E. $Fe(OH)_3$ (-10)

_B__ **2.** (10 pts) Give the complete ionic equation for the reaction (if any) that occurs when aqueous solutions of K_2S and $Fe(NO_3)_2$ are mixed.

A. $K^+(aq) + NO_3^-(aq) \rightarrow KNO_3(s)$ (-10)
B. $2\,K^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\,NO_3^-(aq) \rightarrow FeS(s) + 2\,K^+(aq) + 2\,NO_3^-(aq)$ (correct)
C. $2\,K^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\,NO_3^-(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\,KNO3(s)$ (-5, prod)
D. $Fe^{2+}(aq) + S^{2-}(aq) \rightarrow FeS(s)$ (-5, net ionic)
E. No reaction occurs. (-10)

_A.__ **3.** (15 pts) Determine the oxidizing agent in the following reaction.

$$Ni(s) + 2\,AgClO4(aq) \rightarrow Ni(ClO4)2(aq) + 2\,Ag(s)$$

A. $AgClO4(aq)$ (correct)
B. $Ni(s)$ (-10 reducing agent)
C. $Ag(s)$ (-10 not whole species)
D. $Ni(ClO4)2(aq)$ (-15)
E. This is not an oxidation-reduction reaction (-15)

A. 4. (10 pts) A voltaic cell

A. produces electrical current from a spontaneous chemical reaction. (correct)
B. produces electrical current from electricity. (-10)
C. consumes electrical current to drive a spontaneous chemical reaction. (-5, "consumes")
D. produces electrical current from a nonspontaneous chemical reaction. (-5, "nonspontaneous")
E. consumes electrical current to drive a nonspontaneous chemical reaction. (-10)

B. 5. (15 pts) Which of the following statements correctly explains why the breaking of chemical bonds is an endothermic process?

A. Breaking bonds releases potential energy, making it exothermic. (-15)
B. Breaking bonds requires energy to overcome attractive forces, making it endothermic. (correct)
C. Breaking bonds releases kinetic energy as heat, making it endothermic. (-15)
D. Breaking bonds requires energy to decrease potential energy of the system. (-10)

C. 6. (15 pts) For the generic reactions below, determine the value of ΔH_2 in terms of ΔH_1.

$$2\,A \rightarrow 2\,B + 4\,C \qquad \Delta H_1$$
$$B + 2\,C \rightarrow A \qquad \Delta H_2 = ?$$

A. $-\Delta H_1$ (-5)
B. $2\,\Delta H_1$ (-10)
C. $-\frac{1}{2}\,\Delta H_1$ (correct)
D. $\frac{1}{2}\,\Delta H_1$ (-5)
E. $-2\,\Delta H_1$ (-5)

A. 7. (10 pts) In all electrochemical cells, the electrode where oxidation occurs is the anode.

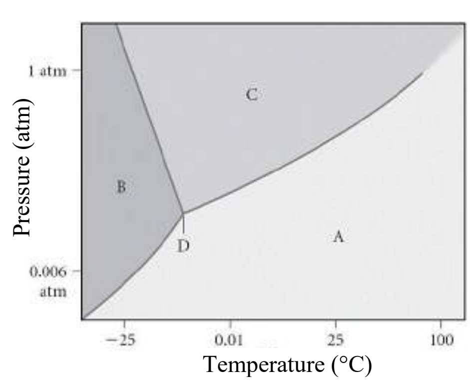
A. True (correct) **B.** False (-10)

B. 8. (10 pts) Consider the heating curve below for a substance being heated from a solid to a gas. Which state of matter exists at 80°C (dashed line)?

A. solid
B. liquid (correct, +10 AON)
C. gas
D. plasma
E. liquid crystal

B. 9. (15 pts) Assign the appropriate labels to the phase diagram shown below.

(-10) **A.** A = liquid, B = solid, C = gas, D = critical point
(crct) **B.** A = gas, B = solid, C = liquid, D = triple point
(-10) **C.** A = gas, B = liquid, C = solid, D = supercritical fluid
(-7) **D.** A = solid, B = gas, C = liquid, D = triple point
(-15) **E.** A = liquid, B = gas, C = solid, D = critical point



13. A Soldier microwaves a rice-filled cloth bag to use as a reusable hand warmer. The rice bag has a mass of 500.0 g, and a specific heat capacity of approximately 1.6 J/g·°C. The Soldier heats the rice bag from 25.0°C to 45.0°C in the microwave, then immediately uses it.

a. (30 pts) Calculate the amount of heat (in J) absorbed by the rice bag from the microwave.

$q_{rice} = m \times C_s \times \Delta T$ +15 concept

+4 mass, +4 C_s, +2 $T_f - T_i$, +1 EA M/U/S

500.0 g	1.6 J	20.0 °C
	g·°C	

= 16000 J (1.6×10^4 J)

+2 correct ΔT

Answer 13a: 16000 J

b. (70 pts) Calculate the final temperature (in °C) of the Soldier's hands if the heated rice bag transfers all its heat energy to the hands. *Assume the following: the mass, initial temperature, and specific heat capacity of the hands is 400.0 g, 25.0°C, and 3.5 J/g·°C, respectively; no heat is lost to the environment.*

$-q_{rice} = q_{hands}$ +25 concept

$-q_{rice} = m_{hands} \times C_{s,hands} \times (T_f - T_i)_{hands}$ +15 ($q = m \times C_s \times \Delta T$)

$T_f = \frac{-q_{rice}}{m_{hands} \times C_{s,hands}} + T_i$ +7 solves for T_f

+5 q_{rice}, +5 T_i

=(

-16000 J		g·°C
	400.0g	3.5 J

) + 25.0 °C

+5 m_{hands}, +5 $C_{s,\text{ hands}}$

Answer 13b: 36.4 °C

= 45 °C +1 EA M/U/S

14. Many commercial hand warmers rely on the following exothermic reaction to generate heat:

$$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$$

a. (50 pts) Calculate the standard enthalpy of reaction, ΔH°_{rxn}, using standard enthalpies of formation and the balanced equation above.

$\Delta H^\circ_{rxn} = \sum n_p \Delta H^\circ_f(products) - \sum n_r \Delta H^\circ_f(reactants)$ +20 concept

-10 S.C. uses alternate method to find H°_{rxn}
-15 S.C."reactants – products"

+4 mol Fe_2O_3, +4 mol Fe, +4 mol O_2

$= [(2\ mol)(-825.5\ kJ/mol)] - [(4mol)(0\ kJ/mol) + (3mol)(0\ kJ/mol)]$ *0 for "reactant" term may also be implied.

+5 $H^\circ_f(Fe_2O_3)$, +5 H°_f(Fe), +5 $H^\circ_f(O_2)$

= –1651 kJ +3 M/U/S (2/3)

Answer 14a: –1651 kJ

b. (25 pts) If 1 mole of iron results in $\Delta H_{rxn} = -412.1$ kJ/mol, calculate the heat, q (in kJ), generated by a handwarmer containing 100.0 g of iron.

$\Delta H_{rxn} = \frac{q_{rxn}}{mol\ Fe}$ +12 concept

$q_{rxn} = \Delta H_{rxn} \times mol\ Fe$ +4 application

Answer 14b: – 737.9 kJ

+2 ΔH_{rxn}, +2 mass Fe, +1 correct MW, +1EA M/U/S

=

- 412.1 kJ	100.0 g Fe	55.85 g Fe
1 mol		1 mol Fe

= – 737.869 kJ

+1 use of MW

Page	Weight
4	175

15. Calcium phosphate is commonly used in bone grafts and dental applications. It can be prepared by mixing aqueous calcium chloride and aqueous sodium phosphate according to the following equation:

$$3\ CaCl_2(aq) + 2\ Na_3PO_4(aq) \rightarrow Ca_3(PO_4)_2(s) + 6\ NaCl(aq)$$

a. (15 pts) Identify whether the reaction above is an acid-base, redox, or precipitation reaction.

Answer 15a: Precipitation Reaction

b. (80 pts) Calculate the theoretical yield (in g) of calcium phosphate if a lab technician mixes 200.0 mL of 0.250 M $CaCl_2$ with 150.0 mL of 0.150 M Na_3PO_4.

+25 concept (uses stoichiometry to determine amount of product)

+3 volume $CaCl_2$	+1 mL → L	+3 conc. $CaCl_2$	+3 uses mol ratio	+1 correct MW	
200.0 mL $CaCl_2$	1 L $CaCl_2$	0.250 mol $CaCl_2$	1 mol $Ca_3(PO_4)_2$	310.18 g $Ca_3(PO_4)_2$	= 5.169 g $Ca_3(PO_4)_2$
	1000 mL $CaCl_2$	L $CaCl_2$	3 mol $CaCl_2$	1 mol $Ca_3(PO_4)_2$	
			+3 correct mol ratio	+2 mol → g	

+3 volume Na_3PO_4	+1 mL → L	+3 conc. $CaCl_2$	+3 uses mol ratio	+1 correct MW	+1 EA M/U/S
150.0 mL Na_3PO_4	1 L Na_3PO_4	0.250 mol Na_3PO_4	1 mol $Ca_3(PO_4)_2$	310.18 g $Ca_3(PO_4)_2$	= 3.489 g $Ca_3(PO_4)_2$
	1000 mL Na_3PO_4	L Na_3PO_4	2 mol Na_3PO_4	1 mol $Ca_3(PO_4)_2$	
			+3 correct mol ratio	+2 mol → g	

+12 chooses lowest yield (FF); still award points if only did stoichiometry for one reactant

Answer 15b: 3.49 g $Ca_3(PO_4)_2$

c. (30 pts) Calculate the percent yield if the recovered mass of calcium phosphate is 2.40 g.

$$\%\ yield = \frac{actual\ yield}{theoretical\ yield} \times 100\%$$ +15 concept

+5 actual yield +1 EA M/U/S

$$\frac{2.40\ g\ Ca_3(PO_4)_2}{3.49\ g\ Ca_3(PO_4)_2} \times 100\% = 68.768\%$$

+5 theoretical yield +2 percentage

Answer 15c: 68.8% $Ca_3(PO_4)_2$

16. Nitric acid is widely used in the production of fertilizers and explosives.

a. (30 pts) **Calculate** the concentration of a nitric acid solution that has a pH is 3.625.

$pH = -\log[H_3O^+]$ +15 concept
$[H_3O^+] = 10^{-pH}$ +5 solves for $[H_3O^+]$
$[H_3O^+] = 10^{-3.625}$ +7 application
$[H_3O^+] = 2.371 \times 10^{-4}$ M +1EA M/U/S

Answer 16a: 2.37×10^{-4} M

b. (20 pts) **Explain** why nitric acid's identity as an Arrhenius acid also explains why it is a strong electrolyte:

Strong electrolytes dissociate completely in aqueous solution. Since sulfuric is an Arrhenius acid, it dissociates completely into H^+ and NO_3^- ions in aqueous solutions.

+10 discusses dissociation into H+ ions (Arrhenius acid)
+10 discusses complete dissociation of strong electrolytes

17. 100. mL of an aqueous hydrobromic acid solution is mixed with 55.0 mL of an aqueous rubidium hydroxide solution to demonstrate an acid-base neutralization reaction. The reaction is as follows:

$$HBr(aq) + RbOH(aq) \rightarrow H_2O(l) + RbBr(aq)$$

a. (15 pts) Write the net-ionic equation which would best describe the reaction above:

$H^+(aq) + OH^-(aq) \rightarrow H2O(l)$ (+15 for correct formula, -5 for missing state symbols)

b. (20 pts) Calculate the pH of the HBr solution if the concentration is 4.13×10^{-2} M.

$[HBr]=[H_3O^+]$ (+5 concept)
$pH=-Log[H_3O^+]$ (+5 concept)
$pH=-Log[4.13 \times 10^{-2}\ M]$ (+7 application)

Answer 17b: 1.384 (+3 M/U/S)

c. (40 pts) When 100. mL of the hydrobromic acid (4.13×10^{-2} M concentration), is mixed with 55.0 mL of the rubidium hydroxide solution (0.100M concentration) is the resulting solution be acidic or basic? ***Show your work.***

Applies Limiting Reactant concepts (+10 for concept)

HBr:

4.13×10^{-2} mol	100 mL	1 L	1 mol H2O
L		1000 mL	1 mol HBr

$=4.13 \times 10^{-3}$ mol H_2O (+10 finding common product/reactant for HBr)

RbOH:

0.100 mol	55 mL	1 L	1 mol H2O
L		1000 mL	1 mol RbOH

$=5.50 \times 10^{-3}$ mol H2O (+10 finding common product/reactant for RbOH)

This problem can be solved multiple ways. Key parts for points:
+10 finds a common reactant or H+ concentration using the HBr
+10 does the same but for OH- and RbOH
A successful answer compares these two values and correctly recognizes that the lesser is what will be consumed, and the solution will have a new pH determined by whatever is excess

Answer 17c: Basic (+10 follows from work)

d. (25 pts) What is the pOH of the resulting 155. mL solution?

Find Excess RbOH (+10, might have done this above):

4.13×10^{-3} mol	1 mol RbOH
	1 mol H_2O

$=4.13 \times 10^{-3}$ mol RbOH consumed

$5.50 \times 10^{-3} - 4.13 \times 10^{-3} = 1.37 \times 10^{-3}$ mol remaining

Find RbOH Concentration (+8):

$$\frac{1.37 \times 10^{-3}\ \text{mol}}{.155\ \text{L}}$$

$=8.84 \times 10^{-3}$ M

Find RbOH pOH:
1 mol RbOH = 1 mol OH^- (+2 concept)
$pOH= -log[OH^-]$ (+2 concept)
$pOH=-log(8.84 \times 10^{-3}\ M)$

Answer 17d: 2.054 (+3 M/U/S)

TAB I TERM END EXAMINATION

Term End Exam - Handouts

Name	**Version** (A) (B) (C) (D) (E)
ID	**Other**
Section	**Marking Instructions** Be sure to completely fill in the appropriate bubble. **Example** ● (B) (C) (D) (E)
Date	

	A	B	C	D	E		A	B	C	D	E		A	B	C	D	E		A	B	C	D	E
1	A	B	C	D	E	26	A	B	C	D	E	51	A	B	C	D	E	76	A	B	C	D	E
2	A	B	C	D	E	27	A	B	C	D	E	52	A	B	C	D	E	77	A	B	C	D	E
3	A	B	C	D	E	28	A	B	C	D	E	53	A	B	C	D	E	78	A	B	C	D	E
4	A	B	C	D	E	29	A	B	C	D	E	54	A	B	C	D	E	79	A	B	C	D	E
5	A	B	C	D	E	30	A	B	C	D	E	55	A	B	C	D	E	80	A	B	C	D	E
6	A	B	C	D	E	31	A	B	C	D	E	56	A	B	C	D	E	81	A	B	C	D	E
7	A	B	C	D	E	32	A	B	C	D	E	57	A	B	C	D	E	82	A	B	C	D	E
8	A	B	C	D	E	33	A	B	C	D	E	58	A	B	C	D	E	83	A	B	C	D	E
9	A	B	C	D	E	34	A	B	C	D	E	59	A	B	C	D	E	84	A	B	C	D	E
10	A	B	C	D	E	35	A	B	C	D	E	60	A	B	C	D	E	85	A	B	C	D	E
11	A	B	C	D	E	36	A	B	C	D	E	61	A	B	C	D	E	86	A	B	C	D	E
12	A	B	C	D	E	37	A	B	C	D	E	62	A	B	C	D	E	87	A	B	C	D	E
13	A	B	C	D	E	38	A	B	C	D	E	63	A	B	C	D	E	88	A	B	C	D	E
14	A	B	C	D	E	39	A	B	C	D	E	64	A	B	C	D	E	89	A	B	C	D	E
15	A	B	C	D	E	40	A	B	C	D	E	65	A	B	C	D	E	90	A	B	C	D	E
16	A	B	C	D	E	41	A	B	C	D	E	66	A	B	C	D	E	91	A	B	C	D	E
17	A	B	C	D	E	42	A	B	C	D	E	67	A	B	C	D	E	92	A	B	C	D	E
18	A	B	C	D	E	43	A	B	C	D	E	68	A	B	C	D	E	93	A	B	C	D	E
19	A	B	C	D	E	44	A	B	C	D	E	69	A	B	C	D	E	94	A	B	C	D	E
20	A	B	C	D	E	45	A	B	C	D	E	70	A	B	C	D	E	95	A	B	C	D	E
21	A	B	C	D	E	46	A	B	C	D	E	71	A	B	C	D	E	96	A	B	C	D	E
22	A	B	C	D	E	47	A	B	C	D	E	72	A	B	C	D	E	97	A	B	C	D	E
23	A	B	C	D	E	48	A	B	C	D	E	73	A	B	C	D	E	98	A	B	C	D	E
24	A	B	C	D	E	49	A	B	C	D	E	74	A	B	C	D	E	99	A	B	C	D	E
25	A	B	C	D	E	50	A	B	C	D	E	75	A	B	C	D	E	100	A	B	C	D	E

	A	B	C	D	E		A	B	C	D	E		A	B	C	D	E		A	B	C	D	E
101	A	B	C	D	E	126	A	B	C	D	E	151	A	B	C	D	E	176	A	B	C	D	E
102	A	B	C	D	E	127	A	B	C	D	E	152	A	B	C	D	E	177	A	B	C	D	E
103	A	B	C	D	E	128	A	B	C	D	E	153	A	B	C	D	E	178	A	B	C	D	E
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105	A	B	C	D	E	130	A	B	C	D	E	155	A	B	C	D	E	180	A	B	C	D	E
106	A	B	C	D	E	131	A	B	C	D	E	156	A	B	C	D	E	181	A	B	C	D	E
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110	A	B	C	D	E	135	A	B	C	D	E	160	A	B	C	D	E	185	A	B	C	D	E
111	A	B	C	D	E	136	A	B	C	D	E	161	A	B	C	D	E	186	A	B	C	D	E
112	A	B	C	D	E	137	A	B	C	D	E	162	A	B	C	D	E	187	A	B	C	D	E
113	A	B	C	D	E	138	A	B	C	D	E	163	A	B	C	D	E	188	A	B	C	D	E
114	A	B	C	D	E	139	A	B	C	D	E	164	A	B	C	D	E	189	A	B	C	D	E
115	A	B	C	D	E	140	A	B	C	D	E	165	A	B	C	D	E	190	A	B	C	D	E
116	A	B	C	D	E	141	A	B	C	D	E	166	A	B	C	D	E	191	A	B	C	D	E
117	A	B	C	D	E	142	A	B	C	D	E	167	A	B	C	D	E	192	A	B	C	D	E
118	A	B	C	D	E	143	A	B	C	D	E	168	A	B	C	D	E	193	A	B	C	D	E
119	A	B	C	D	E	144	A	B	C	D	E	169	A	B	C	D	E	194	A	B	C	D	E
120	A	B	C	D	E	145	A	B	C	D	E	170	A	B	C	D	E	195	A	B	C	D	E
121	A	B	C	D	E	146	A	B	C	D	E	171	A	B	C	D	E	196	A	B	C	D	E
122	A	B	C	D	E	147	A	B	C	D	E	172	A	B	C	D	E	197	A	B	C	D	E
123	A	B	C	D	E	148	A	B	C	D	E	173	A	B	C	D	E	198	A	B	C	D	E
124	A	B	C	D	E	149	A	B	C	D	E	174	A	B	C	D	E	199	A	B	C	D	E
125	A	B	C	D	E	150	A	B	C	D	E	175	A	B	C	D	E	200	A	B	C	D	E

CADET ________________________________ SECTION/HR ________________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 101, AY26-1
Term End Examination (Part II)
December 2025
Time: Not Less Than 100 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Ed
SCOPE: Lessons 01 – 39

<u>Authorized References</u>
TI-36X Pro Calculator or equivalent
Chemistry Reference Data Card (RDC) 13th Edition (**no additional markings**)

1. You may begin work on this portion of the TEE immediately upon receipt from your OIC.

2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly
   b. ***Show all work***
   c. Use complete sentences for all explanations
   d. Answer all questions

3. There are 6 lettered pages in this TEE. Report any illegible or missing pages to the OIC in the hallway. Write your name on every page in the spaces provided.

Sign to Acknowledge: __________________________

(TOTAL WEIGHT 900 POINTS)

THIS SPACE FOR INSTRUCTOR USE ONLY				
PAGE		**MAX**	**CUT**	**POINTS**
PART I		**400**		**/400**
PART II	**A**	**135**		**/135**
	B	**170**		**/170**
	C	**155**		**/155**
	D	**140**		**/140**
	E	**150**		**/150**
	F	**150**		**/150**
			PT II SUB	**/900**
			TEE TOTAL	**/1300**

1. An element has only two naturally occurring isotopes, one with a mass of 68.9256 amu (60.11% abundance) and the other with a mass of 70.9247 amu (39.89% abundance).

   a. (60 pts) ***Calculate*** the average atomic mass of the element. Report your final answer to 4 decimal places.

Answer 1a: ________________________

   b. (20 pts) ***Predict*** the identity of the element. Briefly explain your answer.

   c. (25 pts) ***Explain*** what similarities exist, *with respect to subatomic particles*, between these two isotopes and what causes their atomic masses to be different by approximately 2 amu.

2. (30 pts) ***Calculate*** the mass (**in g**) of **exactly** 1 million ***<u>fluorine</u>* <u>(F)</u>** atoms? *hint: check your significant figures!*

Answer 2: ________________________

1. (50 pts) ***Provide*** the systematic name for the following compounds:

   a. FeS ______________________

   b. NO_2 ______________________

   c. $MgCl_2$ ______________________

   d. $CuSO_4 \cdot 5H_2O$ ______________________

   e. $AgHCO_3$ ______________________

2. (50 pts) ***Provide*** the chemical formula for the following compounds:

   a. tin(II) chloride ______________________

   b. potassium permanganate ______________________

   c. disulfur octafluoride ______________________

   d. cadmium sulfate ______________________

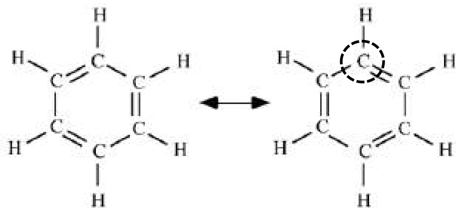
   e. selenium dioxide ______________________

3a. (50 pts) ***Draw*** an appropriate Lewis structure for IO_3^-. ***Indicate*** the total number of valence electrons in the blank provided and be sure to include formal charges, as appropriate.

Total valence electrons: ______________

3b. (20 points) **Draw two additional** resonance structures for IO_3^- as appropriate.

1. (30 pts) Use the resonance hybrid representation of benzene to fill in the following blanks.



Hybridization of circled atom: ____________

Molecular shape at circled atom: ____________

Bond angle around circled atom: ____________

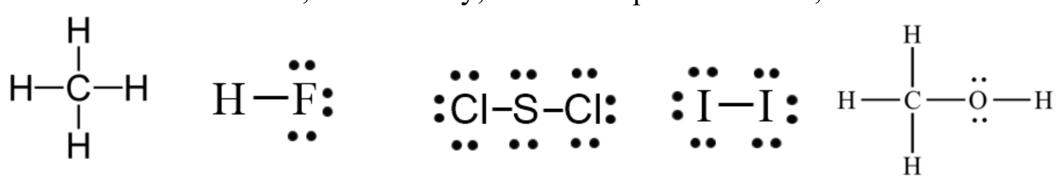
2. (30 pts) ***Circle*** the most appropriate type of bonding present in the molecule or compound.

Ni(s): ***metallic / covalent / ionic***

NaF(s): ***metallic / covalent / ionic***

C_2S(s): ***metallic / covalent / ionic***

3. Use the Lewis structures below, as necessary, to answer questions a – c, below.



Methane (CH_4) *Hydrogen fluoride (HF)* *Sulfur dichloride (SCl_2)* *Iodine (I_2)* *Methanol (CH_3OH)*

a. (60 pts) ***Circle*** all the intermolecular forces (IMF) predicted to occur in a sample of each pure substance depicted above:

i. methane: ***dispersion / dipole-dipole / hydrogen bonding / ion-dipole***

ii. hydrogen fluoride: ***dispersion / dipole-dipole / hydrogen bonding / ion-dipole***

iii. sulfur dichloride: ***dispersion / dipole-dipole / hydrogen bonding / ion-dipole***

iv. iodine: ***dispersion / dipole-dipole / hydrogen bonding / ion-dipole***

v. methanol: ***dispersion / dipole-dipole / hydrogen bonding / ion-dipole***

b. (15 pts) ***Circle*** the correct answer that *predicts* the miscibility or solubility for each pair substances based on their IMF interactions.

i. methane AND hydrogen fluoride: ***miscible / immiscible***

ii. methane AND iodine: ***soluble / insoluble***

iii. methanol AND hydrogen fluoride ***miscible / immiscible***

c. (20 pts) Circle the molecule with the highest predicted boiling point in each of the following pairs:

i. ***methanol / hydrogen fluoride***

ii. ***Methane / sulfur dichloride***

1. (40 pts) ***Balance*** the following chemical reaction by placing the correct coefficients in each space provided (use whole number coefficients):

___ $Al_2O_3(s)$ + ___ $BrF_3(l)$ → ___ $AlF_3(g)$ + ___ $Br_2(l)$ + ___ $O_2(g)$

2. Cyanogen, C_2N_2, has been observed in the atmosphere of Titan (Saturn's largest moon) and in the gases of interstellar nebulas. On Earth, it is used as a welding gas and a fumigant. In its reaction with fluorine gas, it produces carbon tetrafluoride and nitrogen trifluoride. For the following questions, assume 40.0 g of C_2N_2 reacts with 80.0 g of F_2 according to the following balanced chemical equation:

$$C_2N_2(g) + 7\ F_2(g) \rightarrow 2\ CF_4(g) + 2\ NF_3(g)$$

Species	Molar Mass ***(M)***
C_2N_2	$\mathcal{M}$= 52.036 g/mol
F_2	$\mathcal{M}$= 37.996 g/mol
CF_4	$\mathcal{M}$= 88.003 g/mol
NF_3	$\mathcal{M}$= 71.001 g/mol

a. (80 pts) ***Calculate*** the theoretical yield (in g) of carbon tetrafluoride (CF_4)? *Show all work to receive full credit.*

Theoretical Yield: ______________________

b. (20 pts) ***Identify*** the limiting reactant.

Limiting Reactant: ______________________

1. (80 pts) Read the following article excerpt and answer the question below.

Man faces trial on explosives charges

October 22, 2011 **By John Pepin - Staff Writer , Mining Journal**

MARQUETTE - A Sault Ste. Marie man is scheduled to stand trial in federal court in December on six felony charges.

The ATF agent obtained permission to search the informant's property and located **eighty-three (83) 50.0-pound bags** of **ammonium nitrate** sitting on pallets in the shed, according to court records. The following day, a search at the home of Lechner's mother revealed 36 blasting caps, 64 boosters and 2,000 feet of detonator cord. A subsequent search at Lechner's property on Blalock revealed 12 additional blasting caps.

An affidavit stated the informant had asked Lechner about his plans for the explosives. "When the government gets taken over," Lechner allegedly said. "We will be mercenaries."

The balanced detonation reaction for ammonium nitrate is:

$$2\ NH_4NO_3(s) \rightarrow 4\ H_2O(g) + 2\ N_2(g) + O_2(g)$$

Calculate the total volume of gas (in L) created if all 83 bags of the ammonium nitrate were detonated in a parking garage at 27.0º C and 1.00 atm. (molar mass of $NH_4NO_3(s)$ = 80.043 g/mol)

Answer 1: ______________________

2. (70 pts) The products of a reaction generating nitrogen gas (N_2) and hydrogen gas (H_2) gas are bubbled through water and collected. If the total volume of gas collected is 0.3308 L at a temperature of 294 K and pressure of 747 torr, **calculate** the mass (in g) of nitrogen gas produced. The partial pressure of the water vapor is 18.422 torr, and the partial pressure of the hydrogen gas is 546 torr.

Answer 2: ______________________

1. Nitrocellulose or "gun cotton" ($C_{12}H_{14}N_6O_{22}$) is an explosive material that can rapidly decompose into several gases. The balanced decomposition reaction is below.

$$C_{12}H_{14}N_6O_{22}(s) \rightarrow 3CO_2(g) + 7H_2O(g) + 3N_2(g) + 9CO(g)$$

a. (50 pts) ***Calculate*** the standard change in enthalpy ($\Delta H^{\circ}_{\mathbf{rxn}}$) for this reaction given:

Species	$C_{12}H_{14}N_6O_{22}(s)$	$CO_2(g)$	$H_2O(g)$	$N_2(g)$	$CO(g)$
$\Delta H_f°$ (kJ/mol)	-617.0	-393.5	-241.8	0.0	-110.5

b. (20 pts) ***Circle*** one: The above reaction is (***endothermic*** / ***exothermic***).

2. An unknown metal with a mass of 62.9 g initially at 95.0ºC is placed in a coffee-cup calorimeter (C=14.64 J/ºC) containing 85.0 g of water ($C_{s,H2O(l)}$= 4.18J/g°C). The initial temperature of the water and calorimeter was 22.4ºC. After reaching equilibrium, the final temperature is 25.0 ºC.

a. (15 pts) ***Calculate*** the heat, q_{water}, in joules (J) absorbed by the water.

b. (15 pts) ***Calculate*** the heat, q_{cal}, in joules (J) absorbed by the coffee-cup calorimeter.

c. (15 pts) ***Calculate*** the heat, q_{metal}, in joules (J) lost by the metal.

d. (20 pts) ***Calculate*** the specific heat capacity of the unknown metal (in J/g·ºC)?

e. (15 pts) Would the final temperature of the water increase or decrease if a metal with a higher specific heat capacity were used? ***Briefly explain*** your answer.

Term End Exam - Instructor Solution

CH101 TEE Part I Grading Scale (ACS 2015 Exam)							
Questions Correct	Raw Score	Adjusted Percent	Points out of 400	Questions Correct	Raw Score	Adjusted Percent	Points out of 400
1	1.43%	18.45%	74	36	51.43%	78.21%	313
2	2.86%	23.40%	94	37	52.86%	79.20%	317
3	4.29%	27.20%	109	38	54.29%	80.18%	321
4	5.71%	30.40%	122	39	55.71%	81.14%	325
5	7.14%	33.23%	133	40	57.14%	82.09%	328
6	8.57%	35.78%	143	41	58.57%	83.03%	332
7	10.00%	38.12%	152	42	60.00%	83.96%	336
8	11.43%	40.31%	161	43	61.43%	84.88%	340
9	12.86%	42.36%	169	44	62.86%	85.78%	343
10	14.29%	44.30%	177	45	64.29%	86.68%	347
11	15.71%	46.14%	185	46	65.71%	87.56%	350
12	17.14%	47.90%	192	47	67.14%	88.44%	354
13	18.57%	49.59%	198	48	68.57%	89.31%	357
14	20.00%	51.22%	205	49	70.00%	90.17%	361
15	21.43%	52.79%	211	50	71.43%	91.02%	364
16	22.86%	54.31%	217	51	72.86%	91.86%	367
17	24.29%	55.78%	223	52	74.29%	92.69%	371
18	25.71%	57.21%	229	53	75.71%	93.51%	374
19	27.14%	58.60%	234	54	77.14%	94.33%	377
20	28.57%	59.95%	240	55	78.57%	95.14%	381
21	30.00%	61.27%	245	56	80.00%	95.94%	384
22	31.43%	62.56%	250	57	81.43%	96.74%	387
23	32.86%	63.82%	255	58	82.86%	97.53%	390
24	34.29%	65.05%	260	59	84.29%	98.31%	393
25	35.71%	66.26%	265	60	85.71%	99.08%	396
26	37.14%	67.44%	270	61	87.14%	99.85%	399
27	38.57%	68.61%	274	62	88.57%	100.61%	402
28	40.00%	69.75%	279	63	90.00%	101.37%	405
29	41.43%	70.87%	283	64	91.43%	102.12%	408
30	42.86%	71.97%	288	65	92.86%	102.86%	411
31	44.29%	73.05%	292	66	94.29%	103.60%	414
32	45.71%	74.11%	296	67	95.71%	104.33%	417
33	47.14%	75.16%	301	68	97.14%	105.06%	420
34	48.57%	76.19%	305	69	98.57%	105.78%	423
35	50.00%	77.21%	309	70	100.00%	106.50%	426

Cut Scale

CADET ______________________________ SECTION/HR ______________________

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING

CHEMISTRY 101, AY26-1
Term End Examination (Part II)
December 2025
Time: Not Less Than 100 minutes

TEXT: Tro, *Chemistry: Structure and Properties*, 3rd Ed
SCOPE: Lessons 01 – 39

Authorized References
TI-36X Pro Calculator or equivalent
Chemistry Reference Data Card (RDC) 13th Edition (**no additional markings**)

1. You may begin work on this portion of the TEE immediately upon receipt from your OIC.

2. Solve problems in the space provided. Be sure to do the following ***to receive full credit***:
   a. Print legibly
   b. ***Show all work***
   c. Use complete sentences for all explanations
   d. Answer all questions

3. There are 6 lettered pages in this TEE. Report any illegible or missing pages to the OIC in the hallway. Write your name on every page in the spaces provided.

Sign to Acknowledge: ________________________

(TOTAL WEIGHT 900 POINTS)

THIS SPACE FOR INSTRUCTOR USE ONLY				
PAGE		MAX	CUT	POINTS
PART I		400		/400
PART II	A	135		/135
	B	170		/170
	C	155		/155
	D	140		/140
	E	150		/150
	F	150		/150
			PT II SUB	/900
			TEE TOTAL	/1300

Cut Scale

Page	Weight
A	135

1. An element has only two naturally occurring isotopes, one with a mass of 68.9256 amu (60.11% abundance) and the other with a mass of 70.9247 amu (39.89% abundance).

a. (60 pts) ***Calculate*** the average atomic mass of the element. Report your final answer to 4 decimal places.

Atomic Mass = ∑(isotope mass)*(% abundance) +27 concept (weighted average)

Atomic Mass = (68.9256 amu * 0.6011) + (70.9247 amu * 0.3989) +6 application (adds terms)

+6 +6 +6 +6

Atomic Mass = (41.43117 amu) + (28.29186 amu)

= 69.7230 amu +1 EA M/US -15 SC takes normal average (*69.9252 amu*)

Answer 1a: 69.7230 amu

b. (20 pts) ***Predict*** the identity of the element. Briefly explain your answer.

The element must be gallium (Ga), as it is the only one with an average mass between these two isotopes.

+10 gallium +10 reasonable explanation

c. (25 pts) ***Explain*** what similarities exist, *with respect to subatomic particles*, between these two isotopes and what causes their atomic masses to be different by approximately 2 amu.

- *Each isotope has the same number of protons (31) but different the number of neutrons*

  +13 same # of protons

- *Gallium-71 has 2 more neutrons (40) than gallium-69 (38).*

  +12 Ga-71 has 2 more neutrons than Ga-69

2. (30 pts) ***Calculate*** the mass (**in g**) of **exactly** 1 million ***fluorine* (F)** atoms? *hint: check your significant figures!*

$$\frac{1\text{x}10^6 \text{ F atoms}}{} \times \frac{1 \text{ mole F}}{6.022 \text{ x } 10^{23} \text{ F atoms}} \times \frac{19.00 \text{ g F}}{1 \text{ mole F}} = 3.155\text{x}10^{-17} \text{ g F}$$

+7 | +5 value +5 application | +5 value +5 application | +1 EA M/U/S

Answer 2: 3.155×10^{-17} g F

Cut Scale

Page	Weight
B	170

1. (50 pts) ***Provide*** the systematic name for the following compounds:

+4 +2 +4 +1 no extra
a. FeS — iron(II) sulfide

+4 +1 +3 +1 no extra
b. NO_2 — nitrogen dioxide

+4 +3 +1 no extra
c. $MgCl_2$ — magnesium chloride

+4 +1 +4 +1 +1 +1 no extra
d. $CuSO_4 \cdot 5H_2O$ — copper(II) sulfate pentahydrate

+4 +5 (or +5) +1 no extra
e. $AgHCO_3$ — silver hydrogen carbonate_(or bicarbonate)

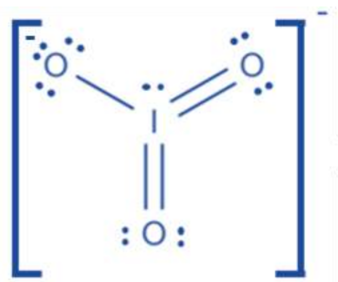
2. (50 pts) ***Provide*** the chemical formula for the following compounds:

a. tin(II) chloride — $SnCl_2$
b. potassium permanganate — $KMnO_4$
c. disulfur octafluoride — S_2F_8
d. cadmium sulfate — $CdSO_4$
e. selenium dioxide — SeO_2

+3 EA atom 1
+2 EA atom 1 subscript
+3 EA atom 2 / polyatomic ion
+2 EA atom 2/ polyatomic subscript

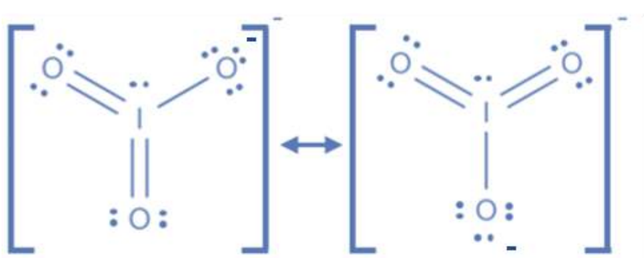
3a. (50 pts) ***Draw*** an appropriate Lewis structure for IO_3^-. ***Indicate*** the total number of valence electrons in the blank provided and be sure to include formal charges, as appropriate.

Total valence electrons: *26* +10 AON



+7 e^- in structure = valence e^- (FF)
+7 skeletal structure
+7 full octet on *all* oxygen atoms
+7 expanded octet on iodine atom
+7 minimizes formal charges
+2 brackets depicted
+3 charge depicted

3b. (20 points) **Draw two additional** resonance structures for IO_3^- as appropriate.

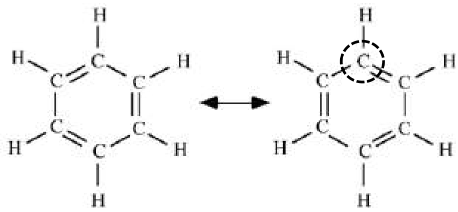


+7 still IO_3^-
+7 correct e^- count for both structures (FF)
+3 one valid resonance structure
+3 second valid resonance structure

Cut Scale

Page	Weight
C	155

1. (30 pts) Use the resonance hybrid representation of benzene to fill in the following blanks.



Hybridization of circled atom: ___sp^2___ +10 AON

Molecular shape at circled atom: ___trigonal planar___ +10 AON (FF)

Bond angle around circled atom: ___120°___ +10 AON (FF)

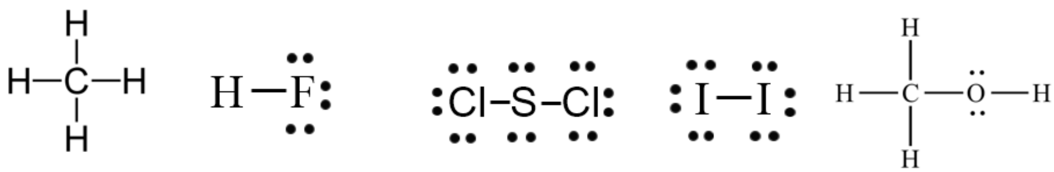
2. (30 pts) ***Circle*** the most appropriate type of bonding present in the molecule or compound.

Ni(s): ***metallic*** / ***covalent*** / ***ionic*** +10 EA AON

NaF(s): ***metallic*** / ***covalent*** / ***ionic***

C_2S(s): ***metallic*** / ***covalent*** / ***ionic***

3. Use the Lewis structures below, as necessary, to answer questions a – c, below.



Methane (CH_4) *Hydrogen fluoride (HF)* *Sulfur dichloride (SCl_2)* *Iodine (I_2)* *Methanol (CH_3OH)*

a. (60 pts) ***Circle*** all the intermolecular forces (IMF) predicted to occur in a sample of each pure substance depicted above: +3 EA: circles correct IMFs +3 EA omits incorrect IMFs

i. methane: ***dispersion*** / ***dipole-dipole*** / ***hydrogen bonding*** / ***ion-dipole***

ii. hydrogen fluoride: ***dispersion*** / ***dipole-dipole*** / ***hydrogen bonding*** / ***ion-dipole***

iii. sulfur dichloride: ***dispersion*** / ***dipole-dipole*** / ***hydrogen bonding*** / ***ion-dipole***

iv. iodine: ***dispersion*** / ***dipole-dipole*** / ***hydrogen bonding*** / ***ion-dipole***

v. methanol: ***dispersion*** / ***dipole-dipole*** / ***hydrogen bonding*** / ***ion-dipole***

b. (15 pts) ***Circle*** the correct answer that *predicts* the miscibility or solubility for each pair substances based on their IMF interactions.

i. methane AND hydrogen fluoride: ***miscible*** / ***immiscible***

ii. methane AND iodine: ***soluble*** / ***insoluble*** +5 EA AON (FF from IMFs)

iii. methanol AND hydrogen fluoride ***miscible*** / ***immiscible***

c. (20 pts) Circle the molecule with the highest predicted boiling point in each of the following pairs:

i. ***methanol*** / ***hydrogen fluoride*** +10 EA AON (FF from IMFs)

ii. ***Methane*** / ***sulfur dichloride***

Cut Scale

Page	Weight
D	140

1. (40 pts) ***Balance*** the following chemical reaction by placing the correct coefficients in each space provided (use whole number coefficients):

$\underline{\ 2\ }\ Al_2O_3(s)\ +\ \underline{\ 4\ }\ BrF_3(l)\ \rightarrow\ \underline{\ 4\ }\ AlF_3(g)\ +\ \underline{\ 2\ }\ Br_2(l)\ +\ \underline{\ 3\ }\ O_2(g)$

Balance (+10 ea):
F
Br
O
Al

-3 SC for not using lowest coefficients

2. Cyanogen, C_2N_2, has been observed in the atmosphere of Titan (Saturn's largest moon) and in the gases of interstellar nebulas. On Earth, it is used as a welding gas and a fumigant. In its reaction with fluorine gas, it produces carbon tetrafluoride and nitrogen trifluoride. For the following questions, assume 40.0 g of C_2N_2 reacts with 80.0 g of F_2 according to the following balanced chemical equation:

$$C_2N_2(g) + 7\ F_2(g) \rightarrow 2\ CF_4(g) + 2\ NF_3(g)$$

Species	Molar Mass ($\mathcal{M}$)
C_2N_2	$\mathcal{M}$= 52.036 g/mol
F_2	$\mathcal{M}$= 37.996 g/mol
CF_4	$\mathcal{M}$= 88.003 g/mol
NF_3	$\mathcal{M}$= 71.001 g/mol

a. (80 pts) ***Calculate*** the theoretical yield (in g) of carbon tetrafluoride (CF_4)? *Show all work to receive full credit.*

+25 concept (uses stoichiometry to determine answer)
+12 chooses lowest yield (FF)

+4 mass C_2N_2 | +3 uses m.m. correctly | +4 uses molar ratio | +3 uses m.m. correctly

$$C_2N_2: \frac{40.0\text{ g } C_2N_2}{} \left| \frac{1\text{ mol } C_2N_2}{52.036\text{ g } C_2N_2} \right| \frac{2\text{ mol } CF_4}{1\text{ mol } C_2N_2} \left| \frac{88.003\text{ g } CF_4}{1\text{ mol } CF_4} \right. = 135\text{ g } CF_4$$

+1 correct m.m | +4 correct molar ratio | +1 correct m.m.

+4 mass C_2N_2 | +3 uses m.m. correctly | +4 uses molar ratio | +3 uses m.m. correctly

$$F_2: \frac{80.0\text{ g } F_2}{} \left| \frac{1\text{ mol } F_2}{37.996\text{ g } F_2} \right| \frac{2\text{ mol } CF_4}{7\text{ mol } F_2} \left| \frac{88.003\text{ g } CF_4}{1\text{ mol } CF_4} \right. = 52.9\text{ g } CF_4$$

+1 correct m.m | +4 correct molar ratio | +1 correct m.m.

Standard Cuts:
-36 uses sum of calculated masses as theoretical yield (191.9 g CF_4)
-8 stops calculation at moles of product
-32 only does stoichiometry for one reactant to product

+1EA M/U/S

Theoretical Yield: 52.9 g

b. (20 pts) ***Identify*** the limiting reactant.

Limiting Reactant: $F_2(g)$ +20 AON (FF)

Cut Scale

Page	Weight
E	150

1. (80 pts) Read the following article excerpt and answer the question below.

Man faces trial on explosives charges

October 22, 2011 **By John Pepin - Staff Writer , Mining Journal**

MARQUETTE - A Sault Ste. Marie man is scheduled to stand trial in federal court in December on six felony charges.

The ATF agent obtained permission to search the informant's property and located **eighty-three (83) 50.0-pound bags** of **ammonium nitrate** sitting on pallets in the shed, according to court records. The following day, a search at the home of Lechner's mother revealed 36 blasting caps, 64 boosters and 2,000 feet of detonator cord. A subsequent search at Lechner's property on Blalock revealed 12 additional blasting caps.

An affidavit stated the informant had asked Lechner about his plans for the explosives. "When the government gets taken over," Lechner allegedly said. "We will be mercenaries."

The balanced detonation reaction for ammonium nitrate is:

$$2\ NH_4NO_3(s) \rightarrow 4\ H_2O(g) + 2\ N_2(g) + O_2(g)$$

Calculate the total volume of gas (in L) created if all 83 bags of the ammonium nitrate were detonated in a parking garage at 27.0º C and 1.00 atm. (molar mass of $NH_4NO_3(s)$ = 80.043 g/mol)

+15 concept (stoichiometry to find moles of gas produced)

+2	+4 bag → lb	+4 lb → g	+4 g → mol	+4 molar ratio	+1 Math
83 bags	50.0 lb NH_4NO_3	453.592 g	1 mol NH_4NO_3	7 mol gas	= $8.2\underline{3} \times 10^4$ mol gas
	bag	1 lb	80.043 g NH_4NO_3	2 mol NH_4NO_3	
				+4 correct ratio	

$PV = nRT$ +15 concept (ideal gas law) +10 application (correctly isolates V)

$$V = \frac{nRT}{P} = \frac{8.2\underline{3} \times 10^4 \text{ mol gas} \times 0.08206\ \text{L}\cdot\text{atm}/(\text{mol}\cdot\text{K}) \times 300.15\ \text{K}}{1.00\ \text{atm}} = 2.03 \times 10^6\ \text{L}$$

+4 (FF from above) +4 T (in K) +1 EA M/U/S +3 R +3 P

Answer 1: 2.03×10^6 L

2. (70 pts) The products of a reaction generating nitrogen gas (N_2) and hydrogen gas (H_2) gas are bubbled through water and collected. If the total volume of gas collected is 0.3308 L at a temperature of 294 K and pressure of 747 torr, **calculate** the mass (in g) of nitrogen gas produced. The partial pressure of the water vapor is 18.422 torr, and the partial pressure of the hydrogen gas is 546 torr.

$P_{N2} = P_{total} - P_{H2O} - P_{H2}$ +10 concept (uses Dalton's Law to find partial pressure of N_2)

P_{N2} = 747 torr – 18.422 torr – 546 torr = $18\underline{2}.578$ torr $N_2(g)$

+3 P_{total} +3 P_{H2O} +3 P_{H2}

$PV = nRT$ +10 concept (ideal gas law to find mass of N_2) $n = PV/RT$ +8 application (correctly isolates n)

+3 (FF from above)		+4 V (in K)		+4 T (in K)	
$18\underline{2}.578$ torr	atm	0.3308 L	mol · K		= $0.0032\underline{9}3$ mol $N_2(g)$
	760 torr		0.08206 L · atm	294 K	
	+4 torr → atm		+4 R		

Mass N_2 = $n \times$ *molar mass* = (0.00329 mol) × (28.014 g/mol) = $0.092\underline{1}7$ g $N_2(g)$

+5 concept (mass) +3 n (FF) +3 m.m. +1EA M/U/S

Answer 2: 0.0922 g $N_2(g)$

Cut Scale

Page	Weight
F	150

1. Nitrocellulose or "gun cotton" ($C_{12}H_{14}N_6O_{22}$) is an explosive material that can rapidly decompose into several gases. The balanced decomposition reaction is below.

$$C_{12}H_{14}N_6O_{22}(s) \rightarrow 3CO_2(g) + 7H_2O(g) + 3N_2(g) + 9CO(g)$$

a. (50 pts) ***Calculate*** the standard change in enthalpy (ΔH°_{rxn}) for this reaction given:

Species	$C_{12}H_{14}N_6O_{22}(s)$	$CO_2(g)$	$H_2O(g)$	$N_2(g)$	$CO(g)$
ΔH_f° (kJ/mol)	-617.0	-393.5	-241.8	0.0	-110.5

$\Delta H^\circ_{rxn} = \Sigma n\Delta H^\circ_{f\,(products)} - \Sigma n\Delta H^\circ_{f\,(reactants)}$ +15 concept (ΔH°_{rxn})

+2 mol CO_2 +2 mol H_2O +2 mol N_2 +2 mol CO

$\Sigma n\Delta H^\circ_{f\,(products)}$ = [(3 mol)(-393.5 kJ/mol) + (7 mol)(-241.8 kJ/mol) + (3 mol)(0.00 kJ/mol) + (9 mol)(-110.5 kJ/mol)] = -3867.6 kJ

+3 $\Delta H^\circ_f(CO_2)$ +3 $\Delta H^\circ_f(H_2O)$ +3 $\Delta H^\circ_f(N_2)$ +3 $\Delta H^\circ_f(CO)$

+2 mol $C_{12}H_{14}N_6O_{22}$ +7 app: prod – react +1 EA M/U/S

$\Sigma n\Delta H^\circ_{f\,(reactants)}$ = [(1 mol)(-617.0 kJ/mol)] = -617.0 kJ ΔH°_{rxn} = [-3867.6 kJ] – [-617.0 kJ] = -3250.6 kJ

+3 $\Delta H^\circ_f(C_{12}H_{14}N_6O_{22})$

b. (20 pts) ***Circle*** one: The above reaction is (***endothermic*** / ***exothermic***). +20 AON

2. An unknown metal with a mass of 62.9 g initially at 95.0ºC is placed in a coffee-cup calorimeter (C=14.64 J/ºC) containing 85.0 g of water ($C_{s,H2O(l)}$= 4.18J/g°C). The initial temperature of the water and calorimeter was 22.4ºC. After reaching equilibrium, the final temperature is 25.0 ºC.

a. (15 pts) ***Calculate*** the heat, q_{water}, in joules (J) absorbed by the water.

+6 concept +2 correct m +2 $T_f - T_i$ = 2.6°C +1 EA M/U/S

$$q_{water} = m \times C_s \times \Delta T = \frac{85.0\text{ g}}{} \left|\frac{4.18\text{ J}}{\text{g}\cdot{}^\circ\text{C}}\right| \frac{(25.0\text{-}22.4)^\circ\text{C}}{} = 9\underline{2}3.78\text{ J}$$

+2 correct C_s

b. (15 pts) ***Calculate*** the heat, q_{cal}, in joules (J) absorbed by the coffee-cup calorimeter.

+6 concept +3 $T_f - T_i$ = 2.6°C +1 EA M/U/S

$$q_{cal} = C \times \Delta T = \frac{14.64\text{ J}}{{}^\circ\text{C}} \left| \frac{(25.0\text{-}22.4)^\circ\text{C}}{} \right. = 38.\underline{0}64\text{ J}$$

+3 correct C_s

c. (15 pts) ***Calculate*** the heat, q_{metal}, in joules (J) lost by the metal.

$q_{metal} = -(q_{water} + q_{cal})$ = – (92$\underline{3}$.78 J + 38.$\underline{0}$64 J) = – 96$\underline{1}$.84 J = – 962 J

+6 concept +3 EA q (FF) +1 EA M/U/S

d. (20 pts) ***Calculate*** the specific heat capacity of the unknown metal (in J/g·ºC)?

+6 concept +2 correct q_{metal} (FF) +2 $T_f - T_i$ = - 70°C +1 EA M/U/S

$$q_{metal} = m \times C_s \times \Delta T \rightarrow C_s = \frac{q_{metal}}{m \times \Delta T} = \frac{-96\underline{1}.84\text{ J}}{} \left| \frac{}{62.9\text{ g}} \right| \frac{}{(25.0\text{-}95.0)\ {}^\circ\text{C}} = 9\underline{2}3.78\text{ J}$$

+5 solves for C_s +2 correct mass

e. (15 pts) Would the final temperature of the water increase or decrease if a metal with a higher specific heat capacity were used? ***Briefly explain*** your answer.

+7 AON "increase"

Increase, the final temperature of the system and surroundings would be higher because the metal would transfer more heat to the surroundings, raising the final temperature of the calorimeter.

+8 logical explanation (definition of C_s, use of $q_{metal} = m \times C_s \times \Delta T$, *1st Law of Thermodynamics, etc)*

TAB J FINAL COURSE GRADES

Student	Current Max Points	Points	Score	Grade
Agres, Josh	5000	4840.09	96.8018	A+
Baggish, Owen	5000	4621.23	92.4246	A-
Baker, Christopher	5000	4741.47	94.8294	A
Balaji, Aditya	5000	4347.47	86.9494	B+
Ballard, Kenyon	5000	4813.54	96.2708	A
Barton, John	5000	4867.97	97.3594	A+
Bauer, Braden	5000	4863.04	97.2608	A+
Ben-Adi, Yael	5000	4673.3	93.466	A
Benjaminson, Ethar	5000	4729.05	94.581	A
Booe, Clare	5000	4851.46	97.0292	A+
Brookes, Jackson	5000	4565.9	91.318	A-
Chew, Madison	5000	4344.03	86.8806	B+
Chiu, Austin	5000	4894.89	97.8978	A+
Contreras, Joaquin	5000	4685.77	93.7154	A
Drewes, Eric	5000	4477.67	89.5534	B+
Driscoll, Andrew	5000	4669.35	93.387	A
Fahey, Neilan	5000	4891.16	97.8232	A+
Forcum, Colby	5000	4650.02	93.0004	A-
Frazier, Quinn	5000	4651.78	93.0356	A-
Gancarz, Jonathan	5000	4746.46	94.9292	A
Harrington, Kaylin	5000	4507.94	90.1588	A-
Henrich, Graham	5000	4820.37	96.4074	A
Hicks, Arianna	5000	4354.74	87.0948	B+
Hicks, Conner	5000	4904.98	98.0996	A+
Howell, Coral	5000	4856.32	97.1264	A+
Iglesias, Marin	5000	4844.32	96.8864	A+
Jiang, Simon	5000	4745.49	94.9098	A
Jo, Paul	5000	4830.52	96.6104	A
Jones, Indiana	5000	4612.22	92.2444	A-
Jung, Joshua	5000	4874.41	97.4882	A+
Juttumahadevan, K	5000	4704.15	94.083	A
Keim, Lilliana	5000	4663.84	93.2768	A-
Kennedy, Margaret	5000	4282.79	85.6558	B
Kim, Ian	5000	4608.75	92.175	A-
Kirk, Robert	5000	4741.32	94.8264	A
Ko, Seokhyun	5000	4811.62	96.2324	A
Lee, Lawrence	5000	4174.13	83.4826	B
Lee, Rebekah	5000	4382.85	87.657	B+
Medallon, Tynisa	5000	4715.35	94.307	A
Meno, Avery	5000	4746.7	94.934	A
Michalopoulos, Ava	5000	4842.67	96.8534	A+
Morrisey, Charles	5000	4661.56	93.2312	A-
Mulholland, James	5000	4753.05	95.061	A
Noonan, Patrick	5000	4831.1	96.622	A
Oh, Hannah	5000	4710.79	94.2158	A
Page, Tyler	5000	4700.68	94.0136	A
Perez, Rafael	5000	4746.73	94.9346	A
Phan, Hieu-Shawn	5000	4714.19	94.2838	A
Qasemi, Al	5000	4755.3	95.106	A
Robertson, Alexand	5000	4774.21	95.4842	A
Rowe, Patrick	5000	4732.02	94.6404	A
Rueda, Christopher	5000	4792.33	95.8466	A
Ryu, Timothy	5000	4545.12	90.9024	A-
Santos, Ethan	5000	4643.19	92.8638	A-
Sikora, William	5000	4786.42	95.7284	A
Skorka, Anson	5000	4815.05	96.301	A
Spaulding, Wyatt	5000	4812.99	96.2598	A
Stephenson, Tyler	5000	4720.48	94.4096	A
Tobler, Charles	5000	4784.05	95.681	A
Tokouete, Kekeli	5000	4468.79	89.3758	B+
Trivino, Brandon	5000	4696.08	93.9216	A
Upadhyay, Harshita	5000	4757.46	95.1492	A
Varma, Abhimanyu	5000	4176.25	83.525	B
Wagner, Colin	5000	4628.47	92.5694	A-
Wagner, Jared	5000	4767.76	95.3552	A
Walsh, Daniel	5000	4587.13	91.7426	A-
Williams, Anna	5000	4802.08	96.0416	A
Wokurka, Asher	5000	4585.41	91.7082	A-
Wu, Patricia	5000	4891.2	97.824	A+
Zeller, Harry	5000	4813.14	96.2628	A

Student	Current Max Points	Points	Score	Grade	OML Rank
Hicks, Conner	5000	4904.98	98.0996	A+	1
Chiu, Austin	5000	4894.89	97.8978	A+	2
Wu, Patricia	5000	4891.2	97.824	A+	3
Fahey, Neilan	5000	4891.16	97.8232	A+	4
Jung, Joshua	5000	4874.41	97.4882	A+	5
Barton, John	5000	4867.97	97.3594	A+	6
Bauer, Braden	5000	4863.04	97.2608	A+	7
Howell, Coral	5000	4856.32	97.1264	A+	8
Booe, Clare	5000	4851.46	97.0292	A+	9
Iglesias, Marin	5000	4844.32	96.8864	A+	10
Michalopoulos, Ava	5000	4842.67	96.8534	A+	11
Agres, Josh	5000	4840.09	96.8018	A+	12
Noonan, Patrick	5000	4831.1	96.622	A	13
Jo, Paul	5000	4830.52	96.6104	A	14
Henrich, Graham	5000	4820.37	96.4074	A	15
Skorka, Anson	5000	4815.05	96.301	A	16
Ballard, Kenyon	5000	4813.54	96.2708	A	17
Zeller, Harry	5000	4813.14	96.2628	A	18
Spaulding, Wyatt	5000	4812.99	96.2598	A	19
Ko, Seokhyun	5000	4811.62	96.2324	A	20
Williams, Anna	5000	4802.08	96.0416	A	21
Rueda, Christopher	5000	4792.33	95.8466	A	22
Sikora, William	5000	4786.42	95.7284	A	23
Tobler, Charles	5000	4784.05	95.681	A	24
Robertson, Alexandra	5000	4774.21	95.4842	A	25
Wagner, Jared	5000	4767.76	95.3552	A	26
Upadhyay, Harshita	5000	4757.46	95.1492	A	27
Qasemi, Al	5000	4755.3	95.106	A	28
Mulholland, James	5000	4753.05	95.061	A	29
Perez, Rafael	5000	4746.73	94.9346	A	30
Meno, Avery	5000	4746.7	94.934	A	31
Gancarz, Jonathan	5000	4746.46	94.9292	A	32
Jiang, Simon	5000	4745.49	94.9098	A	33
Baker, Christopher	5000	4741.47	94.8294	A	34
Kirk, Robert	5000	4741.32	94.8264	A	35
Rowe, Patrick	5000	4732.02	94.6404	A	36
Benjaminson, Ethan	5000	4729.05	94.581	A	37
Stephenson, Tyler	5000	4720.48	94.4096	A	38
Medallon, Tynisa	5000	4715.35	94.307	A	39
Phan, Hieu-Shawn	5000	4714.19	94.2838	A	40
Oh, Hannah	5000	4710.79	94.2158	A	41
Juttumahadevan, Karthikean	5000	4704.15	94.083	A	42
Page, Tyler	5000	4700.68	94.0136	A	43
Trivino, Brandon	5000	4696.08	93.9216	A	44
Contreras, Joaquin	5000	4685.77	93.7154	A	45
Ben-Adi, Yael	5000	4673.3	93.466	A	46
Driscoll, Andrew	5000	4669.35	93.387	A	47
Keim, Lilliana	5000	4663.84	93.2768	A-	48
Morrisey, Charles	5000	4661.56	93.2312	A-	49
Frazier, Quinn	5000	4651.78	93.0356	A-	50
Forcum, Colby	5000	4650.02	93.0004	A-	51
Santos, Ethan	5000	4643.19	92.8638	A-	52
Wagner, Colin	5000	4628.47	92.5694	A-	53
Baggish, Owen	5000	4621.23	92.4246	A-	54
Jones, Indiana	5000	4612.22	92.2444	A-	55
Kim, Ian	5000	4608.75	92.175	A-	56
Walsh, Daniel	5000	4587.13	91.7426	A-	57
Wokurka, Asher	5000	4585.41	91.7082	A-	58
Brookes, Jackson	5000	4565.9	91.318	A-	59
Ryu, Timothy	5000	4545.12	90.9024	A-	60
Harrington, Kaylin	5000	4507.94	90.1588	A-	61
Drewes, Eric	5000	4477.67	89.5534	B+	62
Tokouete, Kekeli	5000	4468.79	89.3758	B+	63
Lee, Rebekah	5000	4382.85	87.657	B+	64
Hicks, Arianna	5000	4354.74	87.0948	B+	65
Balaji, Aditya	5000	4347.47	86.9494	B+	66
Chew, Madison	5000	4344.03	86.8806	B+	67
Kennedy, Margaret	5000	4282.79	85.6558	B	68
Varma, Abhimanyu	5000	4176.25	83.525	B-	69
Lee, Lawrence	5000	4174.13	83.4826	B-	70

SECTION III

ASSIGNMENTS

TAB K QUIZZES

Quiz 1

Quiz 1 - Handouts

CH101 (AY26-1) Cadet Name (Last, First): ______________________________

Version 1 C-Number: ______________________________

200 Points Section: ______________________________

End-of-Hour Concept Quiz #1
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

_____ **1.** (10 pts) How many protons (p^+), neutrons (n^0), and electrons (e^-) are present in ^{37}Cl?

A. 20 p^+, 17 n^0, 17 e^-

B. 20 p^+, 37 n^0, 20 e^-

C. 17 p^+, 20 n^0, 20 e^-

D. 17 p^+, 20 n^0, 17 e^-

_____ **2.** (10 pts) How many protons (p^+), neutrons (n^0), and electrons (e^-) are present in $^{50}Cr^{3+}$?

A. 24 p^+, 26 n^0, 24 e^-

B. 24 p^+, 26 n^0, 21 e^-

C. 50 p^+, 103 n^0, 53 e^-

D. 50 p^+, 26 n^0, 21 e^-

_____ **3.** (15 pts) In certain stars, like our Sun, Argon consists of two isotopes: ^{36}Ar with an abundance of 85.00% and an isotopic mass of 35.9657 amu, and ^{38}Ar with an abundance of 15.00% and an isotopic mass 37.9627 amu. What is the average atomic mass of Argon on the Sun?

A. 36.27 amu

B. 36.97 amu

C. 37.96 amu

D. 37.66 amu

_____ **4.** (15 pts) Given 1 kg samples of the elements below, which one has the most atoms?

A. Silicon

B. Aluminum

C. Beryllium

D. Bromine

_____ **5.** (10 pts) In a chemical reaction, matter is neither created nor destroyed. Which law does this refer to?

A. Law of Definite Proportions

B. Law of the Conservation of Mass

C. Law of Modern Atomic Theory

D. Law of Multiple Proportions

6. (30 pts) The Lyman-α spectral line is a specific emission line in the hydrogen atom's spectrum that is important in characterizing galaxies. In this question, assume the wavelength of the line is 1.27×10^{-7} m. Calculate the frequency (in Hz) of the Lyman-α spectral line.

Answer 6: ____________________

7. A certain electron in a zinc atom has a binding energy of approximately 1.504×10^{-18} J.

a. (40 pts) Does a photon with a wavelength of 4.00 nm have enough energy to eject the electron from the zinc atom? You must show your work to receive full credit.

Circle your answer:

Answer 7a: YES / NO

b. (10 pts) In which region of the electromagnetic spectrum is light with a wavelength of 4.00 nm?

Answer 7b: ____________________

8. Typical silicon wafers used for semiconductor fabrication have a volume of 4.12 cm^3 and consist of approximately 0.341 mol of silicon.

a. (30 pts) Calculate the density (in g/cm^3) of a typical silicon wafer.

Answer 8a: __________________

b. (30 points) If the wafer volume was scaled up to 2060. cm^3 with the same density found above, determine how many atoms of Silicon would be present.

Answer 8b: __________________

Quiz 1 - Instructor Solution

Approved Solutions

CH101 (AY26-1) Cadet Name (Last, First): ____________________

Version 1 C-Number: ____________________

200 Points Section: ____________________

End-of-Hour Concept Quiz #1
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

D. **1.** (10 pts) How many protons (p^+), neutrons (n^0), and electrons (e^-) are present in ^{37}Cl?

A. 20 p^+, 17 n^0, 17 e^-
C. 17 p^+, 20 n^0, 20 e^-
B. 20 p^+, 37 n^0, 20 e^-
D. 17 p^+, 20 n^0, 17 e^-

B. **2.** (10 pts) How many protons (p^+), neutrons (n^0), and electrons (e^-) are present in $^{50}Cr^{3+}$?

A. 24 p^+, 26 n^0, 24 e^-
C. 50 p^+, 103 n^0, 53 e^-
B. 24 p^+, 26 n^0, 21 e^-
D. 50 p^+, 26 n^0, 21 e^-

A. **3.** (15 pts) In certain stars, like our Sun, Argon consists of two isotopes: ^{36}Ar with an abundance of 85.00% and an isotopic mass of 35.9657 amu, and ^{38}Ar with an abundance of 15.00% and an isotopic mass 37.9627 amu. What is the average atomic mass of Argon on the Sun?

A. 36.27 amu
C. 37.96 amu
B. 36.97 amu
D. 37.66 amu

C. **4.** (15 pts) Given 1 kg samples of the elements below, which one has the most atoms?

A. Silicon
C. Beryllium
B. Aluminum
D. Bromine

B. **5.** (10 pts) In a chemical reaction, matter is neither created nor destroyed. Which law does this refer to?

A. Law of Definite Proportions
C. Law of Modern Atomic Theory
B. Law of the Conservation of Mass
D. Law of Multiple Proportions

6. (30 pts) The Lyman-α spectral line is a specific emission line in the hydrogen atom's spectrum that is important in characterizing galaxies. In this question, assume the wavelength of the line is 1.27×10^{-7} m. Calculate the frequency (in Hz) of the Lyman-α spectral line.

$$\nu = \frac{c}{\lambda}$$

$$\nu = \frac{2.998\times10^{8}\ \text{m}}{\text{s}} \left| \frac{}{1.27\times10^{-7}\ \text{m}} \right| = 2.3\underline{6}06 \times 10^{15}\ \text{Hz}$$

Answer 6: 2.36×10^{15} Hz

7. A certain electron in a zinc atom has a binding energy of approximately 1.504×10^{-18} J.

a. (40 pts) Does a photon with a wavelength of 4.00 nm have enough energy to eject the electron from the zinc atom? You must show your work to receive full credit.

$$E_{ph} = \frac{hc}{\lambda}$$

$$E_{ph} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{\text{photon}} \left| \frac{2.998\times10^{8}\ \text{m}}{\text{s}} \right| \frac{}{4.0\ \text{nm}} \left| \frac{1\ \text{nm}}{1\times10^{-9}\ \text{m}} \right. = 4.966 \times 10^{-17}\ \text{J}$$

E_{ph} (4.966×10^{-17} J) > ø (1.504×10^{-18} J)
∴ photon has sufficient energy to eject an electron

Circle your answer:

Answer 7a: YES / NO

b. (10 pts) In which region of the electromagnetic spectrum is light with a wavelength of 4.00 nm?

Answer 7b: X-Ray Region

8. Typical silicon (chemical symbol: Si) wafers used for semiconductor fabrication have a volume of 4.12 cm^3 and consist of approximately 0.341 mol of silicon.

a. (30 pts) Calculate the density (in g/cm^3) of a typical silicon wafer.

$$d = \frac{m}{V}$$

$$d = \frac{0.341\ \text{mol Si}}{} \left| \frac{28.09\ \text{g Si}}{\text{mol Si}} \right| \frac{}{4.12\ \text{cm}^3} = 2.3\underline{2}49\ \text{g/cm}^3$$

Answer 8a: 2.32 g/cm^3

b. (30 points) If the wafer volume was scaled up to 2060. cm^3 with the same density found above, determine how many atoms of Silicon would be present.

Density → mass → moles → atoms

$$\frac{2060\ \text{cm}^3}{} \left| \frac{2.32\ \text{g}}{\text{cm}^3} \right| \frac{1\ \text{mol Si}}{28.09\ \text{g Si}} \left| \frac{6.022\times10^{23}\ \text{atoms Si}}{\text{mol Si}} \right. = 1.0\underline{2}45 \times 10^{26}\ \text{atoms Si}$$

Answer 8b: 1.02×10^{26} atoms Si

Quiz 2

Quiz 2 - Handouts

CH101 (AY26-1) Cadet Name (Last, First): ____________________________

Version 1 C-Number: ____________________________

200 Points Section: ____________________________

End-of-Hour Concept Quiz #2
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

_____ **1.** (10 pts) Which of the following statements best explains how light is emitted from an atom?

A. Photons are excited from lower-energy levels to higher-energy levels.
B. Electrons relax from higher-energy levels to lower-energy levels.
C. Neutrons release photons when they are converted into electrons.
D. Protons in the nucleus gain energy and emit light as they return to a stable state.

_____ **2.** (15 pts) Which of the following is a valid set of quantum numbers for an electron in the 2p subshell (or sublevel)?

A. $n = 1$, $l = 2$, $m_l = +1$, $m_s = +^1/_2$
B. $n = 2$, $l = 1$, $m_l = -1$, $m_s = -^1/_2$
C. $n = 2$, $l = 0$, $m_l = 0$, $m_s = -^1/_2$
D. $n = 1$, $l = 1$, $m_l = +2$, $m_s = +^1/_2$

_____ **3.** (15 pts) $CaCl_2$ is a solid salt that has many uses in the food industry due to its ability to dissociate into Ca^{2+} and Cl^- ions. Calculate the effective nuclear charge experienced by the valence electrons in Ca^{2+} and Cl^-, respectively.

A. $Ca^{2+} = 10$, $Cl^- = 7$
B. $Ca^{2+} = 10$, $Cl^- = -1$
C. $Ca^{2+} = 2$, $Cl^- = 7$
D. $Ca^{2+} = 2$, $Cl^- = -1$

_____ **4.** (10 pts) Give the complete electron configuration for manganese.

A. $1s^22s^22p^63s^23p^64s^24d^5$
B. $1s^22s^22p^63s^23p^64s^33d^4$
C. $1s^22s^22p^63s^23p^64s^23d^5$
D. $1s^22s^22p^33s^23p^44s^24p^5$

_____ **5.** (10 pts) Arrange these elements in order of increasing atomic radius: Sb, Ba, F.

A. Sb < Ba < F
B. Ba < Sb < F
C. F < Sb < Ba
D. F < Ba < Sb

_____ **6.** (10 pts) Metals are generally located on the left side of the periodic table and tend to gain electrons when forming ions.

A. True
B. False

7. (70 pts) Match the most appropriate term with each definition:

___	Energy required to remove the most loosely bound electron from an atom or ion in its gaseous state	A. Ionic Bond
___	Chemical bond characterized by the sharing of valence electrons between two nonmetals	B. Covalent Bond
___	Represents the lowest whole-number ratio of atoms in a compound, determined experimentally	C. Metallic Bond
___	Represents the types and numbers of atoms present in a molecule, reflecting its true molecular composition.	D. Empirical Formula
___	Energy released when gaseous ions assemble into a crystalline solid; serves as a measure of the stability and strength of the ionic bonds in an ionic compound.	E. Molecular Formula
___	Represents the types and numbers of atoms present in a molecule along with how they are bonded to each other	F. Structural Formula
___	Chemical bond formed through the electrostatic attraction between oppositely charged ions of a metal and nonmetal	G. Numerical Formula
		H. Ionization Energy
		I. Lattice Energy
		J. Potential Energy

8. Consider the following elements: Ge, Ga, Se.

A. (5 pts) Place the three elements in order of increasing first ionization energy.

_______ < _______ < _______

B. (15 pts) Explain the general trend (and reason for the trend) in first ionization energy as we move from left to right across a period on the periodic table.

9. (25 pts) Write the full ground state orbital diagram for Ti^{2+}.

10. (15 pts) The ground state electron configuration for cobalt is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$. Explain why cobalt is paramagnetic.

Quiz 2 - Instructor Solution

CH101 (AY26-1) Cadet Name (Last, First): ____________________

Version 1 C-Number: ____________________

200 Points Section: ____________________

End-of-Hour Concept Quiz #2
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

_B._ **1.** (10 pts) Which of the following statements best explains how light is emitted from an atom?

A. Photons are excited from lower-energy levels to higher-energy levels.
B. Electrons relax from higher-energy levels to lower-energy levels.
C. Neutrons release photons when they are converted into electrons.
D. Protons in the nucleus gain energy and emit light as they return to a stable state.

_B._ **2.** (15 pts) Which of the following is a valid set of quantum numbers for an electron in the 2p subshell (or sublevel)?

A.	B.	C.	D.
$n = 1$	$n = 2$	$n = 2$	$n = 1$
$l = 2$	$l = 1$	$l = 0$	$l = 1$
$m_l = +1$	$m_l = -1$	$m_l = 0$	$m_l = +2$
$m_s = +^1/_2$	$m_s = -^1/_2$	$m_s = -^1/_2$	$m_s = +^1/_2$

_A._ **3.** (15 pts) $CaCl_2$ is a solid salt that has many uses in the food industry due to its ability to dissociate into Ca^{2+} and Cl^- ions. Calculate the effective nuclear charge experienced by the valence electrons in Ca^{2+} and Cl^-, respectively.

A. $Ca^{2+} = 10$, $Cl^- = 7$
B. $Ca^{2+} = 10$, $Cl^- = -1$
C. $Ca^{2+} = 2$, $Cl^- = 7$
D. $Ca^{2+} = 2$, $Cl^- = -1$

_C._ **4.** (10 pts) Give the complete electron configuration for manganese.

A. $1s^22s^22p^63s^23p^64s^24d^5$
B. $1s^22s^22p^63s^23p^64s^33d^4$
C. $1s^22s^22p^63s^23p^64s^23d^5$
D. $1s^22s^22p^33s^23p^44s^24p^5$

_C._ **5.** (10 pts) Arrange these elements in order of increasing atomic radius: Sb, Ba, F.

A. Sb < Ba < F
B. Ba < Sb < F
C. F < Sb < Ba
D. F < Ba < Sb

_B._ **6.** (10 pts) Metals are generally located on the left side of the periodic table and tend to gain electrons when forming ions.

A. True
B. False

7. (70 pts) Match the most appropriate term with each definition:

<u>H.</u> Energy required to remove the most loosely bound electron from an atom or ion in its gaseous state

<u>B.</u> Chemical bond characterized by the sharing of valence electrons between two nonmetals

<u>D.</u> Represents the lowest whole-number ratio of atoms in a compound, determined experimentally

<u>E.</u> Represents the types and numbers of atoms present in a molecule, reflecting its true molecular composition.

<u>I.</u> Energy released when gaseous ions assemble into a crystalline solid; serves as a measure of the stability and strength of the ionic bonds in an ionic compound.

<u>F.</u> Represents the types and numbers of atoms present in a molecule along with how they are bonded to each other

<u>A.</u> Chemical bond formed through the electrostatic attraction between oppositely charged ions of a metal and nonmetal

A. Ionic Bond
B. Covalent Bond
C. Metallic Bond
D. Empirical Formula
E. Molecular Formula
F. Structural Formula
G. Numerical Formula
H. Ionization Energy
I. Lattice Energy
J. Potential Energy

8. Consider the following elements: Ge, Ga, Se.

A. (5 pts) Place the three elements in order of increasing first ionization energy.

Ga < Ge < Se

B. (15 pts) Explain the general trend (and reason for the trend) in first ionization energy as we move from left to right across a period on the periodic table.

First ionization energy <u>generally increases</u> as we move from left to right across a period because the <u>effective nuclear charge</u> experienced by the valence electrons <u>increases</u>.

9. (25 pts) Write the full ground state orbital diagram for Ti^{2+}.

1s	2s	2p	3s	3p	4s	3d
⇅	⇅	⇅ ⇅ ⇅	⇅	⇅ ⇅ ⇅	__	↑ ↑ __ __ __

10. (15 pts) The ground state electron configuration for cobalt is $1s^22s^22p^63s^23p^64s^23d^7$. Explain why cobalt is paramagnetic.

Cobalt is paramagnetic because it has <u>unpaired electrons</u>.

Quiz 3

Quiz 3 - Handouts

CH151 (AY26-1) Cadet Name (Last, First): ____________________________

Version 2 Section: ____________________________

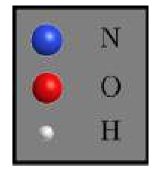
200 Points

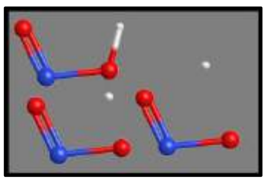
End-of-Hour Concept Quiz #3
(30 Minutes)

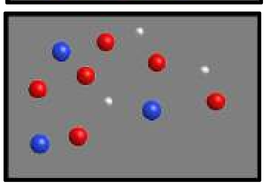
Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

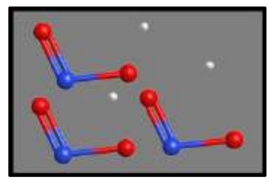
Choose the most appropriate answer for each of the following multiple-choice questions:

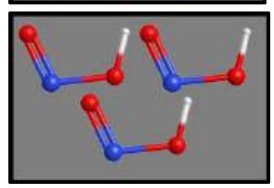
_____ **1.** (15 pts) Which diagram shows the weak electrolyte properties of nitrous acid (HNO_2) in water?



A. 

B. 

C. 

D. 

_____ **2.** (20 pts) Covalent compounds generally do not dissociate in water, whereas ionic compounds generally do dissociate in water.

A. True **B.** False

_____ **3.** (10 pts) If 100.0 mL of a 3.50 M solution of nitric acid is diluted to 700.0 mL, what is the concentration of the diluted solution.

A. 0.438 M **C.** 0.500 M

B. 28.0 M **D.** 24.5 M

_____ **4.** (15 pts) Which of the following best illustrates the law of conservation of mass in a chemical reaction?

A. When hydrogen gas reacts with oxygen gas to form water, the total mass of the water formed is less than the mass of the hydrogen and oxygen gases combined.

B. When solid sodium reacts with chlorine gas to form table salt (NaCl), the mass of the resulting salt is equal to the combined masses of sodium and chlorine used.

C. In a decomposition reaction, mass is destroyed as gas escapes into the air.

D. When vinegar reacts with baking soda, bubbles are produced, indicating that new mass is being created.

5. (25 pts) **Write** the balanced chemical equation to show the reaction of gaseous propane (C_3H_8) with gaseous oxygen to form gaseous carbon dioxide and gaseous water.

6. The core stage of NASA's space launch system used in its Artemis Moon missions is fueled by hydrogen and oxygen. Assume the tank containing liquid oxygen has a volume of 6.01×10^5 L and the tank containing liquid hydrogen has a volume of 1.61×10^6 L.

a. (65 pts) Given the balanced chemical equation below, identify the limiting reactant in core stage of the space launch system. Assume the following: both fuel tanks are full, the density of liquid oxygen is 1.141 g/mL and the density of liquid hydrogen is 0.0708 g/mL.

$$O_2(\ell) + 2\,H_2(\ell) \rightarrow 2H_2O(g)$$

Limiting Reactant: ____________________

b. (15 pts) Calculate the theoretical yield of water (in g).

Theoretical Yield: ____________________

c. (35 pts) Calculate how much of the excess reactant (in mol) remains.

Amount of Excess Reactant: ________________

Quiz 3 - Instructor Solution

Cut Scale

CH101 (AY26-1) Cadet Name (Last, First): ______________________________

Version 2 C-Number: ______________________________

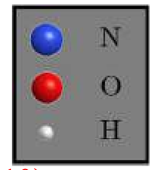
200 Points Section: ______________________________

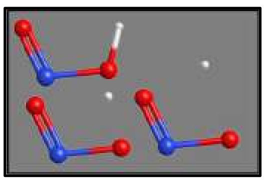
End-of-Hour Concept Quiz #3
(30 Minutes)

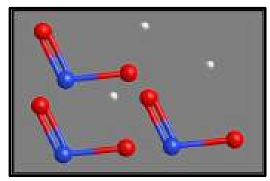
Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

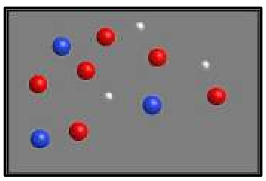
Choose the most appropriate answer for each of the following multiple-choice questions:

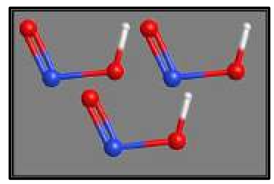
_A._ **1.** (15 pts) Which diagram shows the weak electrolyte properties of nitrous acid (HNO_2) in water?



A.  (correct)

C.  (-10)

B.  (-15)

D.  (-15)

_A._ **2.** (20 pts) Covalent compounds generally do not dissociate in water, whereas ionic compounds generally do dissociate in water.

A. True (correct) **B.** False (-20)

_C._ **3.** (10 pts) If 100.0 mL of a 3.50 M solution of nitric acid is diluted to 700.0 mL, what is the concentration of the diluted solution.

A. 0.438 M (-5, V_2 = 200 mL) **C.** 0.500 M (correct)

B. 28.0 M (-7 confused V_1 and V_2) **D.** 24.5 M (-10)

_B._ **4.** (15 pts) Which of the following best illustrates the law of conservation of mass in a chemical reaction?

(-15) **A.** When hydrogen gas reacts with oxygen gas to form water, the total mass of the water formed is less than the mass of the hydrogen and oxygen gases combined.

(correct) **B.** When solid sodium reacts with chlorine gas to form table salt (NaCl), the mass of the resulting salt is equal to the combined masses of sodium and chlorine used.

(-15) **C.** In a decomposition reaction, mass is destroyed as gas escapes into the air.

(-15) **D.** When vinegar reacts with baking soda, bubbles are produced, indicating that new mass is being created.

5. (25 pts) **Write** the balanced chemical equation to show the reaction of gaseous propane (C_3H_8) with gaseous oxygen to form gaseous carbon dioxide and gaseous water.

$$\underline{1}\ C_3H_8\ (g) + \underline{5}\ O_2\ (g) \rightarrow \underline{3}\ CO_2\ (g) + \underline{4}\ H_2O\ (g)$$

+3 reactants/products on correct sides of equation
+2 EA correct molecular formulas (8 pts total)
+1 EA correct state symbols (4 pts total)
+3 C balanced (FF)
+3 H balanced (FF)
+4 O balanced (FF)

6. The core stage of NASA's space launch system used in its Artemis Moon missions is fueled by hydrogen and oxygen. Assume the tank containing liquid oxygen has a volume of 6.01×10^5 L and the tank containing liquid hydrogen has a volume of 1.61×10^6 L.

a. (65 pts) Given the balanced chemical equation below, identify the limiting reactant in core stage of the space launch system. Assume the following: both fuel tanks are full, the density of liquid oxygen is 1.141 g/mL and the density of liquid hydrogen is 0.0708 g/mL.

$$O_2(\ell) + 2\ H_2(\ell)\ \rightarrow 2H_2O(g)$$

+18 concept (uses stoichiometry to determine amount of water produced)

+3 volume O_2	+1 mL → L	+3 density O_2	+2 MW O_2	+3 uses mol ratio	
6.01×10^5 L O_2	1000 mL O_2	1.141 g O_2	1 mol O_2	2 mol H_2O	$= \underline{4.29 \times 10^7 \text{ mol } H_2O\ (7.72 \times 10^8 \text{ g } H_2O)}$
	1 L O_2	mL O_2	32.00 g O_2	1 mol O_2	
			+1 correct MW	+3 correct mol ratio	

+3 volume H_2	+1 mL → L	+3 density H_2	+2 MW H_2	+3 uses mol ratio	
1.61×10^6 L H_2	1000 mL H_2	0.0708 g H_2	1 mol H_2	2 mol H_2O	$= 5.65 \times 10^7$ mol H_2O (1.02×10^9 g H_2O)
	1 L H_2	mL H_2	2.016 g H_2	2 mol H_2	
			+1 correct MW	+3 correct mol ratio	

+15 chooses the <u>reactant</u> that produces lowest yield/completely consumed (FF);
still award points if only did stoichiometry for one reactant

Limiting Reactant: O_2 (or Oxygen)

b. (15 pts) Calculate the theoretical yield of water (in g).

+5 (FF) uses the lowest yield OR yield produced by their identified limiting reactant

4.29×10^7 mol H_2O	18.016 g H_2O	$= 7.7\underline{21} \times 10^8$ g H_2O
	1 mol H_2O	
	+3 g → mol	+1 EA M/U/S
	+1 correct MW	

+3 provides a yield for water (not oxygen or hydrogen)

Theoretical Yield: 7.72×10^8 g H_2O

c. (35 pts) Calculate how much of the excess reactant (in mol) remains.

+10 concept (stoichiometry to find amount of excess reactant (FF) consumed)

+3 value (LR)	+1 L → mL	+1 mL → g		+2 uses molar ratio	
6.01×10^5 L O_2	1000 mL O_2	1.141 g O_2	1 mol O_2	2 mol H_2	$= 4.29 \times 10^7$ mol H_2 consumed
	1 L O_2	mL O_2	32.00 g O_2	1 mol O_2	
			+2 correct MW	+1 correct molar ratio	

Excess = available – consumed +10 concept (excess)
$= 5.65 \times 10^7$ mol – 4.29×10^7 mol
+1 application (H_2 available)
+1 application (H_2 consumed)

Amount of Excess Reactant: 1.37×10^7 mol H_2

$= 1.3\underline{68} \times 10^7$ mol H_2 remains +1 EA M/U/S

Question 6a Alternate Solution: Compare reactants to each other

+18 concept (uses stoichiometry to determine which reactant is consumed first)

+6 volume O_2	+2 mL → L	+6 density O_2	+4 MW O_2	+6 uses mol ratio	
6.01×10^5 L O_2	1000 mL O_2	1.141 g O_2	1 mol O_2	2 mol H_2	= 4.29×10^7 mol H_2 (7.72×10^8 g H_2)
	1 L O_2	mL O_2	32.00 g O_2	1 mol O_2	
			+2 correct MW	+6 correct mol ratio	

4.29×10^7 mol H_2 > $\underline{2.14 \times 10^7 \text{ mol } O_2}$

+15 chooses the reactant that would be consumed first (lowest value)

Question 6b Alternate Solution: Full stoichiometry from oxygen to water

+5 volume O_2	+1 mL → L	+1 density O_2	+1 MW O_2	+1 mol ratio		+1 EA M/U/S
6.01×10^5 L O_2	1000 mL O_2	1.141 g O_2	1 mol O_2	2 mol H_2O	18.016 g H_2O	= 7.72×10^8 g H_2O
	1 L O_2	mL O_2	32.00 g O_2	1 mol O_2	1 mol H_2O	

+3 provides a yield for water (not oxygen or hydrogen)

Question 6c Alternate Solution: Use of Theoretical Yield to Find Excess

+10 concept (stoichiometry to find amount of excess reactant (FF) consumed)

+3 value (TY)	+3 L → mL	+2 uses molar ratio	
7.72×10^8 g H_2O	1 mol H_2O	2 mol H_2	= 4.29×10^7 mol H_2 consumed
	18.016 g	2 mol H_2O	
		+2 correct molar ratio	

Excess = available – consumed +10 concept (excess)

= 5.65×10^7 mol – 4.29×10^7 mol

+1 application (H_2 available)

+1 application (H_2 consumed)

= 1.37×10^7 mol H_2 remains +3 M/U/S

Question 6c Alternate Solution: Difference in Yield, Then Stoichiometry

Excess = yield from excess reactant – theoretical yield +10 concept (excess)

= 5.65×10^7 mol – 4.29×10^7 mol

+1 application (H_2O from H_2)

+1 application (H_2O from O_2)

= 1.37×10^7 mol H_2O

+10 concept (stoichiometry to find amount of excess reactant (FF) consumed)

+3 value (FF)	+2 uses molar ratio	+1 EA M/U/S
$1.3\underline{68} \times 10^7$ mol H_2O	2 mol H_2	= 1.37×10^7 mol H_2 excess
	2 mol H_2O	
	+2 correct molar ratio	

Quiz 4

Quiz 4 - Handouts

CH151 (AY26-1) Cadet Name (Last, First): ____________________________

Version 1 Section: ____________________________

200 Points

End-of-Hour Concept Quiz #4
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). <u>Show all work</u> and *print clearly* to receive full credit.

<u>Choose the most appropriate answer for each of the following multiple-choice questions:</u>

_____ **1.** (15 pts) Lead(II) iodide crystals play an integral role in X-ray and gamma ray detectors. Choose the most appropriate *precipitation* reaction that produces solid lead(II) iodide.

A. $2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$
B. $2\ NaI(aq) + PbCO_3(s) \rightarrow PbI_2(s) + Na_2CO_3(aq)$
C. $CuI(s) + PbCO_3(s) \rightarrow PbI_2(s) + Cu_2CO_3(s)$
D. $2\ AgI(s) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ AgNO_3(aq)$

_____ **2.** (10 pts) Define an Arrhenius acid.

A. A proton donor
B. A proton acceptor
C. produces H^+ ions in aqueous solution
D. produces OH^- ions in aqueous solution

_____ **3.** (10 pts) Which element is reduced in the following reaction?

$$Zn(s) + 2\ AgNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + 2\ Ag(s)$$

A. Zn
B. Ag
C. N
D. O

_____ **4.** (10 pts) Given the following molecular equation, choose the correct complete ionic equation:

$$AgNO_3(aq) + KBr(aq) \rightarrow AgBr(s) + KNO_3(aq)$$

A. $Ag^+(aq) + NO_3^-(aq) + K^+(aq) + Br^-(aq) \rightarrow Ag^+(s) + Br^-(s) + K^+(aq) + NO_3^-(aq)$
B. $AgNO_3(s) + K^+(aq) + Br^-(aq) \rightarrow Ag^+(s) + Br^-(s) + K^+(aq) + NO_3^-(aq)$
C. $AgNO_3(aq) + KBr(aq) \rightarrow Ag^+(aq) + Br^-(aq) + KNO_3(s)$
D. $Ag^+(aq) + NO_3^-(aq) + K^+(aq) + Br^-(aq) \rightarrow AgBr(s) + K^+(aq) + NO_3^-(aq)$

_____ **5.** (15 pts) Choose the net ionic equation for the reaction (if any) that occurs when aqueous solutions of sodium sulfide and iron(II) nitrate are mixed.

A. $Na^+(aq) + NO_3^-(aq) \rightarrow NaNO_3(aq)$
B. $S^{2-}(aq) + Fe^{2+}(aq) \rightarrow FeS(s)$
C. $2\ Na^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ NaNO_3(aq)$
D. $2\ Na^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow FeS(s) + 2\ Na^+(aq) + 2\ NO_3^-(aq)$
E. No reaction occurs.

6. The following reaction occurs when acetic acid (vinegar) is mixed with ammonia:

$$CH_3COOH(aq) + NH_3(aq) \rightarrow CH_3COO^-(aq) + NH_4^+(aq)$$

a. (15 pts) Identify which species in the reaction is the *conjugate* base.

Answer 6a: ____________________

b. (20 pts) Calculate the pH of vinegar if $[H_3O^+]$ is 3.16×10^{-3} M.

Answer 6b: ____________________

7. One way to recycle aluminum is to treat it with a strong base (KOH) and neutralize it with a strong acid (H_2SO_4). The resulting salt is referred to as "Alum," which is often used in water treatment. The overall reaction is a REDOX reaction as written below:

$$2\ Al(s) + 2\ KOH(aq) + 4\ H_2SO_4(aq) + 22\ H_2O(l) \rightarrow 2\ KAl(SO_4)_2 \bullet 12\ H_2O(s) + H_2(g)$$

___	+1	-2	+1	___	___	___	+1	-2	→	+1	+3	+6	-2	___	___	0
Al	K	O	H	H	S	O	H	O		K	Al	S	O	H	O	H
$Al(s)$	$KOH(aq)$			$H_2SO_4(aq)$			$H_2O(l)$			$KAl(SO_4)_2 \bullet 12\ H_2O(s)$						$H_2(g)$

a. (42 pts) Assign an oxidation number for **each** atom in the reaction above:

b. (10 pts) Identify the reducing agent in the reaction above.

Answer 7b: ________________________

8. (20 pts) In fuel-cell vehicles, hydrogen fuel spontaneously reacts with oxygen gas to produce water, heat, and electricity. Identify what type of cell this exemplifies.

Answer 8: ________________________

9. (33 pts) **Identify** each of the aqueous reactions listed below as either a precipitation, acid-base, or redox reaction.

a. $2\ RbOH(aq) + H_2SO_4(aq) \rightarrow Rb_2SO_4(aq) + 2\ H_2O(l)$ ________________________

b. $Pb(s) + PbO_2(s) + 2\ HSO_4^-(aq) + 2\ H^+(aq) \rightarrow 2\ PbSO_4(s) + 2\ H_2O(l)$ ________________________

c. $MgCl_2(aq) + Na_2CO_3(aq) \rightarrow MgCO_3(s) + 2\ NaCl(aq)$ ________________________

Quiz 4 - Instructor Solution

Cut Scale

CH101 (AY26-1) Cadet Name (Last, First): ____________________________

Version 1 Section: ____________________________

200 Points

End-of-Hour Concept Quiz #4
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

Provide the most appropriate answer in the blanks below for each of the following questions:

A. **1.** (15 pts) Lead(II) iodide crystals play an integral role in X-ray and gamma ray detectors. Choose the most appropriate *precipitation* reaction that produces solid lead(II) iodide.

A. $2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$ (correct)
B. $2\ NaI(aq) + PbCO_3(s) \rightarrow PbI_2(s) + Na_2CO_3(aq)$ (-10)
C. $CuI(s) + PbCO_3(s) \rightarrow PbI_2(s) + Cu_2CO_3(s)$ (-15)
D. $2\ AgI(s) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ AgNO_3(aq)$ (-10)

C. **2.** (10 pts) Choose the most appropriate definition of an Arrhenius acid.

A. a proton donor (-5, B-L acid)
B. a proton acceptor (-10)
C. produces H^+ ions in aqueous solution (correct)
D. produces OH^- ions in aqueous solution (-10)

B. **3.** (10 pts) Which element is reduced in the following reaction?

$$Zn(s) + 2\ AgNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + 2\ Ag(s)$$

A. Zn (-10)
B. Ag (correct)
C. N (-10)
D. O (-10)

D. **4.** (10 pts) Given the following molecular equation, choose the correct complete ionic equation:

$$AgNO_3(aq) + KBr(aq) \rightarrow AgBr(s) + KNO_3(aq)$$

A. $Ag^+(aq) + NO_3^-(aq) + K^+(aq) + Br^-(aq) \rightarrow Ag^+(s) + Br^-(s) + K^+(aq) + NO_3^-(aq)$ (-7)
B. $AgNO_3(s) + K^+(aq) + Br^-(aq) \rightarrow Ag^+(s) + Br^-(s) + K^+(aq) + NO_3^-\ (aq)$ (-10)
C. $AgNO_3(aq) + KBr(aq) \rightarrow Ag^+(aq) + Br^-(aq) + KNO_3(s)$ (-10)
D. $Ag^+(aq) + NO_3^-(aq) + K^+(aq) + Br^-(aq) \rightarrow AgBr(s) + K^+(aq) + NO_3^-(aq)$ (correct)

B. **5.** (15 pts) Choose the net ionic equation for the reaction (if any) that occurs when aqueous solutions of sodium sulfide and iron(II) nitrate are mixed.

A. $Na^+(aq) + NO_3^-(aq) \rightarrow NaNO_3(aq)$ (-10)
B. $S^{2-}(aq) + Fe^{2+}(aq) \rightarrow FeS(s)$ (correct)
C. $2\ Na^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ NaNO_3(aq)$ (-15)
D. $2\ Na^+(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow FeS(s) + 2\ Na^+(aq) + 2\ NO_3^-(aq)$ (-5)
E. No reaction occurs. (-15)

6. The following reaction occurs when acetic acid (vinegar) is mixed with ammonia:

$$CH_3COOH(aq) + NH_3(aq) \rightarrow CH_3COO^-(aq) + NH_4^+(aq)$$

a. (15 pts) Identify which species in the reaction is the *conjugate* base.

Answer 6a: CH_3COO^- (+15 AON)
(-10 S.C. chooses the B-L base [NH_3])

b. (20 pts) Calculate the pH of vinegar if $[H_3O^+]$ is 3.16×10^{-3} M.

pH = -log $[H_3O^+]$ (+12 concept)
= -log [3.16×10^{-3} M] (+5 application)
= 2.5003 (+1 EA M/U/S)

Answer 6b: 2.500

7. One way to recycle aluminum is to treat it with a strong base (KOH) and neutralize it with a strong acid (H_2SO_4). The resulting salt is referred to as "Alum," which is often used in water treatment. The overall reaction is a REDOX reaction as written below:

$$2\ Al(s) + 2\ KOH(aq) + 4\ H_2SO_4(aq) + 22\ H_2O(l) \rightarrow 2\ KAl(SO_4)_2 \bullet 12\ H_2O(s) + H_2(g)$$

a. (42 pts) Assign an oxidation number for each remaining atom in the spaces below.

+7 EA AON

0	+1	-2	+1	+1	+6	-2	+1	-2		+1	+3	+6	-2	+1	-2	0
Al	K	O	H	H	S	O	H	O		K	Al	S	O	H	O	H
$Al(s)$		$KOH(aq)$			$H_2SO_4(aq)$		$H_2O(l)$		$\rightarrow$			$KAl(SO_4)_2 \bullet 12\ H_2O(s)$				$H_2(g)$

b. (10 pts) Identify the reducing agent in the reaction given above.

Answer 7b: $Al(s)$
(+10 AON, state symbol not required)

8. (20 pts) In fuel-cell vehicles, hydrogen fuel spontaneously reacts with oxygen gas to produce water, heat, and electricity. Identify what type of cell this exemplifies.

Answer 8: Voltaic (Galvanic) Cell (+20)
-10 S.C. "electrochemical cell"

9. (33 pts) **Identify** each of the aqueous reactions listed below as either a precipitation, acid-base, or redox reaction.

+11 EA AON

a. $2\ RbOH(aq) + H_2SO_4(aq) \rightarrow Rb_2SO_4(aq) + 2\ H_2O(l)$ Acid-Base

b. $Pb(s) + PbO_2(s) + 2\ HSO_4^-(aq) + 2\ H^+(aq) \rightarrow 2\ PbSO_4(s) + 2\ H_2O(l)$ Redox and/or Acid-Base

c. $MgCl_2(aq) + Na_2CO_3(aq) \rightarrow MgCO_3(s) + 2\ NaCl(aq)$ Precipitation

Quiz 5

Quiz 5 - Handouts

CH151 (AY26-1) Cadet Name (Last, First): ______________________________

Version 1 Section: ______________________________

200 Points

End of Hour Concept Quiz #5
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). Show all work and *print clearly* to receive full credit.

_____ **1.** (10 pts) Which of the following solutes will result in a solution with the lowest vapor pressure when mixed into solution with 1.0 L of water?

A. 0.5 mol NaCl
B. 0.5 mol glucose
C. 0.5 mol $MgCl_2$
D. 0.5 mol ethanol

_____ **2.** (15 pts) Which of the following solids has the lowest melting point?

A. Xe(s)
B. NaCl(s)
C. Fe(s)
D. C(s, graphite)

_____ **3.** (15 pts) Osmosis is best described as:

A. The flow of solvent from high to low solute concentration
B. The flow of solvent from low to high solute concentration
C. The flow of solute across a semi-permeable membrane
D. The flow of gases across a permeable membrane

_____ **4.** (15 pts) What is the osmotic pressure associated with a 1.50M solution of fructose at a temperature of 25 ºC?

A. 3.08 atm
B. 36.7 atm
C. 312 atm
D. 3720 atm

_____ **5.** (10 pts) Which of the following best describes metallic bonding?

A. Metal atoms equally share electrons between them to form stable molecules
B. Metal cations are held together by their attraction to a sea of delocalized electrons
C. Metal atoms transfer electrons among themselves, creating cations and anions
D. Metal atoms stick together through temporary dipoles caused by electron shifts.

6. (40 pts) **Circle** the most appropriate choice in each statement below:

a. Immediately after an M4 is fired, the high-pressure gas created by the firing process fills a sealed gas tube. To eliminate the high-pressure gas and return the weapon to standard conditions, the bolt carrier moves backwards **(*increasing* / *decreasing*)** the volume of the gas inside the M4. (Assume constant temperature and moles)

b. When you fill a HMMWV tire with air, you are increasing the amount of air in the tire, which (***increases / decreases***) the volume of the tire. (Assumes constant temperature and pressure)

c. During summer training, a UH-60 Black Hawk helicopter pilot says it's too hot outside to safely take off with all the Cadets on board. She explains that at hotter temperatures, the air is less dense because the air takes up (***more / less***) volume. (Assumes constant number of moles and pressure)

d. The situation described in part **(*a. / b. / c.*)** is an example of Charles' Law.

7. (30 pts) In lab, a Cadet mixes 7.5 grams of sodium chloride with a beaker filled with pure water. The Cadet realizes they forgot to measure the water before mixing so they weigh the solution and determine that the solution mass is 97.58 grams. What is the boiling point of the new solution?

$K_{b,water}$= 0.512 ºC/m ; $K_{f,water}$= 1.86 ºC/m ; $T_{f,water}$= 0.00 ºC ; $T_{b,water}$= 100.00 ºC

Answer 7: ________________

8. (65 points) A certain ***monatomic*** gas sample has a mass of 0.896 g. Its volume is 0.500 L at a temperature of 41.85°C and a pressure of 866.4 torr.

a. (45 pts) **Calculate** the amount of gas (in mol) in the sample.

Answer 8a: ________________

b. (15 pts) **Calculate** the molar mass of the gas.

Answer 8b: ________________

c. (5 pts) **Identify** the monatomic gas.

Answer 8c: ________________

Quiz 5 - Instructor Solution

CH151 (AY26-1) Cadet Name (Last, First): ____________________________

Version 1 Section: ____________________________

200 Points

End of Hour Concept Quiz #5
(30 Minutes)

Authorized references are a calculator (TI-36X Pro or equivalent) and 13th Edition reference data card (RDC). <u>Show all work</u> and *print clearly* to receive full credit.

__C__ **1.** (10 pts) Which of the following solutes will result in a solution with the lowest vapor pressure when mixed into solution with 1.0 L of water?

A. 0.5 mol NaCl +5 (recognizes "i" has relevance)
B. 0.5 mol glucose +0
C. 0.5 mol $MgCl_2$ +10
D. 0.5 mol ethanol +0

__A__ **2.** (15 pts) Which of the following solids has the lowest melting point?

A. Xe(s) +15
B. NaCl(s) +0
C. Fe(s) +0
D. C(s, graphite) +0

__B__ **3.** (15 pts) Osmosis is best described as:

A. The flow of solvent from high to low solute concentration +5 (backwards)
B. The flow of solvent from low to high solute concentration +15
C. The flow of solute across a semi-permeable membrane +5 (solute vs solvent)
D. The flow of gases across a permeable membrane +0

__B__ **4.** (15 pts) What is the osmotic pressure associated with a 1.50M solution of fructose at a temperature of 25 ºC?

A. 3.08 atm +10 used degrees C and not Kelvin with correct R
B. 36.7 atm +15
C. 312 atm +0 used 8.314 for R and degrees C not Kelvin
D. 3720 atm +10 used 8.314 for R with correct T

__B__ **5.** (10 pts) Which of the following best describes metallic bonding?

A. Metal atoms equally share electrons between them to form stable molecules +0
B. Metal cations are held together by their attraction to a sea of delocalized electrons +10
C. Metal atoms transfer electrons among themselves, creating cations and anions +0
D. Metal atoms stick together through temporary dipoles caused by electron shifts. +0

6. (40 pts) **Circle** the most appropriate choice in each statement below:
+10 points each (AON)

a. Immediately after an M4 is fired, the high-pressure gas created by the firing process fills a sealed gas tube. To eliminate the high-pressure gas and return the weapon to standard conditions, the bolt carrier moves backwards **(** ***increasing*** / ***decreasing*** **)** the volume of the gas inside the M4.

b. When you fill a HMMWV tire with air, you are increasing the amount of air in the tire, which (***increases*** / ***decreases***) the volume of the tire.

c. During summer training, a UH-60 Black Hawk helicopter pilot says it's too hot outside to safely take off with 11 Cadets on board. She explains that at hotter temperatures, the air is less dense because each the air takes up (***more*** / ***less***) volume.

d. The situation described in part **(** ***a.*** / ***b.*** / ***c.*** **)** is an example of Charles' Law.

7. (30 pts) In lab, a Cadet mixes 7.5 grams of sodium chloride with a beaker filled with pure water. The Cadet realizes they forgot to measure the water before mixing so they weigh the solution and determine that the solution mass is 97.58 grams. What is the boiling point of the new solution?

$K_{b,water}$= 0.512 ºC/m ; $K_{f,water}$= 1.86 ºC/m ; $T_{f,water}$= 0.00 ºC ; $T_{b,water}$= 100.00 ºC

+5 concept
$\Delta T = i*Kb*m$

Find Mass Solvent = 97.58g-7.5g = 90.08g ;
+5 solves for mass of solvent (final answer is 101.35 ºC if they skip this)

2	7.5 g	mol		1000 g	0.512 ºC
		58.44 g	90.08 g	kg	m

=1.5 ºC

+10 finds molality

+ 2 Converts g to kg (might be done when finding solvent)

+3 M/U/S

Answer 7: 101.5 ºC

+5 add ΔT to 100.0 ºC

8. A certain ***monatomic*** gas sample has a mass of 0.896 g. Its volume is 0.500 L at a temperature of 41.85°C and a pressure of 866.4 torr.

a. (45 pts) **Calculate** the amount of gas (in mol) in the sample. +12 concept: n=PV/RT

+5 convert T celsius to K

866.4 torr	1 atm	0.500 L	ºC mol	
	760 torr		0.08206 L atm	41.85°C + 273.15 K
+5 application pressure	+5 convert torr to atm	+5 application volume	+5 application R	+5 application T

= 2.21 x 10^{-2} mols

Answer 8a: 2.21 x 10^{-2} mols
+3 M/U/S

b. (15 pts) **Calculate** the molar mass of the gas.

+5 concept: molar mass = mass/ mols

0.896g	
	2.21 x 10^{-2} mols
+5 uses mass from problem	+2 uses moles form 8a.

= 40.6 grams/mol

Answer 8b: 40.6 grams/mol
+3 M/U/S

c. (5 pts) **Identify** the monatomic gas.

Answer 8c: Argon +5 FF

Lsn 33 Gas Laws	40	0.90%	1. Explain the ideal gas law and its component simple gas laws: Boyle's law, Charles' law, and Avogadro's law
	65	1.30%	2. Model ideal gas behavior with the ideal gas law and simple gas laws
	0	0.00%	3. Describe the behavior of a mixture of ideal gases and the behavior of an ideal gas in a mixture
	0	0.00%	4. Conceptually explain and contrast kinetic molecular theory in terms of ideal and real gas behavior
Lsn 36 Col. Prop	45	0.90%	1. Calculate the four colligative properties of solutions: vapor pressure lowering ($P_{solution}$ / Raoult's Law), freezing point depression (ΔT_f), boiling point elevation (ΔT_b), and osmotic pressure (π).
	15	0.20%	2. Explain the physical process of osmosis.
	10	0.20%	3. Understand and apply the van't Hoff factor to all four colligative properties.
Lsn 38 IMF2 / Solids	10	0.20%	1. Distinguish metallic bonding from covalent and ionic bonding.
		0.00%	2. Use electronegativity to classify bonding as covalent, ionic, or metallic.
	15	0.20%	3. Predict properties of substances resulting from the type of bonding involved.

TAB L LABORATORY REPORTS

Lab 1

Lab 1 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

DENSITY AND ERROR CALCULATIONS

Experiment I

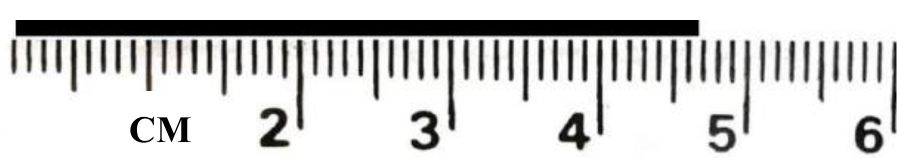
<u>Instructions:</u> Prior to the lab, read pages 1-11 to obtain information and equations needed to complete this lab. **The material covered here is testable.** During the lab, **only turn in pages 11-17 and the works cited page (if applicable).**

Materials

Reagents:	Water		
Equipment:	(2) 50 mL Beaker	(1) 25 mL Burette	Analytical Balance
	(1) 100 mL Beaker	(1) Burette Clamp	Unknown Metal Sample
	(1) 10 mL Graduated Cylinder	(1) Burette Stand	Vernier LabQuest Mini
	(1) 50 mL Graduated Cylinder	(1) Ruler	Vernier Small Thermistor

Measurements and Significant Figures

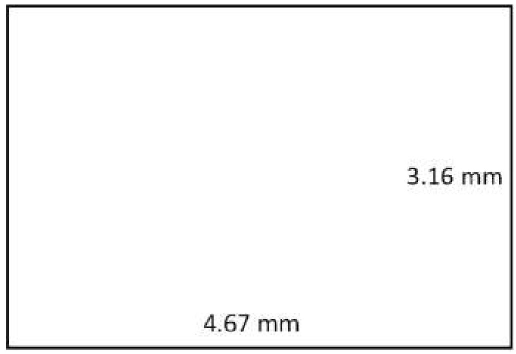
Scientists use significant figures to indicate the certainty they have in a measured quantity. Consider the black rectangle below measured with a ruler. How long is it?



The rectangle is clearly longer than 4.6 cm and shorter than 4.7 cm. It is possible to then say that the tenths place (the number right after the decimal) is known with *certainty*. However, the line is clearly not **exactly** 4.6 cm or 4.7 cm. Therefore, it is up to the observer to make a visual estimation of the value. Any value between the two numbers known

with certainty would be acceptable, 4.67 cm for example. This measurement would be said to have **three significant figures;** note that we include the *uncertain* digit in our measurement.

Significant figures are instrumental in understanding how certain we are about our measurements. For example, consider that you are using this same ruler to compute the area of the rectangle below.



The calculation is simple: $4.67\ mm * 3.16\ mm = 14.7572\ mm^2$. But while that is mathematically correct, the answer has six significant figures, which is a much higher degree of certainty than was measured. A more correct answer would reflect the certainty from the measurements taken to get the data. Therefore, the answer must have three significant figures, meaning that the area is 14.8 mm^2. This is a foundational concept in this course, covered in section E.3 - E.4 of your textbook.

Density

In this experiment, you will determine the densities of a liquid and a solid using procedures that involve the determination of sample mass and volume. Along the way, you will evaluate the precision with which three different volumetric devices can deliver a known volume to a second container. You will judge the precision by massing the volumes of liquid. In this case, the more precise metric—massing—will be used as the reference technique for comparison of the volumetric devices.

One of the fundamental properties of any sample of matter is its density (*d*), defined as the mass (m) per unit of volume (V).

$$d = \frac{m}{V} \qquad \text{(RDC, page 2)}$$

The density of water is exactly 1.0000 g/cm^3 at 4 °C and slightly lower at room temperature (0.9970 g/cm^3 at 25 °C). Densities of liquids and solids span a large range: osmium metal has a density of 22.5 g/cm^3 and is the densest material known at ordinary pressures. Note that cm^3 (cc or cubic centimeters) and mL (milliliters) are equivalent measurements (1 mL is, by definition, 1 cubic centimeter) and are typically used interchangeably (reference the table on page 3 of the RDC titled “Derived SI Units of Measure”).

In density determination, two quantities are measured: the mass and volume of a given quantity of matter. The mass can be determined by massing a sample of the substance on an analytical balance. The quantity we usually think of as “weight” is really the mass of a substance. The mass of a liquid sample in a container can be found by taking the difference between the mass of the liquid-filled container and the mass of the empty container.

The volume of a liquid can be determined by means of a calibrated container. In the laboratory, a graduated cylinder is often used for routine measurements of volume. In titration experiments where it is necessary to add a precise series of volumes of liquid to another container, a burette is employed. However, the most accurate measurements of liquid volume are made by using a pycnometer, which is simply a container having a precisely definable volume. One example of a pycnometer is a 10 mL pipet. The three instruments discussed are shown in Figure 1.

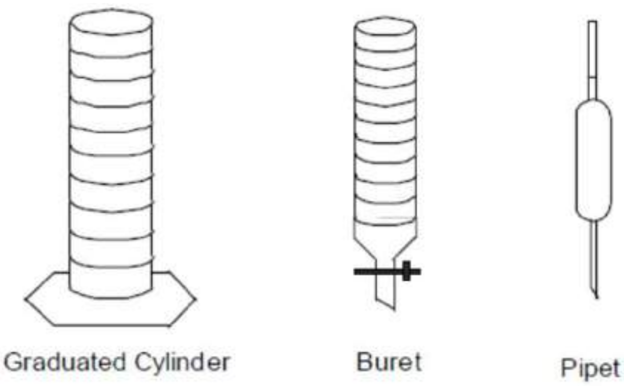


Figure 1: Equipment commonly used to measure volume. Source: http://depts.washington.edu

The volume of a solid can be measured directly if it has a regular geometric shape; however, many solid samples have an irregular shape. In these cases, we measure the volume indirectly by placing the object in a known volume of water. The volume of water that is displaced by the object will equal the volume of the object.

In most scientific experiments we evaluate our results by determining their uncertainty or error. The word "error" does not necessarily imply that one has made an error or been careless. Measured values always contain some amount of uncertainty.

Error Measurement

Errors in experimental measurements may be divided into two classes: (a) systematic errors and (b) random errors (SLAM pg. 3-10). **"Human error" is not a valid class of error.** What may be considered as "human error" falls into one of the other two error types.

Systematic errors: Systematic errors are reproducible and arise from failure or incompleteness of underlying theory, a shortcoming in the instrumentation or procedure, or failure to account for an experimental variable. We characterize the magnitude of systematic errors by specification of the **accuracy**. This term is defined as the difference between the observed value and the true value of the quantity. Thus, accuracy is a measure of the correctness of the result. An **accurate** measurement is one which the systematic errors have been eliminated as much as possible.

Fortunately, in the student laboratory, true values will be provided so that accuracy can be quantified. The goal of any experimental design is to measure values that are synonymous with the true values, but the reason for designing the experiment is that the true values are not yet known. As part of experimental design, researchers aim to reduce systematic error. One method involves repeating the experiment many times, adjusting variables to identify whether they are contributing to error in their measurements. This may include changing location, materials, instruments, time of day, etc. In practice, these manipulations are difficult to carry out, but for crucial experiments, such as testing a fundamental theory of nature, they are always done. More commonly, the researcher analyzes the experiment carefully, checking each step for ‘hidden’ sources of variability.

Accuracy is quantified by calculating the **percent error** of the measured value from the true value. **In this course, a percent error within 5% of the true value is considered accurate.** This quantity is calculated as follows (RDC page 2):

$$\%\, error = \left(\left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right|\right) * 100\% \qquad \textbf{(1)}$$

Random errors: Random errors generate variation in the experimental values from trial to trial or experiment to experiment. We characterize the reproducibility of a measurement as **precision**. The precision of a series of measurements thus reflects how closely each measurement in a series agrees with the others. In many cases, precision is influenced by the consistency of one’s measurement technique—for example, the precision of pipetting a volume of a liquid generally improves with practice. Additionally, precision is influenced by physical effects beyond the control of the scientist. Consider a time that you weighed yourself and chose to do so multiple consecutive times. You likely observed slightly different weights even though the true value isn’t changing. To characterize random error and deduce an experimental value that is as close as possible to the true value, scientists use the patterns from the values they collect after repeating a measurement many times.

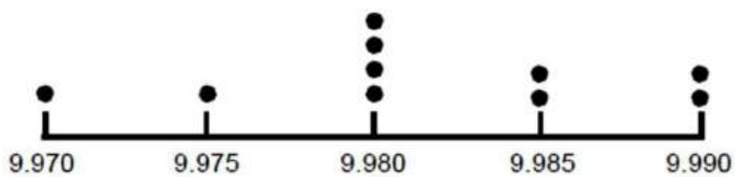


Figure 2: Dot diagram of ten volume measurements dispensed by a 10 mL pipet. Source: http://depts.washington.edu

As an example, consider the ten volume measurements dispensed by a 10 mL pipet represented in Figure 2. The true values are 9.988 mL, 9.973 mL, 9.986 mL, 9.980 mL, 9.975 mL, 9.982 mL, 9.986 mL, 9.982 mL, 9.981 mL, and 9.990 mL. In the figure, we can see how the measurements are distributed: there is a range associated with the values (about 0.025 mL) as well as a cluster of values at 9.980 mL.

Consider the effect of taking many measurements. If there were only three measurements instead of 10, the pattern would not have reflected both the range and the clustering of the possible values. Quantifying precision requires two concepts from statistics: mean and standard deviation. In order to calculate these two values, you use the number of measurements (*N*), and the value of each measurement (*x*). Frequently, subscripts are used to differentiate between measurements. For example, our second measurement would be known as x_2 and have a value of 9.973 mL.

The **mean** $(\bar{x})$ is also known as the average. It is calculated by adding the measurements and then dividing that sum by the number of measurements. N is the total number of measurements (in this case 10), i is the sequential number of the measurement (in this case i ranges from 1 to 10), and x_i is the value of measurement *i* (in this case 9.988 mL, 9.973 mL, 9.986 mL, 9.980 mL, 9.975 mL, 9.982 mL, 9.986 mL, 9.982 mL, 9.981 mL, and 9.990 mL). The mean of the volume samples is 9.9823 mL.

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad \textbf{(2)}$$

The **standard deviation (*s*)** indicates the amount of uncertainty in a set of experimental values. It is calculated by finding the difference between each measurement and the mean, summing the squares of those values, dividing that sum by one less than the total number of measurements, and then calculating the square root of that value. The standard deviation of the volume samples is 0.005438 mL. Note that standard deviation is measured in the same units as the original data.

$$s = \sqrt{\frac{1}{N-1} * \sum_{i=1}^{N}(x_i - \bar{x})^2} \qquad \textbf{(3)}$$

The **coefficient of variation (CV)** is a useful descriptive statistic that reflects the data's dispersion about and in proportion to the mean. A standard deviation of 0.005438 mL seems small, but if the mean value was 0.01 mL, that standard deviation is in fact quite large. The coefficient of variation is calculated by dividing a data set's standard deviation (s) by its mean ($\bar{x}$). **For the purposes of this course, a coefficient of variation that is equal to or less than 0.05 (unitless) or 5% is an indication of precise data.**

$$CV = \frac{s}{\bar{x}} * 100\% \qquad \textbf{(4)}$$

<u>ATTENTION:</u> Equations 1-4 must be used when asked to comment on the accuracy and precision of measurements in this course. You will not receive credit for answers about accuracy that do not show a percent error calculation, and you will not receive credit for answers about precision that do not show a coefficient of variation calculation.

SAMI Checklist

Before beginning any experiment, take time to familiarize yourself with the various glassware and instruments you will use in the laboratory. Reference the CH101 SAMI Layout Guide in Canvas for the location of each item. Inventory your lab station by completing the "check-in" column of the Lab Equipment Inventory Sheet on the next page. Show your instructor when the inventory is complete; save a copy to your computer for use in future labs. **The checklist is worth 5 points of your overall lab grade.**

In the first part of this experiment, you will evaluate the precision in volume measurements using three lab instruments:

- a beaker
- a graduated cylinder
- a burette

In the second part of this experiment, you will find the density of and identify an unknown metal. Note: When measuring the volume of liquid in lab glassware, the bottom of the liquid's meniscus must be level with the mark on the glassware (see Figure 3). Many pieces of volumetric glassware have multiple marks, so it is important to identify the location of the mark you are measuring to before beginning to add any sample.

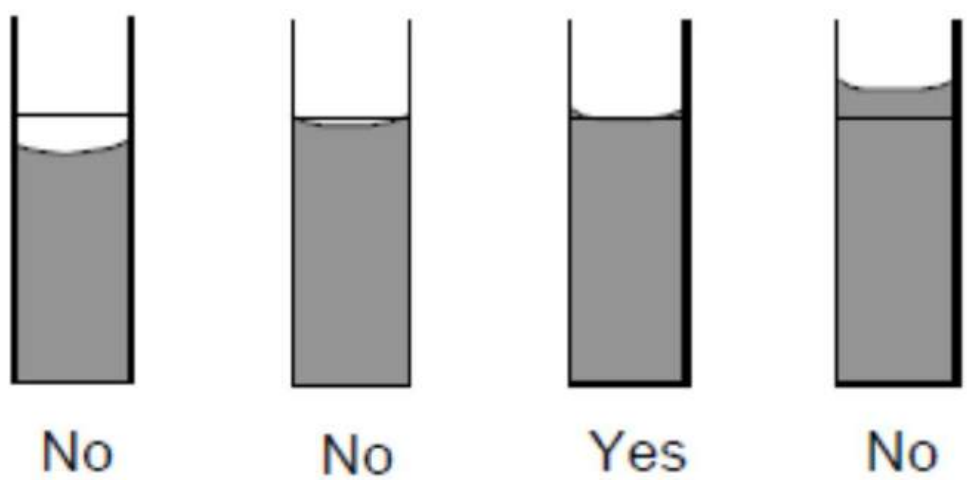


Figure 3: Correct relationship of a meniscus to the measurement mark. Source: http://depts.washington.edu

Part I Procedure: Density of a Known Liquid

Determine the Precision of a Volume Measurement using a 50 mL Beaker

1. Wash **two** 50 mL beakers with soap and rinse thoroughly with distilled water. Use the tape at the instructor bench to label the beakers as Beaker A and Beaker B. *Observe the increments of the graduated marks on the side of the glassware. Determine whether you can estimate the volume to one more decimal place(s).*

2. With one 50 mL beaker (**Beaker A**), obtain ~40 mL of water from the nearest instructor station in the center aisle. The water will be taken from a 1 L bottle using the repipet dispenser already affixed to the bottle. Throughout the course, **NEVER adjust the volume setting on the repipet dispensers**. If the volume is set to 10 mL, obtain four 10 mL pumps. Reference SLAM page 2-7 for instructions on using the repipet.

3. After dispensing ~40mL of distilled water into **Beaker A**, measure the temperature of the water using the small Vernier thermistor (reference Appendix B of the lab introduction for instructions to set up the thermistor).

Water Temperature: ________________________℃

4. Determine the **Water Density** from **Water Temperature** using Appendix A.

Water Density: ________________________ kg/m^3

5. Obtain the mass of the second 50 mL beaker **(Beaker B**) and record it in Table 1.

6. Pour out 10 mL of water from **Beaker A** into **Beaker B**. Then, obtain the new mass of Beaker B and record it in Table 1.

7. Dispose of the water in **Beaker B** and dry **Beaker B**.

8. Conduct a second trial by repeating Steps 5 and 6 with **Beaker B**. Record Trial 2 values in Table 1.

9. Rinse and dry both **Beaker A** and **Beaker B.**

Determine the Precision of a Volume Measurement using a 10 mL Graduated Cylinder

1. Wash your 10 mL graduated cylinder with soap and rinse thoroughly with distilled water. *Observe the increments of the graduated marks on the side of the glassware. Determine whether you can estimate the volume to one more decimal place(s).*
2. Ensure Beaker A is dried. Obtain and record its **dry** mass in line 1 of Table 2.
3. Using your station's distilled water squirt bottle, dispense 10 mL of distilled water into the 10 mL graduated cylinder. Reference the bottom of the meniscus, as shown in Figure 3, to measure this volume correctly. Pour the water from your graduated cylinder into Beaker A and obtain the mass of the water-filled beaker. Record this value in line 2 of Table 2.
4. Dispose of the water in your lab station's sink and repeat steps 2 and 3 for Trial 2.

Determine the Precision of a Volume Measurement using a 25 mL Burette

1. Use your burette clamp (shaped like an 'X') to secure your 25 mL burette to a burette stand (the stand with a white base) as shown in the SLAM (Page 2-8, Figure 2-11). **Verify your setup with your instructor before proceeding.**
2. Thoroughly rinse the burette with distilled water. The distilled water should drain evenly from the inside surface—no droplets should be left behind. If necessary, continue rinsing the burette with distilled water until it drains cleanly.
3. Place Beaker B underneath the 25 mL burette (this is now a waste beaker). Use your station's distilled water bottle to fill the burette to its topmost marking (for some burettes it will be 0 mL; for others it will be 25 mL). Open the stopcock so that water freely flows into the waste beaker until you have drained 2-3 mL into the waste beaker. Close the stopcock and empty the waste beaker. Refill the 25 mL burette so that the bottom of the meniscus meets the topmost marking of the burette.
4. Ensure Beaker A is dried. Obtain and record its **dry** mass in line 2 of Table 3.
5. Dispense 10 mL of water into the beaker. Record the "volume of water delivered" in line 1 of Table 3. This is calculated by subtracting the final reading on the burette from the initial reading.
6. Obtain the mass of the beaker and water and record this value in line 3 on Table 3.
7. Dispose of the water in your lab station's sink and repeat steps 4-6 for Trial 2.

Appendix A: Density of Water Based on Temperature

Table 1A: Density of Water at Various Temperatures (°C)

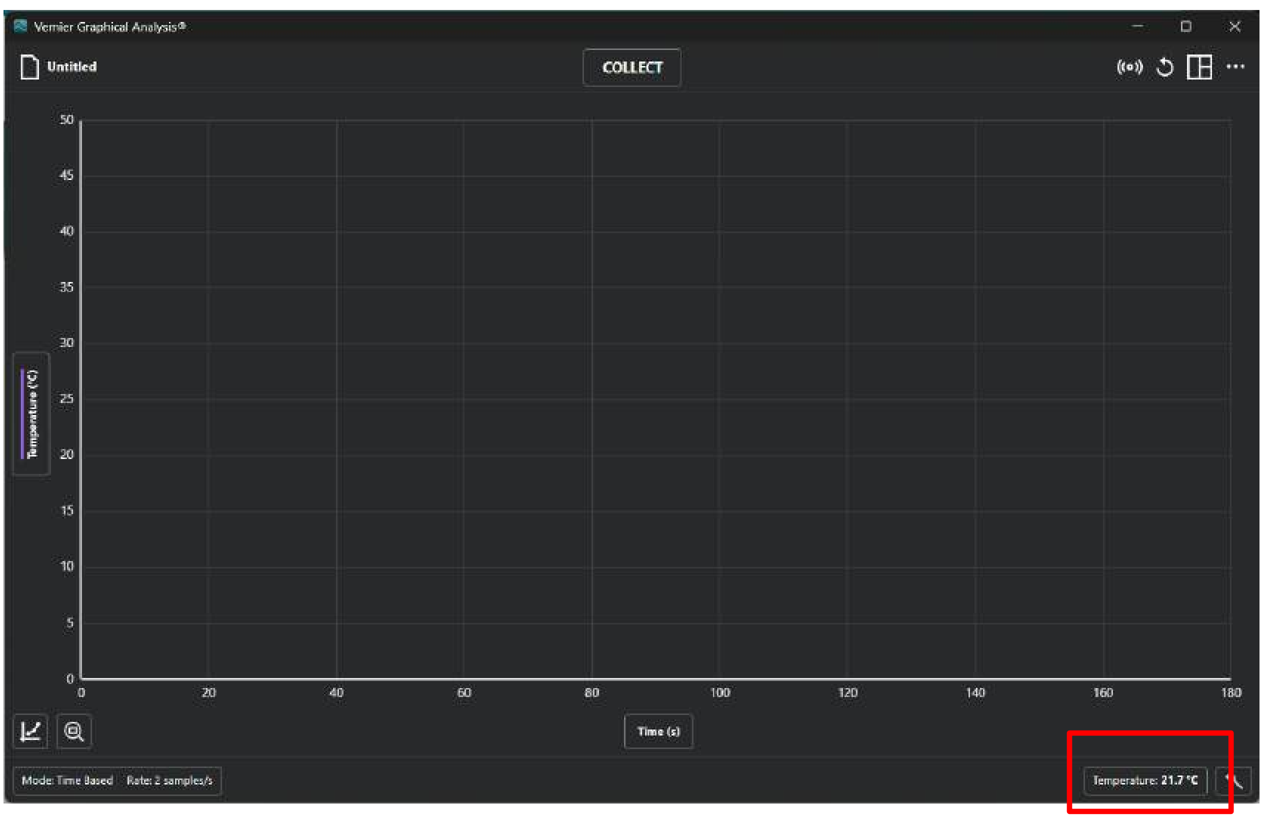
	d, kg/m³									
t,C.	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	999.8426	999.8493	999.8558	999.8622	999.8683	999.8743	999.8801	999.8857	999.8912	999.8964
1	999.9015	999.9065	999.9112	999.9158	999.9202	999.9244	999.9284	999.9323	999.9360	999.9395
2	999.9429	999.9461	999.9491	999.9519	999.9546	999.9571	999.9595	999.9161	999.9636	999.9655
3	999.9672	999.9687	999.9700	999.9712	999.9722	999.9731	999.9738	999.9743	999.9747	999.9749
4	999.9750	999.9748	999.9746	999.9742	999.9736	999.9728	999.9719	999.9709	999.9696	999.9683
5	999.9668	999.9651	999.9632	999.9612	999.9591	999.9568	999.9544	999.9518	999.9490	999.9461
6	999.9430	999.9398	999.9365	999.9330	999.9293	999.9255	999.9216	999.9175	999.9132	999.9088
7	999.9043	999.8996	999.8948	999.8898	999.8847	999.8794	999.8740	999.8684	999.8627	999.8569
8	999.8509	999.8448	999.8385	999.8321	999.8256	999.8189	999.8121	999.8051	999.7980	999.7908
9	999.7834	999.7759	999.7682	999.7604	999.7525	999.7444	999.7362	999.7279	999.7194	999.7108
10	999.7021	999.6932	999.6842	999.6751	999.6658	999.6564	999.6468	999.6372	999.6274	999.6174
11	999.6074	999.5972	999.5869	999.5764	999.5658	999.5551	999.5443	999.5333	999.5222	999.5110
12	999.4996	999.4882	999.4766	999.4648	999.4530	999.4410	999.4289	999.4167	999.4043	999.3918
13	999.3792	999.3665	999.3536	999.3407	999.3276	999.3143	999.3010	999.2875	999.2740	999.2602
14	999.2464	999.2325	999.2184	999.2042	999.1899	999.1755	999.1609	999.1463	999.1315	999.1166
15	999.1016	999.0864	999.0712	999.0558	999.0403	999.0247	999.0090	998.9932	998.9772	998.9612
16	998.9450	998.9287	998.9123	999.8957	998.8791	998.8623	998.8455	998.8285	998.8114	998.7942
17	998.7769	998.7595	998.7419	998.7243	998.7065	998.6886	998.6706	998.6525	998.6343	998.6160
18	998.5976	998.5790	998.5604	998.5416	998.5228	998.5038	998.4847	998.4655	998.4462	998.4268
19	998.4073	998.3877	998.3680	998.3481	998.3282	998.3081	998.2880	998.2677	998.2474	998.2269
20	998.2063	998.1856	998.1649	998.1440	998.1230	998.1019	998.0807	998.0594	998.0380	998.0164
21	997.9948	997.9731	997.9513	997.9294	997.9073	997.8852	997.8630	997.8406	997.8182	997.7957
22	997.7730	997.7503	997.7275	997.7045	997.6815	997.6584	997.6351	997.6118	997.5883	997.5648
23	997.5412	997.5174	997.4936	997.4697	997.4456	997.4215	997.3973	997.3730	997.3485	997.3240
24	997.2994	997.2747	997.2499	997.2250	997.2000	997.1749	997.1497	997.1244	997.0990	997.0735
25	997.0480	997.0223	996.9965	996.9707	996.9447	996.9186	996.8925	996.8663	996.8399	996.8135
26	996.7870	996.7604	996.7337	996.7069	996.6800	996.6530	996.6259	996.5987	996.5714	996.5441
27	996.5166	996.4891	996.4615	996.4337	996.4059	996.3780	996.3500	996.3219	996.2938	996.2655
28	996.2371	996.2087	996.1801	996.1515	996.1228	996.0940	996.0651	996.0361	996.0070	995.9778
29	995.9486	995.9192	995.8898	995.8603	995.8306	995.8009	995.7712	995.7413	995.7113	995.6813

Appendix B: Vernier LabQuest Mini and Thermistor Set-up

1. Log on to your personal computer.

2. Plug the USB cable from the LabQuest Mini unit into one of the USB ports of your computer. Connect the thermistor to the Channel I port of the LabQuest Mini unit (See picture below).

3. Open the Vernier Graphical Analysis software on your computer. The temperature will be displayed in the bottom right corner of the screen, as shown below.



4. You are now ready to start running your experiment and collecting data.

UNITED STATES MILITARY ACADEMY

LAB 1 – DENSITY AND ERROR CALCULATIONS

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:
SIGNATURE: DATE:
SIGNATURE: DATE:

Instructions: Mark the quantity of each item at your lab station. If you are missing anything, notify your instructor. They will assist you in obtaining the required item(s).

Left Cabinet

Qty.	ITEM
	A. 1x Black Ring Stand
	B. 1x White Burette Stand
	C. 1x Braided gas hose
	D. 1x Bunsen burner
	E. 1x Test tube rack (red)
	F. 6x Test tubes

Left Top Drawer

Qty.	ITEM
	A. 2x 250mL Erlenmeyer flasks
	B. 1x Glass funnel (upside down)
	C. 1x Mortar and pestle set
	D. 2x 250mL Beakers
	E. 1x 600mL Beaker
	F. 2x 100mL Beakers
	G. 2x 50mL Beakers
	H. 1x Box of matches
	I. 10x Vernier cuvettes
	J. 2x glass droppers
	K. 3x Cleaning brushes (2 small, 1 large)
	L. 1x Metal spatula
	M. 1x Glass stir rod
	N. 1x Glass stir rod with rubber scraper
	O. 1x Forceps
	P. 1x Test tube holder
	Q. 1x Tweezer

Left Second Drawer

Qty.	ITEM
	A. 2x 50mL volumetric flasks with caps
	B. 2x 100mL volumetric flasks with caps
	C. 1x 250mL volumetric flask with cap
	D. 1x 25mL Burette
	E. 1x 10mL graduated cylinder
	F. 1x 50mL graduated cylinder
	G. 1x 500mL Erlenmeyer flask

Left Third Drawer

Qty.	ITEM
	A. 1x Burette clamp
	B. 1x Large Ring attachment
	C. 1x Medium Ring attachment
	D. 1x Small Ring attachment
	E. 1x Large 3-prong clamp
	F. 1x Medium 3-prong clamp
	G. 1x Small 3-prong clamp
	H. 1x Large tube clamp
	I. 1x Medium tube clamp
	J. 1x Small tube clamp

Left Bottom Drawer

Qty.	ITEM
	A. 1x Heat/stir plate
	B. 1x Magnetic stir bar (on top of hot plate)
	C. 1x Vernier equipment box, including the following:
	a. 2x Temperature Probes: 1x Small and 1x Large
	b. 1x set of conductivity wires
	c. 1x Gas Pressure Sensor
	d. 1x Black Go Direct SpectroVis Plus Box
	e. 1x Gray LabQuest Mini Box
	f. 1x Power Adapter
	g. 1x USB Data Cable

Right Cabinet

Qty.	ITEM
	A. 1x Black Ring Stand
	B. 1x White Burette Stand
	C. 1x Braided gas hose
	D. 1x Bunsen burner
	E. 1x Test tube rack (red)
	F. 6x Test tubes

Right Top Drawer

Qty.	ITEM
	A. 2x 250mL Erlenmeyer flasks
	B. 1x Glass funnel (upside down)
	C. 1x Mortar and pestle set
	D. 2x 250mL Beakers
	E. 1x 600mL Beaker
	F. 2x 100mL Beakers
	G. 2x 50mL Beakers
	H. 1x Box of matches
	I. (EMPTY - no Vernier cuvettes on right side)
	J. 2x glass droppers
	K. 3x Cleaning brushes (2 small, 1 large)
	L. 1x Metal spatula
	M. 1x Glass stir rod
	N. 1x Glass stir rod with rubber scraper
	O. 1x Forceps
	P. 1x Test tube holder
	Q. 1x Tweezer

Right Second Drawer

Qty.	ITEM
	A. 2x 50mL volumetric flasks with caps
	B. 2x 100mL volumetric flasks with caps
	C. 1x 250mL volumetric flask with cap
	D. 1x 25mL Burette
	E. 1x 10mL graduated cylinder
	F. 1x 50mL graduated cylinder
	G. 1x 500mL Erlenmeyer flask
	H. 1x Buchner Funnel (2 pieces) with rubber stopper
	I. 1x Sidearm Flask

Right Third Drawer

Qty.	ITEM
	A. 1x Burette clamp
	B. 1x Large Ring attachment
	C. 1x Medium Ring attachment
	D. 1x Small Ring attachment
	E. 1x Large 3-prong clamp
	F. 1x Medium 3-prong clamp
	G. 1x Small 3-prong clamp
	H. 1x Large tube clamp
	I. 1x Medium tube clamp
	J. 1x Small tube clamp

Right Bottom Drawer

Qty.	ITEM
	A. 2-3x Apron(s)
	B. 1x CH101 Lab SAMI Quick Reference Guide

Top Bench Items

Qty.	ITEM
	1x Distilled Water bottle (water level above blue line)
	1x Soap bottle (soap level above blue line)
	3x Bench valves (Gas, Air, Vacuum) turned off
	Benchtop wiped down

Part I Discussion:

Table 1: 50 mL Beaker Results				
		1	2	3
1	Mass of empty beaker (g)			
2	Mass of beaker and water (g)			
3	Mass of water (g)			
4	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

Table 2: 10 mL Graduated Cylinder Results				
		1	2	3
1	Mass of empty beaker (g)			
2	Mass of beaker and water (g)			
3	Mass of water (g)			
4	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

Table 3: 25 mL Burette Results				
		1	2	3
1	Volume of water delivered (mL)			
2	Mass of empty beaker (g)			
3	Mass of beaker and water (g)			
4	Mass of water (g)			
5	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

1. (5 pts) Use dimensional analysis to convert the water density you determined on page 7 from units of kg/m^3 to g/mL. **Show all work.**

Water Density: ____________________ kg/m^3 Water Density:_______________ g/mL

2. (5 pts) Using the mass of water and your calculated water density, calculate the volume of water obtained for each trial and the mean volume of water. Record your answers in Tables 1-3. **Only show your work below using the data from your Trial 1 Beaker measurements.** *This calculation is detailed in the "Density" section of the introduction.*

3. (5 pts) Using the calculated mean volumes of water for each instrument, **calculate** the percent error of your results given an accepted water volume of 10.000 mL. Record your answers in Tables 1-3. **Only show your work below using the data from your Beaker measurements.**

4. (5 pts) Identify which instrument(s) (beaker, graduated cylinder, and/or burette) produced accurate volume results. Support your choice(s) by analyzing the percent error calculated from the experimental data from each instrument. (*Note: In CH101, percent error should always be used to discuss accuracy*).

5. (5 pts) Using the calculated volumes of water for each trial of each instrument, determine the standard deviation and coefficient of variation and record them in Tables 1-3. *Calculations of mean volume, standard deviation, and coefficient of variation are detailed in the "Error Measurement" section of the introduction*. **Only show your work below using the data from your Beaker measurements.**

6. (5 pts) Identify which instrument(s) produced precise volume results. Support your choice(s) by analyzing the coefficient of variation calculated from the experimental data from each instrument. (*Note: In CH101, coefficient of variation should always be used to discuss precision*).

Part II Procedure and Discussion: Density of an Unknown Solid

Obtain a piece of unknown metal from the instructor bench. For this portion of the lab, you will determine the density of this piece of metal.

1. (10 pts) Write a short, stepwise **PROCEDURE** that you can use to determine the density of your unknown metal. Include the specific type and size of glassware used for each step. Prescribe at least **three trials** in your procedure.

2. (5 pts) Create a table to record the **RESULTS** of your experiment; follow your procedure and record the results of each trial.

1. (5 pts) Using the table provided below, identify the piece of metal. **Explain** your selection and use pertinent, quantitative results to support your identification.

	Aluminum	Copper	Lead	Nickel	Tin	Zinc
Density (g/mL)	2.6989	8.96	11.35	8.902	7.29	7.134

Unknown Metal: ___________________________

Explanation:

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 1 - Instructor Solution

CUT SCALE

UNITED STATES MILITARY ACADEMY

LAB 1 – DENSITY AND ERROR CALCULATIONS

CH101: GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

SIGNATURES: ____

____

____

Instructions: Mark the quantity of each item at your lab station. If you are missing anything, notify your instructor. They will assist you in obtaining the required item(s).

Left Cabinet	
Qty.	**ITEM**
	A. 1x Black Ring Stand
	B. 1x White Burette Stand
	C. 1x Braided gas hose
	D. 1x Bunsen burner
	E. 1x Test tube rack (red)
	F. 6x Test tubes
Left Top Drawer	
Qty.	**ITEM**
	A. 2x 250mL Erlenmeyer flasks
	B. 1x Glass funnel (upside down)
	C. 1x Mortar and pestle set
	D. 2x 250mL Beakers
	E. 1x 600mL Beaker
	F. 2x 100mL Beakers
	G. 2x 50mL Beakers
	H. 1x Box of matches
	I. 10x Vernier cuvettes
	J. 2x glass droppers
	K. 3x Cleaning brushes (2 small, 1 large)
	L. 1x Metal spatula
	M. 1x Glass stir rod
	N. 1x Glass stir rod with rubber scraper
	O. 1x Forceps
	P. 1x Test tube holder
	Q. 1x Tweezer
Left Second Drawer	
Qty.	**ITEM**
	A. 2x 50mL volumetric flasks with caps
	B. 2x 100mL volumetric flasks with caps
	C. 1x 250mL volumetric flask with cap
	D. 1x 25mL Burette
	E. 1x 10mL graduated cylinder
	F. 1x 50mL graduated cylinder
	G. 1x 500mL Erlenmeyer flask
Left Third Drawer	
Qty.	**ITEM**
	A. 1x Burette clamp
	B. 1x Large Ring attachment
	C. 1x Medium Ring attachment
	D. 1x Small Ring attachment
	E. 1x Large 3-prong clamp
	F. 1x Medium 3-prong clamp
	G. 1x Small 3-prong clamp
	H. 1x Large tube clamp
	I. 1x Medium tube clamp
	J. 1x Small tube clamp
Left Bottom Drawer	
Qty.	**ITEM**
	A. 1x Heat/stir plate
	B. 1x Magnetic stir bar (on top of hot plate)
	C. 1x Vernier equipment box, including the following:
	a. 2x Temperature Probes: 1x Small and 1x Large
	b. 1x set of conductivity wires
	c. 1x Gas Pressure Sensor
	d. 1x Black Go Direct SpectroVis Plus Box
	e. 1x Gray LabQuest Mini Box
	f. 1x Power Adapter
	g. 1x USB Data Cable

Right Cabinet	
Qty.	**ITEM**
	A. 1x Black Ring Stand
	B. 1x White Burette Stand
	C. 1x Braided gas hose
	D. 1x Bunsen burner
	E. 1x Test tube rack (red)
	F. 6x Test tubes
Right Top Drawer	
Qty.	**ITEM**
	A. 2x 250mL Erlenmeyer flasks
	B. 1x Glass funnel (upside down)
	C. 1x Mortar and pestle set
	D. 2x 250mL Beakers
	E. 1x 600mL Beaker
	F. 2x 100mL Beakers
	G. 2x 50mL Beakers
	H. 1x Box of matches
	I. (EMPTY - no Vernier cuvettes on right side)
	J. 2x glass droppers
	K. 3x Cleaning brushes (2 small, 1 large)
	L. 1x Metal spatula
	M. 1x Glass stir rod
	N. 1x Glass stir rod with rubber scraper
	O. 1x Forceps
	P. 1x Test tube holder
	Q. 1x Tweezer
Right Second Drawer	
Qty.	**ITEM**
	A. 2x 50mL volumetric flasks with caps
	B. 2x 100mL volumetric flasks with caps
	C. 1x 250mL volumetric flask with cap
	D. 1x 25mL Burette
	E. 1x 10mL graduated cylinder
	F. 1x 50mL graduated cylinder
	G. 1x 500mL Erlenmeyer flask
	H. 1x Buchner Funnel (2 pieces) with rubber stopper
	I. 1x Sidearm Flask
Right Third Drawer	
Qty.	**ITEM**
	A. 1x Burette clamp
	B. 1x Large Ring attachment
	C. 1x Medium Ring attachment
	D. 1x Small Ring attachment
	E. 1x Large 3-prong clamp
	F. 1x Medium 3-prong clamp
	G. 1x Small 3-prong clamp
	H. 1x Large tube clamp
	I. 1x Medium tube clamp
	J. 1x Small tube clamp
Right Bottom Drawer	
Qty.	**ITEM**
	A. 2-3x Apron(s)
	B. 1x CH101 Lab SAMI Quick Reference Guide

Top Bench Items	
Qty.	**ITEM**
	1x Distilled Water bottle (water level above blue line)
	1x Soap bottle (soap level above blue line)
	3x Bench valves (Gas, Air, Vacuum) turned off
	Benchtop wiped down

/5

Part I Discussion:

Table 1: 50 mL Beaker Results				
		1	2	3
1	Mass of empty beaker (g)			
2	Mass of beaker and water (g)			
3	Mass of water (g)			
4	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

Table 2: 10 mL Graduated Cylinder Results				
		1	2	3
1	Mass of empty beaker (g)			
2	Mass of beaker and water (g)			
3	Mass of water (g)			
4	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

Table 3: 25 mL Burette Results				
		1	2	3
1	Volume of water delivered (mL)			
2	Mass of empty beaker (g)			
3	Mass of beaker and water (g)			
4	Mass of water (g)			
5	Calculated volume of water (mL)			
5	Mean volume (mL)			
6	Percent Error			
7	Standard Deviation (mL)			
8	Coefficient of Variation			

1. (5 pts) Use dimensional analysis to convert the water density you determined on page 12 from units of kg/m³ to g/mL. **Show all work.**

Water Density: ______________________ kg/m³ Water Density:________________ g/mL

$$\frac{998.1019\ kg}{m^3} \times \frac{1000\ g}{1\ kg} \times \frac{1 \times 10^{-6}\ m^3}{1\ cm^3} \times \frac{1\ cm^3}{mL} = 0.9981019\ {}^{g}/_{mL}$$

+1 pt: convert kg to g
+1 pt: convert m^3 to cm^3
+1 pt: convert cm^3 to mL
+1 pt: math and units correct
+1 pt: significant figures correct

2. (5 pts) Using the mass of water and your calculated water density, calculate the volume of water obtained for each trial and the mean volume of water and record your answers in Tables 1-3. *This calculation is detailed in the "Density" section of the introduction*. **Only show your work below using the data from your Trial 1 Beaker measurements.**

$$\frac{9.329\ \text{g} * 1\ \text{mL}}{0.9981019\ \text{g}} = 9.347\ mL$$

+1 pt: concept d=m/v
+2 pts: application: manipulates equation correctly
+1 pt: math, units, and significant figures correct (2/3)

$$\bar{x} = \frac{11.722 + 11.715 + 9.347}{3} = 10.928\ mL$$

+1 pt: shows work/calculates mean volume correctly

3. (5 pts) Using the calculated mean volumes of water for each instrument, **calculate** the percent error of your results given an accepted water volume of 10.000 mL. **Only show your work below using the data from your Beaker measurements.**

$$\%\ error = \left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right| \times 100\%$$

+2 pts: concept

+2 pts: application

$$\%\ error = \left|\frac{10.928 - 10.000}{10.000}\right| \times 100\% = 9.2800\%$$

+1 pt: M/U/S (2/3)

/15

4. (5 pts) Identify which instrument(s) (beaker, graduated cylinder, and/or burette) produced accurate volume results. Support your choice(s) by analyzing the percent error calculated from the experimental data from each instrument. (*Note: In CH101, percent error should always be used to discuss accuracy*).

The buret was the most accurate across these trials, as the average volume delivered (10.003 mL) came the closest to the desired 10.00 mL volume. The percent error of the buret was 0.03% while the volumetric pipet's percent error was 1.5%, further supporting that the buret the most accurate glassware.

+2 pts: glassware with a percent error of less than 5% selected
+3 pts: explanation for selection based on percent error

-1 SC: only chooses "most accurate" glassware when other glassware is also accurate

5. (5 pts) Using the calculated volumes of water for each trial of each instrument, determine the standard deviation and coefficient of variation and record them in Tables 1-3. *Calculations of mean volume, standard deviation, and coefficient of variation are detailed in the "Error Measurement" section of the introduction*. **Only show your work below using the data from your Beaker measurements.**

$$\sigma = \sqrt{\frac{(11.722 - 10.928)^2 + (11.715 - 10.928)^2 + (9.329 - 10.928)^2}{3 - 1}} = 1.3796$$

$$CV = \frac{\sigma}{\bar{x}} = \frac{1.3796}{10.928} = .1262 * 100\% = 12.62\%$$

+1 pts: standard deviation formula used
+1 pt: correct values used

+2 pt: coefficient of variation formula used
+1 pt: math, units, significant figures correct (2/3)

6. (5 pts) Identify which instrument(s) produced precise volume results. Support your choice(s) by analyzing the coefficient of variation calculated from the experimental data from each instrument. (*Note: In CH101, coefficient of variation should always be used to discuss precision*).

The graduated cylinder and the buret were both precise. This was evident by the very low (lower than 5%) coefficient of variation shown with both of these pieces of glassware.

+2 pts: glassware with a CV of less than 5% selected
+3 pts: explanation for selection based on data, mentions CV

-1 SC: only chooses "most precise" glassware when other glassware is also precise

$/_{15}$

Part II Procedure and Discussion: Density of an Unknown Solid

Obtain a piece of the unknown metal from the instructor bench. For this portion of the lab, you will determine the density of this piece of metal.

1. (10 pts) Write a short, stepwise procedure that you can use to determine the density of your unknown metal. Include the specific type and size of glassware used for each step. Prescribe at least **three trials** in your procedure.

Possible procedures could involve the following:

- Uses ruler to measure metal's dimensions to determine volume; weighs metal to determine mass
- Uses water displacement to determine volume; weighs metal to determine mass

+3 pts: valid method to determine mass
+3pts: valid method to determine volume
+3 pts: mentions glassware used
+1 pts: repeat trials

2. (5 pts) Create a table to record your data; follow your procedure and record the results of each trial.

Observations of unknown metal: silvery-white color.

	Trial #1 Data:	Trial #2:	Trial #3:
Metal mass:	27.532 g	27.678 g	27.752 g
Initial volume:	30.0 mL	30.1 mL	30.0 mL
Final volume:	40.5 mL	40.7 mL	40.6 mL
Volume displaced:	10.5 mL	10.7 mL	10.6 mL
Calculated density:	2.62 g/mL	2.59 g/mL	2.62 g/mL

Average density: 2.61 g/mL

+4 pts: table has all relevant data
+1 pt: table is neatly presented

/15

1. (5 pts) Using the table provided below, identify the piece of metal. **Explain** your selection and use pertinent, quantitative results to support your identification.

	Aluminum	Copper	Lead	Nickel	Tin	Zinc
Density (g/mL)	2.6989	8.96	11.35	8.902	7.29	7.134

Unknown Metal: Aluminum

Explanation:

Aluminum, our density of 2.61 g/mL most closely matches the aluminum in the table provided.

+2 pts: correct metal is identified
+3 pts: explanation provided using quantitative data

Lab 2

Lab 2 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

ELECTROMAGNETIC SPECTRUM AND SPECTROPHOTOMETRY

Experiment II

Instructions: Prior to the lab, read pages 1-10 to obtain information and equations needed to complete this lab. **The material covered here is testable.** During the lab, **only turn in pages 15-19 and the works cited page (if applicable).**

Materials

Reagents:	0.50 M $Ba(NO_3)_2$ (aq)	0.50 M $Cu(NO_3)_2$ (aq)	Unknown Red Solution
	1.00 M $LiNO_3$ (aq)	1.00 M KNO_3 (aq)	Unknown #1
	0.50 M $Sr(NO_3)_2$ (aq)	1.00 M $NaNO_3$ (aq)	Unknown #2
	2.00 M $CaCl_2$ (aq)	1.33 mM Red Dye Solution	Unknown #3
Equipment:	(11) Scintillation Vials	(1) Bunsen Burner	Vernier Spectrophotometer
	(1) 50 mL Beaker (1) 100 mL Beaker	(2) 10 mL Grad. Cylinder	Wood Splints
	(1) 250 mL Beaker	(8) Cuvettes	Disposable Pipettes
	(1) 600 mL Beaker	Tissue Paper	

Part I: Introduction

A laboratory technician has been sorting waste for disposal. Most of the bottles are properly labeled with their contents, but three bottles are labeled “ionic aqueous solution.” The laboratory technician wants more specific information about which metal ions are present to determine what type of disposal would be most appropriate. The first objective of this lab is to identify the metal ions in these three solutions. The laboratory technician knows that solutions containing metal ions frequently display characteristic colors in the flame of a Bunsen burner (see Figure 1 below).

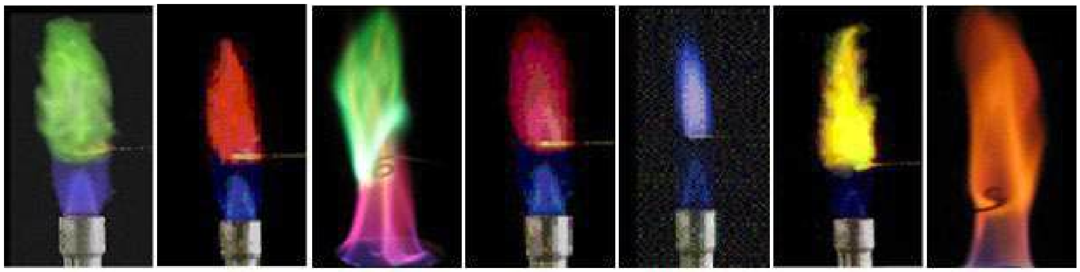


Figure 1: Flame tests of Barium, Calcium, Copper(II), Lithium, Potassium, Sodium, and Strontium (from left to right). Sources: Amazing Rust Website and Volland, Walt.

The heat of the flame excites electrons in the metal ion from a ground state energy level to an excited state with higher energy. When the electrons return to lower energy levels, they release energy as electromagnetic radiation. The color (and wavelength, λ) of this radiation depends on the energy difference between the two levels (Figure 2). Each metal ion has a different number of protons in its nucleus, so the spacing between energy levels in each ion is slightly different. This means that the set of wavelengths (or colors) of light emitted by each ion will be unique. We can observe this light, called an emission spectrum, to identify which atoms or ions are present in a solution.

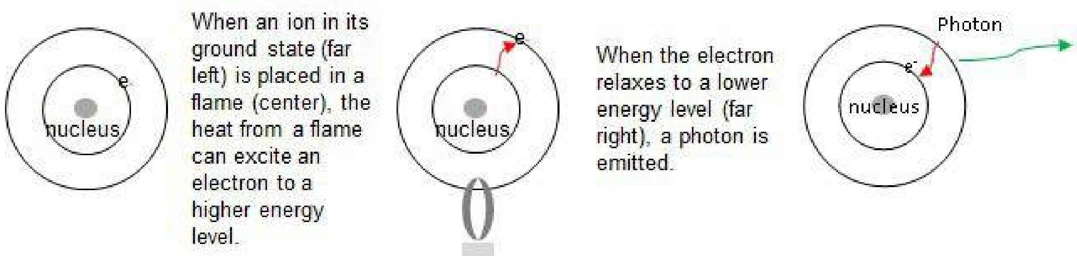


Figure 2: Depiction of absorption and emission of energy by an electron.

The wavelengths emitted by metal atoms or ions are often in the visible spectrum. This means the laboratory technician can use a flame test to make a preliminary identification of which metals are present—the first step in safely disposing of the solutions. The next step will be to determine the concentration of the ions in solution.

Part II: Introduction

The second objective of this laboratory is to determine the concentration of an unknown solution using absorption spectrophotometry. If two solutions of the same substance are placed in the path of a light beam, the more concentrated solution will absorb more light. Therefore, absorbance (A) and concentration (C) are directly proportional ($\propto$), or $A \propto C$. This is shown in Figure 3, where the concentration of the solutions is indicated by the number of dots; denser dots indicate the more concentrated solution.

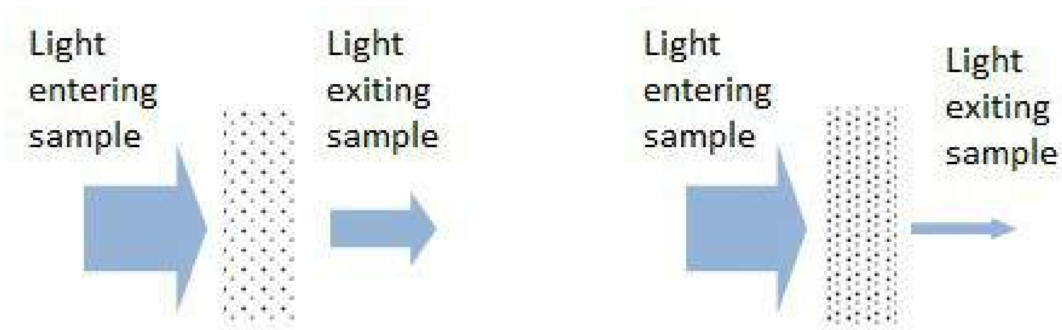


Figure 3: Absorbance of light by a dilute sample (left diagram) and a more concentrated sample (right diagram). The difference in the width of the arrows entering and exiting the sample corresponds (qualitatively) to the amount of light absorbed by the solution.

Concentration is generally measured as molarity, M (in mol/L). This means a specific number of moles are dissolved in a known volume of solution. Recall that moles can be found by dividing a given mass of an element or compound by the molar mass of that element or compound. This number of moles is then dissolved in a solvent divided by a known volume of solution. For example, 0.5 moles of NaCl dissolved in water that makes 1 L of solution will have a molarity of 0.5 M and 0.8 moles of NaCl dissolved in water to make 2. L of solution will have a molarity of 0.4 M.

If the same solution is poured into two vials, a narrow one and a wide one, the solution in the wide vial will absorb more light. The width of the vial is called the path length, b. Path length is also directly proportional to absorbance. Combining these two relationships gives us $A \propto bC$. Scientists can prepare solutions at several known concentrations and then measure the solutions' absorbance values to create a calibration curve. Using the linear fit equation of the calibration curve, y=mx+b, one can determine the concentration (x) of an unknown solution using the absorbance (y) of the unknown. An example of a set of data where the various concentrations of a sample are plotted against their absorbance and correlation is measured using a linear regression is shown below in Figure 4. The equation of this line, the Beer-Lambert Law, is:

$$A = \epsilon * b * C \tag{1}$$

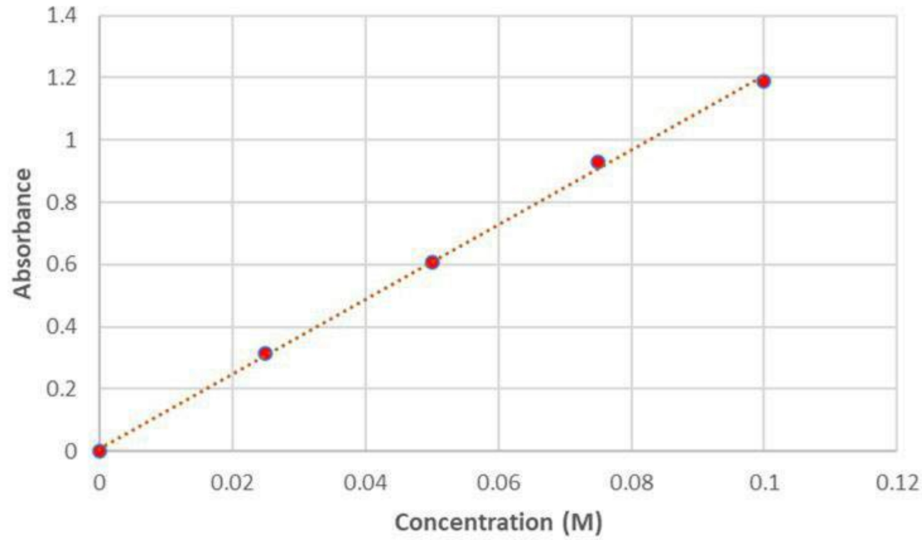


Figure 4: An example of a Beer-Lambert Law plot

In our laboratory, we can create solutions with known concentrations, and we can measure the absorbance of a solution using a spectrophotometer. Thus, we can generate a Beer-Lambert Law plot. The value ϵ (molar absorptivity) is a constant that depends on the wavelength at which absorbance is being measured, the specific substance being analyzed, and the solvent used. Molar absorptivity can range from less than one to over 10,000. **For the purposes of this lab assume the molar absorptivity constant is 1.**

If we compare the Beer-Lambert Law to the standard equation for a straight line, $y = mx + b$, we see that **the slope of the line, m, corresponds to $\epsilon * b$ in the Beer-Lambert Law.** Although we expect zero absorbance for a solution of zero concentration, sometimes in practice a Beer-Lambert plot will have a non-zero y-intercept. When this happens, the y-intercept should still be very close to zero (<0.01).

The linear regression should include all the experimental data points collected. In linear regression analysis, an indicator of the quality of the data is the correlation coefficient, or R value. The closer the R is to a value of 1, the better the fit of all the data to a straight line. If the data is a good fit for the regression line, then the $y = mx + b$ equation will serve as an accurate model for unknown solutions with absorbance values that fall within the limits of the calibration curve.

Part I: Procedure

1. Obtain 11 scintillation vials containing known and unknown solutions (these will be on your lab bench). Place a wood splint in each vial and allow the splints to soak for at least 5 minutes.

2. While the splints soak, set-up a Bunsen burner IAW the SLAM. Once you have reviewed the procedure for lighting a Bunsen burner, request the presence of an instructor. **Do not light the Bunsen burner without first obtaining the approval of your instructor.** Once the Bunsen burner is lit, the instructor will advise you on adjusting your flame; you should aim to have a blue flame that is 4-5 inches tall.

3. Fill a 600-mL beaker with tap water. This will be used to extinguish the wood splints in the next step.

4. Remove a wood splint from the distilled water sample and place the soaked end directly in the blue flame. **CAUTION: Ensure your fingers remain far from the flame.** Note any changes to the color of the flame. Over time, the wood splint will begin to burn. For this lab, we are interested in the flame color prior to the combustion of the wood splint. Once the wood begins to burn, immediately quench the splint in the 600-mL beaker you prepared.

5. Following the distilled water sample, repeat the procedure described in Step 4 with each of the remaining samples of known solutions. Observe and record the color (with specificity) and intensity (light/ medium/ dark) of the flame in Table 1 (page 10). **NOTE: it may take several seconds for a color to fully manifest from the soaked splints**. You will use your descriptions to help you identify the ions in the unknown solutions. Use Appendix B of the SLAM to help you describe these colors in terms of hue, lightness, and saturation.

6. Complete Table 1 by filling in the chemical symbol for the **ion** (i.e. K^+, not potassium, not K) observed with each solution.

7. Follow the same procedure for the three unknown salt solutions and record your results in Table 2. **NOTE: a trace of sodium is found as an impurity in almost all substances stored in glass bottles. Do not report sodium unless that is all you see.**
Hint: If the unknown has a flame color that is similar to two of the known compounds, it may be helpful to do another trial with those known compounds and see which one is most like the unknown.

Part II: Procedure

1. Open the Vernier Spectral Analysis program on your computer.

2. Insert the USB cable from the Vernier SpectroVis Plus spectrophotometer into your computer's USB port and ensure the power and USB lights turn green on the spectrophotometer. An image of the visible region of the electromagnetic spectrum will be shown on the screen. This indicates that the spectrophotometer is working correctly.

3. Rinse the inside of one cuvette with distilled water several times, then fill the cuvette with **distilled water** and cap it. This is the BLANK sample. Wipe the sides of the cuvette with a tech wipe to remove any oils or other particles from the outside. The cuvette has two ridged sides and two flat sides. Insert the cuvette with the flat (clear) side aligned to the arrow as shown in Figure 5 below.

Figure 5: Proper placement of the cuvette in the spectrophotometer

4. Select the "Absorbance" experiment from the options, then select the "vs. Wavelength (Full Spectrum)". The lamp will take 60-90 seconds to warm-up. Once the warm-up is complete click the "Finish Calibration" button on the pop-up window. If you have trouble, refer to Figure 7.

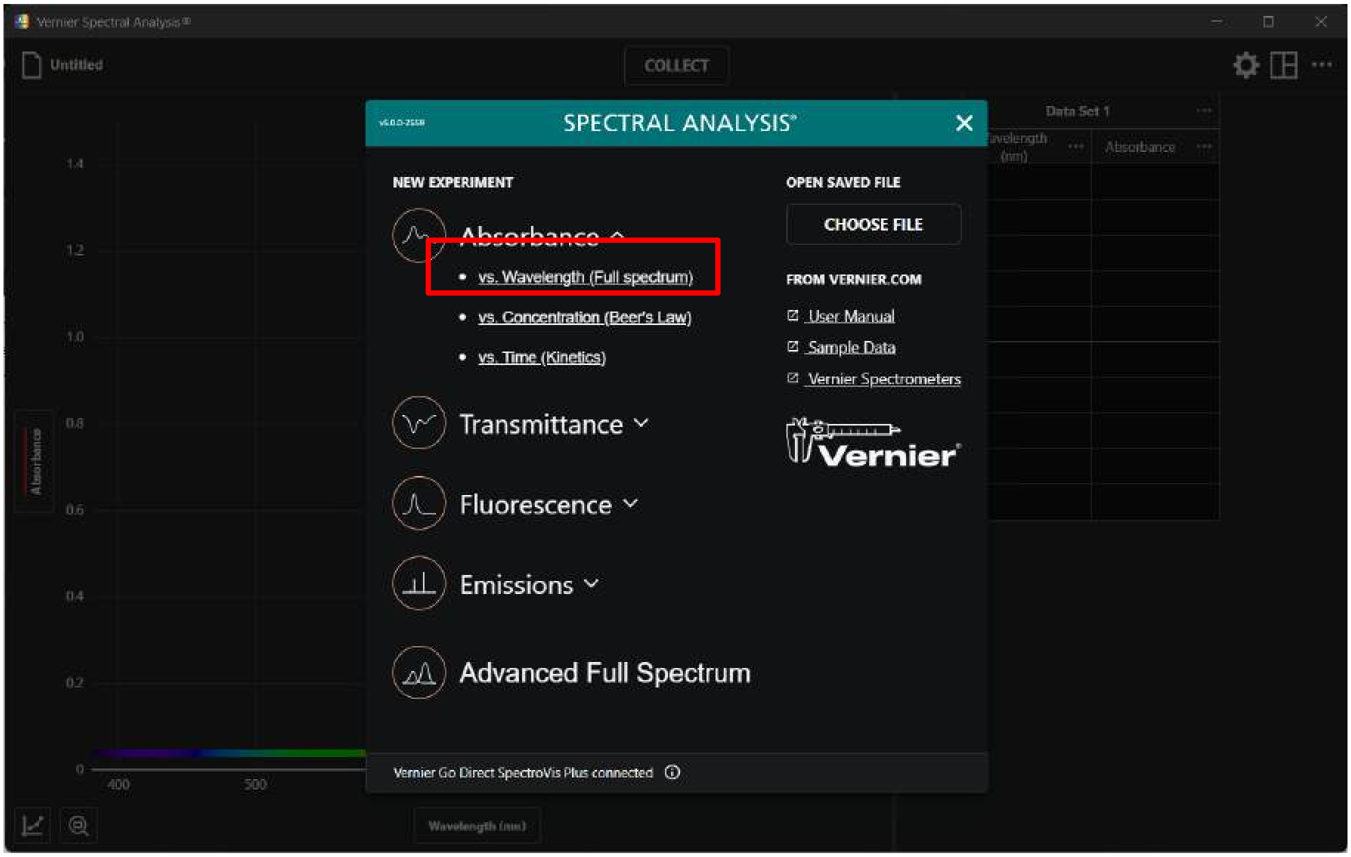


Figure 6: Experiment Selection menu

5. Once the spectrophotometer completes the blank calibration, obtain approximately 20 mL of the **1.33 mM red dye solution** from the instructor bench in a 100 mL beaker. Fill a second cuvette with approximately 3 mL of the **1.33 mM red dye solution** and insert into the spectrophotometer. Click the “Collect” button at the top of the control panel. The spectrum will update with a curve showing the absorbance across all visible wavelengths. Click “Stop” on the tool bar at the top of the control panel.

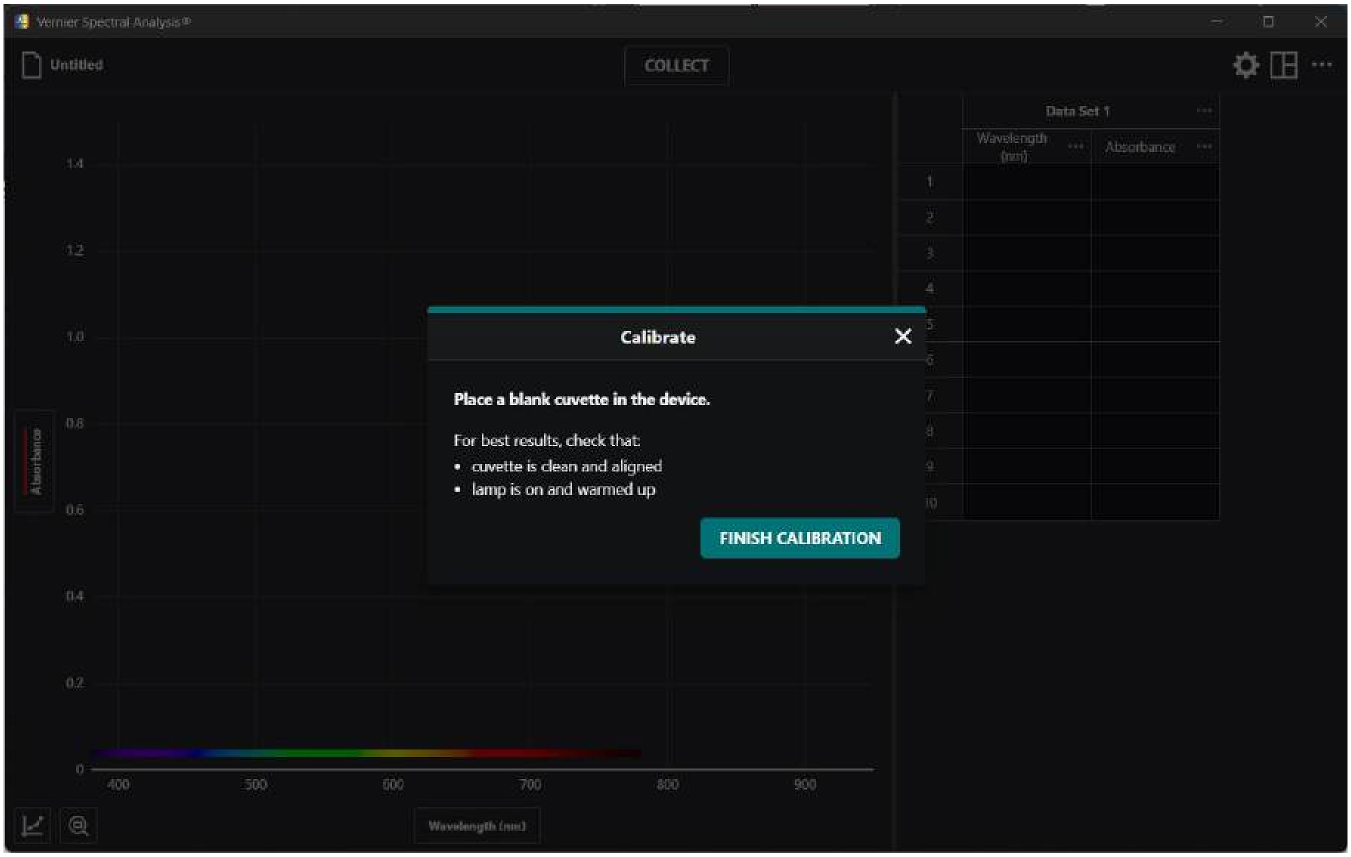


Figure 7: Set up procedures to calibrate the blank cuvette.

6. Select the maximum wavelength by clicking on the plot, as shown in Figure 8. **Ensure that this value is in a *smooth* part of the curve near the maximum absorbance**. Record that value below.

Wavelength of maximum absorbance: ________________________

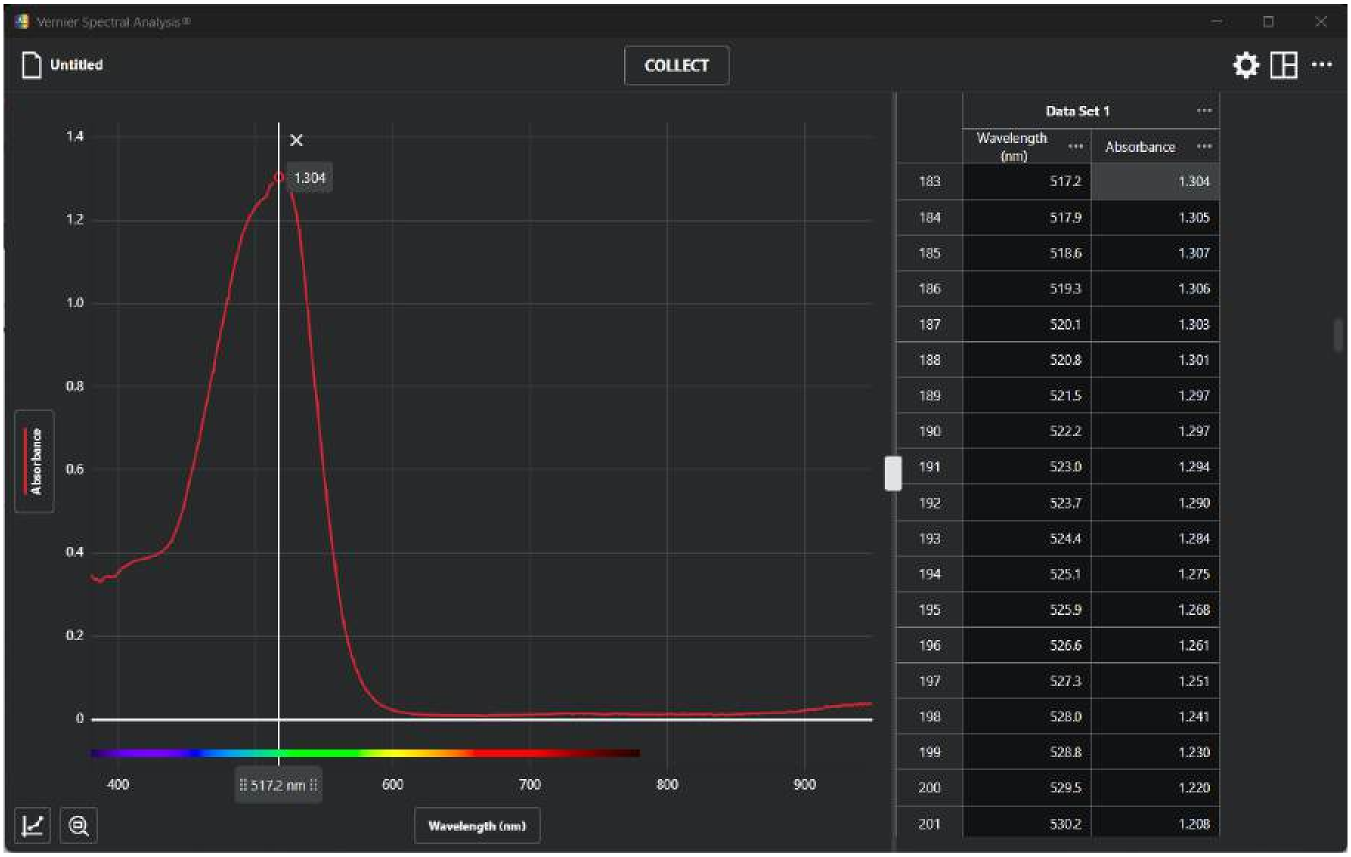


Figure 8: Screen shot of procedure to change spectrophotometer from full spectrum to single wavelength mode.

7. Select “New Experiment” from the file dropdown in the upper left corner of the window. Now select, “vs. Concentration (Beer’s Law)” from the screen, as shown in Figure 9. Enter the wavelength of maximum absorption in the input box. Reinsert your blank cuvette and select “Calibrate Spectrophotometer”. Once calibration is complete, select “Done”. Reference Figure 10 for calibration and experiment setup.

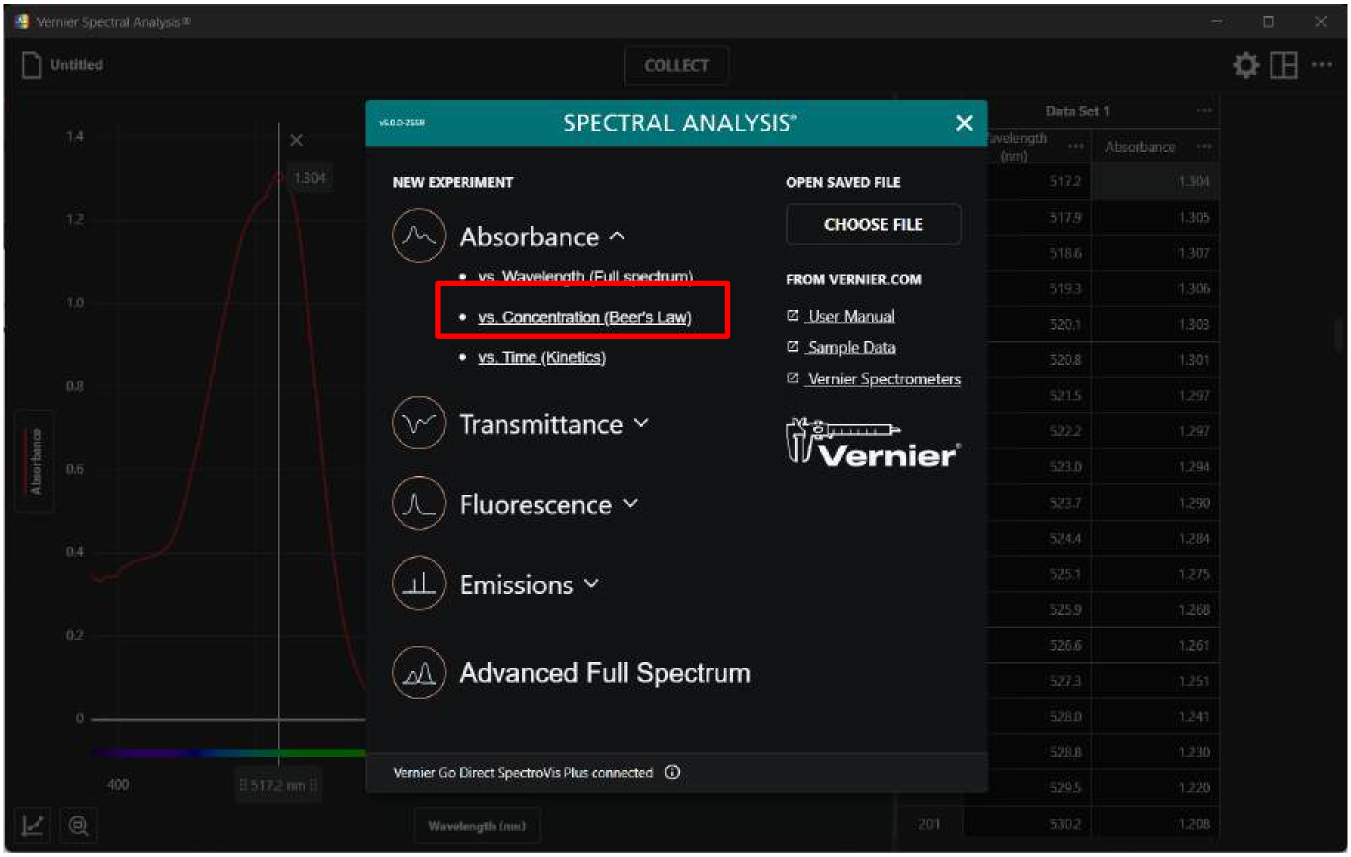


Figure 9: Screen shot of procedure to change spectrophotometer from full spectrum to single wavelength mode.

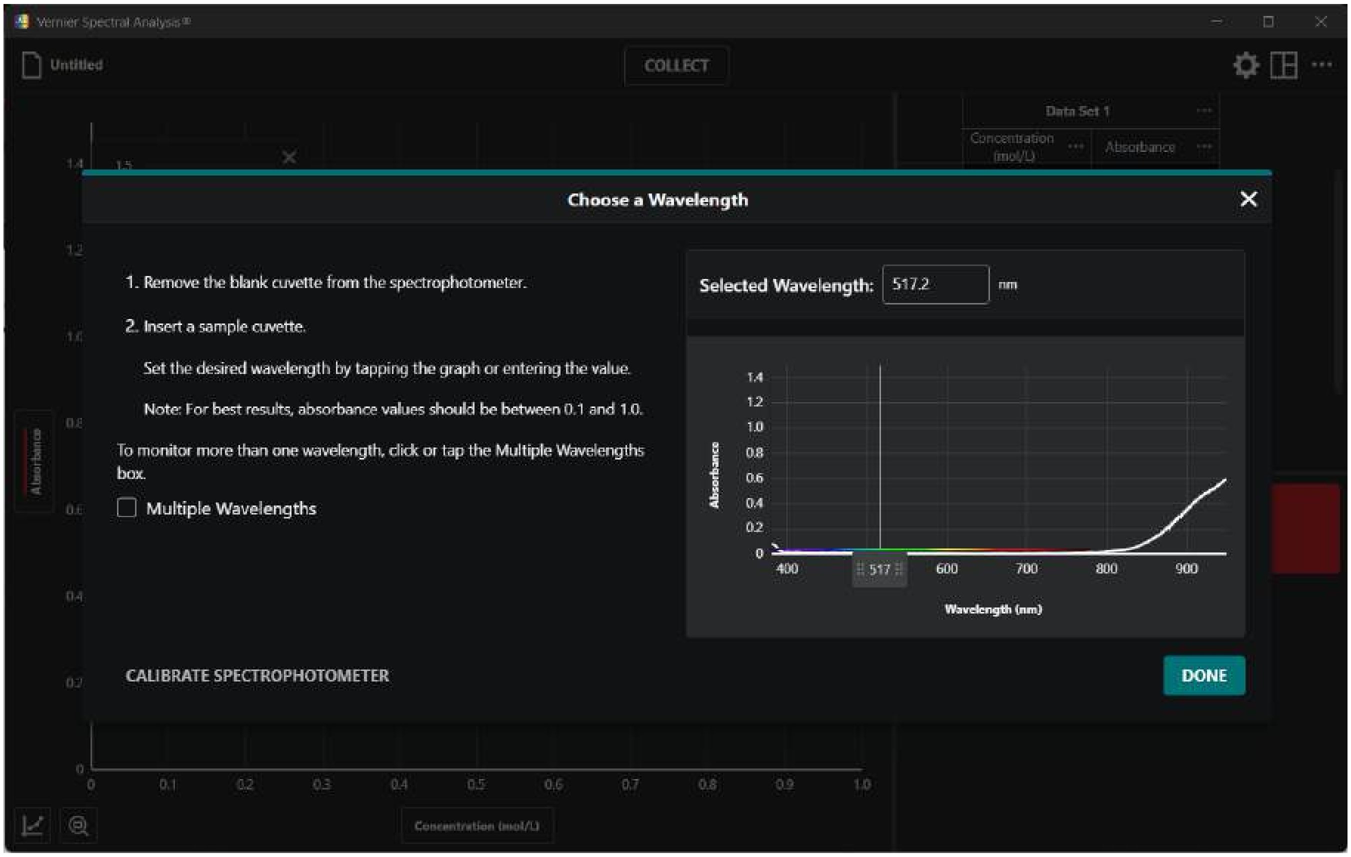


Figure 10: Screen shot of procedure selecting wavelength of maximum absorption

8. Change the units to mmol/L by selecting the ellipsis in the first column and selecting "Column Options". Then type in mmol/L into the "Units" box. Then select "Apply" to exit the menu.

9. The screen will change to a graph with concentration on the X-axis and absorbance on the Y-axis. Sample 1 is your blank cuvette from Step 3. Place the blank sample back into the spectrophotometer and click the "Collect" button. Wait 3-5 seconds while the reading stabilizes, click the "Keep," button and enter a concentration of 0.00 mmol/L. **Do not press the "Stop" button until you have collected data on all your samples**.

10. Prepare each sample as listed in Table 3 (page 16) by adding the appropriate volume (in mL) of the **1.33 mM red dye solution** to a 10 mL graduated cylinder and then diluting with the appropriate volume of distilled water. **Once you have added the 1.33 mM red dye solution and water to the graduated cylinder, place your thumb on top of the cylinder and invert it 2-3 times to ensure proper mixing.** Place each dilute solution in a cuvette and cap the cuvette. **(HINT: Place a piece of paper on your lab bench with a marked spot for each final concentration of sample. As you make the samples, place the filled cuvette in the appropriate spot. This will prevent you from getting your samples mixed up).** Below is a proposed dilution procedure to make Sample 2.

11. **Sample 2**: Using a disposable pipet transfer 1.00 mL of red dye solution from the 50mL beaker to a 10-mL graduated cylinder. Add distilled water to bring the volume of the solution to the 5.00 mL mark exactly. Using a gloved hand, cover the opening to the graduated cylinder and invert the graduated cylinder several times to thoroughly mix the solution. Pour approximately 3 mL of the solution into the cuvette and cap the cuvette. The cuvette should be at least ¾ full. Wipe off the sides of the cuvette with a clean tech wipe, and then place the filled cuvette into the Vernier Spectrophotometer cuvette compartment. Wait approximately 3-5 seconds to allow the absorbance value in the bottom left to stabilize. Select the "Keep" button and enter the concentration of the solution, 0.266 mM for Sample 2. Click on the "OK" button. Record the absorbance value in Table 3 of your lab report for Sample 2 as well. Remove the sample and proceed with the next sample. **Do not click the Stop button until you have gotten the absorbance data for every one of your dilutions.**

12. **Samples 3-6**: Prepare these solutions in the same manner as detailed for Sample 2, using the volumes listed in Table 3. As you prepare each solution, use the Vernier Spectrophotometer to determine the absorbance associated with each solution and record your results in Table 3 of the lab report in the appropriate row. As the concentration of the solute increases, the absorbance should also increase. If your absorbance readings are not increasing, or your data points are not in a generally straight line, notify your instructor. **Be sure to empty and rinse the graduated cylinder with distilled water thoroughly between preparing each sample.**

13. After samples 1-6 are complete, **show your data to your instructor before proceeding**. After instructor approval, select the "Stop" button from the top, then select the "Graph Options" in the bottom left. Then select "Apply Curve Fit" and "Linear" from the options menu. Then select "Apply". Record the values for the equation variables and correlation coefficient (R) on page 16.

Figure 10: Screen shot of spectrophotometer calibration curve with linear fit.

14. Using the repipet, fill a clean cuvette with approximately 3 mL of unknown red dye solution. Place the filled cuvette in the cuvette compartment. Record the absorbance value of the unknown solution in Table 3. You do not need to store this value on your graph, just look at the absorbance in the bottom left of your screen. Record this value above in Table 3.

15. Empty all cuvettes into a 250-mL waste beaker. Rinse all glassware and cuvettes with distilled water and empty into the 250 mL waste beaker. Disposed of all waste in provided hazardous waste container at the instructor station along the center aisle. Ensure to completely rinse all glassware with deionized water before drying and storing.

Appendix A: Using Excel to Create a Line of Best Fit

If your data does not look correct to the instructor, follow the below instructions on creating a best-fit line in Excel rather than using Logger Pro:

1. Record the absorbances of your fixed samples in **Table 3**.

2. Open Microsoft Excel

3. Type the numerical values of your final concentrations in one column. **These are your x values.**

4. Type the numerical values of your absorbances in another column. **These are your y values**.

5. Highlight/select all values from both columns.

6. Click “Insert” from the top row of Excel tabs.

7. Select “Scatter” from the “Charts” options. See **Figure A-1** below.

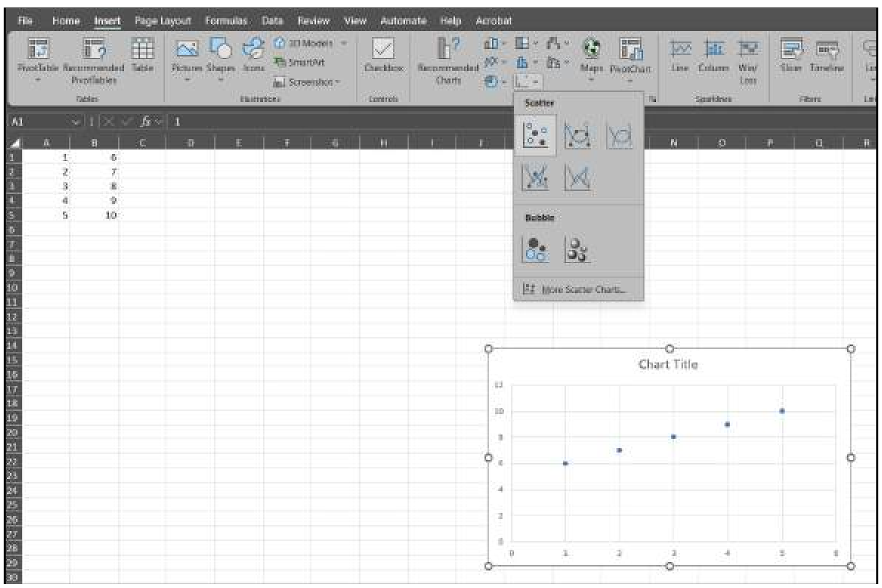


Figure A-1. Finding corrected best-fit line from Microsoft Excel.

8. Click on a single data point from the new scatter plot. This should highlight all dots on the chart.

9. Right click > “Add Trendline”

New options to “Format Trendline” should appear on the right of your screen. Check the block at the bottom of the list for “Display Equation on chart” and “Display R-squared value on chart.”

UNITED STATES MILITARY ACADEMY

LAB 2 - THE ELECTROMAGNETIC SPECTRUM AND SPECTROPHOTOMETRY

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:

SIGNATURE: DATE:
SIGNATURE: DATE:

Part I: Questions

Table 1. Known ionic compounds (salts) flame test results

Element	Flame Color	Intensity
DI Water		
Barium		
Calcium		
Copper		
Lithium		
Potassium		
Sodium		
Strontium		

Table 2. Unknown ionic compounds (salts) flame test results

Unknown	Flame Color	Intensity	Predicted Element
1			
2			
3			

1. (4 pts) In your flame tests of known salts, which ions produced emissions of the highest and lowest energy?

highest energy: _______________

lowest energy: _______________

2. For the ion that produced the highest energy emission:

a. (4 pts) Use the spectrum on page 1 of the RDC to estimate the wavelength of light emitted.

_______________ nm

b. (4 pts) Given the scale on the RDC, how many significant figures should your estimate represent? *Hint: look at the frequencies for visible light.*

_______________ significant figures

3. (10 pts) Using both of your answers to the previous question, calculate the energy (in Joules) of a single photon of this wavelength. **Show all work.**

4. (5 pts) Calculate the energy in a mole of photons of this wavelength in kJ/mol. **Show all work.**

Part II: Questions

Table 3: Dilution Table for Spectrophotometry

Sample	Vol. of Red Dye (mL)	Vol. of Water (mL)	Final Concentration (mM)	Absorbance
1	0	5	0.000	
2	1	4	0.266	
3	2	3	0.532	
4	3	2	0.798	
5	4	1	1.06	
6	5	0	1.33	
Unknown				

1. (8 pts) Record the linear fit variables from Part II, step 11 (page 9).

   m (slope): ____________   b (y-intercept): _____________

   linear equation (y = mx + b): ________________________

   correlation coefficient (R): ______________

2. (5 pts) Use the equation of the linear model from your Beer-Lambert plot to calculate the concentration of the unknown red dye solution. **Show all work.**

3. (5 pts) Use the correlation coefficient from your linear model to explain how well your line fits the data.

4. (5 pts) Imagine that you are asked to determine if a water source is contaminated with an unsafe concentration of an unknown ion. How could you use skills that you have learned in this lab to first determine the ion and then determine the concentration in the contaminated water?

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 2 - Instructor Solution

CUTSCALE

UNITED STATES MILITARY ACADEMY

LAB 2 – ELECTROMAGNETIC SPECTRUM AND SPECTROPHOTOMETRY

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

BY

CADET: __________________________________

CADET: __________________________________

CADET: __________________________________

WEST POINT, NEW YORK

DATE: _______________________

_________ WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________ WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________ WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________ WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

Part I: Questions

Element	Ion	Flame Color
DI Water	N/A	no change
Barium	Ba^{2+}	green-yellow
Calcium	Ca^{2+}	red
Copper	Cu^{2+}	blue-green
Lithium	Li^{+}	red
Potassium	K^{+}	blue-violet
Sodium	Na^{+}	yellow-orange
Strontium	Sr^{2+}	red-orange

Table 2. Unknown ionic compounds (salts) flame test results

Unknown	Flame Color	Intensity	Predicted Ion
1	orange	light	Ba^{2+}
2	red	medium	Li^{+} or Ca^{2+}
3	red-orange	dark/strong	Sr^{2+} or Ca^{2+}

1. (4 pts) In your flame tests of known salts, which ions produced emissions of the highest and lowest energy?

highest energy: Potassium (or Lithium if cadets saw Li^{+} as violet-pink instead of red) // +2 pts FF (AON)

lowest energy: Lithium or Calcium or Strontium // +2 pts FF (AON)

2. For the ion that produced the highest energy emission:

a. (4 pts) Use the spectrum on page 1 of the RDC to estimate the wavelength of light emitted.

420 nm +4 pts: within 30 nm (AON) as long as it's still in the visible spectrum.

b. (4 pts) Given the scale on the RDC, how many significant figures should your estimate represent? *Hint: look at the frequencies for visible light.*

2 significant figures +4pts AON

/12

3. (10 pts) Using both of your answers to the previous question, calculate the energy (in Joules) of a single photon of this wavelength. **Show all work.**

$$E = \frac{hc}{\lambda} \qquad \frac{6.626\text{x}10^{-34}\text{ J s}}{\text{photon}} \cdot \frac{2.998\text{x}10^{8}\text{ m}}{\text{s}} \cdot \frac{}{420\text{ nm}} \cdot \frac{1\text{ x}10^{9}\text{ nm}}{1\text{ m}} = \frac{4.7\text{x}10^{-19}\text{ J}}{\text{photon}}$$

+3 concept +2 constant +2 constant +2 conv +1 M/U/S

4. (5 pts) Calculate the energy in a mole of photons of this wavelength in kJ/mol. **Show all work.**

$$\frac{4.7297\text{x}10^{-19}\text{ J}}{\text{photon}} \cdot \frac{6.022\text{x}10^{23}\text{ photons}}{\text{mol}} \cdot \frac{\text{kJ}}{1000\text{ J}} = \frac{280\text{ kJ}}{\text{mol}}$$

+2 conv +2 conv +1 M/U/S

Part II: Questions

Table 3: Dilution Table for Spectrophotometry

Sample	Vol. of Red Dye (mL)	Vol. of Water (mL)	Final Concentration (mM)	Absorbance
1	0	5	0.000	**0.000**
2	1	4	0.266	**0.173**
3	2	3	0.532	**0.354**
4	3	2	0.798	**0.532**
5	4	1	1.06	**0.703**
6	5	0	1.33	**0.872**
Unknown				**0.443**

+5 Abs measurements complete, AON

1. (8 pts) Record the linear fit variables from Part II, step 11 (page 9).

m (slope): 0.6587 b (y-intercept): 0.001545

linear equation (y = mx + b): y = 0.6587 x + 0.001545

correlation coefficient (R): 0.9987

+1.5 pt: each reported value
+2 pts: equation properly formatted.

/23

2. (5 pts) Use the equation of the linear model from your Beer-Lambert plot to calculate the concentration of the unknown red dye solution. **Show all work.**

$A = \varepsilon\, b\, C$ $\quad C = \frac{A}{(\varepsilon\, b)}$ $\quad y = 0.6587\,x + 0.001545$

$y = mx + b$ $\quad x = \frac{(y-b)}{m}$ $\quad x = \frac{(0.443-0.001545)}{0.6587} = 0.670\ mM$ +1 M/U/S

+4 concept

3. (5 pts) Use the correlation coefficient from your linear model to explain how well your line fits the data.

Because our R value was 0.993, which is very close to 1, our line has a good fit to our data.

+2 pts: Correctly identify how well the line fits the data.
+3 pts: Used R value to explain the fit.

4. (5 pts) Imagine that you are asked to determine if a water source is contaminated with an unsafe concentration of an unknown ion. How could you use skills that you have learned in this lab to first determine the ion and then determine the concentration in the contaminated water?

+2 pts: Mentions burning the sample and using flame color to determine ion.
+3 pts: Mentions using a calibration curve/Beer-Lambert plot to determine concentration.

$/15$

Lab 3

COURE DROP

Lab 4

COURE DROP

Lab 5

Lab 5 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

INTERMOLECULAR FORCES

Experiment V

Instructions: Prior to lab, read pages 1-6 to obtain information needed to complete this lab. **The material covered here is testable.** During lab **only turn in pages 7-12.**

Materials

Reagents:	$C_2H_5OH(aq)$ (ethanol)	$C_4H_8O_2(l)$ (ethyl acetate)	$C_3H_6O(l)$ (acetone)
	$CH_4N_2O(s)$ (urea)	$I_2(s)$ (iodine)	
Equipment:	(1) Hot Plate w/ Stir Bar	(1) Ring Stand	(1) Vernier LabQuest Mini Box
	(1) 600-mL Beaker	(1) Large Ring Support	(1) Vernier Large Thermistor
	(2) 250-mL Beakers	(1) Medium Test Tube Holder	(1) Vernier Small Thermistor
	(1) Glass Rod	(2) Small 3-Prong Clamps	Nitrile Gloves
	(9) 30-mL Test Tubes	Boiling Chips	

Introduction

The purpose of this lab is to explore the effects of intermolecular forces (IMFs) on the physical properties of substances, such as boiling point and solubility. To do this, you will experimentally determine the boiling points of polar and nonpolar liquids. You will apply your knowledge of Lewis structures, molecular shapes, bond polarity, dipole moments, and IMFs to explain your hypotheses and your observations.

Solids that dissolve in certain liquids are referred to as being **soluble** in those liquids. Those that do not dissolve are said to be **insoluble**. Similarly, two liquids that mix in any proportion with each other are referred to as **miscible**. Those that separate are **immiscible**. The polarity of the solvents and solutes will dictate these physical properties. Solvents and solutes that share similar intermolecular forces will tend to mix while those that do not tend to remain separate. This concept is frequently referred to as “like-dissolves-like”. This explains why water is good for cleaning sugary substances (water and sugar both exhibit hydrogen bonding), while gasoline does a far better job solvating tar (as both predominantly exhibit dispersion forces). *NOTE: Procedure Part 1 and Part 2 can be done simultaneously, i.e. start setting up Part 2 while waiting for the water to heat up during Part 1.*

Procedure Part 1: Boiling Point Determination

1. Secure both of your test tube racks and at least nine (9) 30-mL test tubes and ensure that they are free of any cracks. Clean all nine test tubes with a test tube brush (located in your bench top drawer) and soapy water. Rinse the test tubes thoroughly with distilled water and place them back in your test tube rack on the pegs to dry. Three (3) tubes are used in Part 1 and six (6) in Part 2.

2. Using Figure 1 as a guide, construct the beaker-heating apparatus with the 250-mL beaker in the center of the large ring support. Carefully construct the apparatus under the prepositioned fume snorkel as shown. DO NOT move the fume snorkel. Notify the instructor if the fume snorkel is out of position. Secure the **large** thermistor with a small 3-pronged clamp and place it **in the test tube** (be careful not to let the thermistor touch the glass). Secure the **small** thermistor with another small 3-pronged clamp and place it **in the beaker** to monitor water temperature. The test tube should be clamped with a medium test tube clamp. Set up your Vernier LabQuest Mini with a "Temperature vs Time" program as shown in Appendix A. <u>Ensure you set the duration of the experiment to **600 seconds**</u>.

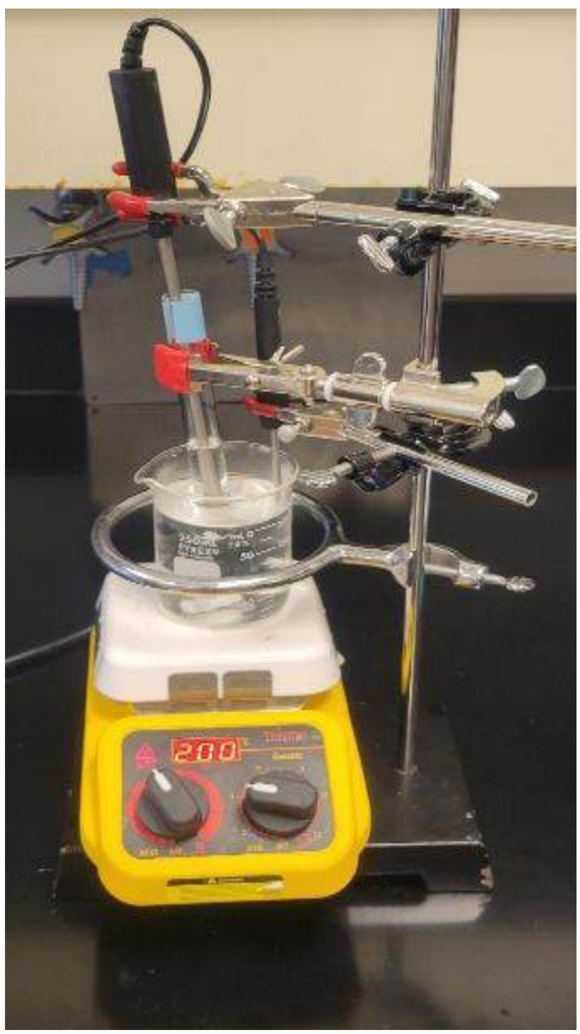


Figure 1. Beaker-heating apparatus

3. Using the hot water bath located in the center station, fill a 250-mL beaker to 200-mL. Place the magnetic stir bar in the beaker. Lower the small thermistor into the water in the 250-mL beaker, being careful that it does not contact the beaker. Turn the hot plate on to 300 ºC. DO NOT exceed 300 ºC on the hot plate unless directed by your instructor. Turn on the magnetic stir control on the hot plate to start rotating the magnetic stir bar. Monitor the water temperature until it reaches 50 ºC. **IMPORTANT:** use nitrile gloves when handling the solvents in the following steps.

4. While the water in the 250-mL beaker is heating to 50ºC, label one 30-mL test tube each with "acetone", "ethyl acetate" and "ethanol". Obtain 10 mL of each reagent from the instructor bench into its labeled test tube. While dispensing each reagent, ensure the test tube is raised near the spout of the repipet dispenser to reduce splash hazards. Using the tweezers, place one or two boiling chips* into each labeled test tube. Set them in the test tube rack (Hint: your 30-mL test tubes should be about $^{1}/_{3}$ full).

   NOTE: Boiling chips are used to provide nucleation sites so the liquid they are placed in boils more calmly without experiencing "bumping" (sudden, violent releases of vapor). The chips do not change the boiling point of the liquid.* ***DO NOT* *place boiling chips in the test tube if the solvent is already hot.*

5. **Boiling point of acetone:** When the water temperature reaches 50 ºC, secure the 30-mL acetone-filled test tube over the water using the medium rubber-gripped test tube clamp. Lower the thermistor into the test tube as far as possible without touching the glass or boiling chips to reduce systematic error. Lower the 30-ml test tube with the acetone and the thermistor into the 50 ºC water. Start your data collection on the Graphical Analysis program. Monitor the temperature of the acetone while observing if any bubbles evolve in the acetone. A bubble may evolve from the tip of the thermistor or from the boiling chips. Wait for the acetone to come to a rolling boil and record the temperature in Table 1. After recording the boiling point for acetone, carefully move the test tube clamp up the ring stand. Then, use a test tube holder to move the hot test tube from the clamp to an empty 250-mL beaker to cool for 5 minutes. **Do not touch the hot test tube with your hands**).

6. Continue to heat the water to 70 ºC. Heating your water bath to 70 ºC will take time. Allow your water bath to heat while beginning **Procedure Part 2**. Once your water bath reaches 70 ºC, **lower the temperature on your hot plate to 200 ºC.** If the water temperature exceeds 70 ºC, inform your instructor. You may need to use ice chips to cool your water bath down to the desired temperature.

7. **Boiling point of ethyl acetate:** When the water bath temperature reaches 70 ºC, repeat step 5 above with the 30-mL test tube containing ethyl acetate. Once the reagent reaches a rolling boil, record the boiling point temperature in Table 1. Remove the test tube from the water and cool the water back to 70 ºC using ice chips.

8. **Boiling point of ethanol:** When the water bath temperature reaches 70 ºC, repeat step 5 above with the 30-mL test tube containing ethanol. Once the reagent reaches a rolling boil, record the boiling point temperature in Table 1. Turn off the hot plate and leave the beaker to cool prior to pouring the water into the sink. Report the boiling point temperatures for each sample to your instructor. Two additional experimental boiling points for each solvent are provided for calculation purposes.

Procedure Part 2: Solubility and Miscibility

1. Secure six (6) clean 30-mL test tubes and label them A, B, C, D, E, and F.

2. Dispense 5 mL of ethyl acetate from the instructor bench into test tubes A, B, and C. Ensure the test tube is raised near the spout of the repipet to reduce splash hazards.

3. Dispense 5 mL of ethanol from the instructor bench into test tube D. Ensure the test tube is raised near the spout of the repipet to reduce splash hazards.

4. Using your distilled water bottle and a graduated cylinder, measure 5 mL of distilled water into test tubes E and F.

5. Using a graduated cylinder, slowly add 5 mL of water to test tubes A and D. Observe the interactions between water and ethyl acetate and ethanol, respectively, and record your observations in Table 2 of the lab report.

   NOTE: In the "Observations" column of Table 2, you should write whether your two reagents are soluble, insoluble, miscible, or immiscible with each other.

6. At your instructor bench, use the metal micro scoop to place two (2) iodine crystals into a weigh boat. Return to your lab bench and place an iodine crystal into test tube B (containing ethyl acetate) and an iodine crystal into test tube E (containing water). *Be careful not to spill the iodine or hold it with your gloves! It is very corrosive and will permanently damage and stain lab benches, equipment, and floors!* Gently stir the iodine crystals in the test tubes with a glass stirring rod from your bench. Wipe off the glass rod before stirring each test tube to avoid cross contamination. Observe and record your observations in Table 2 of the lab report.

7. Locate the urea at the end of the lab bench near the analytical balances. Obtain two scoopfuls of urea in a plastic weigh boat and take it back to your lab station. Evenly divide the urea between test tube C (containing ethyl acetate) and test tube F (containing water). Gently stir the urea in each test tube with the glass stir rod. Wipe off the glass rod before stirring each test tube to avoid cross contamination. Observe and record your observations in Table 2 of the lab report.

8. Conduct a short rinse (1-2 second squeeze of the DI bottle) of each test tube and dispose contents into a 600-mL beaker from your station. DO NOT pour any solutions down the drain. Once all waste is consolidated in the beaker, dispose of it into the large waste containers located at the instructor bench. Clean all test tubes and glassware in the sink with soap and the test tube brush. Shake excess water from each test tube prior to returning them to the lab station cabinet. You do NOT need to completely dry your test tube before returning them to the cabinet. You should not accumulate more than 50 mL of waste from this step.

Appendix A: Vernier LabQuest Mini Set-up Instructions

1. Log on to your personal computer.

2. Plug the USB cable from the LabQuest Mini unit into one of the USB ports of your computer. Connect the small thermistor into the channel 1 port of the LabQuest Mini unit and the large thermistor into the channel 2 port. (See picture below).

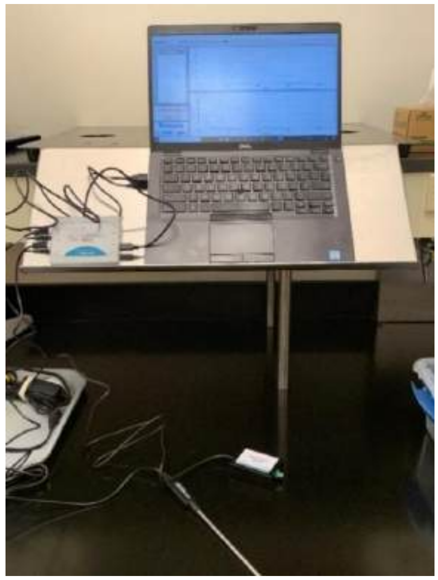


Figure 2: Approved Set-up for Connecting LabQuest Mini and Sensors to Computer

3. Open the Vernier Graphical Analysis software on your computer. The temperature will be displayed in the bottom left corner of the screen as shown below.

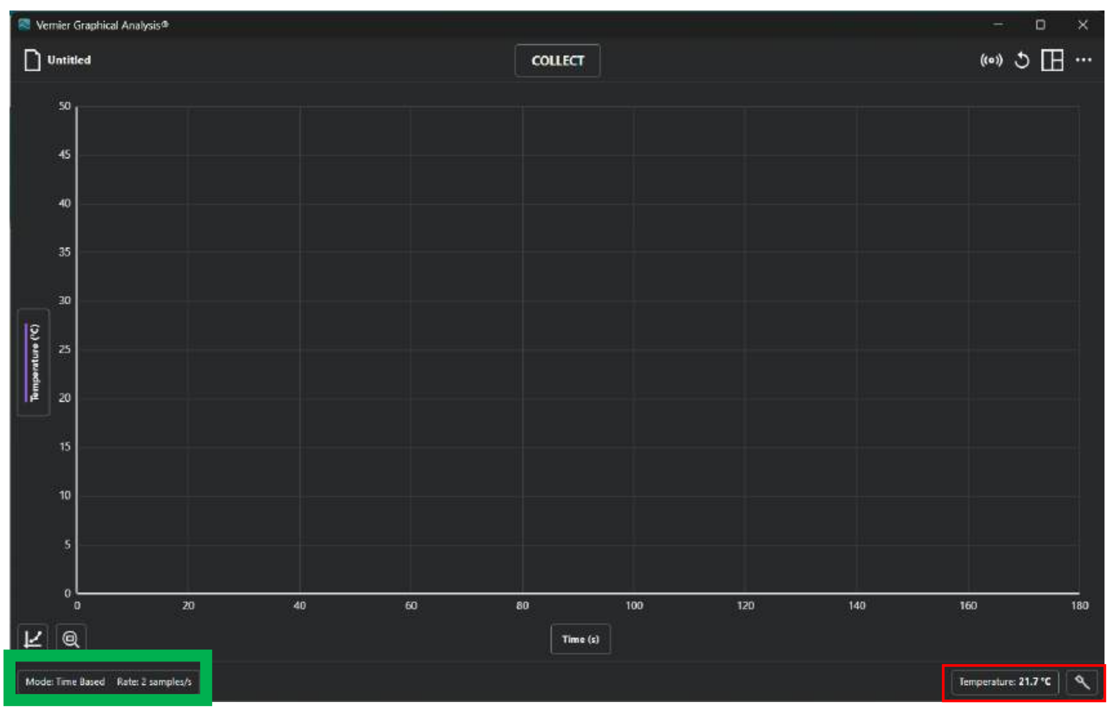


Figure 3: Screen Shot of Graphical Analysis Displaying Temperature Readings

Appendix A: Vernier LabQuest Mini Set-up Instructions

4. Select the “Experiment” drop down menu from the top of the page. Scroll down to the “Data Collection” tab. This will open a new window which will allow you to set up the data collection parameters for the experiment. **Set the duration of the experiment to 600 seconds**. Select “Done” when complete.

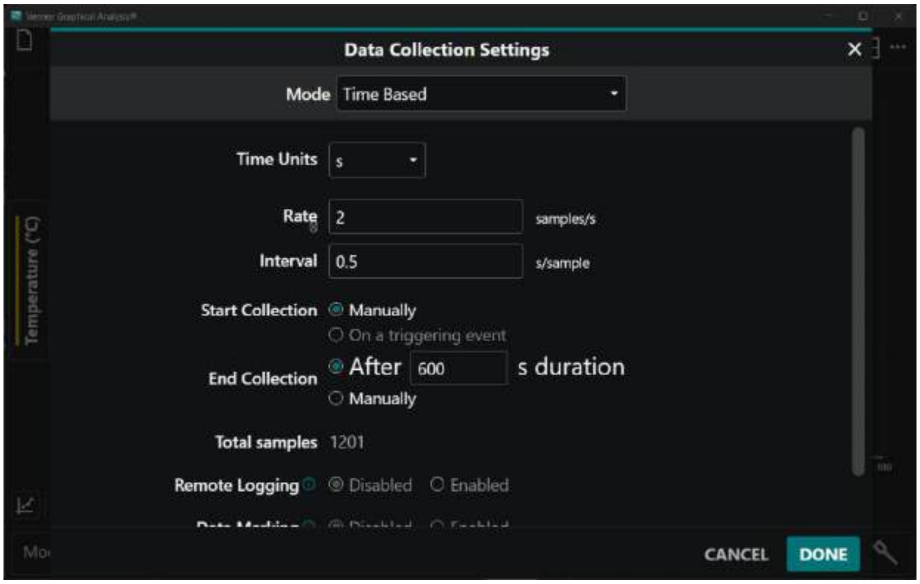


Figure 4: Data collection setting window showing how to adjust duration of data collection.

5. When prepared to start data collection of the samples, click the “Collect” button at the top of the toolbar. Note that the temperature scale initially goes up to 50 °C. As the data approaches this temperature the graph will automatically update to show the data points above 50 °C. Once the experiment is complete use the “Zoom to All Data” feature in the bottom left to scale your graph.

6. Change the name of each data set by selecting the “Temperature” button on the left side of the plot and selecting the pencil image.. Label the data set as the name of the chemical used. You can also change the visibility of the data in the menu. Reference Figure 5 below.

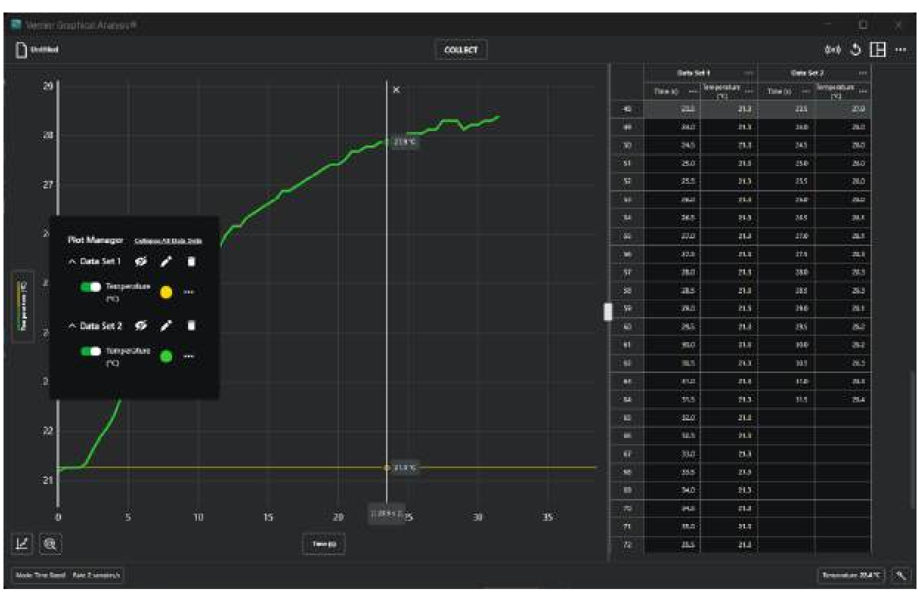


Figure 5: Examples of multiple data sets displayed on a single graph.

UNITED STATES MILITARY ACADEMY

LAB 5 – INTERMOLECULAR FORCES

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

BY

CADET: __________________________________

CADET: __________________________________

CADET: __________________________________

WEST POINT, NEW YORK

DATE: _______________________

_________ WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________ WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________ WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________ WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

Part I: Discussion

Table 1. Experimental Boling Points

Sample	Measured BP (ºC)	Instructor BP 1 (ºC)	Instructor BP 2 (ºC)	Average BP* (ºC)
Acetone, C_3H_6O (*l*)		**50.15**	**58.50**	
Ethyl Acetate, $C_4H_8O_2$ (*l*)		**77.23**	**76.95**	
Ethanol, CH_3CH_2OH(*aq*)		**78.56**	**79.34**	

** Average Boiling Point is a normal average across Measured BP, Instructor BP1, and Instructor BP2*

1. Consider that Table 1 is your collected data from the lab (**including listed blue values**). If the accepted values of acetone, ethyl acetate, and ethanol are 56.00 °C[1], 77.10 °C[1], and 78.37 °C[1] respectively, answer the following questions:

   a. (5 pts) Calculate the percent error for each sample using your average boiling point values. **Only show your work for acetone but list the percent error values for each sample.**

   b. (5 pts) Identify which samples had accurate boiling point measurements. **Use your data to support your conclusions.**

c. (5 pts) Use coefficient of variation to identify which sample(s) had precise boiling point measurements. **Only show your work for acetone but list the CV value for each sample.**

d. (5 pts) Was the error in your lab group's boiling point measurements predominantly systematic or random? Consider both the measured and instructor boiling points. Propose one possible source of error that would result in the error you observed. *Hint: Refer to the SLAM for relating accuracy and precision to error and its sources. "Human error" is never an acceptable answer, as it can either be systematic or random.*

e. (5 pts) Using specific intermolecular forces and their relative strengths, explain why acetone's boiling point is lower than ethanol's boiling point.

Part II: Discussion

Table 2. Observations of Solubility/Miscibility

Test Tube	Solvent	Solute	Observations
A	Ethyl Acetate	Water	
B	Ethyl Acetate	Iodine	
C	Ethyl Acetate	Urea	
D	Ethanol	Water	
E	Water	Iodine	
F	Water	Urea	

Follow the example below for the questions 1-3:

Using **specific** intermolecular forces and their relative strengths, explain your observations regarding iodine's solubility in ethyl acetate.

> We observed iodine was soluble in ethyl acetate. Iodine possesses dispersion forces and ethyl acetate possesses dispersion and dipole-dipole IMFs.
>
> Because iodine has such a large formula mass, it has a relatively strong dispersion force, and because ethyl acetate is mostly nonpolar, it has a relatively weak dipole-dipole interaction. Because iodine is soluble in ethyl acetate, the relative strengths between these IMFs must be comparable.

1. (7 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding ethyl acetate's miscibility with water.

2. (7 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding ethanol's miscibility with water.

3. (7 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding iodine's solubility in water.

4. (4 pts) **Draw** the Lewis structure of urea (NH_2CONH_2), **list** the intermolecular forces present, and use your observations from Table 2 to **explain** whether urea is polar (has a dipole moment)?

Lewis Structure: **IMFs Present:**

Is urea polar? YES / NO

Explanation:

REFERENCES:

1. William E. Acree, Jr., James S. Chickos, "Phase Transition Enthalpy Measurements of Organic and Organometallic Compounds" in NIST Chemistry WebBook, NIST Standard Reference Database Number 69, Eds. P.J. Linstrom and W.G. Mallard, National Institute of Standards and Technology, Gaithersburg MD, 20899, https://doi.org/10.18434/T4D303, (retrieved September 19, 2025).

Cadet Works Cited

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Cadet Notes

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Lab 5 - Instructor Solution

Cut Scale

UNITED STATES MILITARY ACADEMY

LAB 5 – INTERMOLECULAR FORCES

CH151: GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

BY

CADET: __________________________________

CADET: __________________________________

CADET: __________________________________

WEST POINT, NEW YORK

DATE: _______________________

_________ WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________ WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________ WE ACKNOWLEDGE THAT WE USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________ WE ACKNOWLEDGE THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

SIGNATURE: _____________________________ DATE: _______________________

Part I: Discussion

Table 1. Experimental Boiling Points

Sample	Measured BP (ºC)	Instructor BP 1 (ºC)	Instructor BP 2 (ºC)	Average BP (ºC)
Acetone, C_3H_6O (*l*)		**50.15**	**58.50**	
Ethyl Acetate, $C_4H_8O_2$ (*l*)		**77.23**	**76.95**	
Ethanol, CH_3CH_2OH(*aq*)		**78.56**	**79.34**	

Table 1: Average Boiling Point is a normal average across Measured BP, Instructor BP1, and Instructor BP2

1. Consider that Table 1 is your collected data from the lab (**including listed blue values**). If the accepted values of acetone, ethyl acetate, and ethanol are 56.00 °C, 77.10 °C, and 78.37 °C respectively, answer the following questions:

   a. (6 pts) Calculate the percent error for each sample using your average boiling points. **Only show your work for acetone but list the percent error for each sample.**

$$\%\ error = \left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right| \times 100\%$$ +3 Concept

Acetone: $\left|\frac{52.92-56.00}{56.00}\right| \times 100\% = 5.51\%$ +1 Shows % error calculations for acetone

Ethyl Acetate: $\left|\frac{77.11-77.10}{77.10}\right| \times 100\% = 0.013\%$

+1 Shows % error value for ethyl acetate

Ethanol: $\left|\frac{78.65-78.37}{78.37}\right| \times 100\% = 0.36\%$

+1 Shows % error value for ethanol

   b. (6 pts) Identify which samples had accurate boiling point measurements. **Use your data to support your conclusions.**

Our threshold for whether a measurement is accurate is if its percent error is less than 5%.

- Ethanol and Ethyl Acetate had accurate boiling points because their percent errors were 0.36% and 0.013% respectively, which is less than 5%.
- Acetone had an inaccurate boiling point because its percent error is 5.51%, it is inaccurate.

+2 pt ea: correct identification of accuracy using 5% threshold
(i.e. acetone is inaccurate, EtOH and EtAc are)

$/_{21}$

c. (8 pts) Use coefficient of variation to identify which sample(s) had precise boiling point measurements. **Only show work for acetone but list the CV value for each sample.**

Acetone:

$$SD = \sqrt{\frac{(50.10-52.92)^2+(50.15-52.92)^2+(58.50-52.92)^2}{3-1}} = 4.8\ ℃ \qquad CV = \frac{s}{\bar{x}} = \frac{4.8\ ℃}{52.92\ ℃} = 9.1\%$$

+3 pts: STD concept

+3 pts: CV concept

+2 pts: uses 5% threshold for precision

Ethyl Acetate:

$$SD = \sqrt{\frac{(77.15-77.11)^2+(77.23-77.11)^2+(76.95-77.11)^2}{3-1}} = 0.14\ ℃ \qquad CV = \frac{s}{\bar{x}} = \frac{0.14\ ℃}{77.11\ ℃} = 0.19\%$$

Ethanol:

$$SD = \sqrt{\frac{(78.05-78.65)^2+(78.56-78.65)^2+(79.34-78.65)^2}{3-1}} = 0.65\ ℃ \qquad CV = \frac{s}{\bar{x}} = \frac{0.65\ ℃}{78.65\ ℃} = 0.83\%$$

The ethyl acetate and ethanol samples had precise boiling point measurements because coefficients of variation for both samples were less than 5%.

d. (6 pts) Was the error in your lab group's boiling point measurements predominantly systematic or random? Consider both the measured and instructor boiling points. Propose one possible source of error that would result in the error you observed. *Hint: Refer to the SLAM for relating accuracy and precision to error and its sources. "Human error" is never an acceptable answer, as it can either be systematic or random.*

+3 pts: correct error identified

+3 pts: probable source identified

e. (5 pts) Using specific intermolecular forces and their relative strengths, explain why acetone's boiling point is lower than ethanol's boiling point.

Acetone's boiling point is lower than ethanol's boiling point because acetone's strongest IMF is dipole-dipole forces, while ethanol experiences hydrogen bonding. This results in stronger IMFs between the molecules of ethanol and therefore more energy has to be added to the system to cause boiling to occur in an ethanol sample.

+2 pt: Acetone IMFs identified
+2 pt: Ethanol IMFs identified
+1 pt: correct explanation

$/30$

Part II: Discussion

Table 2. Observations of Solubility/Miscibility

Test Tube	Solvent	Solute	Observations
A	Ethyl Acetate	Water	immiscible; two layers form
B	Ethyl Acetate	Iodine	soluble; iodine dissolves; solution turns orange
C	Ethyl Acetate	Urea	insoluble; crystals did not dissolve
D	Ethanol	Water	miscible; no layers
E	Water	Iodine	insoluble
F	Water	Urea	soluble; crystals dissolved

Follow the example below for the questions 1-3:
Using **specific** intermolecular forces and their relative strengths, explain your observations regarding iodine's solubility in ethyl acetate.

We observed Iodine was soluble in ethyl acetate. Iodine possesses dispersion forces and ethyl acetate possesses dispersion and dipole-dipole IMFs.

Because iodine has such a large formula mass, it has a relatively strong dispersion force, and because ethyl acetate is mostly nonpolar, it has a relatively weak dipole-dipole interaction. Because iodine is soluble in ethyl acetate, the relative strengths between these IMFs must be comparable.

1. (5 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding ethyl acetate's miscibility with water.

   We observed that ethyl acetate is immiscible in water. Both ethyl acetate and water experience dipole-dipole forces, but the dipole-dipole force in ethyl acetate is weaker than the strong hydrogen bonding in water. This prevents the two liquids from mixing and results in ethyl acetate being immiscible with water.

   +2 pts: Correct observation
   +2 pts: IMFs of both species correctly identified
   +1 pts: Correct explanation

2. (5 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding ethanol's miscibility with water.

   We observed that ethanol is miscible in water since both ethanol and water experience hydrogen bonding. This allows the two liquids to mix and results in ethanol being miscible with water.

   +2 pts: Correct observation
   +2 pts: IMFs of both species correctly identified
   +1 pts: Correct explanation

/30

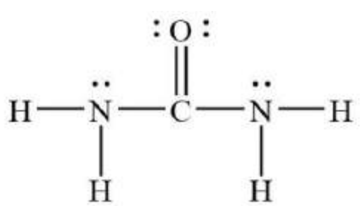
3. (5 pts) Using **specific** intermolecular forces and their relative strengths, explain your observations regarding iodine's solubility in water.

We observed that iodine is insoluble in water. Since iodine primarily experiences dispersion forces and water primarily experiences hydrogen bonding, the iodine is not able to dissolve in water. Because hydrogen bonding is the strongest IMF, and dispersion forces are the weakest, it is expected that iodine would not dissolve in water.

+2 pts: Correct observation
+2 pts: IMFs of both species correctly identified
+1 pts: Correct explanation

4. (4 pts) **Draw** the Lewis structure of urea (NH_2CONH_2), **list** the intermolecular forces present, and use your observations from Table 2 to **explain** whether urea is polar (has a dipole moment)?

Lewis Structure:



+1 pts: correct Lewis structure

IMFs Present:

Dispersion
Dipole-Dipole
Hydrogen Bonding

+2 pts: IMFs identified and correct

Is urea polar? YES / NO

Explanation:

Based on our observations of urea dissolving in water but not in ethyl acetate, we conclude that urea must also have a dipole moment and hydrogen bonding. This would make urea a polar molecule.

+1 pts: correct observation

/19

Lab 6

Lab 6 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

STOICHIOMETRY

Experiment VI

<u>Instructions:</u> Prior to the lab, read pages 1-5 to obtain information and equations needed to complete this lab. **The material covered here is testable.** During the lab, **only turn in pages 6-10 and the works cited page (if applicable).**

Materials

Chemicals:	Distilled Water Zinc Metal	Unknown Cu Sample	HCl (6M)
Equipment:	(2) 250 mL Beaker (1) Stir Plate with Stir Bar (1) Ring Stand and Ring	(1) 500 mL Beaker (1) Fisher Burner (1) Glass stir rod	Analytical Balance (1) Fisher Burner Stand

Objectives

1. Calculate a theoretical yield given a balanced chemical equation and specified amounts of starting material.
2. Determine the percent yield from experimental data and a calculated theoretical yield.
3. Calculate percent error from experimental data and a given accepted value.
4. Use dimensional analysis and unit conversions to determine the mass of a product, given the amounts of reactants.

Introduction

The ability to quantify reactant consumption and product formation is critical in chemical processes. Furthermore, determining the initial amount of a substance before it undergoes reaction is frequently required. Knowledge of stoichiometry enables the calculation of these values, provided accurate laboratory data is obtained.

The Department of Chemical and Biological Science and Engineering was recently contacted by COL(R) Bull (USMA '96), the new owner of Rocky Mountain Mining and Chemicals. During a site assessment, COL(R) Bull observed a substantial, long-standing chemical waste deposit situated behind the main processing facility. Plant personnel offered no specific information regarding its contents, stating only that 'it's always been there.' Given the plant's arid location, the material has remained largely undisturbed. Colonel Bull views this waste as a critical issue, representing either a significant environmental remediation liability or a potential recoverable resource. He has formally requested assistance in determining the composition of the discarded materials.

The initial analysis, conducted by students in the *CH474 Instrumental Methods of Analysis* course, successfully identified the components as $CuSO_4{\cdot}5H_2O$ and NaCl. The crucial step of determining the percent composition was not completed due to scheduling constraints; this task has now been delegated to you. Separately, upon learning the chemical identities, COL Bull commissioned a chemical engineer to evaluate the potential for economic copper reclamation. The resulting viability study established a critical threshold**: the mixture must contain a minimum of 10% copper by mass to offset extraction costs and realize a profit**. Concentrations below this figure would render the recovery process economically unfeasible.

You will be provided with approximately ten grams of the subject mixture. While a direct physical separation of the $CuSO_4{\cdot}5H_2O$ and NaCl followed by mass determination would yield the percent composition, this method is infeasible. Reference checks indicate both compounds are highly water-soluble, eliminating simple physical separation as an option. Therefore, the established analytical procedure requires dissolving the entire sample in water to enable solution-based reaction chemistry.

To facilitate quantitative analysis, a procedure must be employed to recover all copper from the aqueous solution. The displacement reaction: $CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$, is a viable option for this purpose. A critical consideration for achieving complete copper recovery is the introduction of excess solid zinc. This procedure will contaminate the final precipitate, which will consist of both the targeted solid copper and the residual, unreacted zinc.

Since the Cu/Zn mixture prevents the direct weighing of the copper component, a method for removing the excess zinc must be implemented. A possible strategy is to utilize the selective dissolution of zinc in a strong acid, such as hydrochloric acid (HCl), which converts the zinc solid (Zn(s)) into a soluble salt ($ZnCl_2$ (aq)). However, this approach introduces a critical uncertainty: the potential for the acid to also react with the isolated copper, leading to the formation of a soluble copper compound and an inaccurate final measurement.

Research into elemental copper suggests that it will not react with hydrochloric acid. This confirmation justifies the use of HCl as the reagent for the next procedural phase: the selective removal of the residual zinc and the final purification of the copper precipitate.

The presence of soluble $ZnCl_2$ and sulfate salts in the aqueous phase prevents direct gravimetric analysis of the copper, as evaporation would leave the solid copper contaminated. To address this, a review of the page 2-9 in the SLAM will suggest the process of decanting the solution. Decanting the liquid in the solution (after allowing the solid to settle at the bottom of the beaker, will allow for the removal of the excess zinc, chloride, and sodium ions. Additional distilled water rinsing and decanting steps will further reduce the remaining ion presence.

The final steps involve thermal treatment of the purified precipitate to eliminate residual moisture, followed by cooling and measuring the resulting mass. It is understood that heating the elemental copper (Cu) will cause its complete oxidation to copper(II) oxide (CuO). Utilizing the principles of stoichiometry, the original mass of copper can be precisely calculated from the final mass of the oxide, thus completing the analytical plan, and allowing for a recommendation to be made to the Rocky Mountain Mining Company.

New Reactions

While not covered specifically during CH151, a *replacement reaction* is a reaction where elements switch places within compounds. An example of this is equation 1 below:

$$Zn(s) + CuSO_4(aq) \Rightarrow ZnSO_4(aq) + Cu(s) \qquad \textbf{(1)}$$

The other reaction exploited during this lab is the *oxidation-reduction (REDOX) reaction* which involves a reaction where electrons are transferred. An example of this is equation 2 below:

$$Zn(s) + 2HCl(aq) \Rightarrow ZnCl_2(aq) + H_2(g) \qquad \textbf{(2)}$$

Procedure: Determine the Mass Percentage of Copper Sulfate in the Unknown Copper Sample

Form Solid Copper with a replacement reaction

1. Obtain approximately 6.5 grams of the unknown sample and **record** the exact weight in the lab assignment. As you will need the final mass of your dried sample (which will be crusted onto the bottom of the beaker in step 5), **you must weigh your beaker first**. Use a metal ring and ring stand to secure your 250mL beaker onto the stir plate. Completely dissolve the sample in approximately 100 mL of distilled water using a 250 mL beaker.

Unknown Copper Sample Mass:_______________g
Empty Beaker Mass:_______________g

2. Add the least amount of zinc possible (~1.5 grams) while ensuring there is excess zinc present in the final mixture. Stir the solution until the reaction is complete (5-10 minutes).

Remove the excess zinc

3. Have your instructor inspect your work station. If approved, ***<u>use gloves</u>*** and obtain ~60 mL of 6 *M* HCl (*aq*) in a 250mL Beaker. DO NOT BREATHE THE FUMES! Carefully add the acid to your sample and stir the solution until the reaction is complete (15-20 minutes). If necessary, add an additional 5-10 mL of acid.

4. Carefully decant **(see SLAM, page 2-9)** the solution (which is acidic) into a 500 mL aqueous waste beaker, then rinse the sample three times with 100mL distilled water.

Isolate the solid Copper

5. Using a Fisher burner, heat the sample until you observe a pronounced color change. **<u>Have your instructor inspect your apparatus before lighting your Fisher burner.</u>** Continue heating until the new color is <u>stable and uniform</u> throughout your sample. When the beaker has cooled for <u>at least 15 minutes</u>, determine the weight of the product. **Do NOT touch hot glass or metal.** Fisher burners generate more heat than Bunsen burners – be patient. Work on your calculations while you wait for the materials to cool.

6. Find mass of the dry 250mL beaker with the copper residue at the bottom.
Final Beaker Mass:______________g

Appendix A: Equipment Set-up

For procedure steps 1 through 4, use the set-up in figure 1. This set up uses a ring stand and metal ring to safely secure the 250mL glass beaker on top of the magnetic stir plate while mixing your solutions. *Ensure the heat control for the plate is set in the OFF position.*

Figure 1. Stir plate and beaker set-up

For procedure step 5, use the set-up in figure 2. This set up uses a ring stand and the medium sized ring with wire mesh to act as a platform for the 250mL beaker secured with a second ring also attached to the ring stand. This set up should be directly under a fume snorkel and leave enough room for the Fisher Burner to be placed underneath. After sufficient time is spent heating the sample to dry it, shut the gas to the Fisher Burner off and leave the entire apparatus untouched for at least 15 minutes (use a timer).

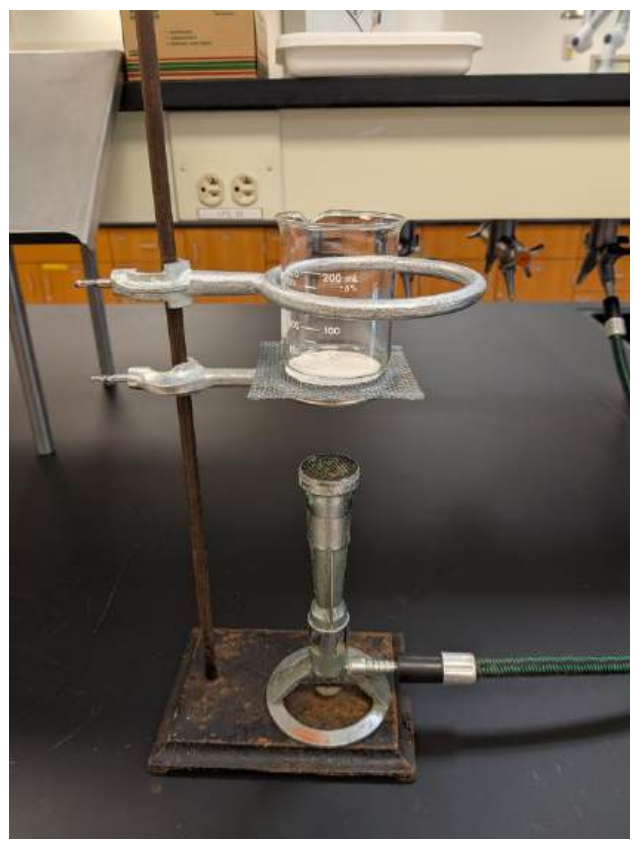


Figure 2. Fisher Burner and Ring Stand set-up

UNITED STATES MILITARY ACADEMY

LAB 6: STOICHIOMETRY

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:

SIGNATURE: DATE:

SIGNATURE: DATE:

Discussion:

Table 1: Results		
1	Mass of Empty Beaker	
2	Mass of Beaker with copper residue	
3	Mass of Copper(II) Oxide	
4	Mass of Copper	
6	% Mass of Copper in Uknown Sample	

1. (5 pts) Write the displacement reaction between zinc and copper(II) sulfate pentahydrate.

2. (5 pts) Write the chemical reaction between zinc and hydrochloric acid which changes excess zinc into a soluble form and releases hydrogen gas.

3. (5 pts) Write the chemical reaction between copper and oxygen to produce the final product (Copper (II) Oxide).

4. (5 pts) Calculate the *mass* and *mass percentage* of copper in the original mixture.

5. (5 pts) If the theoretical mass percentage of copper in the original mixture is 16%, then what is your percent error?

6. (5 pts) Use the provided mass percentage from question 5 and the original mass of the mixture of recorded in step 1 of the procedure for find the theoretical yield of solid copper (not Copper(II) Oxide) in this experiment.

7. (5 pts) Compare the theoretical yield of copper to the actual yield of copper by calculating the percent yield for this experiment.

8. (5 pts) Based on your results, is the unknown material suitable for commercial processing? Explain your answer in the space provided.

9. (5 pts) Use all the resources available to you to explain how you know that Copper(II) Oxide was formed during the drying/heating process. *(You must cite a source for this problem)*

10. (5 pts) If Copper(I) Oxide was formed instead of Copper(II) Oxide, would your results have changed? Explain your reasoning.

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 6 - Instructor Solution

Discussion:

Table 1: Results		
1	Mass of Empty Beaker	
2	Mass of Beaker with copper residue	
3	Mass of Copper(II) Oxide	
4	Mass of Copper	
6	% Mass of Copper in Uknown Sample	

1. (5 pts) Write the displacement reaction between zinc and copper(II) sulfate pentahydrate.

$$Zn(s) + CuSO_4(aq) \Rightarrow ZnSO_4(aq) + Cu(s)$$

-2 missing state symbols
-3 incorrect equation

2. (5 pts) Write the chemical reaction between zinc and hydrochloric acid which changes excess zinc into a soluble form and releases hydrogen gas.

$$Zn(s) + 2HCl(aq) \Rightarrow ZnCl_2(aq) + H_2(g)$$

-2 missing state symbols
-3 incorrect equation

3. (5 pts) Write the chemical reaction between copper and oxygen to produce the final product (Copper (II) Oxide).

$$2Cu(s) + O_2(g) \Rightarrow 2CuO(s)$$

-2 missing state symbols
-3 incorrect equation

4. (5 pts) Calculate the *mass* and *mass percentage* of copper in the original mixture.

-2 solves for wrong species or doesn't find MASS and %Mass -3 M/U/S

5. (5 pts) If the theoretical mass percentage of copper in the original mixture is 16%, then what is your percent error?

+5 correctly compares mass % from Q4 to 16% and reports a % error

6. (5 pts) Use the provided mass percentage from question 5 and the original mass of the mixture of recorded in step 1 of the procedure for find the theoretical yield of solid copper (not Copper(II) Oxide) in this experiment.

+5 correctly applies 16%*total original sample mass to find theoretical yield of solid copper.

6. (5 pts) Compare the theoretical yield of copper to the actual yield of copper by calculating the percent yield for this experiment.

+5 calculates % Yield

7. (5 pts) Based on your results, is the unknown material suitable for commercial processing? Explain your answer in the space provided.

+5 Finds the 10% threshold from the lab introduction and compares that to their calculated Mass %.

8. (5 pts) Use all the resources available to you to explain how you know that Copper(II) Oxide was formed during the drying/heating process. *(You must cite a source for this problem)*

+5 References how copper(II)oxide is black in appearance while copper(I) oxide is brown.

9. (5 pts) If Copper(I) Oxide was formed instead of Copper(II) Oxide, would your results have changed? Explain your reasoning.

+5 Something along the lines of: "Copper(I)Oxide is a heavier molecule resulting in incorrectly predicting a larger mass of copper if I was unaware of the Copper(I) contamination when I did my mass % calculation."

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 7

Lab 7 - Handouts

UNITED STATES MILITARY ACADEMY

LAB 7 – PRACTICAL

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:
SIGNATURE: DATE:
SIGNATURE: DATE:

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(100 points)

PRACTICAL

Part 1 – Determine the Concentration of the Unknown Solution
Note: Correlation (R value) and calculated concentration (question 3) will be assessed for accuracy to evaluate your laboratory techniques.

1. (10 points) Provide the volumes, final concentration, and absorbance of the samples used to make the calibration curve from the 0.250 M $Cu(NO_3)_2$ (aq) solution.(Hint: a table generated in your pre-lab would be a good starting point for you measurements.)

2. (10 points) Provide the linear regression equation from your calibration curve and the correlation coefficient (R) value.

3. (10 points) Record the absorbance of the unknown and calculate the concentration of Cu^{2+} ions in the contaminated sample.

   Absorbance of unknown:______________

4. (5 points) A sperate non-governmental organization (NGO) also determined the concentration of Cu^{2+} ions in the contaminated sample. Obtain the accepted value from your instructor and determine your % error.

   Accepted Value:_____________

5. (10 points) The EPA has a limit for allowable copper in drinking water of 1.3 parts per million (equivalent to $0.00002\underline{0}456$ M concentration) of copper (including copper (II) ions). Based on your experimental results, do we exceed the EPA standard? Use the concentration found in question 3 and the 5.0 mL of the unknown sample.

6. (5 points) The EPA obtained a 55. gallon barrel of the contaminated water for further testing, determine the amount (in grams) of copper (II) ions in the barrel.

Part 2 – Remove the Contaminants

1. (5 points) Write the balanced molecular equation for the reaction which formed a precipitate with the contaminated sample. Assume the only contaminate is $Cu(NO_3)_2$ (aq).

2. (5 points) Write the balanced complete ionic equation for the reaction which formed a precipitate with the contaminated sample. Assume the contaminate is $Cu(NO_3)_2$ (aq).

3. (5 points) Write the balanced net ionic equation for the reaction which formed a precipitate with the contaminated sample. Assume the contaminate is $Cu(NO_3)_2$ (aq).

4. (6 points) Calculate the number of moles of the chosen reagent needed to completely react with the amount of moles of Cu^{2+} in the 5.0 mL contaminate water sample.

5. (6 points) How many mL of the chosen reagent are needed to precipitate all of the Cu^{2+} ions in the 5.00 mL contaminate water sample without further contaminating the water sample.

6. (8 points) Calculate the theoretical yield, in grams, of the solid precipitate containing copper formed from the above reaction.

7. (5 points) Provide the mass of the actual yield of solid precipitate containing copper after performing the separation and allowing the solid to dry.

8. (5 points) Calculate the percent yield of the solid precipitate containing copper.

9. (2 points) Describe if the precipitation reaction was effective at removing the copper (II) ions from the contaminated water supply.

10. (3 points) Describe one method to analytically measure the effectiveness of the precipitation reaction used to remove the Cu^{2+} ions from the contaminate water (i.e. how decontaminated is your water).

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 7 - Instructor Solution

Cut Scale

Date:_____________

Section 1: TITLE *(Provide a descriptive title for this experiment in your own words)*

Section 2: PURPOSE *(Provide the purpose and objectives of this experiment)*

Section 3: MATERIALS

Reagents:	+1 Includes all three required reagents ($Cu(NO_3)_2$, NaOH, contaminated water sample CAN INCLUDE NaCl but not needed) (*List all required reagents and samples for this experiment. Include concentrations of each known solution)*
Equipment:	+1 Includes all required equipment (quantity/size of glassware must be specified) (*List all required equipment for this experiment. Include the size and quantity of each item.)*

SECTION 4: REAGENT SAFETY DATA +1 Completes tables for $Cu(NO_3)_2$, and NaOH from SDS

(This section should contain relevant information from the reagent's Safety Data Sheet (SDS). Cite your sources! Chemical manufacturers, such as Sigma-Aldrich and Fisher Scientific have repositories of SDSs for your use. If you have more than one reagent, copy the provided table for each one.)

Reagent Name:	
Chemical Formula:	**Molar Mass:**
Boiling Point:	**Melting Point:**
Vapor Pressure:	**pH:**
Hazards:	
PPE Requirements:	
Disposal Requirements:	
First Aid:	

(End this section with a 1-paragraph recap of how you will incorporate safety, disposal, and first aid requirements mentioned in the SDSs.)

+1 Reasonable Consolidation of SDS measures

SECTION 5: PROCEDURE

+8 Part 1:
+4 Determines wavelength of maximum absorbance for copper II nitrate
+4 Uses "linear fit" concept to determine linear regression model and correlation coefficient

+8 Part 2:
+2 Determines correct reagent (or provides steps to do so)
+2 Shows mathematical steps to calculate exact volume of reagent needed
+2 Outlines setup of Buchner funnel
+2 Details procedure to determine mass of precipitate

OVERALL Instructor Assessment: Detail
+5 Excellent attention to detail; procedure will effectively accomplish all tasks
+4 Good detail; missing minor components
+3 Good detail; missing one key component
+2 Some detail; missing some key components
+1 Some detail; cannot be executed as written
+0 No detail; cannot be executed as written

SECTION 6: DISCUSSION

Part 1 – Determine the Concentration of the Unknown Solution

1. (10 pts) Provide the volumes, final concentration, and absorbance of the samples used to make the calibration curve from the 0.125 M $Cu(NO_3)_2$ *(aq)* solution. (Hint: copy the table generated in Section 5, Part I.)

Table 1:				
Volume of 0.125 M $Cu(NO_3)_2$ *(aq)* (mL)	Volume of DI water (mL)	Total Volume (mL)	Concentration of new solution	Absorbance
0.00 mL	10.00 mL	10.00 mL	0.00 M	0.001
2.00 mL	8.00 mL	10.00 mL	0.025 M	0.125
4.00 mL	6.00 mL	10.00 mL	0.050 M	0.350
6.00 mL	4.00 mL	10.00 mL	0.075 M	0.525
8.00 mL	2.00 mL	10.00 mL	0.100 M	0.769
10.00 mL	0.00 mL	10.00 mL	0.125 M	1.006
			UNK	0.535

+ 4 for three or more dilutions
+ 4 for correct concentrations
+ 2 AON absorbance recorded to thousandths place

2. (10 pts) Provide the linear regression equation from your calibration curve and the correlation coefficient (R) value. *(Points will be assigned for the R value)*

y = mx + b

Abs = (6.0343) × (Conc) - 0.0038

R=0.9998

+ 3 equation in y=mx+b format
+ 2 R value provided
Variable points for R values:
+ 5 R value > 0.99
+ 4 R value > 0.95
+ 3 R value > 0.90
+ 2 any other R value

3. (7 pts) Record the absorbance of the contaminated water sample and calculate the concentration of Cu^{2+} ions in the contaminated sample.

Absorbance of unknown: + 2 AON

Abs = m * Conc - b

+ 4 for concept

Conc = $\frac{Abs-b}{m}$ = 0.074 M

Concentration of unknown: + 1 M&U (not each)

/27

4. (8 pts) A separate non-governmental organization (NGO) also determined the concentration of Cu^{2+} ions in the contaminated sample. **Obtain the accepted value from your instructor** and determine your % error. *(Pts will be assigned for your % Error value)*

Accepted Value: 0.075 M

% error = $\left(\frac{experimental - accepted}{accepted}\right) * 100$

% error = $\left(\frac{0.074 - 0.075}{0.075}\right) * 100 = 1.33\%$

+ 3 for concept
+ 1 uses concentration from part 3 above
+ 1 M&U (not each)

+ 3 if % error less than 5%
+ 2 if % error greater than 5% but less than 10%
+ 1 if % error greater than 10% but less than 15%
(0 points if % error is greater than 15%)

5. (10 pts) The EPA has a limit for allowable copper in drinking water of 1.3 parts per million (equivalent to 1.02 x 10^{-7} mols in a 5.0 mL sample) of copper (including Cu^{2+} ions). Based on your experimental results, do we exceed the EPA standard? Use the concentration found in question 3 and the 5.0 mL of the unknown water sample.

$\frac{0.074\ mol\ of\ Cu^{2+}}{L} \times \frac{1L}{1000\ mL} \times 5.0\ ml = 3.7 * 10^{-4}$ mol Cu^{2+}

$3.\underline{7} * 10^{-4}$ mol Cu^{2+} > $1.\underline{0}2 * 10^{-7}$ mol Cu^{2+}

Our water example exceeds the EPA standards.

+ 3 for concept
+ 1 uses concentration from part 3
+ 1 each M/U/SF
+ 3 AON correctly identifies and shows that their contaminated water does or does not exceed standards

6. (5 pts) The EPA obtained a 55. gallon barrel of the contaminated water for further testing. Determine the amount (in grams) of Cu^{2+} ions in the barrel. Use the concentration found in question 3 and the 5.0 mL of the unknown sample.

$$\frac{gal}{} * \frac{L}{gal} * \frac{mol}{L} * \frac{g}{mol} = grams\ of\ Cu^{2+}$$

$$\frac{5\underline{5}\ gal}{} * \frac{3.7854\ L}{1\ gal} * \frac{0.074\ mol}{L} * \frac{63.55\ g}{mol} = 9\mathbf{8}0\ grams\ of\ Cu^{2+}$$

+ 2 for concept
+ 1 correct molar mass
+ 1 correct gallon to L conversion
+ 1 M&/SF (not each)

/23

Part 2 – Remove the Contaminants

1. (5 pts) Write the balanced molecular equation for the reaction that formed a precipitate with the contaminated water sample. Assume the only contaminant is $Cu(NO_3)_2$ *(aq)*.

   $Cu(NO_3)_2\,(aq) + 2\,NaOH\,(aq) \rightarrow Cu(OH)_2\,(s) + 2\,NaNO_3\,(aq)$

   + 2 identified NaOH as the proper reagent to form a precipitate
   + 1 molecular equation format
   + 1 balanced
   + 1 state symbols

2. (5 pts) Write the balanced complete ionic equation for the reaction that formed a precipitate with the contaminated water. Assume the only contaminant is $Cu(NO_3)_2$ *(aq)*.

   $Cu^{2+}(aq) + 2NO_3^{-}(aq) + 2\,Na^{+}(aq) + 2OH^{-}(aq) \rightarrow Cu(OH)_2(s) + 2Na^{+}(aq) + 2NO_3^{-}(aq)$

   + 3 complete ionic equation format
   + 1 balanced
   + 1 state symbols

3. (5 pts) Write the balanced net ionic equation for the reaction that formed a precipitate with the contaminated water. Assume the only contaminant is $Cu(NO_3)_2$ *(aq)*.

   $Cu^{2+}(aq) + 2OH^{-}(aq) \rightarrow Cu(OH)_2(s)$

   + 3 complete net ionic equation format
   + 1 balanced
   + 1 state symbols

4. (6 pts) Calculate the number of moles of the chosen reagent needed to completely react with the amount of moles of Cu^{2+} in the 5.0 mL contaminated water sample.

   $$3.\mathbf{7} * 10^{-4}\text{ mol Cu}^{2+} * \frac{2\,mol\ of\ OH^{-}}{1\,mol\ of\ \text{Cu}^{2+}} * \frac{1\,mol\ of\ NaOH}{1\,mol\ of\ OH^{-}} = 7.\mathbf{4} * 10^{-4}\,mol\ NaOH\ (aq)$$

   + 3 concept
   + 2 correct molar ratios
   + 1 M/U/SF (2/3)

$/_{21}$

5. (6 pts) How many mL of the chosen reagent are needed to precipitate all Cu^{2+} ions from the 5.0 mL contaminated water sample without further contaminating the water sample.

$$7.\mathbf{4} * 10^{-4}\ \mathrm{mol}\ NaOH * \frac{1\ L\ NaOH}{1\ mol\ NaOH} * \frac{1000\ mL\ NaOH}{1\ L\ NaOH} = 0.7\mathbf{4}\ mL\ NaOH\ (aq)$$

+ 3 concept
+ 1 correct concentration of NaOH
+ 1 convert to mL
+ 1 M/U/SF (2/3)

6. (8 pts) Calculate the theoretical yield (in grams) of the solid copper-containing precipitate formed from the above reaction.

$$3.\mathbf{7} * 10^{-4}\ \mathrm{mol\ Cu}^{2+} * \frac{1\ mol\ of\ Cu(OH)_2}{1\ mol\ of\ \mathrm{Cu}^{2+}} * \frac{97.55\ g\ Cu(OH)_2}{1\ mol\ of\ Cu(OH)_2} = 0.0\mathbf{3}61\ g\ Cu(OH)_2(s)$$

+ 2 concept
+ 2 correct molar ratios
+ 1 correct molar mass
+ 1 each M/U/SF

7. (5 pts) Provide the actual yield (in grams) of solid copper-containing precipitate after performing the separation and allowing the solid to dry.

Mass of precipitate $Cu(OH)_2$= 0.098 g

+ 4 mass recorded
+ 1 mass recorded to the thousandths place

8. (5 pts) Calculate the percent yield of the solid copper-containing precipitate.

$$\%\ yield = \frac{Actual}{Theoretical} * 100\% = \frac{0.098\ g}{0.0361\ g} * 100\% = 2\mathbf{7}1\%$$

+ 4 Concept (you will have high percent yields)
+ 1 M/U/SF (2/3)

9. (2 pts) Describe if the precipitation reaction was qualitatively effective at removing the copper(II) ions from the contaminated water supply.

+ 1 References observations from lab procedure to identify that a mass of solid was extracted from the water and held in the filter
+ 1 provides statement to the effectiveness of process (i.e. solution not as blue after reaction, some solid lost to flask in process, etc)

10. (3 pts) Describe one method to analytically measure the quantitative effectiveness of the precipitation reaction used to remove the Cu^{2+} ions from the contaminated water (i.e., to determine how decontaminated your water is).

+ 3 AON, any reasonable process (example: conduct the spectrophotometer analysis of the waste after the extraction to observe the new concentration of Cu^{2+})

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Lab 8

Lab 8 - Handouts

DEPARTMENT OF CHEMISTRY & LIFE SCIENCE

United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

THERMODYNAMICS

Experiment VIII

Instructions: Prior to the lab, read pages 1-8 to obtain information needed to complete this lab. **The material covered here is testable.** During the lab **only turn in pages 9-12 and the works cited page (if applicable).**

Materials

Reagents:	2.0 M NaOH (aq)	2.0 M HCl (aq)	
Equipment:	(2) 250-mL Beakers	(1) Stir Plate w/ Stir Bar	Vernier LabQuest Mini
	(2) 50-mL Grad. Cylinders	(1) Ring Stand	Vernier Small Thermistor
	(1) Small 3-Prong Clamp	(1) Styrofoam Calorimeter	
	(1) Medium Ring Attachment	Disposable Pipettes	

Introduction: First Law of Thermodynamics

Thermodynamics is the study of how heat, work, and temperature relate to energy, entropy, and the physical properties of matter and radiation. This is generally expressed in chemistry through the first law of thermodynamics, which states that the change of internal energy of a system is equal to the amount of thermal energy, or heat (q), added plus the work done on the system:

$$\Delta E = q + w \tag{1}$$

This is simply a way of restating the Law of Conservation of Energy - that energy cannot be created or destroyed – in terms of thermodynamic processes (heat and work). Mechanical work (w) is the amount of energy generated by creating a change in volume with a given pressure, or $w = -P \times \Delta V$. For example, you do work when inflating a balloon: the air pressure expelled from your lungs causes a volume change on the balloon. By substituting the equation for work into Equation 1, we can rewrite the first law as follows:

$$\Delta E = q - P\Delta V \tag{2}$$

When considering the amount of energy that is released or required for a chemical reaction, we often want to neglect this mechanical work and purely focus on the energy involved with bonds breaking and reforming, which is often measured by a change in heat (q). We represent this type of energy as **enthalpy**, or ΔH. By combining this definition of enthalpy with our equation for energy, we see that we can determine enthalpy by measuring the heat in a reaction.

$$\Delta H = \Delta E + P\Delta V$$

$$\Delta H = q - P\Delta V + P\Delta V$$

$$\Delta H = q - \cancel{P\Delta V} + \cancel{P\Delta V}$$

$$\Delta H = q \tag{3}$$

In this lab we will use both definitions to determine the amount of energy that is released during a chemical reaction between sodium hydroxide and hydrochloric acid. This acid-base neutralization reaction proceeds as follows:

$$NaOH\ (aq) + HCl\ (aq) \rightarrow H_2O\ (l) + NaCl\ (aq)$$

As stated above, according to the law of conservation of energy, energy cannot be created or destroyed. This is often expressed as stating that energy is simply transferred between a system and its surroundings according to the equation:

$$\Delta E_{sys} = -\Delta E_{surr} \tag{4}$$

In order to calculate the change in energy due to the chemical reaction, we will need to define our system and surroundings. Because we are trying to determine how much energy is being released due to the reaction, we will define the system as our reaction. Our reaction will take place inside a coffee cup calorimeter and both reactants are in an aqueous solution. It therefore follows that we can define our surroundings as both the solution and the coffee cup calorimeter. By substituting this into Equation 4 we arrive at the following equation:

$$\Delta E_{rxn} = -(\Delta E_{sol} + \Delta E_{cal}) \tag{5}$$

To simplify Equation 5, recall Equation 2. Because there will be no gas evolved during this reaction, there will be a negligible amount of volume change to the solution and the calorimeter ($\Delta V \approx 0$). Additionally, because we are trying to find the amount of energy released due to the bonds breaking, we want to use our definition of enthalpy above in Equation 3. By substituting these simplifications into Equation 5 we arrive at our final equation:

$$\Delta H_{rxn} = q_{rxn} = -(q_{sol} + q_{cal}) \tag{6}$$

Heat and Temperature

Heat (q) is the energy transfer due to a difference in temperature across a boundary. The two terms are directly proportional: the greater the **difference in temperature (ΔT)**, the more energy will be transferred across the boundary. By multiplying the temperature difference by a constant called the **heat capacity (C)**, we can calculate the amount of energy transferred using the equation:

$$q = C \times \Delta T \tag{7}$$

Heat capacity is a material property - different materials are better than others at transferring heat. This value is often experimentally determined if there is a mixture of materials that make up a device. The heat capacity for the calorimeter used in this experiment is listed in Table 1.

Parameter	Symbol	Value
Heat capacity of calorimeter	C_{cal}	2.679 J/°C
Specific heat of NaCl *(aq)* solution	$C_{s,sol}$	3.895 J/g·°C
Density of NaCl *(aq)* solution	D_{sol}	1.06 g/mL

Table 1: Thermodynamic data for a neutralization reaction between NaOH*(aq)* and HCl*(aq)*.

The heat capacity of a homogenous material is called a **specific heat capacity (C_s)**. The difference between heat capacity and specific heat capacity is that specific heat capacity depends on the amount of and type of a homogenous material, which leads to the following equation:

$$q = m \times C_s \times \Delta T \tag{8}$$

For example, wood, iron, and water all have different specific heat capacities because the amount of energy transferred scales with how much material there is. A calorimeter, or other composite device, will not have a specific heat capacity because it does not have a variable amount; you will always do your experiment in one calorimeter. The specific heat capacity for the sodium chloride solution used in this experiment is also included in Table 1. Combining Equations 6, 7, and 8 allows us to calculate the amount of heat released by the reaction by measuring the temperature change in the solution.

$$q_{rxn} = -(m \times C_{s,sol} \times \Delta T + C_{cal} \times \Delta T) \tag{9}$$

Recalling how we defined enthalpy in Equation 3, we can then use Equation 9 to determine the experimental enthalpy value for our specific reaction.

Enthalpy of Formation

To verify the accuracy of the experimental enthalpy, a theoretical enthalpy can be calculated by using the **standard heats of formation (ΔH°_f)** for the products and the reactants. The details for this calculation can be found in Section 9.10 of your textbook, the equation for this calculation is on page 2 of your RDC, and the standard enthalpies of formation are listed below in Table 2.

Species	**ΔH°_f (kJ/mol)**
NaOH *(aq)*	-470.1
HCl *(aq)*	-167.2
NaCl *(aq)*	-407.2
H_2O *(l)*	-285.8

Table 2: Thermodynamic data for a neutralization reaction between NaOH*(aq)* and HCl*(aq)*.

Calculating the standard enthalpy of the reaction will provide an accepted (or theoretical) value, to which it will be possible to compare the experimentally obtained value and assess the accuracy of the procedure.

Procedure

1. Check to ensure your lab station has a Styrofoam calorimeter. Ensure your cups are clean, dry, and have no holes. Using Figure 1 as a guide, set-up the calorimeter. **The small thermistor should be used and inserted into the center hole of the calorimeter lid so that the black rubber stopper seals the hole. Rubber stoppers with a hole drilled out are pre-installed onto the small thermistor. The thermistor tip should not touch the stir bar or the bottom of the calorimeter. DO NOT force the thermistor into the lid as to widen the center hole.** The thermistor data cable should be inserted into Channel 1 on the Vernier LabQuest Mini box.

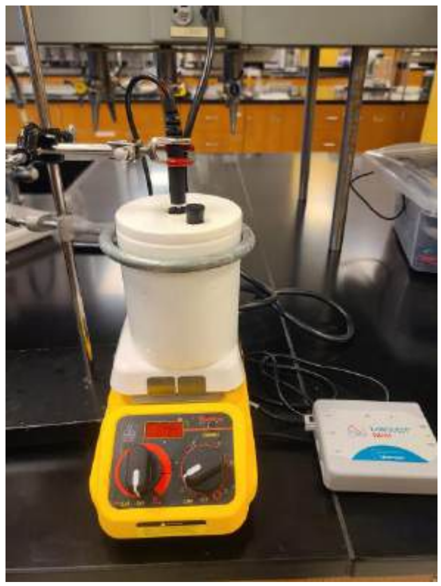


Figure 1: Styrofoam calorimeter set-up

2. Open Vernier Graphical Analysis on your laptop. Select the “Sensor Data Collection” as the experiment type. **Set the duration of the experiment to 600 seconds.** Set the sample rate to 1 sample/second. Click “Done” when complete.

3. Label two 50-mL graduated cylinders with different color strips of tape. Label one “NaOH” and the other “HCl”. There is enough glassware at the lab station to use a different set of glassware for each reagent. It is important to not contaminate the solutions. **Use one tape color for all glassware containing NaOH and another tape color for all glassware containing HCl.** Using a clean, dry 250-mL beaker, obtain about 125 mL of 2.0 M NaOH*(aq)* from the instructor bench. Carefully pour 50.0 mL of the NaOH*(aq)* into the NaOH-labeled graduated cylinder (when you get close to 50.0 mL, use a disposable pipet to be as precise as possible). Record the exact volume of NaOH*(aq)* in Table 3 of the Discussion Section of this lab report.

4. Using the other clean, dry 250-mL beaker, obtain about 125 mL of 2.0 M HCl*(aq)* from the instructor bench. Carefully pour 50.0 mL of the HCl*(aq)* into the HCl-labeled graduated cylinder (when you get close to 50.0 mL, use a disposable pipet to be as precise as possible). Record the exact volume of HCl*(aq)* in Table 3. **DO NOT MIX THE REACTANTS until you have completed step 6.**

5. Place the magnetic stir bar in the calorimeter and ensure it can rotate without hitting the small thermistor.

6. Place the small thermistor into the calorimeter as shown in Figure 1, with the thermistor passing through the rubber stopper. Remove the other rubber stopper and use a glass funnel to pour the 50.0 mL of NaOH*(aq)* solution into the calorimeter. Turn the stir knob to level '6'. The next goal is to minimize the time it takes to start the Vernier collection (by clicking the "Collect" button in the toolbar), add the HCl*(aq)* via glass funnel, and replace the rubber stopper so no heat is lost to the environment as the reaction begins. **Rehearse this process several times before pouring the HCl into the calorimeter!**

7. Once you are confident in your mixing method, you will begin the experiment. **Ensure your system is set to collect for 600 seconds.** Select "Collect" and <u>wait approximately 10 seconds</u> to obtain a baseline temperature. <u>Then, add the HCl*(aq)*</u> from the graduated cylinder using your second glass funnel. Quickly replace the stopper onto the top of the calorimeter and observe as the data is recorded. Collect enough data to show a gradual decline from the peak in temperature, usually about 3 minutes after you observe the peak in temperature.

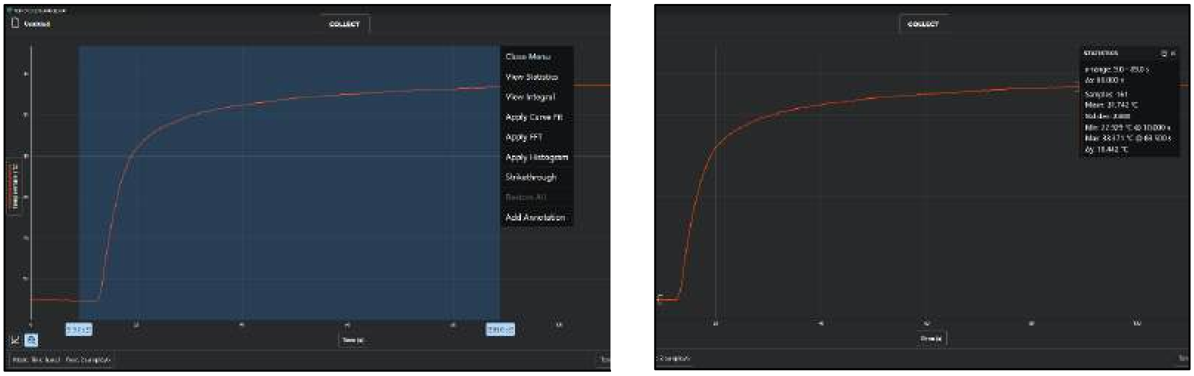
8. After collecting the data, press the "Stop" button and identify the region with the maximum temperature. To accurately find change in temperature follow the steps below.

   (a) Begin by selecting the "Zoom to All Data" button from the bottom left.

   (b) Highlight the entire region starting from your initial temperature (10 second hold from step 7) until the graph begins to decline (see Figure 2a).

   (c) Select the "View Statistics" option from the from the pop-up menu (see Figure 2a). This will open a window that shows the minimum, maximum and change of temperature recorded in the selected region (see Figure 2d). The Δy value is your change in temperature; record this value in Table 3, row 5.

   (d) Show the graph for your first trial to your instructor.



1) Graph highlighted and statistics Function *d) Statistics View*

Figure 2: Graphs depicting heat expelled over time by the reaction of NaOH(aq) with HCl(aq).

9. Retrieve the stir bar from the calorimeter and dispose of the used solutions into a common 600 mL beaker. Rinse the calorimeter **twice** with deionized water and dry the inside with a paper towel.

10. Repeat steps 3-10 for Trial 2. Once all data is collected you can display the data from both iterations on one graph if desired to observe the reproducibility of the experiment. **A third set of trial data has been provided to you. Treat it as your own data.**

11. Once all experiments are complete, take your aqueous waste beaker and discard it into the aqueous waste containers at the instructor bench.

UNITED STATES MILITARY ACADEMY

LAB 8 – THERMODYNAMICS

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:
SIGNATURE: DATE:
SIGNATURE: DATE:

Discussion

For all calculations, assume Trial 3 is your own data (use it in your calculations).

Table 3: Experimental Data Collection

Line	Quantity	Trial 1	Trial 2	Trial 3
1	Volume NaOH *(aq)*, mL			50.0
2	Volume HCl *(aq)*, mL			50.0
3	Solution Volume (line 1 and 2), mL			100.0
4	Solution Mass, g			106.
5	ΔT , °C			13.32
6	Calculated q_{rxn}			-5.535 kJ

1. (10 pts) Calculate the heat evolved from the reaction (q_{rxn}) in kJ for **trial 1 and 2 (Table 3, Line 6)** using your experimental data from Table 3. **Only show your work for Trial 1 below.** *Refer to Equations 6 and 9 of the lab introduction and consider that this is a constant volume and pressure experiment.*

2. (10 pts) In this experiment, both NaOH and HCl are present in 2.00 M concentration and have a 1:1 molar ratio. Therefore, we can use either reactant to calculate the enthalpy change for the reaction (ΔH_{rxn}). Use your average q_{rxn} from all three trials and the concentration of NaOH to calculate the average ΔH_{rxn} per mole of NaOH*(aq)*. **Show your work.**

3. (5 pts) Using the $\Delta H°_{rxn}$ equation from your RDC and Table 2 from the lab introduction, calculate the theoretical (accepted) standard heat of reaction ($\Delta H°_{rxn}$) for this reaction?

4. (4 pts) Calculate the percent error using the average experimental enthalpy change and explain whether your experiment produced accurate results.

5. (4 pts) Calculate the coefficient of variation for your enthalpy change values and explain whether your experiment produced precise results.

6. (5 pts) Were the errors in the measurements of the enthalpy change systematic or random? Explain using the data and any observations and propose one possible source of error that is consistent with your answer.

7. Effective Concentration and Standard States.

   (a) (4 pts) What is the “standard state” for aqueous species (like ions)? *Hint: The answer is in your textbook*).

   (b) (4 pts) After mixing the reactants, but before the reaction begins, what is the effective concentration of NaOH? *Hint: Consider that a dilution occurs upon mixing before the reaction begins.*

   (c) (4 pts) Why is the effective concentration important when comparing your experimental heat of reaction to the theoretical standard heat of reaction from question 3?

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 8 - Instructor Solution

Cut Scale

Discussion

For all calculations, assume Trial 3 is your own data (use it in your calculations).

Table 3: Experimental Data Collection

Line	Quantity	Trial 1	Trial 2	Trial 3
1	Volume NaOH *(aq)*, mL			50.0
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3	Solution Volume (line 1 and 2), mL			100.0
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6	Calculated q_{rxn}			-5.535 kJ

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$q_{sol} = m \times C_{s,sol} \times \Delta T$ +2 concept, +1 application (+1 ea *m, C, ΔT*)

$= (106\text{ g}) \times (3.895\,\frac{J}{g \times °C}) \times (13.59\text{ °C})$

$= 56\underline{1}0.90\text{ J}$

$q_{cal} = C_{cal} \times \Delta T$ +2 concept, +1 application (+1 ea *C, ΔT*)

$= (2.679\,\frac{J}{°C}) \times (13.59°C)$

$= 35.6\underline{8}4\text{ J}$

$q_{rxn} = -(q_{sol} + q_{cal})$ +2 concept, +1 application

$= -(56\underline{1}0.90\text{ J} + 35.6\underline{8}4\text{ J})$

$= -56\underline{4}6.59\text{ J}$

$= -5.65\text{ kJ}$ +1 lists q_{rxn} for Trial 2

/10

2. (10 pts) In this experiment, both NaOH and HCl are present in 2.00 M concentration and have a 1:1 molar ratio. Therefore, we can use either reactant to calculate the enthalpy change for the reaction (ΔH_{rxn}). Use your average q_{rxn} from all three trials and the concentration of NaOH to calculate the average ΔH_{rxn} per mole of NaOH*(aq)*. **Show your work.**

$$\frac{\left(-56.5\frac{kJ}{mol}\right)+\left(-56.1\frac{kJ}{mol}\right)+\left(-55.3\frac{kJ}{mol}\right)}{3} = -55.9\ kJ/mol$$ +3 finds average q_{rxn}

$$50.0\ mL \times \frac{L}{1000\ \mathrm{mL}} \times \frac{2.0\ mol\ NaOH}{\mathrm{L}} = 0.10\ mol$$ +4 determines mol of NaOH (+2 mL to L, +2 L to mol)

$$\frac{\Delta H_{rxn}}{mol\ NaOH} = \frac{-5.59\ kJ}{0.10\ mol} = -5\underline{5}.9\ \frac{kJ}{mol}$$ +3 divides q_{rxn} by mol NaOH to find ΔH_{rxn}

3. (5 pts) Using the $\Delta H°_{rxn}$ equation from your RDC and Table 2 from the lab introduction, calculate the theoretical (accepted) standard heat of reaction ($\Delta H°_{rxn}$) for this reaction?

$$\Delta H°_{rxn} = \sum n_p \Delta H°_f(products) - \sum n_r \Delta H°_f(reactants)$$ +3 correct equation

$$= [\Delta H°_f(H_2O) + \Delta H°_f(NaCl)] - [\Delta H°_f(HCl) + \Delta H°_f(NaOH)]$$

$$= (-285.8 - 407.2) - (-167.2 - 470.1)$$ +1 application

$$= -55.7\ kJ/mol$$ +1 M/U/SF

4. (4 pts) Calculate the percent error using the average experimental enthalpy change and explain whether your experiment produced accurate results.

$$\%\ error = \left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right| \times 100\%$$ +2 Concept

$$= \left|\frac{(-55.9\ kJ)-(-55.7\ kJ)}{-55.7}\right| \times 100\%$$ +1 Application

$$= 0.359\%$$

The measurements were accurate because the percent error was less than 5% of the accepted value.

+1 Correctly uses 0.05 threshold to comment on accuracy of results

5. (4 pts) Calculate the coefficient of variation using experimental enthalpy change values and explain whether your experiment produced precise results.

$$SD(s) = \sqrt{\frac{((-56.5)-(-55.9))^2+((-56.1)-(-55.9))^2+((-55.3)-(-55.9))^2}{3-1}} = 0.61101\ {}^{kJ}/_{mol}$$

$$CV = \frac{s}{\bar{x}} = \frac{0.61101 kJ/mol}{55.9\ kJ/mol} = 0.0109 = 1.093\%$$ +2 Concept

+1 Application

The measurements were precise because the CV was less than 5%.

+1 Correctly uses 0.05 threshold to comment on precision of results

/23

6. (5 pts) Were the errors in the measurements of the enthalpy change systematic or random? Explain using the data and any observations and propose one possible source of error that is consistent with your answer.

The errors in our measurement were systematic because all of our calculated enthalpy change values were above the actual calculated enthalpy change. A source of error could be an extra drop*(s)* beyond exactly 50.0 mL during measurement, resulting in additional heat from the neutralization reaction. The additional reaction of the excess reagents would have resulted in more product formation and therefore more heat produced which is consistent with our results.

+2 Correct form of error (could be either depending on data)
+2 Explanation
+1 Logical source of error

7. Effective Concentration and Standard States.

(a) (4 pts) What is the “standard state” for aqueous species (like ions)? *Hint: The answer is in your textbook*).

All aqueous species have a concentration of 1.0 M. +4 AON

(b) (4 pts) After mixing the reactants, but before the reaction begins, what is the effective concentration of NaOH? *Hint: Consider that a dilution occurs upon mixing before the reaction begins.*

The effective concentration is 1.0 M because mixing the 50.0 mL of a 2.0 M NaOH stock solution of NaOH with 50.0 mL of HCl would bring the total volume to 100.0 mL and halving the original concentration from 2.0 M to 1.0 M.

$M_1V_1 = M_2V_2$

$M_2 = \frac{M_1V_1}{V_2} = \frac{(2.0M)(0.05L)}{0.01L} = 1.0\ M$ +4 AON

(c) (4 pts) Why is the effective concentration important when comparing your experimental heat of reaction to the theoretical standard heat of reaction from question 3?

The theoretical standard heat of reaction calculation is calculated for solutions at standard state conditions and comparing the theoretical to experimental requires that the comparison be at identical conditions.

+4 AON

/17

Lab 9

Lab 9 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY26-1
(50 points)

GAS LAWS AND STOICHIOMETRY

Experiment VII

Instructions: Prior to the lab, read pages 1-6 to obtain information needed to complete this lab. **The material covered here is testable.** During the lab **only turn in pages 7-12 of this lab report.**

Materials

Reagents:	Water	$KClO_3$ (*s*)	Unknown Mixture
Equipment:	(1) 100-mL Beaker (1) 250-mL Beaker (1) 600-mL Beaker (1) Bunsen Burner (1) Spatula	(1) 50-mL Grad. Cylinder (1) 500-mL Erlenmeyer Flask (1) Large Ring Attachment (1) Large 3-Prong Clamp (1) Medium Test Tube Clamp	(4) 30-mLTest Tube Stopper/Tubing Assembly Vernier LabQuest Mini Vernier Thermistor Vernier Pressure Sensor

Introduction

At high temperatures and low pressures, most gases behave **ideally**, meaning that they follow the relationships first codified by Avogadro, Boyle, and Charles. The combination of Avogadro's Law, Boyle's Law, and Charles' Law with the ideal gas constant provides the relationship known as the ideal gas law:

$$PV = nRT \tag{1}$$

Another important relationship is Dalton's law (equation 2) which shows that the total pressure of any given mixture of gases is the sum of the pressure of each individual gas.

$$P_{total} = P_a + P_b + P_c... \tag{2}$$

P_{total} is the total or atmospheric pressure and P_a, P_b, and P_c are all the partial pressures of the various gases. In this experiment, you will determine the amount (in moles) of gas released in an oxidation reaction by using Equation 1 and 2. Solid potassium chlorate when heated decomposes into solid potassium chloride and oxygen gas according to the following equation:

$$2\,KClO_3\,(s) \rightarrow 2\,KCl\,(s) + 3\,O_2\,(g) \tag{3}$$

You will calculate the amount of oxygen gas released through two different methods and compare them to theoretical yield. The first method is by taking the **mass** of $KClO_3$ (*s*) and subtracting from it the mass that remains after the thermal decomposition. The second method is by measuring the amount of gas displaced from a gas displacement device pictured below:

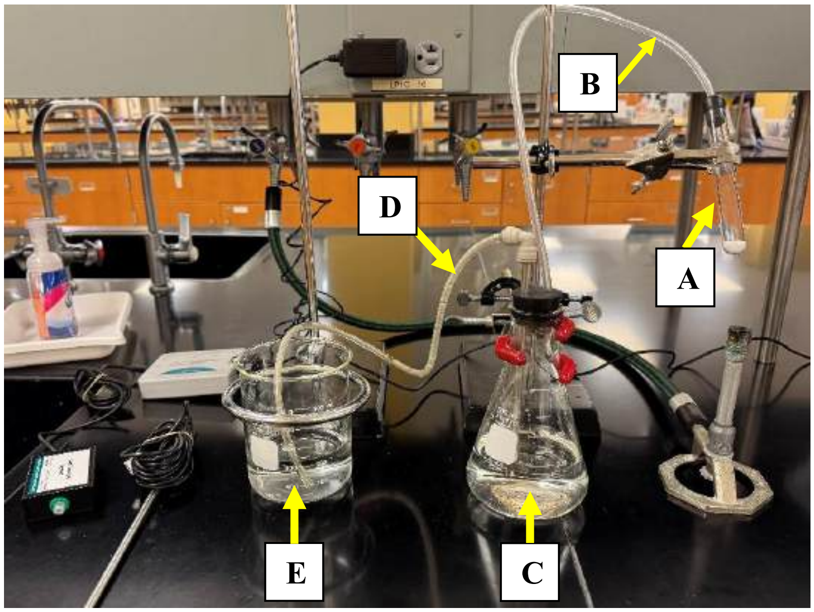

Figure 1: Water Displacement Device. When the sample in test tube A is heated, the sample will undergo thermal decomposition, evolving O_2 gas that will flow through tubing B into the Erlenmeyer flask C. The additional gas in Erlenmeyer flask C will raise the pressure in the Erlenmeyer flask and force water from the Erlenmeyer flask C up the metal straw and through tubing D into the overflow beaker (E), where the volume of water can be measured to determine the volume of gas produced.

In the water displacement device, the thermal decomposition of the $KClO_3$(*s*) will create oxygen gas that will displace the water in the Erlenmeyer flask and push it through the tubing into a beaker where you can then measure it. By rearranging Equation 1, you can solve for the number of moles of oxygen:

$$n_{O_2} = \frac{P_{O_2} * V}{R * T} \tag{4}$$

In the above equation, the volume of water displaced will be equal to the volume of oxygen generated from the reaction. The ideal gas constant (R) is provided on your RDC. The temperature **must be converted to Kelvin**. Because the gas and water are in contact with each other, it is reasonable to assume that the temperature of water that is displaced is equal to the gas temperature. The pressure of the oxygen gas (P_{O_2}) must be calculated using Equation 2. Using the Vernier pressure sensor, you can calculate the atmospheric pressure (which is equal to the total pressure in Equation 2). Inside the flask, there are two gases: the oxygen from the reaction, and the water vapor. The vapor pressure of water can be determined from the table below:

Vapor Pressure Of Water (torr)					
Temp °C	0.0	0.2	0.4	0.6	0.8
16	13.634	13.809	13.967	14.166	14.347
17	14.530	14.715	14.903	15.092	15.284
18	15.477	15.673	15.871	16.071	16.272
19	16.477	16.685	16.894	17.105	17.319
20	17.535	17.753	17.974	18.197	18.422
21	18.650	18.880	19.113	19.349	19.587
22	19.827	20.070	20.316	20.565	20.815
23	21.066	21.324	21.583	21.845	22.110
24	22.377	22.648	22.922	23.198	23.475
25	23.756	24.039	24.326	24.617	24.912
26	25.209	25.509	25.812	26.117	26.426
27	26.739	27.055	27.374	27.696	28.021
28	28.349	28.680	29.015	29.354	29.697
29	30.043	30.392	30.745	31.102	31.461
30	31.824	32.191	32.561	32.934	33.312

Figure 2: Vapor pressures of water as a function of temperature. **Note the units of this pressure**. To use this chart, find the nearest whole degree in the first column, and then locate the closest tenth of a degree in the first row. The intersection of those two values is the vapor pressure. For example, the vapor pressure of 21.2 °C is 18.880 torr.

Once you have found the vapor pressure of water, you can rearrange Equation 2 to solve for the pressure of oxygen gas. Using the pressure of oxygen gas determined from Dalton's Law, the temperature of the liquid (and therefore also the gas), and the volume of water displaced, it is possible to calculate the amount of oxygen that is evolved from the reaction using the water displacement device.

Another method to calculate water displacement is by simply subtracting the mass of the sample before and after heating - the difference in mass can be assumed to be the amount of oxygen gas released. Both experimental values can be compared to the theoretical yield, which can be found using stoichiometry to calculate the amount of oxygen gas that should be produced based on the initial mass of the sample of $KClO_3$(*s*).

In the second part of this lab, you will determine the amount of $KClO_3$(*s*) in an unknown sample where $KClO_3$(*s*) is mixed with an inert substance. This inert substance does not thermally decompose, meaning that all the oxygen evolved must come from the $KClO_3$(*s*). Once you have weighed your initial sample of the unknown and experimentally determined the volume displaced by thermal decomposition, you can calculate the percentage of the initial mass of your unknown that was $KClO_3$(*s*).

Procedure Part I

1. Open the Vernier Graphical Analysis program on your computer. Connect the Vernier Labquest Mini unit to your computer using the provided USB cable. Insert the connection for the Vernier Thermistor in Channel 1. Insert the connection cable for the Vernier Pressure Sensor in Channel 2. The screen will automatically update to display the temperature and pressure in the bottom right. If the units for temperature and pressure are not in °C and torr, select the readout that needs to be changed and change to the appropriate units in the pop-up menu. Place the thermistor and pressure sensor off to the side.

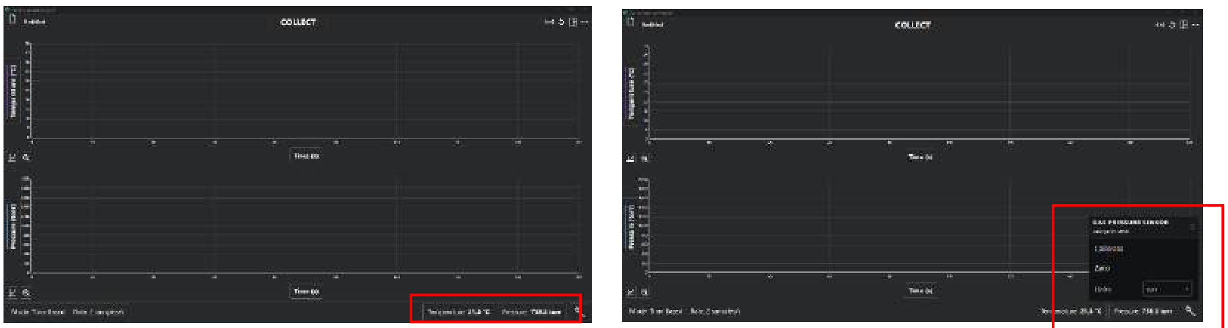

Figure 3. (a) Screen shot indicating area of Vernier Graphical Analysis screen to open menu to switch units for sensors; **(b)** Instructions to change the units for sensors for Vernier unit.

2. Select two 30-mL test tubes and ensure they fit the preassembled stopper/hose securely. Clean the test tubes with soap and deionized water, rinse them thoroughly with distilled water and dry with the Bunsen burner. Use the test tube tongs to hold a test tube over the flame for approximately 3 minutes to thoroughly dry the test tubes. Ensure to pass the entire test tube through the flame to completely remove all water (you should see the water droplets begin to sizzle and evaporate). Place all test tubes in a 250-mL beaker to cool. **Remember, hot glass looks like cold glass - so handle hot test tubes with tongs only.** You will use one tube in Part I and one in Part II.

3. Take a cooled test tube and 100-mL beaker with you to the balance area. Place a weigh boat on the balance, tare the weigh boat, and remove the weigh boat from the balance. Place a small amount of pure $KClO_3$(*s*) into the weigh boat, place the weigh boat back onto the balance, if the balance does not read a mass of **approximately** 0.700 g remove it from the balance and add or remove some of the mixture until the balance reads **approximately** 0.700 g. **Never add reagent to the weigh boat while on the scale to prevent spilling on the balance. Discard excess; do not put excess reagent back in the bottle.** There is no need to record this mass. Remove the weigh boat containing the $KClO_3$(*s*).

4. Place the 100-mL beaker **without the test tube** on the balance and press the tare/zero button. The balance should read 0.000 g.

5. Place the test tube in the empty beaker and record the mass in Line 1 of Table 1.

6. Remove the test tube from the balance and pour the $KClO_3$(*s*) from the weigh boat into the test tube. **Do not try to pour the $KClO_3$(*s*) into the test tube while it is on the balance**. Check that the solid all slides to the bottom of the tube. If the solid smears out along the side of the tube, the test tube is still wet. Try gently tapping the test tube on the bench top to settle it. If that fails, start over with a completely dry tube.

7. Once the $KClO_3$(*s*) is in the test tube, gently place the test tube back into the beaker and record the mass in Line 2 of Table 1. After massing the test tube with the reagent, remove the test tube and beaker from the balance. **Close the reagent jar and clean up the balance area after use.**

8. Set up the water displacement apparatus as shown in Figure 1. Fill a 500-mL Erlenmeyer flask with enough water that the fill line is just below the black stopper (~550 mL). Clamp the flask to a ring stand so that the bottom of the flask rests on the ring stand base or benchtop. Clamp the large ring attachment to another ring stand and place the 600-mL beaker into the ring so that it sits on your benchtop. Place the two-hole rubber stopper with clear tubing in the top of the 500-mL flask and place the free end of the clear tubing into the empty 600-mL beaker. Clamp the test tube to the same ring stand as the flask so it is tilted slightly away from the horizontal position and facing the lab snorkel. This ensures that the stopper will not shoot off the apparatus and injure someone. Also, ensure the test tube is positioned to avoid melting the stopper and rubber pieces on the test tube clamp when heat is applied.

9. Grasp the test tube stopper (the inlet hose) with one hand and the exit hose (where water flows out from) with the other hand. Turn on the air line (orange valve) at your lab station; check to ensure that only a gentle amount of air is flowing. Then, place your hand holding the stopper up to the nozzle of the air valve to start water flowing through the tubing; do this until there are no more air bubbles in the tubing connecting the Erlenmeyer flask to the 600-mL beaker. Immediately seal the system to stop air flow by placing your thumb over the test tube stopper and placing a finger over the end of the exit hose (where the water exits from).

10. Quickly place the rubber stopper with the inlet hose into the 30-mL test tube containing the reagent. Do not take your finger off the exit hose until the sample starts to bubble in step 12. If you must leave your workstation, fill a small 50-mL beaker halfway with tap water and place the exit hose into the water. This ensures the seal remains without air entering the tube.

11. **Have the apparatus checked by the instructor and OBTAIN APPROVAL prior to applying heat.**

12. Light the Bunsen burner and place **the exit hose** into the 600-mL beaker. Apply heat slowly to the test tube until the solid melts and bubbles, then apply steady heat. You should observe the water level in your Erlenmeyer flask decrease and the water level in your collection beaker increase. **DO NOT** allow the test tube to get red-hot. Ensure the entire sample is heated by moving the flame back and forth along the bottom of the test tube. **CAUTION: Do not burn yourself by grasping the barrel of the Bunsen burner – IT IS HOT! Hold the burner at its base plate.** Continue heating until the evolution of oxygen stops. Then, heat for one more minute but do not allow the water level in the Erlenmeyer flask to go below the bottom of the water inlet tube.

13. After the reaction is complete, allow the apparatus to cool. While the apparatus is cooling, measure both the temperature of the water in the beaker and the atmospheric pressure and record both values in Lines 10 and 11, respectively of Table 1.

14. Determine the volume of water displaced by carefully using a 50-mL graduated cylinder to measure the water collected in the beaker. You may have to empty and refill the graduated cylinder several times. Record the volume of water displaced in Line 7 of Table 1.

15. Remove the stopper from the test tube. Be careful: when you lose the seal, water will pour out of the outlet hose that was in the beaker. Using the test tube tongs, remove the test tube from the water displacement apparatus, weigh the test tube, and record the mass in Line 4 of Table 1. Place the test tube containing the heated sample in a 250-mL beaker to cool while continuing to Part II.

Procedure Part II

1. Take a cooled test tube and 100-mL beaker with you to the balance area. Place a weigh boat on the balance, tare the weigh boat, and remove the weigh boat from the balance. Place a small amount of **unknown sample** into the weigh boat, place the weigh boat back onto the balance, if the balance does not read a mass of **approximately** 1.000 g remove it from the balance and add or remove some of the mixture until the balance reads **approximately** 1.000 g. **Never add reagent to the weigh boat while on the scale to prevent spilling on the balance. Discard excess; do not put excess reagent back in the bottle.** Record the mass of the sample. Remove the weigh boat containing the unknown sample and transfer the sample to a clean, dry test tube.

2. Using the same water displacement apparatus as Part I, clamp the test tube containing the mixture to the ring stand. Repeat the procedures in steps 8-15 of Part I.

3. After all test tubes have cooled, scrape the dry residue from all test tubes to loosen it and dispose of residue in the **solid waste beaker** at the instructor bench. Next wash the tube thoroughly at the closest sink with a soapy test tube brush and water.

UNITED STATES MILITARY ACADEMY

LAB 9 – GAS LAWS AND STOICHIOMETRY

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: __________________________________

Cadet: __________________________________

Cadet: __________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:
SIGNATURE: DATE:
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Discussion

Table 1: Experimental Data Collection

	Line	Quantity	Part I	Part II (Unk)
Mass Difference Method	1	Mass of empty tube, g		
	2	Mass of sample and test tube before heating, g		
	3	Mass of the sample before heating, g (Line 2 - Line 1)		
	4	Mass of sample and test tube after heating, g		
	5	Mass of the sample after heating, g (Line 4- Line 1)		
	6	Mass of O_2 (g), g (Line 3 - Line 5)		
Water Displacement Method	7	Volume of water collected, mL		
	8	Water Temperature, K		
	9	Water Vapor Pressure, torr (ref. Figure 2)		
	10	Atmospheric Pressure, torr		
	11	Oxygen partial pressure, torr (ref. Equation 2)		
	12	Oxygen partial pressure, atm		

Part I Discussion

1. (5 pts) Calculate the theoretical yield of oxygen gas produced in grams using the mass of $KClO_3$(*s*) (Table 1, Line 3) used in Part I.

2. (5 pts) Mass Difference Method: Using the theoretical yield from question 1 as the accepted value and the mass of oxygen collected by the change in mass of the sample (Table 1, Line 6) as the experimental value, **calculate** the percent error and **explain** whether or not the "mass difference" calculation method provides accurate data.

3. Water Displacement Method:

   a. (5 pts) Calculate the actual yield of oxygen gas in grams produced using the value for the volume of water displaced during the reaction. *Hint: Use the ideal gas law and values that you recorded in Table 1 (Lines 7, 8, 12).*

   b. (5 pts) Using the actual yield of oxygen as the experimental value, **calculate** the percent error and explain whether the "water displacement" calculation method provides accurate data.

4. (5 pts) **Determine** which of the two calculation methods used was more accurate and **provide** a reason why the data collected by one method may be more accurate than the other.

Part II Discussion

1. For this question, you will use the water displacement method of determining the amount of oxygen gas produced to find the mass percent of $KClO_3$(*s*) in the unknown sample.

    a. (5 pts) Calculate the actual yield of oxygen gas in grams produced using the value for the volume of water displaced during the reaction.

    b. (5 pts) Calculate the mass (in grams) of $KClO_3$ (s) that must have been present in the mixture to produce the amount of oxygen calculated above.

    c. (5 pts) Given the mass of $KClO_3$(s) calculated above, determine the mass percent of $KClO_3$(s) in the unknown sample mixture.

d. (5 pts) Given the accepted value for the mass percent of $KClO_3(s)$ in the mixture as 72.11% and your experimental value calculated above, calculate the percent error and explain whether your data is accurate.

2. (5 pts) What is one military use of potassium chlorate or sodium chlorate? You may use your textbook or the internet to find a reputable source. **Cite your source(s) properly for this question, even if the source is your instructor or the textbook.** If you cannot find anything, ask your instructor for a hint.

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 9 - Instructor Solution

Cut Scale

Discussion

Table 1: Experimental Data Collection

	Line	Quantity	Part I	Part II (Unk)
Mass Difference Method	1	Mass of empty tube, g		
	2	Mass of sample and test tube before heating, g		
	3	Mass of the sample before heating, g (Line 2 - Line 1)		
	4	Mass of sample and test tube after heating, g		
	5	Mass of the sample after heating, g (Line 4- Line 1)		
	6	Mass of O_2 (g), g (Line 3 - Line 5)		
Water Displacement Method	7	Volume of water collected, mL		
	8	Water Temperature, K		
	9	Water Vapor Pressure, torr (ref. Figure 2)		
	10	Atmospheric Pressure, torr		
	11	Oxygen partial pressure, torr (ref. Equation 2)		
	12	Oxygen partial pressure, atm		

Part I Discussion

1. (5 pts) Calculate the theoretical yield of oxygen gas produced in grams using the mass of $KClO_3(s)$ (Table 1, Line 3) used in Part I.

+ 1 correct mm for $KClO_3$ +1 mol → g O_2

$$\begin{array}{l|l|l|l} 0.70\underline{1}\text{ g }KClO_3\ _{(T1,\ L3)} & 1\text{ mol }KClO_3 & 3\text{ mol }O_2 & 32.00\text{ g }O_2 \\ \hline & 122.548\text{ g }KClO_3 & 2\text{ mol }KClO_3 & 1\text{ mol }O_2 \end{array} = 0.27\underline{5}\text{ g }O_2$$

+ 1 g → mol $KClO_3$ + 1 molar ratio $KClO_3 : O_2$ + 1 M/U/S (2/3) /5

2. (5 pts) Mass Difference Method: Using the theoretical yield from question 1 as the accepted value and the mass of oxygen collected by the change in mass of the sample (Table 1, Line 6) as the experimental value, **calculate** the percent error and **explain** whether or not the "mass difference" calculation method provides accurate data.

+ 1 concept

$$\% \text{ error} = \left| \frac{\text{Experimental (T1, L6)} - \text{Accepted (from question 1)}}{\text{Accepted (from question 1)}} \right| \times 100\%$$

$$\% \text{ error} = \left| \frac{0.201 \text{ g} - 0.27\underline{5}}{0.27\underline{5}} \right| \times 100\% = 26.\underline{8}\%$$

+ 1 application + 1 M/U/S (2/3)

The mass difference method does **not** provide accurate data for gas evolution of oxygen gas because the % error of 26.8% is greater than 5%. + 2 conclusion (uses data)

3. Water Displacement Method:

   a. (5 pts) Calculate the actual yield of oxygen gas in grams produced using the value for the volume of water displaced during the reaction. *Hint: Use the ideal gas law and values that you recorded in Table 1 (Lines 7, 8, 12).*

$PV = nRT \rightarrow n = \frac{PV}{RT}$ + 1 concept

+1 Pressure (in atm) + 1 mol → g of O_2

$$\frac{205.\underline{4} \text{ mL } O_2}{} \left| \frac{\text{L}}{1000 \text{ mL}} \right| \frac{0.9570\underline{7} \text{ atm}}{294.05 \text{ K}} \left| \frac{\text{mol K}}{0.08206 \text{ L atm}} \right| \frac{32.00 \text{ g } O_2}{\text{mol } O_2} = 0.260\underline{7} \text{ g } O_2$$

+1 Temp (in K) + 1 M/U/S (2/3)

   b. (5 pts) Using the actual yield of oxygen as the experimental value, **calculate** the percent error and explain whether the "water displacement" calculation method provides accurate data.

+ 1 concept

$$\% \text{ error} = \left| \frac{\text{Experimental (T1, L6)} - \text{Accepted (from 1a)}}{\text{Accepted (from 1a)}} \right| \times 100\%$$

$$\% \text{ error} = \left| \frac{0.2607 \text{ g} - 0.27\underline{5}}{0.27\underline{5}} \right| \times 100\% = 5.052\%$$

+ 1 application + 1 M/U/S (2/3)

The water displacement method does **not** provide accurate data for gas evolution of oxygen gas because the % error of 5.052% is greater than 5%. + 2 conclusion (uses data)

/15

4. (5 pts) **Determine** which of the two calculation methods used was more accurate and **provide** a reason why the data collected by one method may be more accurate than the other.

The water displacement method is more accurate predictor of % yield because the error was only 5.052% compared to 26.8% for the mass difference method. Water displacement accounts for all of the oxygen gas collected, versus the mass difference in the pre-heating and post-heating mass does not capture if there is an incomplete reaction or any additional moisture in the test tube.

+ 1 FF water displacement more accurate
+ 2 uses both % errors as supporting evidence
+ 2 logical explanation why water displacement is better

Part II Discussion

1. For this question, you will use the water displacement method of determining the amount of oxygen gas produced to find the mass percent of $KClO_3(s)$ in the unknown sample.

   a. (5 pts) Calculate the actual yield of oxygen gas in grams produced using the value for the volume of water displaced during the reaction.

$PV = nRT \rightarrow n = \frac{PV}{RT}$ + 1 concept

+1 Pressure (in atm) + 1 mol → g of O_2

$$\frac{108.5\text{ mL }O_2}{} \left| \frac{\text{L}}{1000\text{ mL}} \right| \frac{0.97280\text{ atm}}{291.55\text{K}} \left| \frac{\text{mol}\cdot\text{K}}{0.08206\text{ L}\cdot\text{atm}} \right| \frac{32.00\text{ g }O_2}{\text{mol }O_2} = 0.234\underline{8}\text{ g }O_2$$

+1 Temp (in K) + 1 M/U/S (2/3)

   b. (5 pts) Calculate the mass (in grams) of $KClO_3$ (s) that must have been present in the mixture to produce the amount of oxygen calculated above.

+ 1 molar mass of $KClO_3$

$$\frac{0.234\underline{8}\text{ g }O_2}{} \left| \frac{1\text{ mol }O_2}{32.00\text{ g }O_2} \right| \frac{2\text{ mol }KClO_3}{3\text{ mol }O_2} \left| \frac{122.55\text{ g }KClO_3}{1\text{ mol }KClO_3} \right. = 0.599\underline{5}\text{ g }KClO_3$$

+ 1g to mol O_2 + 1 correct mm for O_2 + 1 molar ratio $KClO_3 : O_2$ + 1 M/U/S (2/3)

   c. (5 pts) Given the mass of $KClO_3(s)$ calculated above, determine the mass percent of $KClO_3(s)$ in the unknown sample mixture.

+ 2 concept

$$\frac{\text{Mass of Part}}{\text{Mass of Whole}} \times 100\% = \%\text{ mass}$$

$$\frac{0.71\underline{7}\text{ g }KClO_3}{0.95\underline{8}\text{ g }KClO_3} \times 100\% = 74.\underline{9}\ \%$$

+ 2 application + 1 M/U/S (2/3)

/20

d. (5 pts) Given the accepted value for the mass percent of $KClO_3(s)$ in the mixture as 72.11% and your experimental value calculated above, calculate the percent error and explain whether your data is accurate.

+ 1 concept

$$\% \text{ error} = \left| \frac{\text{Experimental}_{\text{(from 2c)}} - \text{Accepted}}{\text{Accepted}} \right| \times 100\%$$

$$\% \text{ error} = \left| \frac{74.9\% - 72.11\%}{72.11\%} \right| \times 100\% = 3.8\underline{7}\%$$

+1 application + 1 M/U/S (2/3)

Our data is accurate because the % error of 3.87% is less than 5%. + 2 conclusion (uses data)

2. (5 pts) What is one military use of potassium chlorate or sodium chlorate? You may use your textbook or the internet to find a reputable source. **Cite your source(s) properly for this question, even if the source is your instructor or the textbook.** If you cannot find anything, ask your instructor for a hint.

Answers may vary, but possible answers include: both can be used for emergency oxygen source in submarines, airplanes and tunneling/mining operations, and both can be used for homemade explosives as well

+ 3 Reasonable answer
+ 1 Proper in-line citation of reputable source
+ 1 Proper citation of reputable source in works cited

/10

Lab 10

Lab 10 - Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCE AND ENGINEERING
United States Military Academy
West Point, New York
CH151, AY25-2
(75 points)

COLLIGATIVE AND PHASE PROPERTIES

Experiment X

<u>Instructions:</u> Prior to the lab, read pages 1-8 to obtain information needed to complete this lab. **The material covered here is testable.** During the lab **only turn in pages 9-16.**

Materials

Reagents	DI Water	t-butanol (tert-butanol)	
	NaCl (s)	Ethylene glycol	
Equipment	(1) Plastic cup	(1) Forceps	(1) Test tube holder
	(1) 50-mL test tube	(2) 10-mL grad. cylinders	Vernier LabQuest Mini
	(1) Blue test tube stopper	(2) Small 3-prong clamps	Vernier Pressure Sensor
	(1) Waste centrifuge tube	(1) Med. test tube clamp	Vernier Small Thermistor
	(2) 50-mL beakers	(1) Med. test tube clamp	Vernier Large Thermistor
	(1) 600-mL beaker	(1) Hotplate w/ stir bar	
	(1) Glass stir rod	(2) Ring Stands	

Background

Phase changes occur around us every day, from water boiling in the kitchen to ice forming in the freezer. At their core, temperature and phase changes for any substance involve the gain or loss of heat. To illustrate this process, the heating curve for water serves as a relatable example. In the case of very cold ice, the solid must first gain heat before it can reach its melting point. Initially, added thermal energy steadily increases the temperature, or average kinetic energy, of the water molecules within the crystalline solid. The amount of heat gained over a range of temperatures can be calculated from Equation 1. In this equation, m is the mass of the ice, C_s is the specific heat capacity of the ice (remember that in different phases, molecules have different specific heat capacities), and ΔT is the change in temperature.

$$q = \mathrm{m} * C_s * \Delta T \qquad (1)$$

In the case of ice, water molecules in the crystalline lattice progressively vibrate faster as more heat is added until the energy gain is sufficient to overcome the intermolecular forces holding the molecules in place. During the ensuing transition, melted portions of liquid water contain enough kinetic energy for the molecules to tumble past one another, but not enough for molecules to completely break free of the surrounding intermolecular forces. As heat transfer continues during the phase change, all added energy is entirely utilized to overcome intermolecular forces associated with the departing phase. Consequently, the temperature remains constant throughout the phase change of a pure substance even as more energy is added. The amount of energy required for the entire amount of a pure substance to undergo a phase change can be determined via Equation 2. In this equation, n is the number of moles of ice and ΔH_{fus} is the heat of fusion for water, which is the amount of energy that it takes to change between solid and liquid.

$$q = n * \Delta H_{fus} \tag{2}$$

In contrast to its pure substance, adding solute particles to a solvent will change certain physical properties of the resulting solution. Specifically, colligative properties in solutions emerge from the number of particles, or solute, dissolved within a solvent. It is important to note that **only the total number of solute particles added** to the solvent is critical to calculating the effect of the mixture – the identity of the particles is not important. Increasing the concentration of solute has the effect of lowering the solution vapor pressure when compared against the original pure solvent. This occurrence is largely a function of **entropy**, or nature's tendency toward mixing. While more advanced chemistry classes will cover entropy in greater detail, it is useful to establish a preliminary understanding of this interrelated thermodynamic property.

Consider a solution which contains a solvent and a nonvolatile solute. Since vapor pressure equates to a dynamic process in which molecules escape from a liquid to enter the gaseous state, then the same process of vaporization from a solution would leave a more ordered system (the most order occurring with separated, pure substances). However, increasing the order of a system is generally energetically unfavorable and takes energy to accomplish. Therefore, nature tends toward mixing substances into a more disordered state, making the solvent particles in the example solution increasingly less likely to transition from the liquid phase into the vapor phase. We can physically observe this phenomenon by measuring the vapor pressure of solutions as solute concentrations increase. Alternatively, we can also use Raoult's Law (Equation 3) to calculate the solution vapor pressure since it is directly proportional to the mole fraction of solvent to the total solution. In Equation 3, $X_{solvent}$ is the mol fraction (as a percentage) of the solvent and $P^{\circ}_{solvent}$ is the vapor pressure of the solvent. For example, if the mole fraction of a solvent is 95%, the solution vapor pressure should be 95% of the pure solvent.

$$P_{solution} = X_{solvent} * P^{\circ}_{solvent} \tag{3}$$

Since boiling occurs when the vapor pressure of a liquid equals the external pressure, the reduced vapor pressure of a solution results in a lower vaporization curve (Figure 1). Likewise, the fusion curve must also shift to join with the vaporization curve at a new triple point. When pressure remains constant (like atmospheric pressure in a lab), we can now see that the solution has both a lower freezing point and a higher boiling point when compared to the original, pure solvent.

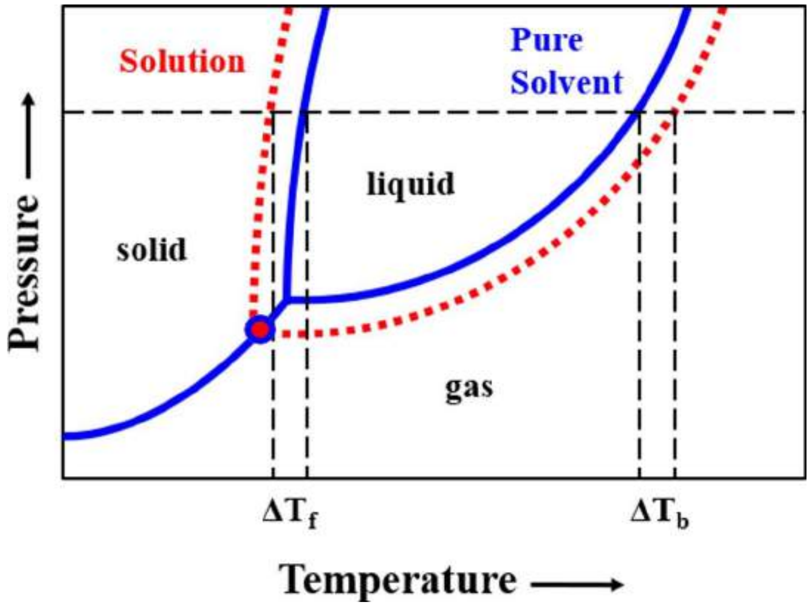


Figure 1: Phase diagram of colligative properties resulting from vapor pressure lowering.

As with the effect of vapor pressure lowering, we could empirically determine numerous data points for adjusted phase change temperatures over a range of pressures. Fortunately, chemists have charted this data for numerous substances, providing the scientific community with colligative property constants (K_f and K_b) that are specific to many common solvents.

To determine the colligative property changes of a solution, we use the equations listed below to calculate the difference in freezing and boiling points, ΔT_f and ΔT_b, respectively. Note that these calculations incorporate the solution concentration in terms of **molality** (m), or moles of solute per kilogram of solvent.

$$\Delta T_f = m * K_f \tag{4}$$

$$\Delta T_b = m * K_b \tag{5}$$

In this lab we explore the addition of both a nonelectrolyte and an electrolyte as solutes. As stated above, the amount of solute (number of particles) is what causes vapor pressure lowering. In the case of nonelectrolytes, each molecule provides one dissolved particle. However, if we instead use salts to explore colligative properties of aqueous solutions, we need to account for the difference between moles of dissociated ions and initial number of solid particles. To account for strong electrolytes, colligative property calculations (shown below) incorporate a van't Hoff Factor (i), which multiplies the change in freezing/boiling point by the ratio of moles of particles in solution to moles of formula units dissolved. For example, a solution with sodium chloride multiplies the freezing/boiling point change by a factor of two, since two moles of separate ions (Na^+ and Cl^-) result from the dissociation of one mole of the NaCl formula unit. This expected van't Hoff factor for a salt occurs as solutions approach infinite dilution. Deviation from expected values occur as the concentration of salt solution increases since there is more opportunity for solute-solute interactions (cation-anion), causing the actual value to fall below the expected van't Hoff factor. For this lab **assume complete dissociation**, resulting in an expected van't Hoff factor.

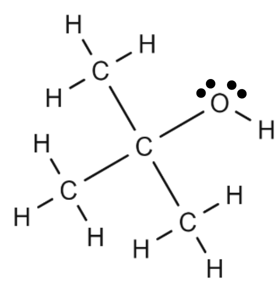
$$\Delta T_b = i * m * K_b \tag{6}$$

$$\Delta T_f = i * m * K_f \tag{7}$$

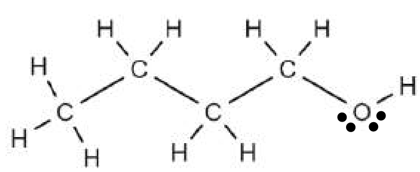
Many people employ colligative properties every day, often without even realizing why. Cities and their residents use salt in the winter to coat highways and driveways alike. Some of the more cost-effective salts possess the smallest van't Hoff factor, like NaCl, and yet prove more effective at preventing ice buildup than more expensive alternatives. Not only is sodium chloride relatively easy to obtain, but its relatively small molar mass provides more dissociated particles (ions) in solution than a much larger salt of the same bag weight. Since colligative properties principally depend on the number of particles in solution, the higher concentration of ions delivers a larger freezing point depression for when winter conditions arrive.

Introduction

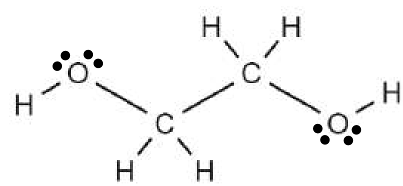
In this lab, you will measure the effect of adding ethylene glycol to t-butanol in order to lower the freezing point. Since ethylene glycol is a nonelectrolyte the van't Hoff factor will be 1. Additionally, you will add sodium chloride to DI water in order to raise the boiling point. In this case, you may assume that sodium chloride completely dissociates into sodium and chlorine ions, creating a van't Hoff factor of 2. The constants in Table 1 of the Discussion section are for your reference as you solve for how much solute you need to add in order to affect the desired temperature change.



tert-butanol



1-butanol



ethylene glycol

Figure 2. Lewis Structures for tert-butanol (left), 1-butanol (center) and ethylene glycol (right)

As a reminder, the K_b and K_f values used in Equations 6 and 7 respectively are for the **solvent**. Additionally, remember that the definition of molality is the number of moles of **solute** divided by the kilograms of **solvent.**

Procedure Part I: Boiling Point of Water

1. 1.

   Open the Vernier Graphical Analysis program on your computer. Connect the Vernier Labquest Mini unit to your computer using the provided USB cable. Insert the connection for the Vernier Thermistor in Channel 1. Insert the connection cable for the Vernier Pressure Sensor in Channel 2. The screen will automatically update to display the temperature and pressure in the bottom right. If the units for temperature and pressure are not in °C and torr, select the readout that needs to be changed and change to the appropriate units in the pop-up menu. **Ensure you set the duration of the experiment to 600 seconds**. Select "Done" when complete.

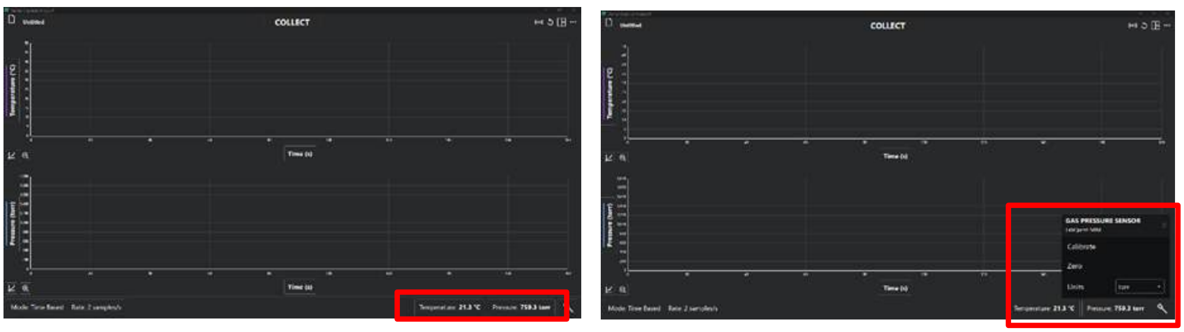


Figure 3: How to change units in Vernier Graphical Analysis

2. Using a 50 mL beaker, obtain 50 mL of hot **DI water** from the hot water bath. Using your 10 mL graduated cylinder, add 17.5 mL of the hot DI water to another 50 mL beaker along with a magnetic stir bar. **This water must be DI water and cannot come from the sinks.** Place the beaker on top of the heating plate and set up the rest of the apparatus as shown in Figure 4 below.

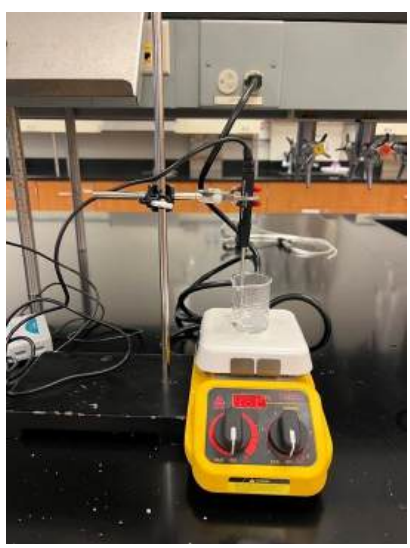


Figure 4: Water boiling point apparatus

3. Once the device is established, lower your thermistor into the water ensuring that you do not contact the edges or bottom of the beaker. Record the temperature and atmospheric pressure in **Table 2, Lines 1 and 2 in Section 1, Lab Data (pg 10).**

4. Turn the magnetic stir bar on to a medium level and double check that the thermistor is not in the way of the stir bar. **After checking to ensure that there are no cables near the hotplate surface, turn the hotplate on to 300 °C.**
5. Once you reach a temperature on 80 °C, press collect on Vernier Graphical Analysis.
6. Once the water comes to a rolling boil, record the temperature of the boiling point in **Table 2, Line 3 in Section 1, Lab Data (pg 10).** Turn off the heating plate and remove the 50 mL beaker using forceps. **Do not touch the hot beaker.**
7. Fill the provided plastic cup with ice. Pour a couple of ice cubes into the 50 mL beaker of hot water to rapidly cool it. After a minute, remove the magnetic stir bar with beaker tongs from the 50 mL beaker and dry it off.
8. (8 pts) While your water sample is cooling, calculate the mass of sodium chloride that will need to be added to 17.5 mL of water in order to increase the boiling point by 3.0 °C. Remember, sodium chloride is a strong electrolyte, and you can assume complete dissociation. **Complete this calculation in Section 2, Laboratory Calculations (pg 11). Check with your instructor before moving on.**
9. Locate the sodium chloride at one of the analytical balances. Place a weigh boat on the balance and tare the balance. Using the scoops provided, measure out the amount of sodium chloride calculated above. Once you have the desired amount of sodium chloride, carry it back to your lab bench. **Record the amount of sodium chloride acquired in Table 2, Line 4 in Section 1, Lab Data (pg 10).**
10. In a new 50 mL beaker, use the 10 mL graduated cylinder to add 17.5 mL of fresh hot of DI water to the beaker. After you have added the water, add your sodium chloride as well. Ensure that none of the sodium chloride is stuck to the sides.
11. Place the beaker on the hotplate and turn on the magnetic stir bar to medium. **Do not turn on the heat until all of the salt has dissolved.** Lower the thermometer into the water at this time.
12. Once the salt has dissolved completely, turn the hotplate on to 300 °C. As in step 3, ensure that there are no cables near the heating surface. Once the temperature reaches 80 °C, click collect on the Vernier Graphical Analysis program.
13. Observe the solution carefully. As soon as the solution begins to boil, record the temperature. This data must be collected as soon as the water begins to boil as there will be a slight increase in temperature during the boiling (it won't be a perfect plateau). **Record the boiling point of the DI water/NaCl solution in Table 2, Line 5 in Section 1, Lab Data (pg 10).**

14. Remove the hot water using forceps as detailed earlier in the procedure. Turn off the hotplate

and allow time for it to cool down before storing it.

15. Dispose of the water-sodium chloride solution (salt water) in the sink drain at your lab station.

Procedure Part II: Freezing Point of t-butanol

For this part of the lab, you will be using t-butanol. **Keep the t-butanol capped when not in use.**

1. Secure a prefilled centrifuge tube from the fume hood at the back of the lab (blue lid). The t-butanol may be either solid or liquid depending on ambient conditions in the lab. It should be at least half full.
2. Fill up a 600 mL beaker with hot water from the sink at your lab station (not from the hot water bath).
3. Place the t-butanol in the 600 mL beaker of warm water. While the t-butanol is warming, set up the freezing point apparatus as shown in Figure 5 below using a 50-mL test tube with rubber stopper, a large thermistor, and a plastic cup filled with ice and water.

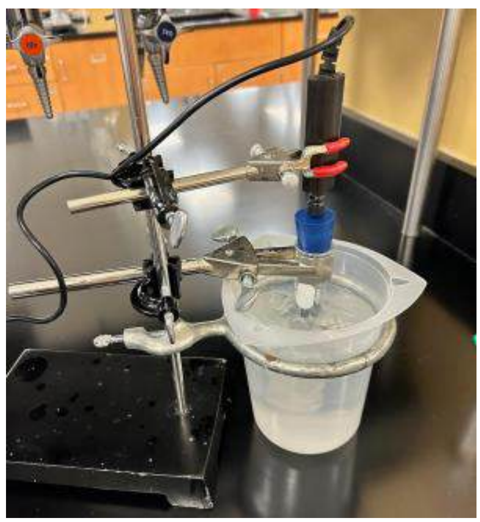


Figure 5: T-butanol freezing point apparatus

4. Remove the centrifuge tube with the t-butanol from the warming beaker (it should be liquid by now) along with an empty 250 mL beaker and bring them to an analytical balance. Place the centrifuge tube with the t-butanol inside of it in the 250 mL beaker, then place the beaker on the balance and record the mass. **Record the mass of the full centrifuge tube and the beaker in Table 3, Line 1.**
5. Pour the t-butanol into the 50 mL test tube. With your empty centrifuge tube and 250 mL beaker, return to the analytical balance. Place the empty centrifuge tube in the 250 mL beaker, then place the beaker on the balance and **record the mass in Table 3, Line 2.**

6. Calculate the mass of t-butanol in the test tube and **record it in Table 3, Line 3.**
7. Secure the test tube in a test tube clamp on a ring stand as shown in Figure 5 above.
8. Insert the **large** thermistor probe through the small hole in the rubber stopper until the tip is submerged midway in the t-butanol sample.
9. Use the small thermistor to confirm that your ice bath is at or below 18 °C. Lower your test tube with the t-butanol sample into the cold-water bath and record the temperature when the sample freezes. *Note: your freezing point occurs when the sample begins to freeze. Remember, during the freezing phase transition, your temperature should remain constant. This should confirm your freezing point.* **Record your value in Table 3, Line 4.**
10. Lift the t-butanol out of the ice bath and place the test tube in the warm water by replacing the ice water beaker with the warm water beaker.
11. (8 pts) While the t-butanol is melting, calculate the volume of ethylene glycol that will need to be added to lower the freezing point by 5.0 °C. You should be able to solve for a mass of ethylene glycol and then convert that to volume using density. **Complete this calculation in Discussion Section 2: Laboratory Calculations. Check with your instructor before moving on.**
12. Once your instructor has approved your value, obtain that amount from an ethylene glycol dispenser from the fume hoods in the back using a 10 mL graduated cylinder. Remove the rubber stopper and large thermistor probe and add the ethylene glycol to the test tube with the t-butanol. Make sure to stir the mixture well using a glass stir rod. **Record the amount added in Table 3, Line 5.**
13. Return rubber stopper and large thermistor probe and place the test tube back in the ice bath. If you have run out of ice, secure some more. Press collect and observe the temperature change.
14. When the temperature begins to level off, record the new freezing point in **Table 3, Line 6.**
15. Raise the solution from the ice bath. Pour the t-butanol and ethylene glycol mixture into the centrifuge tube labeled **"WASTE"** (orange lid) and use a funnel to pour the solution into the waste containers under the fume hood, returning the "WASTE" centrifuge tube back to your lab station. Note: you may need to warm the t-butanol/ethylene glycol sample in hot water again in order to dispose of it into the waste centrifuge tube. Do **NOT** put the solution back into the original centrifuge tube.
16. Return the empty blue-capped centrifuge tube that held the pure t-butanol to the container under the fume hood. As materials cool down, you can return them and proceed to clean up.

UNITED STATES MILITARY ACADEMY

DEPARTMENT OF CHEMISTRY AND LIFE SCIENCES

LAB 10 – COLLIGATIVE PROPERTIES

CH151: ADVANCED GENERAL CHEMISTRY I

SECTION: ______________________

INSTRUCTOR: ______________________

By

Cadet: ___________________________________

Cadet: ___________________________________

Cadet: ___________________________________

West Point, New York

Date: _______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

_________WE CERTIFY THAT USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: DATE:
SIGNATURE: DATE:
SIGNATURE: DATE:

Discussion

Table 1. Useful constants for calculations required for this lab

Chemical	Molecular Weight (g/mol)	Density (g/cm^3)	ΔH_{fus} (kJ/mol)	ΔH_{vap} (kJ/mol)	$C_{S,\ solid}$ (J/(g*K))	$C_{S,\ liquid}$ (J/(g*K))	$C_{S,\ gas}$ (J/(g*K))	K_f (°C/*m*)	K_b (°C/*m*)	*VP* (torr)
water	18.016	0.997	6.02	40.7	2.09	4.184	2.01	1.858	0.512	23.8
t-butanol	74.122	0.7886	6.7	39.07	1.971	2.906		8.19	1.99	40.7
ethylene glycol	62.068	1.113	9.958	65.0		2.426	1.607	3.53	1.76	0.092

Section 1. Lab Data

Table 2. Lab Data from Procedure Part 1

	Description	Value
1.	Initial Temperature of Water	
2.	Atmospheric Pressure	
3	Boiling Point of pure DI Water	
4.	Amount of sodium chloride added to DI Water	
5.	Boiling Point of DI Water/NaCl solution	

Table 3. Lab Data from Procedure Part 2

	Description	Value
1.	Mass of full centrifuge tube and 250 mL beaker	
2.	Mass of empty centrifuge tube and 250 mL beaker	
3.	Mass of t-butanol	
4.	Freezing Point of pure t-butanol (assume at 760 torr)	
5.	Amount of ethylene glycol added to t-butanol	
6.	Freezing Point of t-butanol/ethylene glycol solution	
7.	Given Boiling Point of t-butanol at 760 torr	**82.3 °C**

Section 2. Laboratory Calculations

1. (5 pts) While your water sample is cooling, calculate the mass of sodium chloride that will need to be added to 17.5 mL of water in order to increase the boiling point by 3.0 °C. Remember, sodium chloride is a strong electrolyte, and you can assume complete dissociation. Show your work and write your answer on the answer line below. **Check with your instructor before moving on.**

Amount of sodium chloride needed (g): ____________

2. (5 pts) While the t-butanol is melting, calculate the volume of ethylene glycol required to lower the freezing point by 5.0 °C. You should be able to solve for a mass of ethylene glycol and then convert that to volume using density. Show your work and write your answer on the answer line below. **Check with your instructor before moving on.**

Amount of ethylene glycol (mL): _____________

Section 3. Discussion

1. This problem draws directly from your data collected during the lab. Assume that you ran two additional trials of both boiling point elevation and freezing point depression.

   (a) (4 pts) Calculate the accuracy and precision of your group's boiling point elevation if the boiling point temperature values of the water-NaCl solution from your additional trials were 104.132 °C and 104.975 °C and the accepted T_b value is 103.1 °C. Use these results to quantifiably explain whether your data is accurate and/or precise. **Show your work.**

   (b) (4 pts) Calculate the accuracy and precision of your group's freezing point depression if the freezing point temperatures of the t-butanol-ethylene glycol solution for your additional trials were 22.864 °C and 19.152 °C and the accepted solution T_f value was 20.4°C. Use these results to quantifiably explain whether your data is accurate and/or precise. **You do NOT need to show your work.**

2. This problem concerns the heat required to create a phase change.

   (a) (3 pts) **Draw a heating/cooling curve** for the process of heating your water sample from the starting temperature (Table 2, line 1) to 120.0 °C. Ensure you **label** the following: **axes (with units), phase(s), and phase change(s).**

   (b) (4 pts) Calculate the amount of heat (q) in Joules required to heat your water sample from the starting temperature (Table 2, line 1) to 120.0 °C. All required thermodynamic values are located in Table 1. *Hint: each segment of your curve above requires a separate heat calculation.*

   (c) (2 pts) Consider your heating curve and its associated calculations above. If you reversed the process and cooled the sample from 120 °C to your starting temperature, what is the value of q (heat absorbed/lost) required to bring the water back down to your original starting temperature? Is this process endothermic or exothermic?

3. (4 pts) For your final solution of water and sodium chloride, calculate the solution vapor pressure, in torr, under atmospheric conditions using Raoult's Law.

4. (4 pts) In this experiment you may have noticed that the temperature of pure water remained constant while boiling (aside from minor equipment fluctuations), whereas you may have noticed that the temperature of the solution with sodium chloride did gradually change. **Explain** why we might expect the solution temperature to vary during a phase change. *Hint – look at the equation used to calculate the change in freezing point/boiling point.*

5. (2 pts) **Explain** why we had to change the water between the initial boiling point measurement of pure water (Part 1, Step 3) and the salt solution (Part 1, Step 11). **Predict** what would have happened if you did not change the water between those steps.

6. (9 pts) Using the information from the pre-lab, Table 2, Table 3, and the lab demo video, complete the phase diagram for pure t-butanol by **drawing and labeling** the three points (freezing point at 760 torr, boiling point at 760 torr, and triple point), three phase transition curves, and three phase regions.

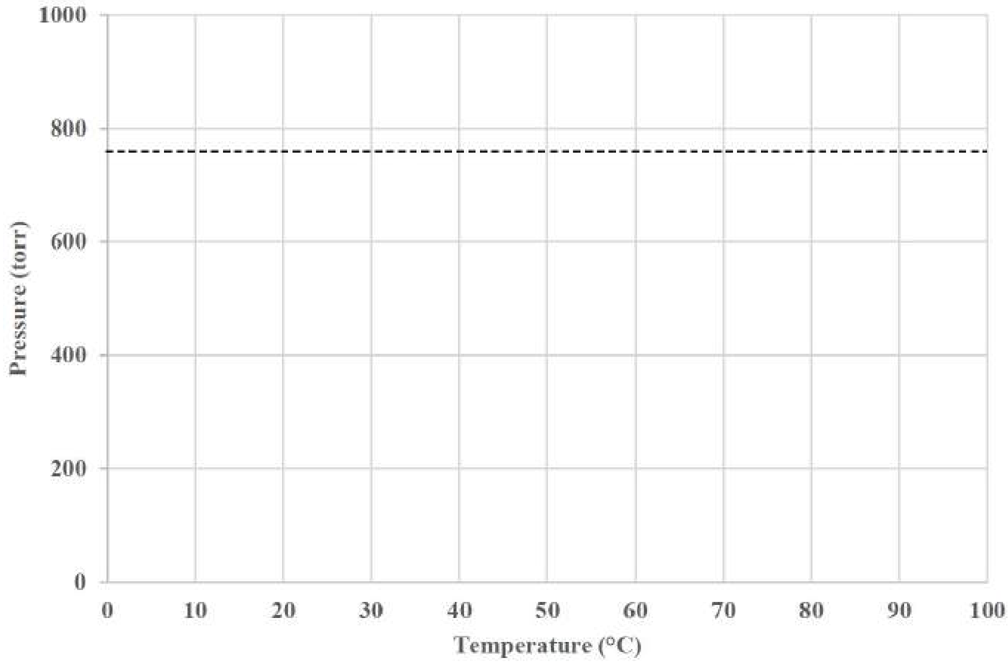


Figure 6: Phase diagram for pure t-butanol

7. (4 pts) Suppose you needed to boil a sample of t-butanol as part of an experiment where the temperature cannot increase past 50°C. Use the phase diagram constructed above to explain how you could boil the sample of t-butanol?

REFERENCES:

Cadet Works Cited

Cadet Notes

Lab 10 - Instructor Solution

Cut Scale

UNITED STATES MILITARY ACADEMY

DEPARTMENT OF CHEMISTRY AND LIFE SCIENCES

LAB 8 – COLLIGATIVE PROPERTIES

CH101: GENERAL CHEMISTRY I

SECTION: ____________

INSTRUCTOR: _____________________

By

CADET: _________________________________

CADET: _________________________________

CADET: _________________________________

WEST POINT, NEW YORK

Date: ______________________

_________WE CERTIFY THAT WE HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT WE USED TO COMPLETE THIS ASSIGNMENT AND THAT WE ACKNOWLEDGED ALL ASSISTANCE WE RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

_________WE CERTIFY THAT WE DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

SIGNATURES: ____

____

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Discussion

Table 1. Useful constants for calculations required for this lab

Chemical	Molecular Weight (g/mol)	Density (g/cm^3)	ΔH_{fus} (kJ/mol)	ΔH_{vap} (kJ/mol)	$C_{S,\ solid}$ (J/(g*K))	$C_{S,\ liquid}$ (J/(g*K))	$C_{S,\ gas}$ (J/(g*K))	K_f (°C/m)	K_b (°C/m)	VP (torr)
water	18.016	0.997	6.02	40.7	2.09	4.184	2.01	1.858	0.512	23.8
t-butanol	74.122	0.7886	6.7	39.07	1.971	2.906		8.19	1.99	40.7
ethylene glycol	62.068	1.113	9.958	65.0		2.426	1.607	3.53	1.76	0.092

Section 1. Lab Data

Table 2. Lab Data from Procedure Part 1

	Description	Value
1.	Initial Temperature of Water	**60.000 °C**
2.	Atmospheric Pressure	**758.674 torr**
3	Boiling Point of pure DI Water	**100.318 °C**
4.	Amount of sodium chloride added to DI Water	**2.998 g**
5.	Boiling Point of DI Water/NaCl solution	**103.116 °C**

Table 3. Lab Data from Procedure Part 2

	Description	Value
1.	Mass of full centrifuge tube and 250 mL beaker	**43.831 g**
2.	Mass of empty centrifuge tube and 250 mL beaker	**20.173 g**
3.	Mass of t-butanol	**23.658 g**
4.	Freezing Point of pure t-butanol (assume at 760 torr)	**25.418 °C**
5.	Amount of ethylene glycol added to t-butanol	**0.76 mL**
6.	Freezing Point of t-butanol/ethylene glycol solution	**21.894 °C**
7.	Given Boiling Point of t-butanol at 760 torr	**82.3 °C**

Section 2. Laboratory Calculations

1. (5 pts) While your water sample is cooling, calculate the mass of sodium chloride that will need to be added to 17.5 mL of water in order to increase the boiling point by 3.0 °C. Remember, sodium chloride is a strong electrolyte, and you can assume complete dissociation. Show your work and write your answer on the answer line below. **Check with your instructor before moving on.**

$T_b = m \times K_b \times i \quad \rightarrow \quad T_b = \frac{mol\ NaCl}{kg\ Water} \times K_b \times i$ +1 concept

+1 application

$$mol\ NaCl = \frac{T_b \times kg\ water}{K_b \times i} = \frac{3.0\ °C \times 0.0174\ kg \times mol}{0.512\ °C{\cdot}kg \times 2} = 0.05\underline{0}98\ mol\ NaCl$$

+1 correct K_b +1 correct i

$mass\ NaCl = mol\ NaCl \times molar\ mass\ NaCl = 0.05\mathbf{0}98\ mol \times 58.44 \frac{g}{mol} = 2.\mathbf{98} g\ NaCl$ +1 M/U/S (2/3)

$$^{*}kg\ water = \frac{17.5\ mL \times 0.997\ g \times 1\ kg}{mL \times 1000\ g} = 0.0174\ kg$$

Amount of sodium chloride needed (g): ____________

2. (5 pts) While the t-butanol is melting, calculate the volume of ethylene glycol required to lower the freezing point by 5.0 °C. You should be able to solve for a mass of ethylene glycol and then convert that to volume using density. Show your work and write your answer on the answer line below. **Check with your instructor before moving on.**

$T_f = m \times K_f \times i \quad \rightarrow \quad T_f = \frac{mol\ EG}{kg\ t{-}butanol} \times K_f \times i$ +1 concept

$$mol\ EG = \frac{T_f \times kg\ solvent}{K_f \times i} = \frac{5.0\ °C \times 0.0237\ kg \times mol}{8.19\ °C{\cdot}kg \times 1} = 0.01\underline{4}47\ mol\ EG$$

+1 correct K_f +1 correct i

$$volume\ EG = \frac{mol\ EG \times m.m\ EG}{density\ EG} = \frac{0.01447\ mol\ EG \times 62.068\ g\ EG \times mL\ EG}{mol\ EG \times 1.113\ g\ EG} = 0.8\underline{0}69\ mL\ EG$$

+1 g → mL EG +1 M/U/S (2/3)

Amount of ethylene glycol (mL): ____________

/10

Section 3. Discussion

1. This problem draws directly from your data collected during the lab. Assume that you ran two additional trials of both boiling point elevation and freezing point depression.

   (a) (4 pts) Calculate the accuracy and precision of your group's boiling point elevation if the boiling point temperature values of the water-NaCl solution from your additional trials were 104.132 °C and 104.975 °C and the accepted T_b value is 103.1 °C. Use these results to quantifiably explain whether your data is accurate and/or precise. **Show your work.**

$$\%\ error = \left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right| \times 100\%$$ +1 % error calc.

$$\left|\frac{104.074-103.1}{103.1}\right| \times 100\% = 0.9359\%$$ use average to only calculate %error once

$$SD = \sqrt{\frac{(104.975-104.074)^2+(104.132-104.074)^2+(103.116-104.074)^2}{3-1}} = 0.90845\ ℃$$

$$CV = \frac{s}{\bar{x}} = \frac{0.90845\ ℃}{104.074\ ℃} * 100 = 0.8729\%$$ +1 CV calc

Since percent error and CV are below 5 %, the measurements are accurate and precise.

+1 correctly identifies accuracy
+1 correctly identifies precision

   (b) (4 pts) Calculate the accuracy and precision of your group's freezing point depression if the freezing point temperatures of the t-butanol-ethylene glycol solution for your additional trials were 22.864 °C and 19.152 °C and the accepted solution T_f value was 20.4°C. Use these results to quantifiably explain whether your data is accurate and/or precise. **You do NOT need to show your work.**

$$\%\ error = \left|\frac{experimental\ value - accepted\ value}{accepted\ value}\right| \times 100\%$$ +1 % error calc.

$$\left|\frac{21.303-20.4}{20.4}\right| \times 100\% = 4.428\%$$ use average to only calculate %error once

$$SD = \sqrt{\frac{(21.894-21.303)^2+(22.864-21.303)^2+(19.152-21.303)^2}{3-1}} = 1.9252\ ℃$$

$$CV = \frac{s}{\bar{x}} = \frac{1.9252\ ℃}{21.303\ ℃} * 100 = 9.037\%$$ +1 CV calc

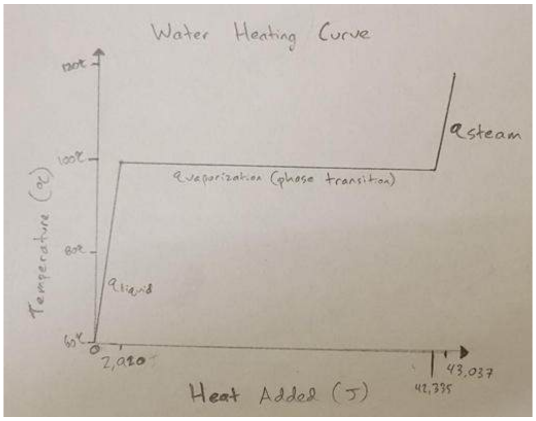
Since percent error is below 5 %, the measurements are accurate. Because the CV is above 5 %, the measurements are not precise.

+1 correctly identifies accuracy
+1 correctly identifies precision

/8

2. This problem concerns the heat required to create a phase change.

(a) (3 pts) **Draw a heating/cooling curve** for the process of heating your water sample from the starting temperature (Table 2, line 1) to 120.0 °C. Ensure you **label** the following: **axes (with units), phase(s), and phase change(s).**



+1 shape (2 flats and 3 slopes) correct

+1 correctly labeled axis

+1 pt phase transition labeled

(b) (4 pts) Calculate the amount of heat (q) in Joules required to heat your water sample from the starting temperature (Table 2, line 1) to 120.0 °C. All required thermodynamic values are located in Table 1. *Hint: each segment of your curve above requires a separate heat calculation.*

+1 concept (q_{liquid})

q_{liquid} =

17.5. mL H_2O	1 cm^3	0.997 g	4.184 J	(100. – 60.000) °C	1 K
	1 mL	1 cm^3	g °K		1 °C

= 2,920.0 J

+1 concept (q_{vap})

q_{vap} =

17.5. mL H_2O	1 cm^3	0.997 g	1 mol	40.7 kJ	1000 J
	1 mL	1 cm^3	18.016 g	mol	1 kJ

= 39,415.70 J

+1 concept (q_{gas})

q_{steam} =

17.5. mL H_2O	1 cm^3	0.997 g	2.01 J	(120.0-100.0) °C	1 K
	1 mL	1 cm^3	g °K		1 °C

= 701.39 J

$q_{total} = q_{liquid} + q_{vap} + q_{gas}$

+1 added 3 heats together

q_{total} = 2,920.0 J + 39,415.70 J + 701.39 J = 43,037.1 J

(c) (2 pts) Consider your heating curve and its associated calculations above. If you reversed the process and cooled the sample from 120 °C to your starting temperature, what is the value of q (heat absorbed/lost) required to bring the water back down to your original starting temperature? Is this process endothermic or exothermic?

-43,037.1 J, it would be an exothermic process

+1 same value but negative
+1 AON exothermic

/9

3. (4 pts) For your final solution of water and sodium chloride, calculate the solution vapor pressure, in torr, under atmospheric conditions using Raoult's Law.

Raoult's Law: $P_{solution} = (X_{solvent})(P°_{solvent})$ +2 pts concept

$$\text{mol } H_2O = \frac{17.5 \text{ mL } H_2O}{} \left| \frac{1 \text{ cm}^3 H_2O}{1 \text{ mL } H_2O} \right| \frac{0.997 \text{ g } H_2O}{1 \text{ cm}^3 H_2O} \left| \frac{\text{mol } H_2O}{18.016 \text{ g } H_2O} \right. = 0.96\underline{8}4 \text{ mol } H_2O$$

$$\text{mol NaCl} = \frac{2.998 \text{ g NaCl}}{} \left| \frac{\text{mol NaCl}}{58.44 \text{ g NaCl}} \right. = 0.0513\underline{0} \text{ mol NaCl}$$

+1 pts correct mol ratio

$$X_{solvent} = \frac{\text{moles water}}{\text{moles of water} + i \bullet \text{mol NaCl}} = \frac{0.968 \text{ mol}}{0.968 \text{ mol} + 2(0.05130 \text{ mol})} = 0.904$$

$$P_{solution} = (0.904) \times 23.8\, torr = 21.\mathbf{5}2\, torr$$

+1 pts correct $P°$

4. (4 pts) In this experiment you may have noticed that the temperature of pure water remained constant while boiling (aside from minor equipment fluctuations), whereas you may have noticed that the temperature of the solution with sodium chloride did gradually change. **Explain** why we might expect the solution temperature to vary during a phase change. *Hint – look at the equation used to calculate the change in freezing point/boiling point.*

As you boil the solution, water gets converted to steam and leaves the solution. This changes the molality by reducing the mass of the solvent. As a result, the temperature increase continues to increase.

+2 molality
+2 pt explanation

5. (2 pts) **Explain** why we had to change the water between the initial boiling point measurement of pure water (Part 1, Step 3) and the salt solution (Part 1, Step 11). **Predict** what would have happened if you did not change the water between those steps.

Because we raised the water to a boil, we would have lost some of the mass of the water. This would have affected the molality similarly to problem 5.

+2 AON correct explanation

/10

6. (9 pts) Using the information from the pre-lab, Table 2, Table 3, and the lab demo video, complete the phase diagram for pure t-butanol by **drawing and labeling** the three points (freezing point at 760 torr, boiling point at 760 torr, and triple point), three phase transition curves, and three phase regions.

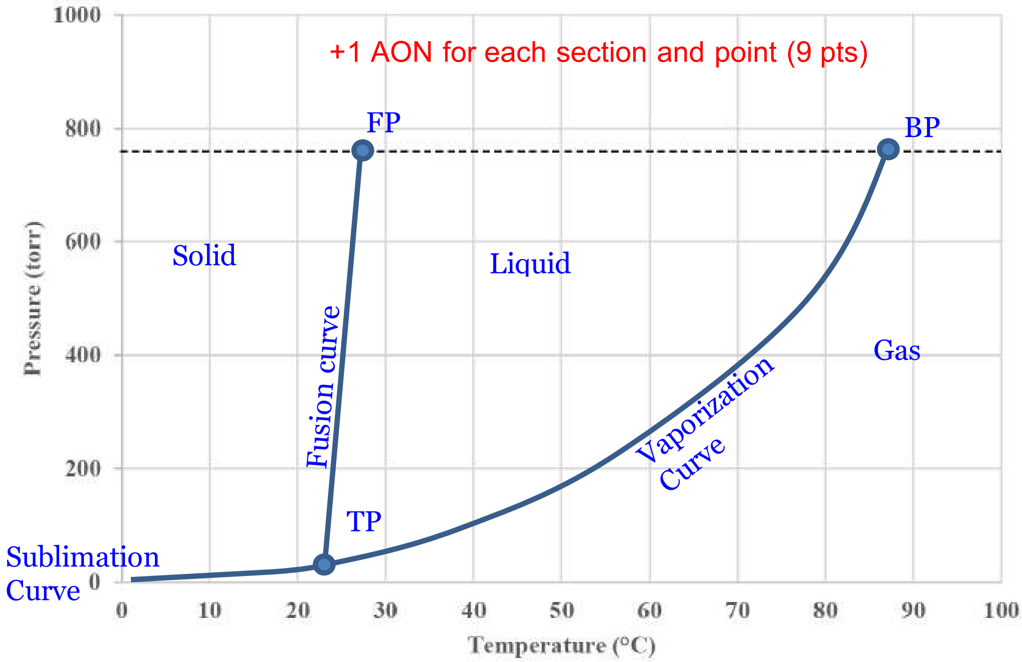


Figure 6: Phase diagram for pure t-butanol

7. (4 pts) Suppose you needed to boil a sample of t-butanol as part of an experiment where the temperature cannot increase past 50°C. Use the phase diagram constructed above to explain how you could boil the sample of t-butanol?

At a lower temperature than the accepted boiling point of ~82.3°C (or below the observed experimental boiling point), a researcher could apply a vacuum to the sample and lower the pressure until the sample begins to boil at the temperature of ≤ 50°C.

+4 AON

/13